

DERIVATIVES PRICING

A COMPUTATIONAL & THEORETICAL FRAMEWORK

©:Zac Kienzle[†]

[†] University of Queensland, Brisbane, Australia

Contents

I	Mathematical Foundations	9
1	Measure Theory	10
2	Probability	18
3	Convergence	29
4	Conditioning	35
5	Martingales	42
6	Wiener and Brownian Motion Processes	51
7	Itô Calculus	57
7.1	Itô Integral	57
7.2	Itô Calculus	62
7.3	Stochastic Differential Equations	64
7.4	Lévy Processes	68
II	Jump Processes	71
8	Levy Processes	72
9	Integro-differential equations and numerical methods	73
III	Financial Underpinnings	74

10 Assets, Portfolios and Arbitrage	75
10.1 Portfolio Allocation and Short Selling	75
10.2 Arbitrage	77
10.3 Risk-neutral Probability Measures	77
10.4 Hedging of Contingent Claims	78
10.5 Market Completeness	78
11 Discrete-time Market Model	79
11.1 Discrete Time Compounding	79
11.2 Arbitrage and Self-Financing Portfolios	82
12 Continuous-time Market Model	83
12.1 Asset Price Modelling	83
12.2 Arbitrage and Risk-Neutral Measures	84
12.3 Self-financing Portfolio Strategies	85
IV Risk-Neutral Valuation	87
V Martingale Methods	88
13 The Black-Scholes Model	89
13.1 Risk-free Bond	89
13.2 Common Stock Price	90
13.3 Self-financing Trading Strategies	95
13.4 Martingale Measure for the Black-Scholes Model	95
13.5 Black-Scholes Option Pricing Formula	100
13.6 Put-Call Parity for Spot Options	102
13.7 Black-Scholes PDE	103

14 A Dividend-paying Stock	105
14.1 Case of a Constant Dividend Yield	105
14.2 Case of Known Dividends	106
15 Robustness of the Black-Scholes Approach	109
15.1 Uncertain Volatility	109
15.2 European Call and Put Options	109
16 Discretisation Schemes	110
16.1 Stochastic Taylor Expansion	110
16.2 Strong Convergence Criterion	117
16.3 Weak Convergence Criterion	118
16.4 Euler-Maruyama	119
16.5 Milstein	123
VI Monte Carlo Methods	127
17 Generating Random Numbers and Random Variables	129
17.1 Simulating Random Variables	132
17.1.1 Inverse-Transform Method	132
17.1.2 Acceptance-Rejection Method	133
18 Simulating Random Vectors and Processes	135
18.0.1 Normal Random Variables and Vectors	135
19 Quasi Monte Carlo	138
19.1 Discrepancy Measures	138
19.2 Low-discrepancy Sequences	138
19.2.1 Halton and Hammersley	138

19.2.2	Faure	138
19.2.3	Sobol	138
19.3	Koksma-Hlawka	138
20	Variance Reduction	139
20.0.1	Antithetics	139
20.0.2	Common Random Numbers	143
20.0.3	Control Variates	146
20.0.4	Importance Sampling	150
VII	Grid & Lattice Methods	159
21	Lattice Methods	160
22	Finite Difference Methods	164
23	Contract Sensitivities	165
23.0.1	Finite Differences	165
23.0.2	Likelihood Ratio	166
23.0.3	Pathwise Derivatives	167
24	Finite-Difference Approximations	168
24.1	Bias and Variance	168
24.2	Optimal Mean Square Error	171
VIII	Financial Contracts	172
25	European Options	173
26	Digital Options	174

27 Barrier Options	175
27.1 Knock-out Barrier	175
27.1.1 Up-and-out barrier call option	175
27.1.2 Up-and-out barrier put option	178
27.1.3 Down-and-out barrier call option	180
27.1.4 Down-and-out barrier put option	182
27.2 Knock-in Barrier	184
28 American Options	185
28.1 Approximate Continuation Values	185
29 Lookback Options	189
29.1 Maximum of Brownian Motion	189
29.2 The Lookback Put Option	192
30 Asian Options	194
31 Basket Options	197
32 Quantile Options	202
IX Hedging	203
33 Delta Hedging	204
Appendices	207
Appendices	208
A Existence of Lebesgue Measure	208

B	Linear Algebra and Functional Analysis	209
B.1	Vector Spaces, Bases, and Matrices	209
B.2	Inner Product	213
B.3	Complex Vectors and Matrices	215
B.4	Orthogonal Projections	216
B.5	Eigenvalues and Eigenvectors	217
B.6	Matrix Decompositions	218
B.6.1	(P)LU Decomposition	219
B.6.2	Woodburry Identity	220
B.6.3	Cholesky Decomposition	222
B.7	Functional Analysis	224
B.8	Fourier Transforms	228
B.8.1	Discrete Fourier Transform	232
B.8.2	Fast Fourier Transform	235
C	Multivariate Differentiation and Optimisation	237
C.1	Multivariate Differentiation	237
C.2	Optimisation Theory	239
C.3	Numerical Root-Finding and Minimisation	241
C.3.1	Newton-Like Methods	242
C.4	Constrained Minimisation via Penalty Functions	244
D	Probability and Statistics	245
D.1	Random Experiments and Probability Spaces	245
D.2	Random Variables and Probability Distributions	247
D.3	Expectation	249
D.4	Joint Distributions	252
D.5	Conditioning and Independence	252

D.5.1	Conditional Probability	253
D.5.2	Independence	253
D.5.3	Expectation and Covariance	254
D.6	Markov Chains	257
E	Python Primer	260

Part I

Mathematical Foundations

1 Measure Theory

Definition 1.1: σ -Algebra

A σ -algebra $\mathcal{E}$ on E is a collection of subsets of E that contains E itself, and that is closed under *complements* and *countable unions*; that is:

1. $E \in \mathcal{E}$.
2. If $A \in \mathcal{E}$, then also $A^c \in \mathcal{E}$.
3. If $A_1, A_2, \dots \in \mathcal{E}$, then also $\bigcup_n A_n \in \mathcal{E}$.

Definition 1.2: p-System

A collection $\mathcal{C}$ of subsets of E is called a *p-system* (or π -system) if it is closed under finite intersections.

Definition 1.3: d-System

A collection $\mathcal{D}$ of subsets of E is called a *d-system* (or λ -system) if it satisfies the following three conditions:

1. $E \in \mathcal{D}$.
2. If $A, B \in \mathcal{D}$, with $A \supseteq B$, then $A \setminus B \in \mathcal{D}$.
3. If $(A_n) \in \mathcal{D}$, with $A_1 \subseteq A_2 \subseteq A_3 \subseteq \dots \subseteq \bigcup_n A_n := A$, then $A \in \mathcal{D}$.

Theorem 1.1: σ -Algebra as a d- and p-System

A collection of subsets of E is a σ -algebra if and only if it is both a p-system and a d-system on E .

Theorem 1.2: Monotone Class Theorem

If a d-system contains a p-system, then it also contains the σ -algebra generated by that p-system.

Definition 1.4: Product Space

The *product space* of two measurable spaces $(E, \mathcal{E})$ and $(F, \mathcal{F})$ is the measurable space $(E \times F, \mathcal{E} \otimes \mathcal{F})$, where $E \times F := \{(x, y) : x \in E, y \in F\}$ is the Cartesian product of E and F , and $\mathcal{E} \otimes \mathcal{F}$ is the *product σ -algebra* on $E \times F$ that is generated by the rectangle sets

$$A \times B, \quad A \in \mathcal{E}, B \in \mathcal{F}.$$

Definition 1.5: Measurable Function

Let $(E, \mathcal{E})$ and $(F, \mathcal{F})$ be two measurable spaces, and let f be a function from E to F . The function f is called $\mathcal{E}/\mathcal{F}$ -*measurable* if for all $B \in \mathcal{F}$ it holds that

$$\{x \in E : f(x) \in B\} \in \mathcal{E}. \quad (1.1)$$

Theorem 1.3: Measurable Function on a Generated σ -Algebra

A function $f : E \rightarrow F$ is $\mathcal{E}/\mathcal{F}$ -measurable if and only if [eq. \(1.1\)](#) holds for all $B \in \mathcal{F}_0$, with $\mathcal{F}_0$ generating $\mathcal{F}$.

Theorem 1.4: Composition of Measurable Functions

Let $(E, \mathcal{E})$, $(F, \mathcal{F})$, and $(G, \mathcal{G})$ be measurable spaces. If $f : E \rightarrow F$ and $g : F \rightarrow G$ are respectively $\mathcal{E}/\mathcal{F}$ - and $\mathcal{F}/\mathcal{G}$ -measurable, then $g \circ f : E \rightarrow G$ is $\mathcal{E}/\mathcal{G}$ -measurable.

Definition 1.6: Numerical Function

Let $(E, \mathcal{E})$ be a measurable space.

- A function $f : E \rightarrow \mathbb{R}$ is called a *real-valued function* on E .
- A function $f : E \rightarrow \bar{\mathbb{R}} := [-\infty, +\infty]$ is called a *numerical function* on E .

- An $\mathcal{E}/\overline{\mathcal{B}}$ -measurable numerical function f is said to be $\mathcal{E}$ -measurable; we write $f \in \mathcal{E}$. Also, $\mathcal{E}_+$ is the class of *positive* $\mathcal{E}$ -measurable functions.

Definition 1.7: Infimum and Supremum

Let (a_n) be a sequence of real numbers.

1. The *infimum*, $\inf a_n$, is the greatest element in $\overline{\mathbb{R}}$ that is less than or equal to all a_n . It can be $-\infty$.
2. The *supremum*, $\sup a_n$, is the smallest element in $\overline{\mathbb{R}}$ that is greater than or equal to all a_n . It can be ∞ .

Definition 1.8: Liminf and Limsup

Let (a_n) be a sequence of real numbers.

1. The *limit inferior*, $\liminf a_n$, is the eventual lower bound for the sequence (a_n) ; that is, $\liminf a_n := \lim_{m \rightarrow \infty} \inf_{n \geq m} a_n = \sup_m \inf_{n \geq m} a_n$.
2. The *limit superior*, $\limsup a_n$, is the eventual upper bound for the sequence (a_n) ; that is, $\limsup a_n := \lim_{m \rightarrow \infty} \sup_{n \geq m} a_n = \inf_m \sup_{n \geq m} a_n$.

Proposition 1.1: Limits of Measurable Functions

Let (f_n) be a sequence of $\mathcal{E}$ -measurable numerical functions. Then, the numerical functions $\sup f_n$, $\inf f_n$, $\limsup f_n$, and $\liminf f_n$ are also $\mathcal{E}$ -measurable. In particular, limits of measurable functions (when $\limsup f_n = \liminf f_n$) are measurable as well.

Theorem 1.5: Approximation from Below

A *positive* numerical function on E is $\mathcal{E}$ -measurable if and only if it is the (pointwise) limit of an increasing sequence of simple functions.

Theorem 1.6: Positive and Negative Part of a Function

A function f is $\mathcal{E}$ -measurable if and only if both f^+ and f^- are.

Definition 1.9: Monotone Class of Functions

Let $(E, \mathcal{E})$ be a measurable space. A collection $\mathcal{M}$ of numerical functions on E is called a *monotone class* if it satisfies:

1. $\mathbb{1}_E \in \mathcal{M}$.
2. If $f, g \in \mathcal{M}$ are bounded and $a, b \in \mathbb{R}$, then $af + bg \in \mathcal{M}$.
3. If (f_n) is a sequence of positive functions in $\mathcal{M}$ that increases to some f , then $f \in \mathcal{M}$.

Theorem 1.7: Monotone Class Theorem for Functions

Let $\mathcal{M}$ be a monotone class of functions on E and let $\mathcal{C}$ be a p-system that generates $\mathcal{E}$. If $\mathbb{1}_A \in \mathcal{M}$ for every $A \in \mathcal{C}$, then $\mathcal{M}$ includes

- all positive $\mathcal{E}$ -measurable functions,
- all bounded $\mathcal{E}$ -measurable functions.

Definition 1.10: Measure

Let $(E, \mathcal{E})$ be a measurable space. A *measure* μ on $\mathcal{E}$ is a set function

$$\mu : \mathcal{E} \rightarrow [0, \infty]$$

with $\mu(\emptyset) = 0$, such that for every sequence (A_n) of disjoint sets in $\mathcal{E}$,

$$\mu(\cup_n A_n) = \sum_n \mu(A_n). \quad (1.2)$$

Theorem 1.8: Extension Theorem (Carathéodory)

Let E be a set with an algebra $\mathcal{E}_0$ thereon. Every pre-measure $\mu_0 : \mathcal{E}_0 \rightarrow [0, \infty]$ can be extended to a measure μ on $(E, \sigma(\mathcal{E}_0))$.

Proposition 1.2: Properties of a Measure

Let μ be a measure on a measurable space $(E, \mathcal{E})$, and let A, B and $A_1, A_2, \dots$ be measurable sets. Then, the following hold:

1. (*Monotonicity*): $A \subseteq B \implies \mu(A) \leq \mu(B)$.
2. (*Continuity from below*): If $A_1, A_2, \dots$ is increasing (i.e., $A_1 \subseteq A_2 \subseteq \dots$), then

$$\lim_n \mu(A_n) = \mu(\cup_n A_n).$$

3. (*Countable subadditivity*): $\mu(\cup_n A_n) \leq \sum_n \mu(A_n)$.

Theorem 1.9: Uniqueness of Finite Measures

Let $\mathcal{C}$ be a p-system that generates $\mathcal{E}$. If μ and ν are two measures on $(E, \mathcal{E})$ with $\mu(E) = \nu(E) < \infty$, then

$$\mu(A) = \nu(A) \text{ for all } A \in \mathcal{C} \implies \mu(A) = \nu(A) \text{ for all } A \in \mathcal{E}.$$

Definition 1.11: Product Measure

Let $(E, \mathcal{E}, \mu)$ and $(F, \mathcal{F}, \nu)$ be two measure spaces. The *product measure* of μ and ν on the product space $(E \times F, \mathcal{E} \otimes \mathcal{F})$ is defined as the measure π with

$$\pi(A \times B) := \mu(A)\nu(B), \quad A \in \mathcal{E}, B \in \mathcal{F}. \quad (1.3)$$

This measure π is often written as $\mu \otimes \nu$.

Proposition 1.3: Insensitivity of the Integral

Let $(E, \mathcal{E}, \mu)$ be a measure space and let f and g be positive $\mathcal{E}$ -measurable functions that are equal μ -almost everywhere. Then, $\mu f = \mu g$.

Theorem 1.10: Properties of an Integral

Below a, b are in $\mathbb{R}_+$ and f, g, f_n are in $\mathcal{E}_+$.

1. (Positivity and Monotonicity): $\mu f \geq 0$, and $f = 0 \implies \mu f = 0$. Also, $f \leq g \implies \mu f \leq \mu g$.
2. (Linearity): $\mu(af + bg) = a\mu f + b\mu g$.
3. (Monotone convergence): If $f_n \uparrow f$, then $\mu f_n \uparrow \mu f$.

Theorem 1.11: Measures from Linear Functionals

Let $L : \mathcal{E}_+ \rightarrow \overline{\mathbb{R}}_+$. Then, there exists a unique measure μ on $(E, \mathcal{E})$ such that $L(f) = \mu f$ if and only if

1. $f = 0 \implies L(f) = 0$,
2. $f, g \in \mathcal{E}_+$ and $a, b \in \mathbb{R}_+ \implies L(af + bg) = aL(f) + bL(g)$,
3. $(f_n) \in \mathcal{E}_+$ and $f_n \uparrow f \implies L(f_n) \uparrow L(f)$.

Theorem 1.12: Integrals with Densities

Let $\nu := p\mu$. Then, for every $f \in \mathcal{E}_+$, we have $\nu f = \mu(pf)$.

Theorem 1.13: Radon–Nikodym

Let ν and μ be two measures on $(E, \mathcal{E})$ such that μ is σ -finite and $\nu \ll \mu$. Then, there exists a $p \in \mathcal{E}_+$ such that $\nu = p\mu$. Moreover, p is unique up to equivalence; that is, if there is another $\tilde{p} \in \mathcal{E}_+$ such that $\nu = \tilde{p}\mu$, then $p = \tilde{p}$, μ -almost everywhere.

Theorem 1.14: Integration with Respect to Image Measures

Let $\nu = \mu \circ h^{-1}$ be the image measure of μ under h . Then, for every positive $\mathcal{F}$ -measurable function f ,

$$\nu f = (\mu \circ h^{-1})f = \int_F f d\nu = \int_E f \circ h d\mu = \mu(f \circ h).$$

Definition 1.12: Transition Kernel

Let $(E, \mathcal{E})$ and $(F, \mathcal{F})$ be measurable spaces. A *transition kernel* from $(E, \mathcal{E})$ into $(F, \mathcal{F})$ is a mapping $K : E \times \mathcal{F} \rightarrow \overline{\mathbb{R}}_+$ such that:

1. $K(\cdot, B)$ is $\mathcal{E}$ -measurable for every $B \in \mathcal{F}$.
2. $K(x, \cdot)$ is a measure on $(F, \mathcal{F})$ for every $x \in E$.

If $(F, \mathcal{F}) = (E, \mathcal{E})$ and $K(x, \cdot)$ is a probability measure for every $x \in E$, K is said to be a *probability transition kernel* on $(E, \mathcal{E})$.

Theorem 1.15: Measures and Functions from Kernels

Let K be a transition kernel from $(E, \mathcal{E})$ into $(F, \mathcal{F})$.

1. For all $f \in \mathcal{F}_+$ the function Kf defined by $(Kf)(x) := \int_F K(x, dy)f(y)$, $x \in E$, is in $\mathcal{E}_+$.
2. For every measure μ on $(E, \mathcal{E})$ the set function μK defined by $(\mu K)(B) := \int_E \mu(dx)K(x, B)$, $B \in \mathcal{F}$, is a measure on $(F, \mathcal{F})$.
3. For every measure μ on $(E, \mathcal{E})$ and $f \in \mathcal{F}_+$,

$$(\mu K)f = \mu(Kf) = \int_E \mu(dx) \int_F K(x, dy)f(y).$$

Theorem 1.16: Fubini

Let μ and ν be Σ -finite measures on $(E, \mathcal{E})$ and $(F, \mathcal{F})$, respectively.

1. There exists a unique Σ -finite measure π on the product space $(E \times F, \mathcal{E} \otimes \mathcal{F})$ such that for every positive f in $\mathcal{E} \otimes \mathcal{F}$,

$$\pi f = \int_E \mu(dx) \int_F \nu(dy) f(x, y) = \int_F \nu(dy) \int_E \mu(dx) f(x, y). \quad (1.4)$$

2. If $f \in \mathcal{E} \otimes \mathcal{F}$ and is π -integrable, then $f(x, \cdot)$ is ν -integrable for μ -almost every x , and $f(\cdot, y)$ is μ -integrable for ν -almost every y and [eq. \(1.4\)](#) holds again.

2 Probability

Definition 2.1: Probability Measure

A *probability measure* is a measure on $(\Omega, \mathcal{H})$ with $\mathbb{P}(\Omega) = 1$. Thus, $\mathbb{P}$ is a mapping from $\mathcal{H}$ to $[0, 1]$ with the following properties:

1. $\mathbb{P}(\emptyset) = 0$ and $\mathbb{P}(\Omega) = 1$.
2. For any sequence $A_1, A_2, \dots$ of disjoint events,

$$\mathbb{P}(\cup_n A_n) = \sum_n \mathbb{P}(A_n). \quad (2.1)$$

Theorem 2.1: Properties of a Probability Measure

Let $\mathbb{P}$ be a probability measure on a measurable space $(\Omega, \mathcal{H})$, and let A, B and $A_1, A_2, \dots$ be events. Then, the following hold:

1. (*Monotonicity*): $A \subseteq B \implies \mathbb{P}(A) \leq \mathbb{P}(B)$.
2. (*Continuity from below*): If $A_1, A_2, \dots$ is an increasing sequence of events, i.e., $A_1 \subseteq A_2 \subseteq \dots$, then

$$\lim \mathbb{P}(A_n) = \mathbb{P}(\cup_n A_n). \quad (2.2)$$

3. (*Countable subadditivity*): $\mathbb{P}(\cup_n A_n) \leq \sum_n \mathbb{P}(A_n)$.

Theorem 2.2: Additional Properties of a Probability Measure

Let $\mathbb{P}$ be a probability measure on a measurable space $(\Omega, \mathcal{H})$, and let A, B and $A_1, A_2, \dots$ be events. Then, the following hold:

1. (*Complement*): $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
2. (*Union*): $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

3. (*Continuity from above*): If $A_1, A_2, \dots$ is a decreasing sequence of events, i.e., $A_1 \supseteq A_2 \supseteq \dots$, then

$$\lim \mathbb{P}(A_n) = \mathbb{P}(\cap_n A_n). \quad (2.3)$$

Definition 2.2: Random Variable

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and $(E, \mathcal{E})$ a measurable space. A *random variable* X taking values in E is an $\mathcal{H}/\mathcal{E}$ -measurable function; that is, a mapping $X : \Omega \rightarrow E$ that satisfies

$$X^{-1}A = \{X \in A\} = \{\omega \in \Omega : X(\omega) \in A\} \in \mathcal{H} \quad \text{for all } A \in \mathcal{E}. \quad (2.4)$$

Definition 2.3: Stochastic Process

A collection of random variables $X := \{X_t, t \in \mathbb{T}\}$, where $\mathbb{T}$ is any index set, is called a *stochastic process*. When $\mathbb{T}$ is finite, X is called a *random vector*.

Theorem 2.3: Stochastic Process

$X := \{X_t, t \in \mathbb{T}\}$ is a stochastic process taking values in $\times_{t \in \mathbb{T}} E_t$ with σ -algebra $\otimes_{t \in \mathbb{T}} \mathcal{E}_t$ if and only if each X_t is a random variable taking values in E_t with σ -algebra $\mathcal{E}_t$.

Definition 2.4: Probability Distribution

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and let X be a random variable taking values in $(E, \mathcal{E})$. The (*probability*) *distribution* of X is the image measure μ of $\mathbb{P}$ under X . That is,

$$\mu(B) := \mathbb{P}(X \in B), \quad B \in \mathcal{E}. \quad (2.5)$$

Definition 2.5: Cumulative Distribution Function

The (*cumulative*) *distribution function* of a numerical random variable X is the function $F : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F(x) := \mathbb{P}(X \leq x), \quad x \in \mathbb{R}. \quad (2.6)$$

Proposition 2.1: Properties of a Cumulative Distribution Function

1. (*Bounded*): $0 \leq F(x) \leq 1$.
2. (*Increasing*): $x \leq y \implies F(x) \leq F(y)$.
3. (*Right-continuous*): $\lim_{h \downarrow 0} F(x + h) = F(x)$.

Definition 2.6: Discrete Distribution

We say that X has a *discrete distribution* μ on $(E, \mathcal{E})$ if μ is a discrete measure; that is, it is of the form

$$\mu = \sum_{x \in D} f(x) \delta_x$$

for some countable set D and positive masses $\{f(x), x \in D\}$.

Definition 2.7: Distribution with a Density

We say that X has an *absolutely continuous* distribution μ on $(E, \mathcal{E})$ with respect to a measure λ on $(E, \mathcal{E})$ if there exists a positive $\mathcal{E}$ -measurable function f such that $\mu = f\lambda$; that is,

$$\mu(B) = \mathbb{P}(X \in B) = \int_B \lambda(dx) f(x), \quad B \in \mathcal{E}.$$

Definition 2.8: Independent Random Variables

Let X and Y be random variables taking values in $(E, \mathcal{E})$ and $(F, \mathcal{F})$, respectively. They are said to be *independent* if for any $A \in \mathcal{E}$ and $B \in \mathcal{F}$ it holds that

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A) \mathbb{P}(Y \in B). \quad (2.7)$$

Definition 2.9: Independency

We say that a collection of random variables $\{X_t, t \in \mathbb{T}\}$, with each X_t taking values in $(E_t, \mathcal{E}_t)$, is an *independency* if for any finite choice of indexes $t_1, \dots, t_n \in \mathbb{T}$ and sets

$A_{t_1} \in \mathcal{E}_{t_1}, \dots, A_{t_n} \in \mathcal{E}_{t_n}$ it holds that

$$\mathbb{P}(X_{t_1} \in A_{t_1}, \dots, X_{t_n} \in A_{t_n}) = \mathbb{P}(X_{t_1} \in A_{t_1}) \cdots \mathbb{P}(X_{t_n} \in A_{t_n}).$$

Definition 2.10: Expectation

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and X a numerical random variable. The *expectation* of X is defined as the integral

$$\mathbb{E}X := \mathbb{P}X = \int X d\mathbb{P}.$$

Theorem 2.4: Properties of an Expectation

Let $a, b \in \mathbb{R}$ and let X, Y , and (X_n) be numerical random variables for which the expectations below are well-defined. Then, the following hold:

1. (*Monotonicity*): If $X \leq Y$, then $\mathbb{E}X \leq \mathbb{E}Y$.
2. (*Linearity*): $\mathbb{E}(aX + bY) = a\mathbb{E}X + b\mathbb{E}Y$.
3. (*Monotone convergence*): If $X_n \uparrow X$, then $\mathbb{E}X_n \uparrow \mathbb{E}X$.

Lemma 2.1: Fatou

For any sequence (X_n) of positive numerical random variables,

$$\mathbb{E} \liminf X_n \leq \liminf \mathbb{E}X_n.$$

Theorem 2.5: Dominated and Bounded Convergence

Let (X_n) be a sequence of numerical random variables for which $\lim X_n = X$ exists. Then, the following hold:

1. (*Dominated convergence*): If, for each n , $|X_n| \leq Y$ for some random variable Y with $\mathbb{E}Y < \infty$, then $\mathbb{E}X_n \rightarrow \mathbb{E}X < \infty$.

2. (Bounded convergence): If $|X_n| \leq c$ for all n for some $c \in \mathbb{R}$, then $\mathbb{E}X_n \rightarrow \mathbb{E}X < \infty$.

Theorem 2.6: Expectation and Image Measure

Let X be a random variable taking values in $(E, \mathcal{E})$ with distribution μ and let h be an $\mathcal{E}$ -measurable function. Then, provided the integral exists:

$$\mathbb{E}h(X) = \int h(X) d\mathbb{P} = \int_{\mathbb{R}} \mu(dx) h(x) = \mu h.$$

Theorem 2.7: Transformation Rule

Let X be a random vector with density f_X with respect to the Lebesgue measure on $(\mathbb{R}^n, \mathcal{B}^n)$. Let $Z := g(X)$, where $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is invertible with inverse g^{-1} and Jacobian $|\frac{\partial z}{\partial x}|$. Then, at $z = g(x)$ the random vector Z has the following density with respect to the Lebesgue measure on $(\mathbb{R}^n, \mathcal{B}^n)$:

$$f_Z(z) := \frac{f_X(x)}{|\frac{\partial z}{\partial x}|} = f_X(g^{-1}(z)) \left| \frac{\partial x}{\partial z} \right|, \quad z \in \mathbb{R}^n. \quad (2.8)$$

Proposition 2.2: Expectation of a Quadratic Form

Let A be an $n \times n$ matrix and X an n -dimensional random vector with expectation vector μ and covariance matrix Σ . The random variable $Y := X^\top A X$ has expectation $\text{tr}(\Sigma A) + \mu^\top A \mu$.

Definition 2.11: L^p Space

For each $p \in [1, \infty]$ the space L^p is comprised of all numerical random variables X for which $\|X\|_p < \infty$.

Lemma 2.2: Jensen's Inequality

Let $h : \mathcal{X} \rightarrow \mathbb{R}$ be a convex function and let X be a random variable taking values in $\mathcal{X}$, with expectation vector $\mathbb{E}X$. Then,

$$\mathbb{E}h(X) \geq h(\mathbb{E}X).$$

Theorem 2.8: Properties of the L^p Norm

Let X and Y be numerical random variables on $(\Omega, \mathcal{H}, \mathbb{P})$. Then, the following hold:

1. (Positivity): $\|X\|_p \geq 0$, and $\|X\|_p = 0 \iff X = 0$ almost surely.
2. (Multiplication by a constant): $\|cX\|_p = |c|\|X\|_p$.
3. (Minkowski's (triangle) inequality):

$$|\|X\|_p - \|Y\|_p| \leq \|X + Y\|_p \leq \|X\|_p + \|Y\|_p.$$

4. (Hölder's inequality): For $p, q, r \in [1, \infty]$ with $\frac{1}{p} + \frac{1}{q} = \frac{1}{r}$,

$$\|XY\|_r \leq \|X\|_p \|Y\|_q. \quad (2.9)$$

5. (Monotonicity): If $1 \leq p < q \leq \infty$, then $\|X\|_p \leq \|X\|_q$.

Definition 2.12: Moment Generating Function

The *moment generating function* (MGF) of a numerical random variable X with distribution μ is the function $M : \mathbb{R} \rightarrow [0, \infty]$, given by

$$M(s) := \mathbb{E}e^{sX} = \int_{-\infty}^{\infty} \mu(dx) e^{sx}, \quad s \in \mathbb{R}.$$

Theorem 2.9: Uniqueness of the MGF

A measure μ is determined by its MGF M if the latter is finite in a neighborhood of 0.

Theorem 2.10: Taylor's Theorem for MGFs

Suppose $M(s) := \mathbb{E}e^{sX}$ is finite in the neighborhood $(-s_0, s_0)$ of 0. Then, M has the Taylor expansion

$$M(s) = \sum_{k=0}^{\infty} \frac{s^k}{k!} \mathbb{E}X^k, \quad |s| < s_0.$$

In particular, all moments $\mathbb{E}X^k, k \in \mathbb{N}$ are finite.

Theorem 2.11: Product of MGFs

If X and Y are independent, then $M_{X+Y}(s) = M_X(s)M_Y(s)$.

Definition 2.13: Characteristic Function

The *characteristic function* of a numerical random variable X with distribution μ is the function $\psi : \mathbb{R} \rightarrow \mathbb{C}$, defined by

$$\psi(r) := \mathbb{E}e^{irX} = \mathbb{E} \cos(rX) + i \mathbb{E} \sin(rX) = \int_{-\infty}^{\infty} \mu(dx) e^{irx}, \quad r \in \mathbb{R}.$$

Theorem 2.12: Inversion and Uniqueness of the Characteristic Function

If the probability measure μ has characteristic function ψ , and if $\mu\{a\} = \mu\{b\} = 0$, then

$$\mu(a, b] = \lim_{t \rightarrow \infty} \frac{1}{2\pi} \int_{\mathbb{R}} dr \mathbb{1}_{[-t, t]}(r) \frac{e^{-ira} - e^{-irb}}{ir} \psi(r) := \lim_{t \rightarrow \infty} I_t. \quad (2.10)$$

In particular, the characteristic function determines μ uniquely. If, in addition, $\int_{\mathbb{R}} dr |\psi(r)| < \infty$, then μ has a density f with respect to the Lebesgue measure, given by

$$f(x) := \frac{1}{2\pi} \int_{-\infty}^{\infty} dr e^{-irx} \psi(r). \quad (2.11)$$

Proposition 2.3: Properties of the Characteristic Function

1. (*Symmetry*): The random variables X and $-X$ are identically distributed if and only if the characteristic function of X is real-valued.
2. (*Convolution*): If X and Y are independent, then $\psi_{X+Y} = \psi_X \psi_Y$.
3. (*Affine transformation*): If a and b are real numbers, then

$$\psi_{aX+b}(r) = e^{irb} \psi_X(ar), \quad r \in \mathbb{R}.$$

4. (*Taylor's theorem*): If $\mathbb{E}|X^n| < \infty$, then

$$\psi(r) = \sum_{k=0}^n \frac{\mathbb{E}X^k}{k!} (ir)^k + O(r^n), \quad r \in \mathbb{R}.$$

Definition 2.14: σ -Algebra Generated by a Random Variable

Let X be a random variable taking values in $(E, \mathcal{E})$. The sets $\{X \in A\}, A \in \mathcal{E}$ form a σ -algebra, called the σ -algebra generated by X , and we write σX .

Theorem 2.13: σ -Algebra Generated by a Stochastic Process

For the stochastic process $X := \{X_t, t \in \mathbb{T}\}$, the σ -algebra generated by X is the smallest σ -algebra on Ω that is generated by the union of the σ -algebras $\sigma X_t, t \in \mathbb{T}$. We write $\bigvee_{t \in \mathbb{T}} \sigma X_t$ or $\sigma\{X_t, t \in \mathbb{T}\}$.

Theorem 2.14: σX -Numerical Random Variables

Let X be a random variable taking values in a measurable space $(E, \mathcal{E})$. A mapping $V : \Omega \rightarrow \overline{\mathbb{R}}$ belongs to σX if and only if $V = h \circ X$ for some $h \in \mathcal{E}$.

Definition 2.15: Filtration

Let $\mathbb{T}$ be a subset of $\mathbb{R}$. A *filtration* is an increasing collection $\{\mathcal{F}_t : t \in \mathbb{T}\}$ of sub- σ -algebras of $\mathcal{H}$; that is, $\mathcal{F}_s \subseteq \mathcal{F}_t$ whenever $s < t$.

Theorem 2.15: Stochastic Process with Arbitrary Index Set

For each t in an arbitrary set $\mathbb{T}$, let X_t be a random variable taking values in a measurable set $(E_t, \mathcal{E}_t)$. Then, the mapping $V : \Omega \rightarrow \bar{\mathbb{R}}$ belongs to $\sigma\{X_t, t \in \mathbb{T}\}$ if and only if there exists a sequence (t_n) in $\mathbb{T}$ and a function h in $\bigotimes_n \mathcal{E}_{t_n}$ such that

$$V = h(X_{t_1}, X_{t_2}, \dots).$$

Definition 2.16: Independence for σ -Algebras

Let $\{\mathcal{F}_i, i = 1, \dots, n\}$ be a collection of sub- σ -algebras of $\mathcal{H}$. The $\{\mathcal{F}_i\}$ are said to be (*mutually*) *independent* or form an *independency* if

$$\mathbb{E}V_1 \cdots V_n = \mathbb{E}V_1 \cdots \mathbb{E}V_n \quad (2.12)$$

for all positive random variables $V_i \in \mathcal{F}_i, i = 1, \dots, n$.

Theorem 2.16: Independence of σ -Algebras

Sub- σ -algebras $\mathcal{F}_1, \dots, \mathcal{F}_n$ of $\mathcal{H}$ generated by p-systems $\mathcal{C}_1, \dots, \mathcal{C}_n$ are independent if and only if for all $H_i \in \mathcal{C}_i \cup \{\Omega\}, i = 1, \dots, n$, it holds that:

$$\mathbb{P}(H_1 \cap \cdots \cap H_n) = \mathbb{P}(H_1) \cdots \mathbb{P}(H_n). \quad (2.13)$$

Theorem 2.17: Independence of Random Variables

Random variables X and Y taking values in $(E, \mathcal{E})$ and $(F, \mathcal{F})$ are independent if and only if

$$\mathbb{E}g(X)h(Y) = \mathbb{E}g(X)\mathbb{E}h(Y), \quad g \in \mathcal{E}_+, h \in \mathcal{F}_+. \quad (2.14)$$

Definition 2.17: Gaussian Process

A real-valued stochastic process $\{X_t, t \in \mathbb{T}\}$ is said to be *Gaussian* if all its finite-dimensional distributions are Gaussian (normal); that is, if the vector $[X_{t_1}, \dots, X_{t_n}]^\top$ is multivariate normal for any choice of n and $t_1, \dots, t_n \in \mathbb{T}$.

Definition 2.18: Poisson Random Measure

A random measure N on $(E, \mathcal{E})$ is said to be a *Poisson random measure* with mean measure μ if the following properties hold:

1. $N(A) \sim \text{Poi}(\mu(A))$ for any set $A \in \mathcal{E}$.
2. For any selection of disjoint sets $A_1, \dots, A_n \in \mathcal{E}$, the random variables $N(A_1), \dots, N(A_n)$ are independent.

Proposition 2.4: Poisson Random Measure on a Product Space

Let $X := \{X_i, i \in I\}$ be the atoms of a Poisson random measure N on $(E, \mathcal{E})$ with mean measure μ , and let $\{Y_i, i \in \mathbb{N}\}$ be iid random variables on $(F, \mathcal{F})$ with distribution π , independent of X . Then, $\{(X_i, Y_i), i \in I\}$ form the atoms of a Poisson random measure M on $(E \times F, \mathcal{E} \otimes \mathcal{F})$ with mean measure $\mu \otimes \pi$.

Theorem 2.18: Conditioning on the Total Number of Points

Let N be a Poisson random measure on $(E, \mathcal{E})$ with mean measure μ satisfying $\mu(E) := c < \infty$. Then, conditional on $N(E) = k$, the k atoms of N are iid with probability distribution μ/c .

Definition 2.19: Compound Poisson Process

The process $(X_t, t \geq 0)$ defined by

$$X_t := \int_{[0,t] \times \mathbb{R}^d} N(ds, dx) x, \quad t \geq 0,$$

is the *compound Poisson process* corresponding to the measure ν .

Definition 2.20: Lévy Process

The process X is said to be a *Lévy process* if

1. the paths of X are almost surely right-continuous and left-limited, with $X_0 = 0$, and
2. for every $t_1 < t_2 < t_3 < t_4$, the increments $X_{t_2} - X_{t_1}$ and $X_{t_4} - X_{t_3}$ are independent, and for every $t, u \in \mathbb{R}_+$, the increment $X_{t+u} - X_t$ has the same distribution as X_u .

Theorem 2.19: Lévy – It Decomposition

The process $X := (X_t, t \geq 0)$ is a Lévy process if and only if there exists a d -dimensional Brownian motion process $B := (B_t, t \geq 0)$ starting at $\mathbf{0}$ and a Poisson random measure N on $\mathbb{R}_+ \times \mathbb{R}^d$ that is independent of B and has mean measure $\text{Leb} \otimes \nu$, with ν satisfying

$$\int_{\mathbb{R}^d} \nu(d\mathbf{x}) (|\mathbf{x}|^2 \wedge 1) < \infty, \quad (2.15)$$

such that $X = B + U + Z$, where the summands are independent, $U := (U_t, t \geq 0)$ is the compound Poisson process

$$U_t := \int_0^t \int_{\|\mathbf{x}\| > 1} N(ds, d\mathbf{x}) \mathbf{x}, \quad t \geq 0,$$

and $Z := (Z_t, t \geq 0)$ is the limit of compensated compound Poisson processes:

$$Z := \lim_{\delta \downarrow 0} Z^\delta \quad \text{with} \quad Z_t^\delta := \int_0^t \int_{\delta \leq \|\mathbf{x}\| \leq 1} [\mathbf{x} N(ds, d\mathbf{x}) - ds \mathbf{x} \nu(d\mathbf{x})], \quad t \geq 0.$$

3 Convergence

Proposition 3.1: Cauchy Criteria for Convergence

The following statements satisfy (1) $\iff$ (2) and (3) $\implies$ (2):

1. (x_n) converges to some real $x := \lim x_n$.
2. (x_n) is Cauchy convergent, that is, $\lim_{m,n \rightarrow \infty} |x_m - x_n| = 0$.
3. $\sum_n \mathbb{1}_{\{|x_{n+1} - x_n| > \varepsilon_n\}} < \infty$ for any (ε_n) such that $c := \sum_n \varepsilon_n < \infty$.

Definition 3.1: Almost Sure Convergence

The sequence of random variables (X_n) converges *almost surely* (a.s.) to a random variable X if

$$\mathbb{P}(\lim X_n = X) = 1.$$

We denote the almost sure convergence by $X_n \xrightarrow{\text{a.s.}} X$.

Theorem 3.1: Almost Sure Convergence

The sequence of random variables (X_n) converges almost surely to a random variable X if and only if for every $\varepsilon > 0$

$$\lim_{m \rightarrow \infty} \mathbb{P} \left(\sup_{n \geq m} |X_n - X| > \varepsilon \right) = 0. \quad (3.1)$$

Proposition 3.2: Cauchy Criteria for Almost Sure Convergence

The following statements are equivalent:

1. $X_n \xrightarrow{\text{a.s.}} X$.
2. $\lim_{n \rightarrow \infty} \mathbb{P}(\sup_{k \geq 0} |X_{n+k} - X_n| > \varepsilon) = 0$ for any choice of $\varepsilon > 0$.
3. $\lim_{k \rightarrow \infty} \mathbb{P}(\sup_{m, n \geq k} |X_m - X_n| > \varepsilon) = 0$ for any choice of $\varepsilon > 0$.

4. $\mathbb{P}(\lim_{m,n \rightarrow \infty} |X_m - X_n| = 0) = 1$; that is, (X_n) is a.s. a Cauchy sequence.

Proposition 3.3: Markov's and Chebyshev's Inequality

For any random variable Y and $\varepsilon > 0$, Markov's inequality holds:

$$\mathbb{P}(|Y| > \varepsilon) \leq \frac{\mathbb{E}|Y|}{\varepsilon}. \quad (3.2)$$

As a consequence, we have Chebyshev's inequality, where $\mu := \mathbb{E}X$:

$$\mathbb{P}(|X - \mu| > \varepsilon) \leq \frac{\text{Var } X}{\varepsilon^2}. \quad (3.3)$$

Theorem 3.2: Kolmogorov's Inequality

Let $S_n := \sum_{i=1}^n X_i$, where $X_1, X_2, \dots$ are independent with 0 mean. Then, for every $\varepsilon > 0$:

$$\mathbb{P}\left(\sup_{k \leq n} |S_k| > \varepsilon\right) \leq \frac{\text{Var } S_n}{\varepsilon^2}. \quad (3.4)$$

Lemma 3.1: Borel–Cantelli

Let (H_n) be a sequence of events. Then, the following hold:

1. $\sum_n \mathbb{P}(H_n) < \infty$ implies that $\mathbb{P}(\sum_n \mathbf{1}_{H_n} < \infty) = 1$.
2. $\sum_n \mathbb{P}(H_n) = \infty$ implies that $\mathbb{P}(\sum_n \mathbf{1}_{H_n} < \infty) = 0$, provided that the events (H_n) are pairwise independent.

Definition 3.2: Convergence in Probability

The sequence of random variables (X_n) converges in probability to a random variable X if, for all $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| > \varepsilon) = 0.$$

We denote the convergence in probability by $X_n \xrightarrow{\mathbb{P}} X$.

Proposition 3.4: Convergence of a Subsequence

If (X_n) converges to X in probability, there is a subsequence (X_{n_k}) that converges to X almost surely.

Proposition 3.5: Cauchy Criterion for Convergence in Probability

The sequence of random variables (X_n) converges in probability if and only if for every $\varepsilon > 0$,

$$\lim_{m,n \rightarrow \infty} \mathbb{P}(|X_m - X_n| > \varepsilon) = 0. \quad (3.5)$$

Definition 3.3: Convergence in Distribution

The sequence of random variables (X_n) is said to *converge in distribution* to a random variable X with cdf F provided that:

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n \leq x) = F(x) \quad \text{for all } x \text{ such that } \lim_{a \rightarrow x} F(a) = F(x). \quad (3.6)$$

We denote the convergence in distribution by $X_n \xrightarrow{d} X$.

Theorem 3.3: Characteristic Function and Convergence in Distribution

Suppose that $\psi_{X_1}(\mathbf{r}), \psi_{X_2}(\mathbf{r}), \dots$ are the characteristic functions of the sequence of d -dimensional random vectors $X_1, X_2, \dots$ and $\psi_X(\mathbf{r})$ is the characteristic function of X . Then, the following three statements are equivalent:

1. $\lim_n \psi_{X_n}(\mathbf{r}) = \psi_X(\mathbf{r})$ for all $\mathbf{r} \in \mathbb{R}^d$.
2. $X_n \xrightarrow{d} X$.
3. $\lim_n \mathbb{E}h(X_n) = \mathbb{E}h(X)$ for all bounded continuous functions $h : \mathbb{R}^d \rightarrow \mathbb{R}$.

Definition 3.4: Convergence in L^p Norm

The sequence of random variables (X_n) converges in L^p norm (for some $p \in [1, \infty]$) to a random variable X , if $\|X_n\|_p < \infty$ for all n , $\|X\|_p < \infty$, and

$$\lim_{n \rightarrow \infty} \|X_n - X\|_p = 0.$$

We denote the convergence in L^p norm by $X_n \xrightarrow{L^p} X$.

Proposition 3.6: Cauchy Criterion for Convergence in L^p Norm

The sequence of random variables (X_n) converges in L^p norm for $p \in [1, \infty]$ if and only if $\|X_n\|_p < \infty$ for all n and

$$\lim_{m, n \rightarrow \infty} \|X_m - X_n\|_p = 0.$$

Proposition 3.7: Integrable Random Variable

A real-valued random variable X is integrable if and only if

$$\lim_{b \rightarrow \infty} \mathbb{E}|X| \mathbb{1}_{\{|X| > b\}} = 0. \quad (3.7)$$

Definition 3.5: Uniform Integrability

A collection $\mathcal{K}$ of random variables is said to be *uniformly integrable* (UI) if

$$\lim_{b \rightarrow \infty} \sup_{X \in \mathcal{K}} \mathbb{E}|X| \mathbb{1}_{\{|X| > b\}} = 0.$$

Proposition 3.8: Conditions for Uniform Integrability

Let $\mathcal{K}$ be a collection of random variables.

1. If $\mathcal{K}$ is a finite collection of integrable random variables, then it is UI.
2. If $|X| \leq Y$ for all $X \in \mathcal{K}$ and some integrable Y , then $\mathcal{K}$ is UI.

3. $\mathcal{K}$ is UI if and only if there is an increasing convex function f such that

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x} = \infty \quad \text{and} \quad \sup_{X \in \mathcal{K}} \mathbb{E}f(|X|) < \infty.$$

4. $\mathcal{K}$ is UI if $\sup_{X \in \mathcal{K}} \mathbb{E}|X|^{1+\varepsilon} < \infty$ for some $\varepsilon > 0$.

5. $\mathcal{K}$ is UI if and only if

(a) $\sup_{X \in \mathcal{K}} \mathbb{E}|X| < \infty$, and

(b) for every $\varepsilon > 0$ there is a $\delta > 0$ such that for every event H :

$$\mathbb{P}(H) \leq \delta \implies \sup_{X \in \mathcal{K}} \mathbb{E}|X| \mathbb{1}_H \leq \varepsilon. \quad (3.8)$$

Theorem 3.4: L^1 Convergence and Uniform Integrability

A sequence (X_n) of real-valued integrable random variables converges in L^1 if and only if it converges in probability and is uniformly integrable.

Theorem 3.5: Weak Law of Large Numbers

If $X_1, X_2, \dots$ are iid with finite expectation μ , then for all $\varepsilon > 0$

$$\lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) = 0.$$

In other words, $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$.

Theorem 3.6: Strong Law of Large Numbers

If $X_1, X_2, \dots$ are iid with expectation μ and $\mathbb{E}X^2 < \infty$, then for all $\varepsilon > 0$

$$\lim_{m \rightarrow \infty} \mathbb{P} \left(\sup_{n \geq m} |\bar{X}_n - \mu| > \varepsilon \right) = 0.$$

In other words, $\bar{X}_n \xrightarrow{\text{a.s.}} \mu$.

Theorem 3.7: Central Limit Theorem

If $X_1, X_2, \dots$ are iid with finite expectation μ and finite variance σ^2 , then for all $x \in \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq x \right) = \Phi(x),$$

where Φ is the cdf of the standard normal distribution. In other words, $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} \mathcal{N}(0, 1)$.

4 Conditioning

Definition 4.1: Conditional Expectation

The conditional expectation of X given $\mathcal{F}$, denoted $\mathbb{E}_{\mathcal{F}}X$, is defined in two steps:

1. For $X \geq 0$, $\mathbb{E}_{\mathcal{F}}X$ is defined as any random variable $\widehat{X}$ with
 - (a) $\widehat{X} \in \mathcal{F}_+$.
 - (b) $\mathbb{E}VX = \mathbb{E}V\widehat{X}$ for every $V \in \mathcal{F}_+$ (*projection property*).
2. For arbitrary X , if $\mathbb{E}X$ exists (possibly $\pm\infty$), $\mathbb{E}_{\mathcal{F}}X := \mathbb{E}_{\mathcal{F}}X^+ - \mathbb{E}_{\mathcal{F}}X^-$. Otherwise, the conditional expectation is left undefined.

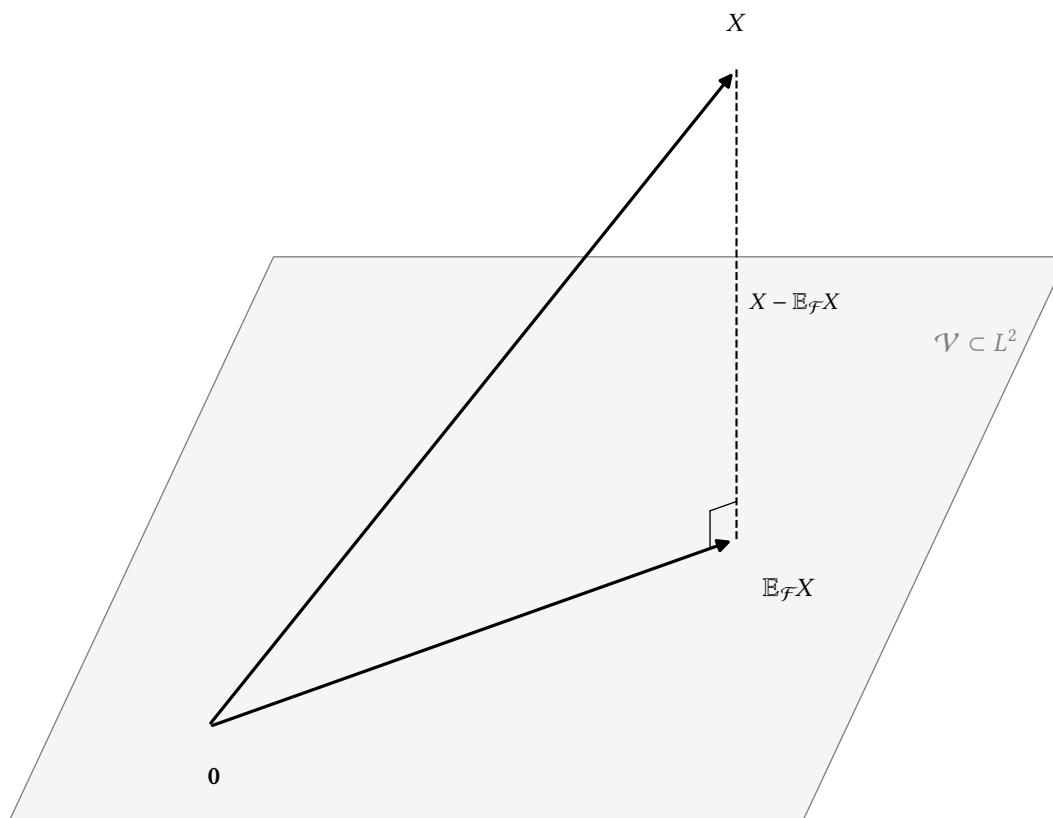


Figure 1: Geometric interpretation of Conditional Expectation $\mathbb{E}_{\mathcal{F}}X$ as an orthogonal projection.

Theorem 4.1: Existence and Uniqueness of the Conditional Expectation

Let $X \geq 0$ be a random variable and let $\mathcal{F}$ be a sub- σ -algebra of $\mathcal{H}$. Then, $\mathbb{E}_{\mathcal{F}}X$ exists and is unique up to equivalence.

Theorem 4.2: Additional Properties of the Conditional Expectation

Let $\mathcal{F}$ and $\mathcal{G}$ be sub- σ -algebras of $\mathcal{H}$. Let W and X be random variables such that $\mathbb{E}X$ and $\mathbb{E}WX$ exist. Then, the following hold:

1. (*Taking out what is known*): If $W \in \mathcal{F}$, then $\mathbb{E}_{\mathcal{F}}WX = W\mathbb{E}_{\mathcal{F}}X$.

2. (Repeated conditioning): If $\mathcal{F} \subseteq \mathcal{G}$, then

$$\mathbb{E}_{\mathcal{F}}\mathbb{E}_{\mathcal{G}}X = \mathbb{E}_{\mathcal{G}}\mathbb{E}_{\mathcal{F}}X = \mathbb{E}_{\mathcal{F}}X. \quad (4.1)$$

Theorem 4.3: Conditional Expectation in L^2

For $X \in L^2$, the conditional expectation $\mathbb{E}_{\mathcal{F}}X$ is the orthogonal projection of X onto the subspace $\mathcal{V}$ of $\mathcal{F}$ -measurable random variables in L^2 .

Definition 4.2: Conditional Probability for Events

Let A and B be two events with $\mathbb{P}(B) > 0$. The *conditional probability* that A occurs given that B occurs is denoted by

$$\mathbb{P}(A \mid B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}. \quad (4.2)$$

Theorem 4.4: Properties Involving Conditional Probabilities

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and $A, A_1, A_2, \dots, A_n, B_1, B_2, \dots$ events such that $\mathbb{P}(A_1 \cap \dots \cap A_n) > 0$ and $\{B_i\}$ form a partition of Ω , with $\mathbb{P}(B_i) > 0$ for all i . Then, the following hold:

1. *Law of total probability*: $\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)$.

2. *Bayes' rule*: $\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(B_j)\mathbb{P}(A \mid B_j)}{\sum_i \mathbb{P}(B_i)\mathbb{P}(A \mid B_i)}$.

3. *Chain or product rule*:

$$\mathbb{P}(\cap_{i=1}^n A_i) = \mathbb{P}(A_1)\mathbb{P}(A_2 \mid A_1) \cdots \mathbb{P}(A_n \mid A_1 \cap \dots \cap A_{n-1}).$$

Definition 4.3: Conditional Distribution

Let Y be a random variable on the probability space $(\Omega, \mathcal{H}, \mathbb{P})$ taking values in the measurable space $(E, \mathcal{E})$. Let $\mathcal{F}$ be a sub- σ -algebra of $\mathcal{H}$. The *conditional distribution of Y given*

$\mathcal{F}$ is any probability transition kernel L from $(\Omega, \mathcal{F})$ to $(E, \mathcal{E})$ such that

$$\mathbb{P}_{\mathcal{F}}(Y \in B) = L(\cdot, B), \quad B \in \mathcal{E}. \quad (4.3)$$

Theorem 4.5: Existence of the Conditional Distribution

For every standard measurable space $(E, \mathcal{E})$ there exists a version of the conditional distribution of Y given $\mathcal{F}$.

Theorem 4.6: Regular Version of a Conditional Probability

Let $(\Omega, \mathcal{H})$ be a standard measurable space and $\mathcal{F}$ a sub- σ -algebra of $\mathcal{H}$. Then, there *exists* a regular version of the conditional probability $\mathbb{P}_{\mathcal{F}}$.

$$\pi(dx, dy) = \mu(dx)K(x, dy). \quad (4.4)$$

Theorem 4.7: Conditional Distribution

If π is of the form [eq. \(4.4\)](#), then the kernel L defined by

$$L(\omega, B) := K(X(\omega), B), \quad \omega \in \Omega, B \in \mathcal{E} \quad (4.5)$$

is a version of the conditional distribution of Y given X .

Theorem 4.8: Disintegration

Let π be a probability measure on the product space $(D \times E, \mathcal{D} \otimes \mathcal{E})$, where $(E, \mathcal{E})$ is a standard measurable space. Then, there exists a probability measure μ on $(D, \mathcal{D})$ and a transition kernel K from $(D, \mathcal{D})$ to $(E, \mathcal{E})$ such that

$$\pi(dx, dy) = \mu(dx)K(x, dy), \quad x \in D, y \in E. \quad (4.6)$$

Definition 4.4: Conditional Independence

Let $\mathcal{F}, \mathcal{F}_1, \dots, \mathcal{F}_n$ be sub- σ -algebras of $\mathcal{H}$. The $\{\mathcal{F}_i\}$ are said to be *conditionally independent* given $\mathcal{F}$ if

$$\mathbb{E}_{\mathcal{F}} V_1 \dots V_n = \mathbb{E}_{\mathcal{F}} V_1 \dots \mathbb{E}_{\mathcal{F}} V_n \quad (4.7)$$

for all positive random variables $V_1, \dots, V_n$ in $\mathcal{F}_1, \dots, \mathcal{F}_n$, respectively.

Proposition 4.1: Conditional Independence

$\mathcal{F}_1$ and $\mathcal{F}_2$ are conditionally independent given $\mathcal{F}$ if and only if

$$\mathbb{E}_{\mathcal{F} \vee \mathcal{F}_1} V_2 = \mathbb{E}_{\mathcal{F}} V_2 \quad \text{for all positive } V_2 \in \mathcal{F}_2. \quad (4.8)$$

$$\mathbb{P}(B_0 \times \dots \times B_n \times E_{n+1} \times \dots) = \pi_n(B_0 \times \dots \times B_n), \quad B_i \in \mathcal{E}_i, i = 0, \dots, n. \quad (4.9)$$

Theorem 4.9: Ionescu–Tulcea’s Existence Theorem

There exists a unique probability measure $\mathbb{P}$ on $(\Omega, \mathcal{H})$ such that [eq. \(4.9\)](#) holds.

Theorem 4.10: Kolmogorov’s Extension Theorem

Let $(E, \mathcal{E})$ be a standard measurable space and K an index set. For each finite subset J of K , let π_J be a probability measure on the product space $(E^J, \mathcal{E}^J)$. If the (π_J) are *consistent* in the sense that

$$\pi_I = \pi_J \circ \text{proj}_{JI}^{-1}, \quad I \subseteq J, \quad |J| < \infty,$$

then there exists a unique probability measure $\mathbb{P}$ on $(\Omega, \mathcal{H})$ and a stochastic process X taking values in $(E, \mathcal{E})^K$ such that, with $X_J := (X_j)_{j \in J}$:

$$\mathbb{P}(X_J \in A) = \pi_J(A), \quad A \in \mathcal{E}^J, \quad |J| < \infty.$$

Definition 4.5: Hölder Continuity

A function $f : \mathbb{R}_+ \rightarrow \mathbb{R}$ is said to be *Hölder continuous* of order $\alpha > 0$ on $B \subseteq \mathbb{R}_+$ if there is a constant k such that

$$|f(t) - f(s)| \leq k|t - s|^\alpha, \quad s, t \in B.$$

Theorem 4.11: Hölder Continuous Modification

Let $X := (X_t, t \in [0, 1])$ be a stochastic process with state space $\mathbb{R}$. If there exist constants c, p, q in $(0, \infty)$ such that

$$\mathbb{E}|X_t - X_s|^p \leq c|t - s|^{1+q}, \quad s, t \in [0, 1], \quad (4.10)$$

then for every $\alpha \in (0, q/p)$ there is a modification $\tilde{X}$ of X that is almost surely Hölder continuous of order α on $[0, 1]$.

Definition 4.6: Adaptedness

A stochastic process $X := (X_t, t \in \mathbb{T})$ taking values in some measurable space $(E, \mathcal{E})$ is said to be *adapted* to a filtration $\mathcal{F} := (\mathcal{F}_t, t \in \mathbb{T})$ if for all $t \in \mathbb{T}$, X_t is $\mathcal{F}_t/\mathcal{E}$ -measurable.

Definition 4.7: Markov Property

X is said to be *Markovian* relative to $\mathcal{F}$ if for every t , the past $\mathcal{F}_t$ and the future $\mathcal{G}_t$ are conditionally independent, given the present state X_t .

Definition 4.8: Markovian Transition Function

A *Markovian transition function* on $(E, \mathcal{E})$ is a family $(P_{t,u}, t, u \in \mathbb{T}, t < u)$ of probability transition kernels on $(E, \mathcal{E})$ that satisfy the *Chapman–Kolmogorov equations*:

$$P_{s,t}P_{t,u} = P_{s,u}, \quad s < t \leq u. \quad (4.11)$$

Theorem 4.12: Generator of a Markov Jump Process

The infinitesimal generator of a Markov jump process with Lévy kernel L is given by

$$(Gf)(x) = \int_{\mathbb{R}^d} L(x, dy) [f(y) - f(x)], \quad f \in \mathcal{B}^d(\text{bounded}). \quad (4.12)$$

Proposition 4.2: Poisson Integrals

Let N be a Poisson random measure on $\mathbb{R}_+ \times E$ with mean measure μ satisfying $\mu(\{0\} \times E) = 0$. Let $R := (R(t, z), t \geq 0, z \in E)$ be a positive stochastic process in $\mathcal{F}^P \otimes \mathcal{E}$. Then,

$$\mathbb{E}NR = \mathbb{E}_\mu R. \quad (4.13)$$

5 Martingales

Definition 5.1: Stopping Time

A random time $T : \Omega \rightarrow \bar{\mathbb{T}}$ is called a *stopping time* of $\mathcal{F}$ if

$$\{T \leq t\} \in \mathcal{F}_t \quad \text{for each } t \in \mathbb{T}.$$

Definition 5.2: Past until T

Let $\mathcal{F}$ be a filtration on $\bar{\mathbb{T}}$, and let T be a stopping time of it. The σ -algebra of *past information until T* is defined as

$$\mathcal{F}_T := \{H \in \mathcal{H} : H \cap \{T \leq t\} \in \mathcal{F}_t \text{ for every } t \in \bar{\mathbb{T}}\}.$$

Theorem 5.1: $\mathcal{F}_T$ Random Variables

A random variable V belongs to $\mathcal{F}_T$ if and only if

$$X_t := V \mathbb{1}_{\{T \leq t\}} \in \mathcal{F}_t \text{ for every } t \in \bar{\mathbb{T}}.$$

Definition 5.3: (Sub/Super)Martingale

Let $X := (X_t, t \in \mathbb{T})$ be a real-valued stochastic process that is adapted to a filtration $\mathcal{F}$, and for which each X_t is integrable. The process X is called an $\mathcal{F}$ -(sub/super)martingale if for all $t > s$:

$$\begin{aligned} \mathbb{E}_s[X_t - X_s] &\geq 0 \quad (\text{submartingale}), \\ \mathbb{E}_s[X_t - X_s] &= 0 \quad (\text{martingale}), \\ \mathbb{E}_s[X_t - X_s] &\leq 0 \quad (\text{supermartingale}). \end{aligned} \tag{5.1}$$

Theorem 5.2: Properties of (Sub)Martingales

1. When $\mathbb{T} = \mathbb{N}$, the martingale equality $\mathbb{E}_s[X_t - X_s] = 0$ holds if and only if $\mathbb{E}_k[X_{k+1} - X_k] = 0, k \in \mathbb{N}$.
2. If X and Y are $\mathcal{F}$ -submartingales, then so is $aX + bY$ for $a, b \geq 0$.
3. If X and Y are $\mathcal{F}$ -submartingales, then so is $X \vee Y$.
4. Let f be a convex function on $\mathbb{R}$ and X an $\mathcal{F}$ -martingale. If $f(X_t)$ is integrable for all t , then $f(X) := (f(X_t))$ is an $\mathcal{F}$ -submartingale.

Definition 5.4: Predictable Process

Let $\mathcal{F} := (\mathcal{F}_t, t \geq 0)$ be a filtration. A real-valued stochastic process F is said to be $\mathcal{F}$ -predictable (or simply *predictable*) if it is measurable with respect to the σ -algebra

$$\mathcal{F}^P := \sigma(H \times (a, b] : a, b \in \mathbb{R}_+, H \in \mathcal{F}_a) \vee \sigma(H \times \{0\} : H \in \mathcal{F}_0). \quad (5.2)$$

Definition 5.5: Discrete Predictable Process

Given a filtration $\mathcal{F} := (\mathcal{F}_n, n \in \mathbb{N})$, a process $F := (F_n, n \in \mathbb{N})$ is said to be *predictable* if $F_0 \in \mathcal{F}_0$ and $F_n \in \mathcal{F}_{n-1}, n = 1, 2, \dots$

Theorem 5.3: Integral Transform of a Martingale

Let F be a bounded predictable process and X a martingale adapted to a filtration $\mathcal{F}$. Then, the integral transform $Z_n := \int_0^n F_k dX_k, n \in \mathbb{N}$ is an $\mathcal{F}$ -martingale as well.

Definition 5.6: Stopped Stochastic Process

Let $X := (X_t, t \in \mathbb{T})$ be a stochastic process and T a random time with values in $\overline{\mathbb{T}}$. By the process X *stopped at time T* we mean the process $(X_t^\dagger, t \in \mathbb{T})$ defined by

$$X_t^\dagger(\omega) := X_{T(\omega) \wedge t}(\omega) = \begin{cases} X_t(\omega) & \text{if } t \leq T(\omega), \\ X_{T(\omega)}(\omega) & \text{if } t > T(\omega). \end{cases} \quad (5.3)$$

Theorem 5.4: Stopped Martingale

Let $M := (M_n, n \in \mathbb{N})$ be a martingale and T a stopping time with values in $\overline{\mathbb{N}}$. Then, the process M stopped at time T is again a martingale.

Proposition 5.1: Stochastic Integral for Predictable Integrands

Let N be defined as above. Then, for any positive $\mathcal{F}$ -predictable process (F_t) ,

$$\mathbb{E} \int_{\mathbb{R}_+} F_t dN_t = \mathbb{E} \int_{\mathbb{R}_+} F_t d\nu_t. \quad (5.4)$$

Theorem 5.5: Integral Transform of a Continuous Martingale

Let $\overline{N}$ be an $\mathcal{F}$ -martingale, as defined above. For any bounded $\mathcal{F}$ -predictable process F , the integral transform

$$Z_t := \int_0^t F_s d\overline{N}_s, \quad t \geq 0, \quad (5.5)$$

is a martingale.

Theorem 5.6: Doob's Stopping Theorem

A stochastic process $M := (M_n, n \in \mathbb{N})$ is a martingale if and only if for every pair of *bounded* stopping times S and T with $S \leq T$ the random variables M_S and M_T are integrable, and

$$\mathbb{E}_S(M_T - M_S) = 0. \quad (5.6)$$

Lemma 5.1: Maximum Inequality for Positive Submartingales

Let $(Y_k, k \in \mathbb{N})$ be a positive submartingale and define $Y_n^* := \max_{k \leq n} Y_k$. Then, for any $b > 0$,

$$b\mathbb{P}(Y_n^* \geq b) \leq \mathbb{E}[Y_n \mathbb{1}_{\{Y_n^* \geq b\}}] \leq \mathbb{E}Y_n. \quad (5.7)$$

Theorem 5.7: Doob–Kolmogorov Inequality

Let $M := (M_k)$ be a martingale in L^p for some $p \in [1, \infty)$. Then, for $b > 0$,

$$\mathbb{P}(\max_{k \leq n} |M_k| > b) \leq \frac{\mathbb{E}|M_n|^p}{b^p}. \quad (5.8)$$

Theorem 5.8: Expected Upcrossings of a Submartingale

Suppose that X is a submartingale. Then, the expected number of upcrossings $U_n(a, b)$ of (a, b) satisfies:

$$(b - a)\mathbb{E}U_n(a, b) \leq \mathbb{E}[(X_n - a)^+ - (X_0 - a)^+]. \quad (5.9)$$

Theorem 5.9: (Sub)Martingale Convergence

Let $X := (X_n, n \in \mathbb{N})$ be a submartingale. Suppose that $\sup_n \mathbb{E}X_n^+ < \infty$. Then, X converges almost surely to an integrable random variable X_∞ .

Theorem 5.10: (Sub)Martingale Convergence and Uniform Integrability

Let $X := (X_n, n \in \mathbb{N})$ be a submartingale. Then, X converges almost surely and in L^1 if and only if it is uniformly integrable. Moreover, if X converges, setting $X_\infty := \lim X_n$ extends X to a submartingale $\bar{X} := (X_n, n \in \bar{\mathbb{N}})$.

Theorem 5.11: Uniformly Integrable Martingale

A process $M := (M_n, n \in \mathbb{N})$ is a UI martingale if and only if there exists an integrable random variable Z such that

$$M_n = \mathbb{E}_n Z, \quad n \in \mathbb{N}. \quad (5.10)$$

Moreover, then M converges almost surely and in L^1 to an integrable random variable

$$M_\infty := \mathbb{E}_\infty Z, \quad (5.11)$$

and $(M_n, n \in \bar{\mathbb{N}})$ is again a UI martingale. If $Z \in \mathcal{F}_\infty$, then $M_\infty = Z$.

Theorem 5.12: Convergence for Martingales in Reversed Time

For $\mathbb{T} = \{\dots, -2, -1, 0\}$, let $M := (M_n, n \in \mathbb{T})$ be a martingale with respect to the filtration $(\mathcal{F}_n, n \in \mathbb{T})$. Then, M is UI and, moreover, as $n \downarrow -\infty$ it converges almost surely and in L^1 to the integrable random variable $M_{-\infty} := \mathbb{E}_{-\infty} M_0$, where $\mathcal{F}_{-\infty} := \bigcap_{n \in \mathbb{T}} \mathcal{F}_n$.

Theorem 5.13: Stopping for Uniformly Integrable Martingales

If $M := (M_n, n \in \bar{\mathbb{N}})$ is a uniformly integrable martingale, then for *every* pair of stopping times S and T with $S \leq T$ it holds that

$$\mathbb{E}_S M_T = M_S$$

In particular,

$$\mathbb{E} M_T = \mathbb{E} M_0.$$

Theorem 5.14: Kolmogorov's 0–1 Law

If $H \in \mathcal{T}$, then $\mathbb{P}(H)$ is either 0 or 1. Consequently, a numerical random variable Z that is $\mathcal{T}$ -measurable must be almost surely constant.

Definition 5.7: Separable σ -Algebra

A σ -algebra on Ω is said to be *separable* if it is generated by a sequence (H_n) of subsets of Ω .

Theorem 5.15: Radon–Nikodym

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space, where $\mathcal{H}$ is separable, and let Q be a finite measure on $(\Omega, \mathcal{H})$ with $Q \ll \mathbb{P}$. Then, there exists a positive random variable $Z \in \mathcal{H}$, written $Z = dQ/d\mathbb{P}$, such that $Q = Z\mathbb{P}$; that is,

$$Q(H) = \int_H Z d\mathbb{P} = \mathbb{E}1_H Z, \quad H \in \mathcal{H}. \quad (5.12)$$

Definition 5.8: Right-continuous Filtration

A filtration $\mathcal{F} := (\mathcal{F}_t)$ is said to be *right-continuous* if for all t ,

$$\bigcap_{\varepsilon > 0} \mathcal{F}_{t+\varepsilon} = \mathcal{F}_t.$$

Definition 5.9: Local Martingale

A process $(Z_t, t \geq 0)$ is called a *local martingale* if there exists a sequence of stopping times $(T_n, n \in \mathbb{N})$, called a *localizing sequence*, such that $T_n \rightarrow \infty$ almost surely and each stopped process $(Z_{t \wedge T_n}, t \geq 0)$ is a martingale.

Theorem 5.16: Properties of Local Martingales

Let $Z := (Z_t, t \geq 0)$ be a local martingale with localizing sequence $(T_n, n \in \mathbb{N})$. Then, the following hold:

1. If $Z \geq 0$ and $\mathbb{E}Z_0 < \infty$, then Z is a supermartingale.
2. If for each $t \geq 0$ the sequence $(Z_{t \wedge T_n}, n \in \mathbb{N})$ is uniformly integrable, then Z is a martingale.

Definition 5.10: Doob Martingale

Let ζ be a stopping time. A process $M := (M_t, t \in \mathbb{T})$ is called a *Doob martingale* on $[0, \zeta]$ if for all stopping times $0 \leq S \leq T \leq \zeta$ it holds that

$$\mathbb{E}_S M_T = M_S, \quad (5.13)$$

where M_S and M_T are integrable.

Theorem 5.17: Characterization of a Doob Martingale

Let ζ be a stopping time. A process $M := (M_t, t \in \mathbb{T})$ is a Doob martingale on $[0, \zeta]$ if and only if for every stopping time $T \leq \zeta$,

$$\mathbb{E} M_T = \mathbb{E} M_0. \quad (5.14)$$

Theorem 5.18: Doob's Stopping Theorem for Continuous Martingales

A martingale $M := (M_t, t \in \mathbb{R}_+)$ is a Doob martingale on $[0, b]$ for every $b \in \mathbb{R}_+$ and satisfies

$$\mathbb{E}_S M_T = M_S \quad (5.15)$$

for every pair (S, T) of bounded stopping times with $S \leq T$.

Proposition 5.2: Doob's Maximal Inequality

Let $X := (X_t, t \geq 0)$ be a submartingale that is positive and continuous. Then, for $p \geq 1$ and $b > 0$,

$$\mathbb{P}(\max_{0 \leq s \leq t} X_s \geq b) \leq \frac{\mathbb{E} X_t^p}{b^p}. \quad (5.16)$$

Proposition 5.3: Doob's Norm Inequality

Let $M := (M_t, t \geq 0)$ be a continuous martingale in L^p for some $p > 1$ and let $Z_t := \max_{s \leq t} |M_s|$. Then,

$$\mathbb{E}Z_t^p \leq q^p \mathbb{E}|M_t|^p, \text{ with } q := p/(p-1). \quad (5.17)$$

Lemma 5.2: Continuous Martingales with Finite Variation

Let $X := (X_t, t \geq 0)$ be a martingale with continuous paths and total variation $V_t < \infty$ on each interval $[0, t]$. Then, X is almost surely constant.

Proposition 5.4: Doob Martingale on $\overline{\mathbb{R}}_+$

A process M is a Doob martingale on $\overline{\mathbb{R}}_+$ if and only if it is a martingale on $\overline{\mathbb{R}}_+$. If so, then

$$\mathbb{E}_S M_T = M_S \quad (5.18)$$

for arbitrary stopping times with $S \leq T$.

Theorem 5.19: Extension of UI Martingales

A martingale M on $\mathbb{R}_+$ can be extended to a Doob martingale $\overline{M}$ on $\overline{\mathbb{R}}_+$ if and only if it is uniformly integrable; that is, if and only if

$$M_t = \mathbb{E}_t Z, \quad t \in \mathbb{R}_+ \quad (5.19)$$

for some integrable random variable Z . Moreover, then, it converges almost surely and in L^1 to an integrable random variable M_∞ and can be extended to a UI martingale on $\overline{\mathbb{R}}_+$.

Proposition 5.5: Sufficient Condition for Being Doob

Let M be a martingale on $\mathbb{R}_+$ and ζ a stopping time. If, almost surely,

$$\sup_{t \in [0, \zeta] \cap \mathbb{R}_+} |M_t| \leq Z, \tag{5.20}$$

where $\mathbb{E}Z < \infty$ (i.e., M is dominated by an integrable random variable Z), then $M_\zeta = \lim M_{\zeta \wedge t}$ exists and is integrable. Moreover, M is a Doob martingale on $[0, \zeta]$.

6 Wiener and Brownian Motion Processes

We defined the Wiener process as a Gaussian process as follows:

Definition 6.1: Wiener Process

The *Wiener process* $(W_t, t \geq 0)$ is a zero-mean Gaussian process with continuous sample paths and covariance function

$$\gamma_{s,t} = \min\{s, t\} =: s \wedge t, \quad s, t \geq 0.$$

A Wiener process is a Lévy process with *continuous* sample paths, as shown in the following result, which can also serve as an equivalent definition of the Wiener process.

Theorem 6.1: Wiener Process as a Lévy Process

$W = (W_t)_{t \geq 0}$ is a Wiener process if and only if the following three properties hold:

1. (*Independent increments*): For any $t_1 < t_2 \leq t_3 < t_4$, the increments $W_{t_4} - W_{t_3}$ and $W_{t_2} - W_{t_1}$ are independent.
2. (*Gaussian stationarity*): For all $t, u \geq 0$, $W_{t+u} - W_t \sim \mathcal{N}(0, u)$.
3. (*Continuity of paths*): W has continuous paths, with $W_0 = 0$.

The following algorithm, [algorithm 1](#) uses the same ideas as in the previous proof to simulate the Wiener process at specific time points $t_1, \dots, t_n$

Algorithm 1 Simulating a Wiener Process

- 1: Let $0 = t_0 < t_1 < \dots < t_n$ be the times at which the process needs to be simulated.
 - 2: Simulate $Z_1, \dots, Z_n \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$.
 - 3: Return $W_0 := 0$ and $W_{t_k} := \sum_{i=1}^k Z_i \sqrt{t_i - t_{i-1}}$, $k = 1, \dots, n$.
-

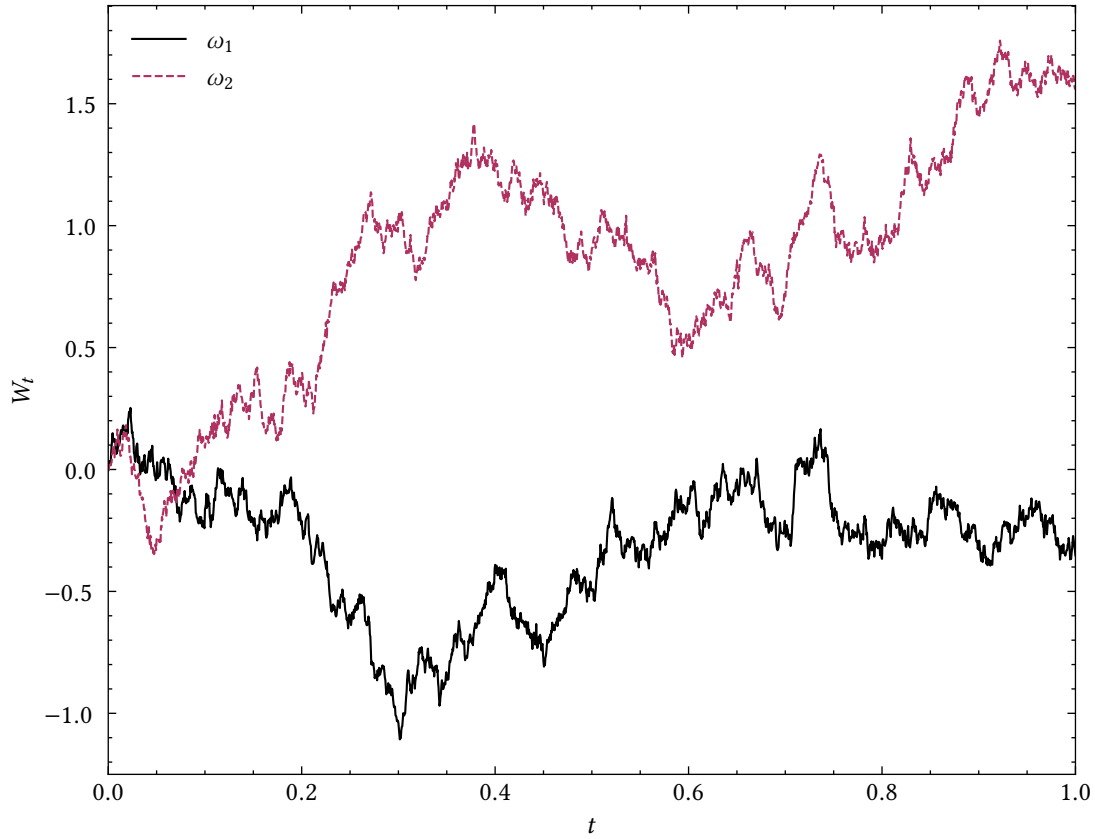


Figure 2: Two realisations of the Wiener process on the interval $[0, 1]$.

Definition 6.2: Brownian Motion

A stochastic process $(B_t, t \geq 0)$ satisfying

$$B_t = B_0 + \mu t + \sigma W_t, \quad t \geq 0,$$

where (W_t) is a Wiener process independent of B_0 , is called a *Brownian motion with drift μ and diffusion coefficient σ* .

A standard Brownian motion is one where $\mu = 0$ and $\sigma = 1$. The only difference between a Wiener process and a Brownian motion is its (random) starting position.

Algorithm 2 Simulating a Brownian Motion

- 1: Simulate the starting position B_0 .
 - 2: Simulate outcomes $W_{t_1}, \dots, W_{t_n}$ of a Wiener process at times $t_1, \dots, t_n$.
 - 3: Return $B_{t_i} := B_0 + \mu t_i + \sigma W_{t_i}$, $i = 1, \dots, n$ as the outcomes of the Brownian motion at times $t_1, \dots, t_n$.
-

Multidimensional Brownian motions are likewise obtained from an affine transformation $B_t = B_0 + \mu t + \sigma W_t$ of a multidimensional Wiener process, where σ is a matrix.

Definition 6.3: Multidimensional Wiener Process

Let $(W_{t,i}, t \geq 0)$, $i = 1, \dots, d$ be independent Wiener processes and let $W_t := [W_{t,1}, \dots, W_{t,d}]^\top$. The process $(W_t, t \geq 0)$ is called a d -dimensional Wiener process.

If (W_t) is a d -dimensional Wiener process, then W_t has a $\mathcal{N}(\mathbf{0}, t\mathbf{I}_d)$ multivariate normal distribution, where $\mathbf{I}_d$ denotes the d -dimensional identity matrix.

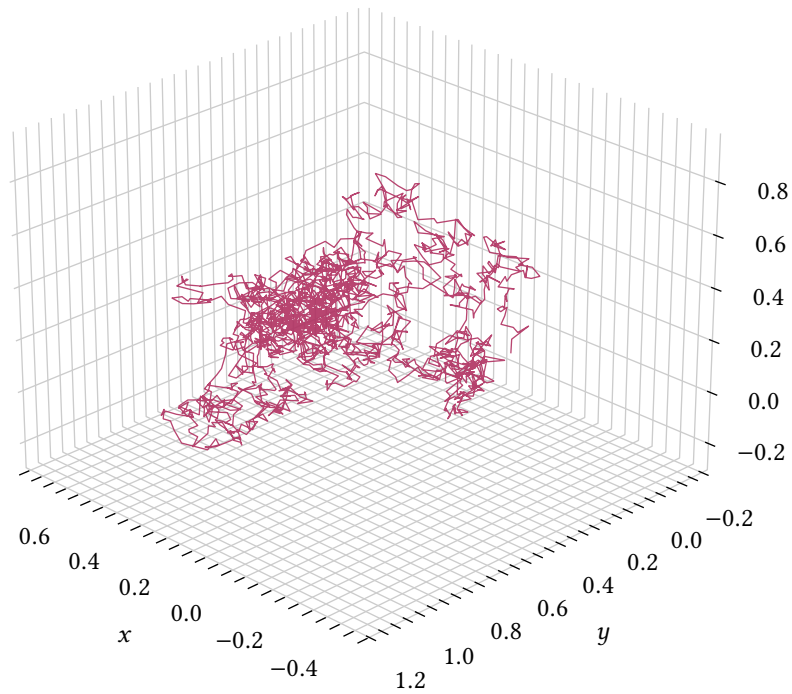


Figure 3: Three-dimensional Wiener process.

Theorem 6.2: Time-reversal and Time-shifting

If $(W_s, s \in [0, t])$ is a Wiener process on $[0, t]$, then so is the time-reversed process $(\widehat{W}_s, s \in [0, t])$ defined by $\widehat{W}_s := W_{t-s} - W_t$. Similarly, if $(W_s, s \geq 0)$ is a Wiener process, then for any $t \geq 0$ the process $(\widetilde{W}_s, s \geq 0)$ defined by $\widetilde{W}_s := W_{t+s} - W_t$ is also a Wiener process.

Theorem 6.3: Scaling

If (W_t) is a Wiener process, then so is (X_t) , with $X_t := W_{at}/\sqrt{a}$, $t \geq 0$ for any $a > 0$.

Lemma 6.1: Probability Bound on the Supremum

For a Wiener process (W_t) , it holds for every $\varepsilon > 0$ that

$$\mathbb{P} \left(\sup_{0 \leq t \leq 1} |W_t| > \varepsilon \right) \leq 2e^{-\varepsilon^2/2}.$$

Theorem 6.4: Reciprocal Time

If (W_t) is a Wiener process, then so is (X_t) , with $X_t := tW_{1/t}$, $t > 0$, $X_0 := 0$.

Theorem 6.5: Existence of the Wiener Process

There exists a probability space $(\Omega, \mathcal{H}, \mathbb{P})$ and a zero-mean Gaussian process $W := (W_t, t \in [0, 1])$ with covariance function $\mathbb{E}W_s W_t = s \wedge t$ and uniformly continuous sample paths.

Theorem 6.6: Right-continuous Filtration

For the Wiener process, the augmentation $(\overline{\mathcal{F}}_t)$ of the natural filtration is right-continuous; that is,

$$\overline{\mathcal{F}}_t = \overline{\mathcal{F}}_t^+ := \bigcap_{\varepsilon > 0} \overline{\mathcal{F}}_{t+\varepsilon}.$$

Theorem 6.7: Wiener Process is Markov w.r.t $\mathcal{F}^+$

Let W be a Wiener process. For every $t \geq 0$, the process $(W_{t+u} - W_t, u \geq 0)$ is independent of $\mathcal{F}_t^+$. Consequently, W is Markovian with respect to $\mathcal{F}^+$.

Theorem 6.8: Strong Markov Property

For any finite stopping time T with respect to $\mathcal{F}^+$, the process $(W_{T+u} - W_T, u \geq 0)$ is a Wiener process independent of $\mathcal{F}_T^+$.

Theorem 6.9: Reflection Principle

Let T be a stopping time. Then, $(\tilde{W}_t, t \geq 0)$, defined by

$$\tilde{W}_t := W_t \mathbb{1}_{\{t \leq T\}} + (2W_T - W_t) \mathbb{1}_{\{t > T\}}, \quad t \geq 0, \quad (6.1)$$

is a Wiener process.

Theorem 6.10: Transition Kernel of the Wiener Process

The Wiener process W admits the transition function

$$P_{t,u}(x, A) = \mathbb{P}(W_u \in A \mid W_t = x) = \int_A dy p_{u-t}(y \mid x),$$

where the *transition density* $p_t(y \mid x)$ is given by

$$p_t(y \mid x) := \frac{1}{\sqrt{2\pi t}} e^{-\frac{(y-x)^2}{2t}}, \quad t \geq 0, \quad x, y \in \mathbb{R}. \quad (6.2)$$

Theorem 6.11: Exponential Martingale

A continuous process $X := (X_t, t \geq 0)$ is a Wiener process if and only if for each $r \in \mathbb{R}$ the process $S := (S_t)$ defined by

$$S_t := e^{rX_t - \frac{1}{2}r^2 t}, \quad t \geq 0, \quad (6.3)$$

is a martingale.

Theorem 6.12: Martingales from Functions of a Brownian Motion

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be twice continuously differentiable and let B be d -dimensional standard Brownian motion. If for all $t > 0$ and $\mathbf{x} \in \mathbb{R}^d$,

$$\mathbb{E}^{\mathbf{x}}|f(\mathbf{B}_t)| < \infty \quad \text{and} \quad \mathbb{E}^{\mathbf{x}} \int_0^t |\Delta f(\mathbf{B}_s)| ds < \infty,$$

then the process $(X_t, t \geq 0)$ defined by

$$X_t := f(\mathbf{B}_t) - \frac{1}{2} \int_0^t \Delta f(\mathbf{B}_s) ds \tag{6.4}$$

is a martingale.

7 Itô Calculus

7.1 Itô Integral

We will work with a Wiener process W on a complete probability space $(\Omega, \mathcal{H}, \mathbb{P})$ with a right-continuous filtration $\mathcal{F} := (\mathcal{F}_s, s \geq 0)$ and such that W is adapted to $\mathcal{F}$. We also use the following definition.

Definition 7.1: Simple Processes

A process $F := (F_s, s \in [0, t])$ is called *simple* if it is of the form

$$F_s := \xi_0 \mathbb{1}_{\{0\}}(s) + \sum_{k=0}^{n-1} \xi_k \mathbb{1}_{(s_k, s_{k+1}]}(s), \quad s \in [0, t], \quad (7.1)$$

where $\{s_0, \dots, s_n : 0 = s_0 < \dots < s_n = t\} =: \Pi_n, n \in \mathbb{N}$, is a segmentation of $[0, t]$ and $\xi_k \in \mathcal{F}_{s_k}$ for all k .

Note that the simple process in (7.1) is right-continuous at $s = 0$ and is left-continuous at $s \in (0, t]$. Figure 4 shows a realisation of a simple process on $[0, 1]$ for the segmentation

$$\{s_0, \dots, s_{10}\} = \{0, 0.07, 0.13, 0.19, 0.41, 0.49, 0.52, 0.67, 0.71, 0.76, 1\}.$$

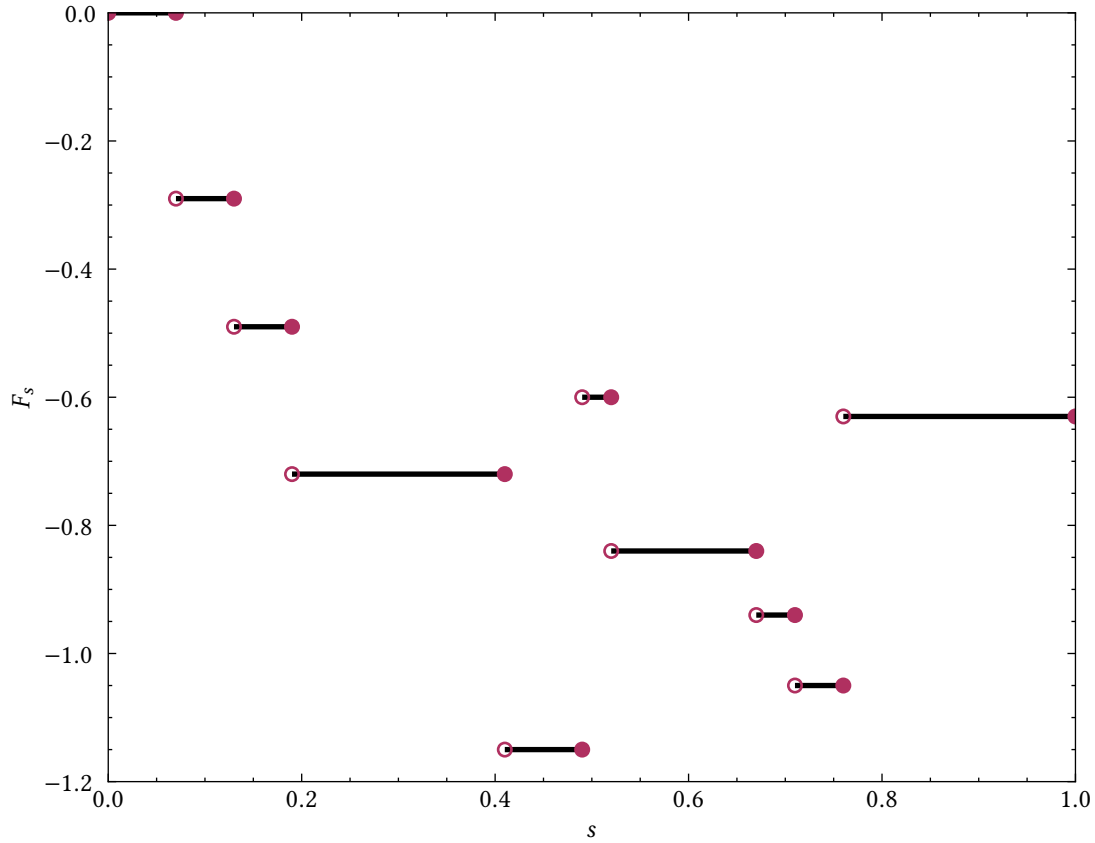


Figure 4: A realisation of a simple process.

Definition 7.2: Itô Integral for Simple Processes

Suppose that F is a simple process of the form [eq. \(7.1\)](#), with $\int_0^t \mathbb{E} F_s^2 ds < \infty$. Then, the *Itô integral* of F with respect to W over $[0, r]$ is defined as

$$\int_0^r F_s dW_s := \sum_{k=0}^{n-1} \xi_k (W_{s_{k+1} \wedge r} - W_{s_k \wedge r}), \quad r \in [0, t]. \quad (7.2)$$

Theorem 7.1: Properties of the Itô Integral

Assume that the Itô integral is well-defined. Then, for $\alpha, \beta \in \mathbb{R}$ and $r \in [0, t]$:

1. (Linearity): $\int_0^r [\alpha F_s + \beta G_s] dW_s = \alpha \int_0^r F_s dW_s + \beta \int_0^r G_s dW_s$.
2. (Zero mean): $\mathbb{E} \int_0^r F_s dW_s = 0$.

3. (Isometry): $\mathbb{E} \left[\int_0^r F_s dW_s \times \int_0^r G_s dW_s \right] = \int_0^r \mathbb{E}[F_s G_s] ds.$

4. (Martingale): $(\int_0^r F_s dW_s, r \in [0, t])$ is an L^2 -martingale.

Definition 7.3: Canonical Approximation

For any $\mathcal{F}$ -adapted process $F := (F_s, s \in [0, t])$, we define its *canonical approximation* as

$$F_s^{(n)} := F_0 \mathbb{1}_{\{0\}}(s) + \sum_{k=0}^{n-1} F_{s_k} \mathbb{1}_{(s_k, s_{k+1}]}(s), \quad s \in [0, t]. \quad (7.3)$$

Theorem 7.2: Properties of Left-continuous Adapted Processes

Suppose that $F := (F_s, s \in [0, t])$ is a process on $(\Omega, \mathcal{H}, \mathbb{P})$ that is adapted to the filtration $\mathcal{F}$ and is left-continuous on $(0, t]$. Then, the following hold:

1. (Predictable): The process F is $\mathcal{F}$ -predictable (i.e., $\mathcal{F}^P$ -measurable).
2. (Progressively Measurable): For each t , the mapping $F : [0, t] \times \Omega \rightarrow \mathbb{R}$ is jointly measurable with respect to $\mathcal{B}_{[0, t]} \otimes \mathcal{F}_t$.
3. (Indistinguishable): If $\tilde{F}$ is a modification of F and almost surely left-continuous, then $\mathbb{P}(\cap_{s \leq t} \{F_s = \tilde{F}_s\}) = 1$.

Definition 7.4: The Class of Processes $\mathcal{P}_t$

Let $\mathcal{P}_t$ be the class of stochastic processes $F := (F_s, s \in [0, t])$ that are left-continuous on $(0, t]$, $\mathcal{F}$ -adapted, and satisfy the norm condition:

$$\|F\|_{\mathcal{P}_t}^2 := \mathbb{E} \int_0^t F_s^2 ds < \infty.$$

Theorem 7.3: Approximating the Class $\mathcal{P}_t$ via Simple Processes

Suppose that $F \in \mathcal{P}_t$. Then, there exists a sequence $(F^{(n)})$ of simple processes of the form [eq. \(7.1\)](#) such that $\text{mesh}(\Pi_n) \rightarrow 0$ implies that $F_s^{(n)} \xrightarrow{\text{a.s.}} F_s$ for all $s \in [0, t]$ and

$$\lim_{n \rightarrow \infty} \|F^{(n)} - F\|_{\mathcal{P}_t} = 0. \quad (7.4)$$

Proposition 7.1: Deterministic Integrands and Gaussian Processes

Let $f : \mathbb{R}_+ \times \mathbb{R}_+ \rightarrow \mathbb{R}$ be left-continuous in the first argument and such that $\|f(\cdot, t)\|_{\mathcal{P}_t}^2 = \int_0^t f^2(s, t) ds < \infty$. Then, the stochastic integral process

$$Z_t := \int_0^t f(s, t) dW_s, \quad t \geq 0, \quad (7.5)$$

is a zero-mean Gaussian process with covariance function

$$\text{Cov}(Z_{t_1}, Z_{t_2}) = \int_0^{t_1 \wedge t_2} f(s, t_1) f(s, t_2) ds, \quad t_1, t_2 \geq 0. \quad (7.6)$$

$$\lim_{m \rightarrow \infty} \sup_{s \in [0, t]} \mathbb{E} F_s^2 \mathbf{1}_{\{F_s^2 > m\}} = 0. \quad (7.7)$$

Lemma 7.1: Quadratic Variation Integral

Suppose that $F \in \mathcal{P}_t$ satisfies the condition [eq. \(7.7\)](#). Then, as $\text{mesh}(\Pi_n) \rightarrow 0$:

$$\sum_{k=0}^{n-1} F_{s_k} (W_{s_{k+1}} - W_{s_k})^2 \xrightarrow{L^2} \int_0^t F_s ds. \quad (7.8)$$

Theorem 7.4: Enlarging the Class $\mathcal{P}_t$ via Localization

Let $F := (F_s, s \in [0, t])$ be an $\mathcal{F}$ -adapted process, left-continuous on $(0, t]$, and such that (7.34) holds. Then, there exists a localizing sequence (T_n) such that the stopped process $(\int_0^{r \wedge T_n} F_s dW_s, r \in [0, t])$ is a square-integrable martingale. In addition, the Itô integral of

F with respect to W on $[0, t]$ is defined as the almost sure limit:

$$\int_0^t F_s dW_s := \lim_{n \rightarrow \infty} \int_0^{t \wedge T_n} F_s dW_s. \quad (7.9)$$

Proposition 7.2: Continuous Modification and Stopping Times

If $F \in \mathcal{P}_t$ and $T \leq t$ is a stopping time with respect to $\mathcal{F}$, then almost surely:

$$\int_0^T F_s dW_s = \int_0^t \mathbb{1}_{[0, T]}(s) F_s dW_s. \quad (7.10)$$

Theorem 7.5: Continuous Modification of the Itô Integral

Suppose that $F \in \mathcal{P}_t$ and $(F^{(n)})$ is a sequence of simple processes satisfying (7.17). Denote the limit $\int_0^r F_s^{(n)} dW_s =: Z_r^{(n)} \xrightarrow{L^2} Z_r$ as the Itô integral on $[0, r]$ for $r \leq t$. Then, there exists a modification $\tilde{Z}$ of $(Z_r, r \in [0, t])$ such that the mapping $r \mapsto \tilde{Z}_r$ is almost surely continuous. Moreover, this modification is unique up to indistinguishability.

Theorem 7.6: Existence of the Itô Integral with Integrand $F \in \mathcal{P}_t$

Suppose that $F \in \mathcal{P}_t$ and $(F^{(n)})$ is a sequence of simple processes satisfying (7.17). Then, the following limit exists in L^2 norm:

$$\int_0^t F_s dW_s := \lim_{n \rightarrow \infty} \int_0^t F_s^{(n)} dW_s, \quad r \in [0, t], \quad (7.11)$$

and it defines the *Itô integral* of F with respect to W on $[0, t]$.

7.2 Itô Calculus

Theorem 7.7: Itô Formula for the Wiener Process

Let W be a Wiener process and $f : \mathbb{R} \rightarrow \mathbb{R}$ be twice continuously differentiable with derivatives f' and f'' . Then, almost surely

$$f(W_t) - f(W_0) = \int_0^t f'(W_s) dW_s + \frac{1}{2} \int_0^t f''(W_s) ds.$$

Definition 7.5: Integral With Respect To An Itô Process

Let X be an Itô(μ, σ) process and F a left-continuous $\mathcal{F}$ -adapted process such that almost surely $\int_0^t (|F_s \mu_s| + (F_s \sigma_s)^2) ds < \infty$ for all t . Then, the integral of F with respect to X is defined as the Itô($F\mu, F\sigma$) process with:

$$\int_0^t F_s dX_s := \int_0^t F_s \mu_s ds + \int_0^t F_s \sigma_s dW_s. \quad (7.12)$$

Definition 7.6: Quadratic Variation On $[0, t]$

Let $\Pi_n := \{s_k, k = 0, \dots, n\}$ be a segmentation of $[0, t]$ and let $(X^{(n)})$ be a sequence of simple processes with almost sure limit $X_s := \lim_n X_s^{(n)}$ for all $s \in [0, t]$ as $\text{mesh}(\Pi_n) \rightarrow 0$. The *quadratic variation* of $X := (X_s, s \in [0, t])$ over $[0, t]$, denoted as $\langle X \rangle_t$, is defined as the limit in probability of

$$\langle X^{(n)} \rangle_t := \sum_{k=0}^{n-1} (X_{s_{k+1}}^{(n)} - X_{s_k}^{(n)})^2. \quad (7.13)$$

Theorem 7.8: Quadratic Variation Of An Itô Process

If X is an Itô(μ, σ) process, then

$$\langle X^{(n)} \rangle_t \xrightarrow{L^2} \langle X \rangle_t = \int_0^t \sigma_s^2 ds. \quad (7.14)$$

Furthermore, if $F \in \mathcal{P}_t$ satisfies the condition (7.19) and $\text{mesh}(\Pi_n) \rightarrow 0$, then:

$$\sum_{k=0}^{n-1} F_{s_k} (X_{s_{k+1}} - X_{s_k})^2 \xrightarrow{L^2} \int_0^t F_s \sigma_s^2 ds. \quad (7.15)$$

Theorem 7.9: Itô's Formula For Itô Processes

Let X be an Itô process and let f be twice continuously differentiable. Then,

$$f(X_t) = f(X_0) + \int_0^t f'(X_s) dX_s + \frac{1}{2} \int_0^t f''(X_s) d\langle X \rangle_s. \quad (7.16)$$

Definition 7.7: Covariation Of Two Itô Processes

Let X and Y be two Itô processes adapted to the same filtration $\mathcal{F}$. Then, the *covariation* between X and Y is defined as the quantity:

$$\langle X, Y \rangle_t := \frac{\langle X + Y \rangle_t - \langle X \rangle_t - \langle Y \rangle_t}{2}. \quad (7.17)$$

Theorem 7.10: Itô's Formula In $\mathbb{R}^d$

Let (X_t) be a d -dimensional Itô process, and let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be twice continuously differentiable in all variables. Then,

$$df(X_t) = \sum_{i=1}^d \partial_i f(X_t) dX_{t,i} + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d \partial_{ij} f(X_t) d\langle X_{\cdot,i}, X_{\cdot,j} \rangle_t. \quad (7.18)$$

Theorem 7.11: Itô Processes And Martingales

An Itô process in $\mathbb{R}^d$ is a local martingale if and only if its drift is 0.

7.3 Stochastic Differential Equations

Stochastic differential equations are based on the same principle as ordinary differential equations, relating an unknown function to its derivatives, but with the additional feature that part of the unknown function is driven by randomness.

Definition 7.8: Stochastic Differential Equation

A *stochastic differential equation* (SDE) for a stochastic process $(X_t, t \geq 0)$ is an expression of the form

$$dX_t = a(X_t, t)dt + b(X_t, t)dW_t, \quad (7.19)$$

where $(W_t, t \geq 0)$ is a Wiener process and a and b are deterministic functions, called the *drift* and *diffusion* functions. The resulting process $(X_t, t \geq 0)$ is called an *(Itô) diffusion*.

Consider the autonomous SDE

$$dX_t = a(X_t)dt + b(X_t)dW_t. \quad (7.20)$$

As already mentioned, the meaning of this SDE is that X is a stochastic process that satisfies the integral equation

$$X_t = X_0 + \int_0^t a(X_s)ds + \int_0^t b(X_s)dW_s. \quad (7.21)$$

However, it is not clear that such a process indeed exists. We show in this section that under mild conditions on a and b we can obtain a unique solution X of the SDE via a method of successive approximations, similar to *Picard iteration* for ordinary differential equations.

The condition that is placed on a and b is that they be *Lipschitz continuous*; that is, for all $x \in \mathbb{R}$ there exists a constant c such that

$$|a(y) - a(x)| + |b(y) - b(x)| \leq c|y - x|. \quad (7.22)$$

The successive approximation idea is very simple. Suppose that $X_0 = x$. Define $X_t^{(0)} := x$ for all t , and let

$$X_t^{(n+1)} := x + \int_0^t a(X_s^{(n)}) ds + \int_0^t b(X_s^{(n)}) dW_s \quad (7.23)$$

for $n \in \mathbb{N}$. The following is the main existence result for SDEs:

Theorem 7.12: Existence of SDE Solutions

For any $t \in [0, v]$, the sequence $(X^{(n)})$ in (7.23) converges almost surely in the uniform norm on $[0, t]$ to a process X that satisfies the integral equation (??).

Lemma 7.2: Integral Bound

Let $X^{(n)}$, $n = 1, 2, \dots$ be defined in (7.23), with $X^{(0)} = x$. Then, for every $v > 0$ there exists a constant α such that

$$\mathbb{E} \sup_{s \leq t} \left(X_s^{(n+1)} - X_s^{(n)} \right)^2 \leq \alpha \|X^{(n)} - X^{(n-1)}\|_{\mathcal{P}_t}^2, \quad t \leq v.$$

Definition 7.9: Itô Stratonovich Integral

Let X and Y be Itô processes such that $\int_0^t Y_s dX_s$ and $\int_0^t X_s dY_s$ are well-defined stochastic integrals. Then, the *Itô–Stratonovich integral* of Y with respect to X is defined as

$$\int_0^t Y_s \circ dX_s := \int_0^t Y_s dX_s + \frac{1}{2} \langle X, Y \rangle_t. \quad (7.24)$$

Proposition 7.3: Itô Stratonovich Formula

Let (X_t) be a d -dimensional Itô process, and let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be three times continuously differentiable in all variables. Then,

$$df(X_t) = \sum_{i=1}^d \partial_i f(X_t) \circ dX_{t,i}. \quad (7.25)$$

$$\tilde{\mathbb{P}}^t(A) := \mathbb{E} M_t 1_A, \quad A \in \mathcal{F}_t. \quad (7.26)$$

$$M_t = \exp \left(\int_0^t \mu_s dW_s - \frac{1}{2} \int_0^t \mu_s^2 ds \right) = e^{Z_t - \frac{1}{2} \langle Z \rangle_t}, \quad t \geq 0, \quad (7.27)$$

$$\lim_{\varepsilon \downarrow 0} \varepsilon \ln \mathbb{E} \exp((1 - \varepsilon) \langle Z \rangle_t / 2) = 0. \quad (7.28)$$

Theorem 7.13: Girsanov's Change of Measure

Suppose that μ satisfies the Krylov condition [eq. \(7.28\)](#) and M is the exponential martingale [eq. \(7.27\)](#). Then, under the change of measure [eq. \(7.26\)](#) induced by M , the Itô process

$$\tilde{W}_t := - \int_0^t \mu_s ds + W_t, \quad t \geq 0$$

is a Wiener process.

GBM.py

```

1 N, T = 1000, 1.0
2 mu, sigma = 0.5, 0.2
3 dt = T / N
4 t = np.linspace(0, T, N + 1)
5
6 rng = np.random.default_rng(4)
7 dW = rng.normal(0, np.sqrt(dt), N)
8 W = np.r_[0, np.cumsum(dW)]
9
10 drift = (mu - 0.5 * sigma**2) * t
11 diffusion = sigma * W
12 S_t = np.exp(drift + diffusion)
13 E_St = np.exp(mu * t)

```

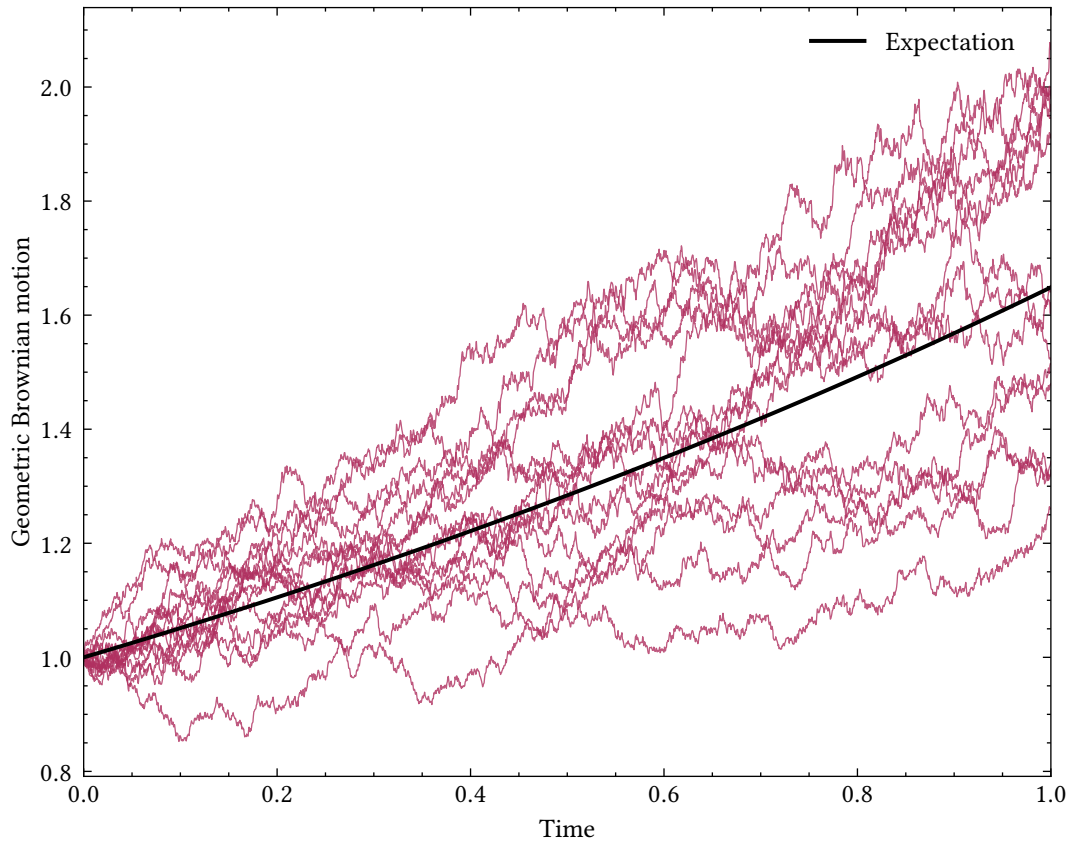


Figure 5: Geometric Brownian motion sample path and expectation for $\mu = 0.5$, $\sigma = 0.2$ and $S_0 = 1$.

7.4 Lévy Processes

Consider now a jump-diffusion process

$$X_t = \sigma W_t + \mu t + J_t = X^c(t) + J_t, \quad (7.29)$$

where J is a compound Poisson process and X^c is the continuous part of X :

$$J_t = \sum_{i=1}^{N_t} \Delta X_i, \quad X_t^c = \mu t + \sigma W_t. \quad (7.30)$$

Define $Y_t = f(X_t)$ where $f \in C^2(\mathbb{R})$ and denote by $T_i, i = 1 \dots N_T$ the jump times of X . On $]T_i, T_{i+1}[$, X evolves according to

$$dX_t = dX_t^c = \sigma dW_t + \mu dt, \quad (7.31)$$

so by applying the Itô formula in the Brownian case we obtain

$$\begin{aligned} Y_{T_{i+1}-} - Y_{T_i} &= \int_{T_i}^{T_{i+1}-} \frac{\sigma^2}{2} f''(X_t) dt + \int_{T_i}^{T_{i+1}-} f'(X_t) dX_t \\ &= \int_{T_i}^{T_{i+1}-} \left\{ \frac{\sigma^2}{2} f''(X_t) dt + f'(X_t) dX_t^c \right\} \end{aligned} \quad (7.32)$$

since $dX_t = dX^c(t)$ on this interval. If a jump of size ΔX_t occurs then the resulting change in Y_t is given by $f(X_{t-} + \Delta X_t) - f(X_{t-})$. The total change in Y_t can therefore be written as the sum of these two contributions:

$$\begin{aligned} f(X_t) - f(X_0) &= \int_0^t f'(X_s) dX_s^c + \int_0^t \frac{\sigma^2}{2} f''(X_s) ds \\ &\quad + \sum_{0 \leq s \leq t, \Delta X_s \neq 0} [f(X_{s-} + \Delta X_s) - f(X_{s-})]. \end{aligned} \quad (7.33)$$

Replacing dX_s^c by $dX_s - \Delta X_s$ one obtains an equivalent expression:

$$\begin{aligned} f(X_t) - f(X_0) = & \int_0^t f'(X_{s-}) dX_s + \int_0^t \frac{\sigma^2}{2} f''(X_s) ds \\ & + \sum_{0 \leq s \leq t, \Delta X_s \neq 0} [f(X_{s-} + \Delta X_s) - f(X_{s-}) - \Delta X_s f'(X_{s-})]. \end{aligned} \quad (7.34)$$



When the number of jumps is finite, this form is equivalent to [eq. \(7.33\)](#). However, as we will see below, the form [eq. \(7.34\)](#) is more general, wherein both the stochastic integral and the sum over the jumps in [eq. \(7.34\)](#) are well-defined for any semi-martingale, even if the number of jumps is infinite, while the sum in [eq. \(7.33\)](#) may not converge if jumps have infinite variation.

Here we have only used the Itô formula for diffusions, which is of course still valid if σ is replaced by a process $(\sigma_t)_{t \in [0, T]}$ belonging to the class $\mathcal{P}_T$.

Proposition 7.4: Itô Formula for Jump-Diffusion Processes

Let X be a diffusion process with jumps, defined as the sum of a drift term, a Brownian stochastic integral and a compound Poisson process:

$$X_t = X_0 + \int_0^t b_s ds + \int_0^t \sigma_s dW_s + \sum_{i=1}^{N_t} \Delta X_i,$$

where b_t and σ_t are continuous adapted processes with

$$\mathbb{E} \left[\int_0^T \sigma_t^2 dt \right] < \infty.$$

Then, for any $C^{1,2}$ function $f : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$, the process $Y_t = f(t, X_t)$ can be represented as

$$\begin{aligned} f(t, X_t) - f(0, X_0) &= \int_0^t \left[\frac{\partial f}{\partial s}(s, X_s) + \frac{\partial f}{\partial x}(s, X_s) b_s \right] ds \\ &\quad + \frac{1}{2} \int_0^t \sigma_s^2 \frac{\partial^2 f}{\partial x^2}(s, X_s) ds + \int_0^t \frac{\partial f}{\partial x}(s, X_s) \sigma_s dW_s \\ &\quad + \sum_{\{i \geq 1, T_i \leq t\}} [f(X_{T_i-} + \Delta X_i) - f(X_{T_i-})]. \end{aligned}$$

Expressed in differential notation

$$\begin{aligned} dY_t &= \frac{\partial f}{\partial t}(t, X_t) dt + b_t \frac{\partial f}{\partial x}(t, X_t) dt + \frac{\sigma_t^2}{2} \frac{\partial^2 f}{\partial x^2}(t, X_t) dt \\ &\quad + \frac{\partial f}{\partial x}(t, X_t) \sigma_t dW_t + [f(X_{t-} + \Delta X_t) - f(X_{t-})]. \end{aligned}$$

Definition 7.10: Infinite Divisibility

A probability distribution F on $\mathbb{R}^d$ is said to be *infinitely divisible* if for any integer $n \geq 2$, there exist n iid random variables $Y_1, \dots, Y_n$ such that $Y_1 + \dots + Y_n$ has distribution F .

Proposition 7.5: Infinite Divisibility and Lévy Processes

Let $X := (X_t)_{t \geq 0}$ be a Lévy process. Then for every t , X_t has an infinitely divisible distribution. Conversely, if F is an infinitely divisible distribution, there exists a Lévy process X such that the distribution of X_1 is given by F .

Part II

Jump Processes

8 Levy Processes

9 Integro-differential equations and numerical methods

An approximate answer to the right problem is worth a good deal more than an exact answer to an approximate problem.

John Tukey

Part III

Financial Underpinnings

10 Assets, Portfolios and Arbitrage

10.1 Portfolio Allocation and Short Selling

We will use the following notation. An element $\bar{x}$ of $\mathbb{R}^{d+1}$ is a vector

$$\bar{x} = \left(x^{(0)}, x^{(1)}, \dots, x^{(d)} \right) \quad (10.1)$$

made of $d + 1$ components. The scalar product $\bar{x} \cdot \bar{y}$ of two vectors $\bar{x}, \bar{y} \in \mathbb{R}^{d+1}$ is defined by

$$\langle \bar{x}, \bar{y} \rangle := x^{(0)}y^{(0)} + x^{(1)}y^{(1)} + \dots + x^{(d)}y^{(d)}. \quad (10.2)$$

The vector

$$\bar{S}_0 = \left(S_0^{(0)}, S_0^{(1)}, \dots, S_0^{(d)} \right) \quad (10.3)$$

denotes the prices at time $t = 0$ of $d + 1$ assets. Namely, $S_0^{(i)} > 0$ is the price at time $t = 0$ of asset $n^o i = 0, 1, \dots, d$.

The asset values $S_1^{(i)} > 0$ of assets No $i = 0, 1, \dots, d$ at time $t = 1$ are represented by the vector

$$\bar{S}_1 = \left(S_1^{(0)}, S_1^{(1)}, \dots, S_1^{(d)} \right), \quad (10.4)$$

whose components $\left(S_1^{(1)}, \dots, S_1^{(d)} \right)$ are random variables defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

In addition we will assume that asset $n^o 0$ is a riskless asset (of savings account type) that yields an interest rate $r > 0$, *i.e.* we have

$$S_1^{(0)} = (1 + r)S_0^{(0)}. \quad (10.5)$$

Definition 10.1: Portfolio

A portfolio based on the assets $0, 1, \dots, d$ is a vector

$$\bar{\xi} = \left(\xi^{(0)}, \xi^{(1)}, \dots, \xi^{(d)} \right) \in \mathbb{R}^{d+1}, \quad (10.6)$$

in which $\xi^{(i)}$ represents the (possibly fractional) quantity of asset $n^o i$ owned by an investor, $i = 0, 1, \dots, d$.

The *price* of such a portfolio, or the cost of the corresponding investment, is given by

$$\bar{\xi} \cdot \bar{S}_0 = \sum_{i=0}^d \xi^{(i)} S_0^{(i)} = \xi^{(0)} S_0^{(0)} + \xi^{(1)} S_0^{(1)} + \dots + \xi^{(d)} S_0^{(d)} \quad (10.7)$$

at time $t = 0$. At time $t = 1$, the *value* of the portfolio has evolved into

$$\bar{\xi} \cdot \bar{S}_1 = \sum_{i=0}^d \xi^{(i)} S_1^{(i)} = \xi^{(0)} S_1^{(0)} + \xi^{(1)} S_1^{(1)} + \dots + \xi^{(d)} S_1^{(d)}. \quad (10.8)$$

There are various ways to construct a portfolio allocation $(\xi^{(i)})_{i=0,1,\dots,d}$.

- i) If $\xi^{(0)} > 0$, the investor puts the amount $\xi^{(0)} S_0^{(0)} > 0$ on a savings account with interest rate r .
- ii) If $\xi^{(0)} < 0$, the investor borrows the amount $-\xi^{(0)} S_0^{(0)} > 0$ with the same interest rate r .
- iii) For $i = 1, 2, \dots, d$, if $\xi^{(i)} > 0$, the investor purchases a (possibly fractional) quantity $\xi^{(i)} > 0$ of the asset $n^o i$.
- iv) If $\xi^{(i)} < 0$, the investor borrows a quantity $-\xi^{(i)} > 0$ of asset i and sells it to obtain the amount $-\xi^{(i)} S_0^{(i)} > 0$.

In the latter case one says that the investor *short sells* a quantity $-\xi^{(i)} > 0$ of the asset $n^o i$, which lowers the cost of the portfolio.

Definition 10.2: Short Selling Ratio

The *short selling ratio*, or percentage of daily turnover activity related to short selling, is defined as the ratio of the number of daily short sold shares divided by daily volume.

Profits are typically generated by first buying at a low price and then selling at a higher price. Short sellers apply the same rule, but in reverse chronological order: first sell high and then buy low, if possible, by following the procedure below.

1. Borrow the asset $n^o i$.
2. At time $t = 0$, sell the asset $n^o i$ on the market at the price $S_0^{(i)}$ and invest the amount $S_0^{(i)}$ at the interest rate $r > 0$.
3. Buy back the asset $n^o i$ at time $t = 1$ at the price $S_1^{(i)}$, with hopefully $S_1^{(i)} < (1 + r)S_0^{(i)}$.
4. Return the asset to its owner, with possibly a (small) fee $p > 0$.¹

At the end of the operation, the profit made on share $n^o i$ equals

$$(1 + r)S_0^{(i)} - S_1^{(i)} - p > 0, \quad (10.9)$$

which is positive provided that $S_1^{(i)} < (1 + r)S_0^{(i)}$ and $p > 0$ is sufficiently small.

10.2 Arbitrage

10.3 Risk-neutral Probability Measures

¹The cost p of short selling will not be taken into account in later calculations.

10.4 Hedging of Contingent Claims

Definition 10.3: Contingent Claim

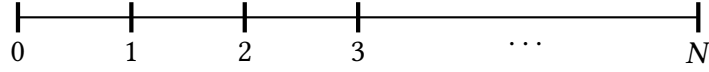
A contingent claim is a financial derivative whose payoff $C : \Omega \rightarrow \mathbb{R}$ is a random variable depending on the realisation of uncertain events.

10.5 Market Completeness

11 Discrete-time Market Model

11.1 Discrete Time Compounding

We work in a finite-horizon discrete-time model indexed by $\{0, 1, \dots, N\}$



We invest a cash flow C each year in an investment plan that carries a constant interest rate r . At the end of the N -th year, the value of the amount C invested at the beginning of year $k = 1, 2, \dots, N$ has turned into $C(1+r)^{N-k+1}$ and the total value of the plan V_N at the end of the N -th year becomes

$$\begin{aligned} V_N &= C \sum_{k=1}^N (1+r)^{N-k+1} \\ &= C \sum_{k=1}^N (1+r)^k \\ &= C(1+r) \frac{(1+r)^N - 1}{r}, \end{aligned} \tag{11.1}$$

hence the duration N of the plan satisfies

$$N+1 = \frac{1}{\ln(1+r)} \ln \left(1+r + \frac{rV_N}{C} \right). \tag{11.2}$$

At time $t = 0$ one borrows an amount L over a period of N years at the constant interest rate r per year.

Proposition 11.1: Constant Repayment

Assuming the loan is fully amortized at the beginning of year $N + 1$, the constant annual repayment C is uniquely determined by

$$C = \frac{r(1+r)^N L}{(1+r)^N - 1} = \frac{r}{1 - (1+r)^{-N}} L. \quad (11.3)$$

Proof. Let L_k denote the outstanding principal balance at the beginning of year k , with $L_1 = L$.

The evolution of the debt is governed by the linear recurrence relation

$$L_{k+1} = (1+r)L_k - C, \quad \forall k \geq 1. \quad (11.4)$$

Iteratively expanding this relation yields the value of the debt at the start of any arbitrary year k

$$\begin{aligned} L_k &= (1+r)^{k-1} L_1 - C \sum_{j=0}^{k-2} (1+r)^j \\ &= (1+r)^{k-1} L - C \frac{(1+r)^{k-1} - 1}{r}. \end{aligned} \quad (11.5)$$

The amortization condition requires that the debt is extinguished at the horizon N , such that $L_{N+1} = 0$. Substituting $k = N + 1$ into eq. (11.5) implies

$$0 = (1+r)^N L - C \frac{(1+r)^N - 1}{r}.$$

Rearranging for C yields the result in 11.1. Furthermore, substituting this expression for C back into eq. (11.5) provides the closed-form outstanding balance

$$L_k = \frac{C}{r} [1 - (1+r)^{k-1-N}] \equiv \frac{C - (1+r)^{k-1}(C - rL)}{r}. \quad (11.6)$$

□

Rearranging the result in 11.1 yields the annuity factor

$$\frac{L}{C} = \frac{1 - (1 + r)^{-N}}{r}, \quad (11.7)$$

and solving for the duration N implies

$$N = \frac{1}{\ln(1 + r)} \ln \left(\frac{C}{C - rL} \right) = -\frac{\ln(1 - \frac{rL}{C})}{\ln(1 + r)}. \quad (11.8)$$



For N to be finite, we require the condition $C > rL$. This means the annual repayment must exceed the interest accrued on the initial principal; otherwise, the principal balance will never decrease.

Proposition 11.2: Repayment Decomposition

The k -th repayment C decomposes into an interest component rL_k and a principal repayment component $C - rL_k$ given by

$$rL_k = C \left(1 - \frac{1}{(1 + r)^{N-k+1}} \right) = Cr \sum_{l=1}^{N-k+1} (1 + r)^{-l} \quad (11.9)$$

$$C - rL_k = \frac{C}{(1 + r)^{N-k+1}}, \quad k = 1, 2, \dots, N. \quad (11.10)$$

Proof. Recall from the proof of 11.1 that the outstanding balance L_k represents the present value of the remaining $N - k + 1$ payments. We express L_k as

$$L_k = \frac{C}{r} \left[1 - (1 + r)^{-(N-k+1)} \right]. \quad (11.11)$$

Multiplying by r immediately yields the interest component in eq. (11.9). Consequently, the principal repayment is determined by the residual

$$\begin{aligned} C - rL_k &= C - C \left[1 - (1 + r)^{-(N-k+1)} \right] \\ &= C(1 + r)^{-(N-k+1)}, \end{aligned} \quad (11.12)$$

which corresponds to eq. (11.10). □

We verify that the sum of discounted payments at rate r recovers the initial loan amount

$$\sum_{l=1}^N \frac{C}{(1 + r)^l} = C \frac{1 - (1 + r)^{-N}}{r} = L. \quad (11.13)$$

In particular, the first interest repayment satisfies

$$rL_1 = Cr \sum_{l=1}^N \frac{1}{(1 + r)^l} = C \left(1 - (1 + r)^{-N} \right), \quad (11.14)$$

and the first principal repayment is

$$C - rL = \frac{C}{(1 + r)^N}. \quad (11.15)$$

11.2 Arbitrage and Self-Financing Portfolios

12 Continuous-time Market Model

The continuous-time market model enables the incorporation of portfolio reallocation algorithms within a stochastic dynamic programming framework.

12.1 Asset Price Modelling

The prices at time $t \geq 0$ of $d + 1$ assets numbered $0, 1, \dots, d$ is denoted by the random vector

$$\bar{S}_t = \left(S_t^{(0)}, S_t^{(1)}, \dots, S_t^{(d)} \right) \quad (12.1)$$

which forms a stochastic process $(\bar{S}_t)_{t \in \mathbb{R}_+}$. As in discrete time, the asset $n^o 0$ is a riskless asset (of savings account type) yielding an interest rate r , *i.e.* we have

$$S_t^{(0)} = S_0^{(0)} e^{rt}, \quad t \geq 0. \quad (12.2)$$

Definition 12.1: Discounting

Let

$$\bar{X}_t := \left(\tilde{S}_t^{(0)}, \tilde{S}_t^{(1)}, \dots, \tilde{S}_t^{(d)} \right), \quad t \in \mathbb{R}, \quad (12.3)$$

denote the vector of discounted asset prices, defined as:

$$\tilde{S}_t^{(k)} = e^{-rt} S_t^{(k)}, \quad t \geq 0, \quad k = 0, 1, \dots, d. \quad (12.4)$$

We can also write

$$\bar{X}_t := e^{-rt} \bar{S}_t, \quad t \geq 0. \quad (12.5)$$

Definition 12.2: Portfolio Strategy

A portfolio strategy is a stochastic process $(\bar{\xi}_t)_{t \in \mathbb{R}_+} \subset \mathbb{R}^{d+1}$, where $\xi_t^{(k)}$ denotes the (possibly fractional) quantity of asset $n^o k$ held at time $t \geq 0$.

The value at time $t \geq 0$ of the portfolio strategy $(\bar{\xi}_t)_{t \in \mathbb{R}_+} \subset \mathbb{R}^{d+1}$ is defined by

$$V_t := \bar{\xi}_t \cdot \bar{S}_t = \sum_{k=0}^d \bar{\xi}_t^{(k)} S_t^{(k)}, \quad t \geq 0. \quad (12.6)$$

The discounted value of the portfolio is defined by

$$\begin{aligned} \widetilde{V}_t &:= e^{-rt} V_t \\ &= e^{-rt} \bar{\xi}_t \cdot \bar{S}_t \\ &= e^{-rt} \sum_{k=0}^d \bar{\xi}_t^{(k)} S_t^{(k)} \\ &= \sum_{k=0}^d \bar{\xi}_t^{(k)} \widetilde{S}_t^{(k)} \\ &= \bar{\xi}_t \cdot \bar{X}_t, \quad t \geq 0. \end{aligned} \quad (12.7) \quad (12.8)$$

The effect of discounting from time t to time 0 is to divide prices by e^{rt} , making all prices comparable at time 0.

12.2 Arbitrage and Risk-Neutral Measures

In continuous-time, the definition of arbitrage follows the lines of its analogs in the one-step and discrete-time models. In what follows we will only consider *admissible* portfolio strategies whose total value V_t remains nonnegative for all times $t \in [0, T]$.

Definition 12.3: Admissible Portfolio Strategy

A portfolio strategy $(\xi_t^{(k)})_{t \in [0, T], k=0,1,\dots,d}$ with value

$$V_t = \bar{\xi}_t \cdot \bar{S}_t = \sum_{k=0}^d \bar{\xi}_t^{(k)} S_t^{(k)}, \quad t \in [0, T], \quad (12.9)$$

constitutes an arbitrage opportunity if all three following conditions are satisfied:

- | | |
|--|---|
| i) $V_0 \leq 0$ at time $t = 0$, | [Start from a zero-cost portfolio, or with a debt.] |
| ii) $V_T \geq 0$ at time $t = T$, | [Finish with a nonnegative amount.] |
| iii) $\mathbb{P}(V_T > 0) > 0$ at time $t = T$. | [Profit is made with nonzero probability.] |

Roughly speaking, (ii) means that the investor wants no loss, (iii) means that he wishes to sometimes make a strictly positive gain, and (i) means that he starts with zero capital or even with a debt.

Next, we turn to the definition of risk-neutral probability measures (or martingale measures) in continuous time, which states that under a risk-neutral probability measure $\mathbb{P}^*$, the return of the risky asset over the time interval $[u, t]$ equals the return of the riskless asset given by

$$S_t^{(0)} = e^{(t-u)r} S_u^{(0)}, \quad 0 \leq u \leq t. \quad (12.10)$$

Recall that the filtration $(\mathcal{F}_t)_{t \in \mathbb{R}_+}$ is generated by Brownian motion $(B_t)_{t \in \mathbb{R}_+}$, i.e.

$$\mathcal{F}_t = \sigma(B_u : 0 \leq u \leq t), \quad t \geq 0. \quad (12.11)$$

12.3 Self-financing Portfolio Strategies

Let $\xi_t^{(i)}$ denote the (possibly fractional) quantity invested at time t over the time interval $[t, t + dt)$, in the asset $S_t^{(k)}$, $k = 0, 1, \dots, d$, and let

$$\bar{\xi}_t = \left(\xi_t^{(k)} \right)_{k=0,1,\dots,d}, \quad \bar{S}_t = \left(S_t^{(k)} \right)_{k=0,1,\dots,d}, \quad t \geq 0, \quad (12.12)$$

denote the associated portfolio value and asset price processes. The portfolio value V_t at time $t \geq 0$ is given by

$$V_t = \bar{\xi}_t \cdot \bar{S}_t = \sum_{k=0}^d \xi_t^{(k)} S_t^{(k)}, \quad t \geq 0. \quad (12.13)$$

Definition 12.4: Self-Financing Portfolio Update

The portfolio strategy $(\bar{\xi}_t)_{t \in \mathbb{R}_+}$ is self-financing if the portfolio value remains constant after updating the portfolio from $\bar{\xi}_t$ to $\bar{\xi}_{t+dt}$, i.e.

$$\bar{\xi}_t \cdot \bar{S}_{t+dt} = \sum_{k=0}^d \bar{\xi}_t^{(k)} S_{t+dt}^{(k)} = \sum_{k=0}^d \bar{\xi}_{t+dt}^{(k)} S_{t+dt}^{(k)} = \bar{\xi}_{t+dt} \cdot \bar{S}_{t+dt}, \quad (12.14)$$

which is the continuous-time analog of the self-financing condition already encountered in the discrete setting of [section 10](#), see [10.1](#). A major difference with the discrete-time case of [10.1](#), however, is that the continuous-time differentials dS_t and $d\xi_t$ do not make pathwise sense as continuous-time stochastic integrals are defined by L^2 limits, cf. [3.6](#), or by convergence in probability.

Part IV

Risk-Neutral Valuation

Part V

Martingale Methods

13 The Black-Scholes Model

We start by describing processes that model the prices of primary securities, a savings account (equivalently, a risk-free bond), and a common stock.

13.1 Risk-free Bond

The first primary security represents in our model an accumulation factor corresponding to a *savings account* (a *money market account*). We assume that the *short-term interest rate* r is constant (but not necessarily nonnegative) over the trading interval $[0, T^*]$. The risk-free security is assumed to continuously compound in value at the rate r ; that is, $dB_t = rB_t dt$. We adopt the usual convention that $B_0 = 1$, so that its price equals $B_t = e^{rt}$ for every $t \in [0, T^*]$ (Merton, 1973). When dealing with the Black-Scholes model, we may equally well replace the savings account by the *risk-free bond*. A unit zero-coupon bond maturing at time T is a security that pays its holder 1 unit of cash at a predetermined date T in the future, known as the bond's *maturity date*. Let $B(t, T)$ stand for the price at time $t \in [0, T]$ of a bond maturing at time T (Brennan and Schwartz, 1979). It is easily seen that to replicate the payoff 1 at time T it suffices to invest B_t/B_T units of cash at time t in the savings account B . This shows that, in the absence of arbitrage opportunities, the price of the bond satisfies

$$B(t, T) = e^{-r(T-t)}, \quad \forall t \in [0, T]. \quad (13.1)$$

Note that for any fixed T , the bond price solves the ordinary differential equation

$$dB(t, T) = rB(t, T)dt, \quad B(0, T) = e^{-rT}. \quad (13.2)$$

We consider here a *risk-free bond*, meaning that its issuer will not default on its obligation to pay to the bondholder the face value at maturity date (Bielecki, Jeanblanc, and Rutkowski, 2004).

13.2 Common Stock Price

Let us fix the horizon date $T^* > 0$. The continuous-time evolution of the stock price process $(S_t)_{t \in [0, T^*]}$ is described by the following linear *stochastic differential equation* (SDE)

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_0 > 0, \quad (13.3)$$

where $\mu \in \mathbb{R}$ is a constant *appreciation rate* of the stock price, $\sigma > 0$ is a constant *volatility*, and S_0 is the initial value of the stock price (Black and Scholes, 1973). As we shall see in what follows, the restrictive condition that μ and σ are constant can be weakened substantially (Heston, 1993; Bensoussan, Crouhy, and Dan Galai, 1994).

The driving noise W_t , $t \in [0, T^*]$, is a one-dimensional standard Brownian motion defined on a filtered probability space $(\Omega, \mathbb{F}, \mathbb{P})$.

We refer to eq. (13.3) as an SDE; however, expression eq. (13.3) is merely a shorthand notation for the following *stochastic integral equation* (see section 7.3)

$$S_t = S_0 + \int_0^t \mu S_u du + \int_0^t \sigma S_u dW_u, \quad \forall t \in [0, T^*]. \quad (13.4)$$

In other words, by a stochastic differential equation of the form eq. (13.3) we mean in fact the stochastic integral equation eq. (13.4). The definition of the second integral in eq. (13.4), interpreted here as the *Itô stochastic integral*, can be found in section 7.3. Let us only mention here that if X is a continuous, $\mathcal{F}$ -adapted process that satisfies suitable integrability conditions, then

$$\int_0^t X_u dW_u = \lim_{n \rightarrow +\infty} \sum_{j=0}^{n-1} X_{t_j^n} (W_{t_{j+1}^n} - W_{t_j^n}), \quad (13.5)$$

where the limit is in fact taken over an arbitrary sequence of finite partitions $\{t_0^n = 0 < t_1^n < \dots < t_n^n = t\}$ of the interval $[0, t]$ such that $\lim_{n \rightarrow \infty} \max_{k=0, \dots, n-1} (t_{k+1}^n - t_k^n) = 0$.

Proposition 13.1: Solution of the SDE

The process S given by the formula

$$S_t = S_0 \exp \left(\sigma W_t + \left(\mu - \frac{1}{2} \sigma^2 \right) t \right), \quad \forall t \in [0, T^*], \quad (13.6)$$

is the unique solution of the stochastic differential equation [eq. \(13.3\)](#), or equivalently, the stochastic integral equation [eq. \(13.4\)](#).

Proof. We will first check that the process given by [eq. \(13.6\)](#) satisfies [eq. \(13.3\)](#). As customary, we use the differential notation, rather than the integral notation. Let us introduce the auxiliary process X by setting

$$dX_t = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t$$

with $X_0 = 0$. Then the process S given by [eq. \(13.6\)](#) satisfies $S_t = g(X_t)$ for $t \in [0, T^*]$, where the function $g : \mathbb{R} \rightarrow \mathbb{R}$ is defined as $g(x) = S_0 e^x$. Obviously, $g(x) = g'(x) = g''(x)$ and thus Itô's formula yields

$$\begin{aligned} dS_t &= dg(X_t) \\ &= g'(X_t) \left(\mu - \frac{1}{2} \sigma^2 \right) dt + g'(X_t) \sigma dW_t + \frac{1}{2} g''(X_t) \sigma^2 dt \\ &= g(X_t) (\mu dt + \sigma dW_t) \\ &= S_t (\mu dt + \sigma dW_t), \end{aligned}$$

as expected. Uniqueness of solutions to [eq. \(13.4\)](#) is an immediate consequence of the general result, due to Itô, stating that an SDE with Lipschitz continuous coefficients has a unique solution.

Alternatively, one can prove the uniqueness by applying the Itô formula to the ratio of two solutions to [eq. \(13.4\)](#). Let us assume that Y is another solution to [eq. \(13.3\)](#) with the initial value $Y_0 = S_0$. We wish to show that the Itô differential of the ratio Y/S is equal to zero. To this

end, we will first compute the Itô differential of the process $1/S$. Using Itô's formula again, we obtain (note that now we take $g(s) = 1/s$)

$$\begin{aligned} d\left(\frac{1}{S_t}\right) &= -\frac{1}{S_t^2}dS_t + \frac{1}{S_t^3}d\langle S \rangle_t = -\frac{1}{S_t^2}(\mu S_t dt + \sigma S_t dW_t) + \frac{1}{S_t^3}\sigma^2 S_t^2 dt \\ &= -\frac{1}{S_t}((\mu - \sigma^2)dt + \sigma dW_t), \end{aligned}$$

where we used the fact that the quadratic variation of S equals

$$\langle S \rangle_t = \int_0^t \sigma^2 S_u^2 d\langle W \rangle_u = \int_0^t \sigma^2 S_u^2 du. \quad (13.7)$$

By applying Itô's integration by parts formula to processes Y and $1/S$, we get

$$\begin{aligned} d\left(\frac{Y_t}{S_t}\right) &= \frac{1}{S_t}dY_t + Y_t d\left(\frac{1}{S_t}\right) + d\langle Y, 1/S \rangle_t \\ &= \frac{Y_t}{S_t}(\mu dt + \sigma dW_t) - \frac{Y_t}{S_t}((\mu - \sigma^2)dt + \sigma dW_t) - \frac{Y_t}{S_t}\sigma^2 dt \\ &= 0, \end{aligned}$$

since the cross-variation of Y and $1/S$ equals (recall that $\langle W \rangle_t = t$)

$$\langle Y, 1/S \rangle_t = - \int_0^t \sigma^2 \frac{Y_u}{S_u} d\langle W \rangle_u = - \int_0^t \sigma^2 \frac{Y_u}{S_u} du. \quad (13.8)$$

Since the Itô differential of the ratio Y/S is zero, this process is constant, and thus it equals its initial value, which is 1. We conclude that $Y_t = S_t$ for every $t \in [0, T^*]$, and thus the uniqueness of a solution to [eq. \(13.4\)](#) is established. $\square$

We make the standing assumption that the underlying filtration $\mathcal{F} = (\mathcal{F}_t)_{t \in [0, T^*]}$ is the natural filtration $\mathcal{F}^W$ of the driving Brownian motion. More precisely, $\mathcal{F}$ is the standard augmentation of the natural filtration of W , but it will still be denoted by $\mathcal{F}^W$. For any fixed $t \in [0, T^*]$, we have $S_t = f(W_t)$ for some function $f : \mathbb{R} \rightarrow \mathbb{R}_+$ that determines the one-to-one

correspondence between the values S_t of the stock price and the values W_t of the Brownian motion. Specifically, the strictly increasing function f is given by the formula

$$f(x) = S_0 \exp \left(\sigma x + \left(\mu - \frac{1}{2} \sigma^2 \right) t \right), \quad \forall x \in \mathbb{R}. \quad (13.9)$$

Consequently, for any natural n , any dates $0 \leq t_1 < t_2 < \dots < t_n \leq T^*$, and any real numbers $x_1, x_2, \dots, x_n$, we have

$$\omega \in \Omega \mid S_{t_1} < x_1, \dots, S_{t_n} < x_n = \omega \in \Omega \mid W_{t_1} < y_1, \dots, W_{t_n} < y_n, \quad (13.10)$$

where we set $y_i = f^{-1}(x_i)$. It is thus clear that we have, for every $t \in [0, T^*]$,

$$\mathcal{F}_t^W = \sigma W_u \mid u \leq t = \sigma S_u \mid u \leq t = \mathcal{F}_t^S. \quad (13.11)$$

We conclude that the filtration generated by the stock price coincides with the natural filtration of the driving noise W , and thus $\mathcal{F}^S = \mathcal{F}^W = \mathcal{F}$. This means that the information structure of the model is based on observations of the stock price process only.

It is readily verified that the Brownian motion W is a martingale with respect to the underlying filtration $\mathcal{F}$. Indeed, for any t , the random variable W_t is $\mathbb{P}$ -integrable, and for any $u \leq t$:

$$\mathbb{E}_{\mathbb{P}}(W_t \mid \mathcal{F}_u) = \mathbb{E}_{\mathbb{P}}(W_t - W_u \mid \mathcal{F}_u) + \mathbb{E}_{\mathbb{P}}(W_u \mid \mathcal{F}_u) = W_u. \quad (13.12)$$

Define an auxiliary process M by setting:

$$M_t = \exp \left(\sigma W_t - \frac{1}{2} \sigma^2 t \right), \quad \forall t \in [0, T^*]. \quad (13.13)$$

Lemma 13.1: Martingale Property

The process M is a martingale under $\mathbb{P}$ with respect to the filtration $\mathcal{F}$.

Proof. In order to show that M is a martingale under $\mathbb{P}$ with respect to $\mathbb{F}$, it suffices to note that

$$\mathbb{E}_{\mathbb{P}}(e^{\sigma(W_t - W_u)} \mid \mathcal{F}_u) = \mathbb{E}_{\mathbb{P}}(e^{\sigma(W_t - W_u)}) = \exp\left(\frac{1}{2}\sigma^2(t - u)\right), \quad (13.14)$$

where the first equality is a consequence of the independence of the increment $W_t - W_u$ of the σ -field $\mathcal{F}_u$ and the second follows from the fact that $W_t - W_u$ has the Gaussian distribution $\mathcal{N}(0, t - u)$. $\square$

Since $S_t = e^{\mu t} M_t$, we have, for every $u \leq t \leq T^*$,

$$\mathbb{E}_{\mathbb{P}}(S_t \mid \mathcal{F}_u) = \mathbb{E}_{\mathbb{P}}(e^{\mu t} M_t \mid \mathcal{F}_u) = e^{\mu t} M_u = S_u e^{\mu(t-u)}. \quad (13.15)$$

Corollary 13.1: Martingale Property of Stock Price

The stock price S follows a martingale under the actual probability $\mathbb{P}$ with respect to the filtration $\mathbb{F}$ if and only if its appreciation rate $\mu = 0$. If $\mu > 0$ (respectively $\mu < 0$) then $\mathbb{E}_{\mathbb{P}}(S_t \mid \mathcal{F}_u) > S_u$ (respectively $\mathbb{E}_{\mathbb{P}}(S_t \mid \mathcal{F}_u) < S_u$) for every $t > u$ and thus the stock price S is a submartingale (respectively a supermartingale) under the actual probability $\mathbb{P}$.

It is apparent from eq. (13.6) that the logarithmic stock returns are Gaussian, meaning that the random variable $\ln(S_t/S_u)$ has under $\mathbb{P}$ a Gaussian probability distribution for any choice of dates $u < t \leq T^*$. Moreover, S is a *time-homogeneous Markov process* under $\mathbb{P}$ with respect to the filtration $\mathbb{F}$ with the transition probability density function, for $x, y > 0$ and $t > u$,

$$p_S(u, x, t, y) = \frac{1}{\sqrt{2\pi(t-u)}\sigma y} \exp\left\{-\frac{(\ln(y/x) - \mu(t-u) + \frac{1}{2}\sigma^2(t-u))^2}{2\sigma^2(t-u)}\right\} \quad (13.16)$$

where formally $p_S(u, x, t, y) = \mathbb{P}\{S_t = y \mid S_u = x\}$. It is important to observe that even if the appreciation rate μ is replaced by an adapted stochastic process μ_t and the stock price S is no longer Markovian under the actual probability $\mathbb{P}$, it will always be a time-homogeneous Markov process under the martingale measure $\mathbb{P}^*$ for the discounted stock price.

13.3 Self-financing Trading Strategies

By a *trading strategy* we mean a pair $\phi = (\phi^1, \phi^2)$ of $\mathcal{F}$ -progressively measurable stochastic processes on the underlying probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The concept of a self-financing trading strategy in the Black–Scholes market is formally based on the notion of the Itô integral. Intuitively, this choice of stochastic integral is supported by the fact that, in the case of the Itô integral (as opposed, for instance, to the Fisk–Stratonovich integral), the underlying process is integrated in a predictable way, meaning that we take its values on the left-hand end of each (infinitesimal) time interval. Unless explicitly stated otherwise, we will assume that a stock S does not pay dividends (at least during the option's lifetime).

Definition 13.1: Self-Financing Trading Strategy

We say that a trading strategy $\phi = (\phi^1, \phi^2)$ over the time interval $[0, T]$ is *self-financing* if its wealth process $V(\phi)$, which is set to equal

$$V_t(\phi) = \phi_t^1 S_t + \phi_t^2 B_t, \quad \forall t \in [0, T], \quad (13.17)$$

satisfies the following condition

$$V_t(\phi) = V_0(\phi) + \int_0^t \phi_u^1 dS_u + \int_0^t \phi_u^2 dB_u, \quad \forall t \in [0, T], \quad (13.18)$$

where the first integral is understood in the Itô sense and the second is the path-wise Riemann (or Lebesgue) integral.

13.4 Martingale Measure for the Black-Scholes Model

We find it convenient to introduce the concept of the *admissibility* of a trading strategy directly in terms of a martingale measure. For this reason, our next goal is to study the concept of a martingale measure within the Black-Scholes set-up.

Let $S^* = S/B$ denote the discounted stock price. Using the geometric Brownian motion solution [eq. \(13.6\)](#) and $B_t = e^{rt}$, we obtain for every $t \in [0, T^*]$

$$S_t^* = S_0^* \exp \left(\sigma W_t + \left(\mu - r - \frac{1}{2} \sigma^2 \right) t \right). \quad (13.19)$$

Equivalently, the process S^* is the unique solution to the SDE

$$dS_t^* = (\mu - r) S_t^* dt + \sigma S_t^* dW_t, \quad S_0^* = S_0. \quad (13.20)$$

The martingale property of S^* under $\mathbb{P}$ allows for analysis identical to the stock price S .

Corollary 13.2: Martingale Property of Discounted Stock

The discounted stock price S^* is a martingale under the actual probability $\mathbb{P}$ with respect to the filtration $\mathcal{F}$ if and only if $\mu = r$. If $\mu > r$ (respectively $\mu < r$), then $\mathbb{E}_{\mathbb{P}}(S_t^* | \mathcal{F}_u) > S_u^*$ (respectively $\mathbb{E}_{\mathbb{P}}(S_t^* | \mathcal{F}_u) < S_u^*$) for every $t > u$, implying S^* is a sub-martingale (respectively a super-martingale) under $\mathbb{P}$.

The subsequent definition directly addresses the martingale property of the discounted stock price S^* . We seek a probability measure under which the drift term in the governing equation [eq. \(13.20\)](#) disappears.

Definition 13.2: Martingale Measure for S^*

A probability measure $\mathbb{Q}$ on $(\Omega, \mathcal{F}_{T^*})$, equivalent to $\mathbb{P}$, is called a *martingale measure* for S^* if S^* is a local martingale under $\mathbb{Q}$.

To analyse the no-arbitrage property of the model and derive the risk-neutral valuation formula, we focus on the martingale property of the discounted wealth process.

Definition 13.3: Spot Martingale Measure

A probability measure $\mathbb{P}^*$ on $(\Omega, \mathcal{F}_{T^*})$, equivalent to $\mathbb{P}$, is called a *spot martingale measure* if the discounted wealth $V_t^*(\phi) = V_t(\phi)/B_t$ of any self-financing trading strategy ϕ is a local martingale under $\mathbb{P}^*$.

These notions coincide within the present setup, provided the stock S pays no dividends during the period $[0, T^*]$. The case of a dividend-paying stock is analysed in subsequent sections.

Lemma 13.2: Equivalence of Martingale Measures

A probability measure is a spot martingale measure if and only if it is a martingale measure for the discounted stock price S^* .

Proof. Let ϕ be a self-financing strategy and let $V^* = V^*(\phi)$. Using Itô's integration by parts formula, we get (note that $dB_t^{-1} = -rB_t^{-1} dt$)

$$\begin{aligned} dV_t^* &= d(V_t B_t^{-1}) = V_t dB_t^{-1} + B_t^{-1} dV_t \\ &= (\phi_t^1 S_t + \phi_t^2 B_t) dB_t^{-1} + B_t^{-1} (\phi_t^1 dS_t + \phi_t^2 dB_t) \\ &= \phi_t^1 (B_t^{-1} dS_t + S_t dB_t^{-1}) = \phi_t^1 dS_t^*. \end{aligned}$$

We thus obtain the following equality

$$V_t^*(\phi) = V_0^*(\phi) + \int_0^t \phi_u^1 dS_u^*, \quad \forall t \in [0, T^*].$$

It suffices to make use of the local martingale property of the Itô integral. □

In the Black-Scholes setting, the martingale measure for the discounted stock price is unique and can be explicitly derived using the Girsanov theorem.

Lemma 13.3: Unique Martingale Measure

(i) The unique martingale measure $\mathbb{Q}$ for the discounted stock price process S^* is given by the Radon-Nikodým derivative

$$\frac{d\mathbb{Q}}{d\mathbb{P}} \stackrel{\text{a.s.}}{=} \exp \left(\frac{r - \mu}{\sigma} W_{T^*} - \frac{1}{2} \left(\frac{r - \mu}{\sigma} \right)^2 T^* \right) \quad (13.21)$$

(ii) Under the martingale measure $\mathbb{Q}$, the discounted stock price S^* satisfies

$$dS_t^* = \sigma S_t^* dW_t^*, \quad (13.22)$$

where the continuous, $\mathcal{F}$ -adapted process W^* defined by

$$W_t^* = W_t - \frac{r - \mu}{\sigma} t, \quad \forall t \in [0, T^*], \quad (13.23)$$

is a standard Brownian motion on the probability space $(\Omega, \mathcal{F}, \mathbb{Q})$.

Proof. Let us first prove part (i).

$$\left. \frac{d\mathbb{Q}}{d\mathbb{P}} \right|_{\mathcal{F}_t} = \mathcal{E}_t \left(\int_0^t \gamma_u dW_u - \frac{1}{2} \int_0^t \gamma_u^2 du \right), \quad \mathbb{P} \text{ a.s.}$$

Moreover, by Girsanov's theorem, the process W^* defined by

$$W_t^* = W_t - \int_0^t \gamma_u du, \quad \forall t \in [0, T^*],$$

is a standard Brownian motion under $\mathbb{Q}$. Substituting $dW_t = dW_t^* + \gamma_t dt$ into the dynamics of S^* in [eq. \(13.20\)](#), we obtain

$$dS_t^* = (\mu - r + \gamma_t \sigma) S_t^* dt + \sigma S_t^* dW_t^*, \quad S_0^* = S_0.$$

□

Definition 13.4: Admissible Trading Strategy

A trading strategy $\phi \in \Phi$ is called $\mathbb{P}^*$ -admissible if the discounted wealth process

$$V_t^*(\phi) = B_t^{-1} V_t(\phi), \quad \forall t \in [0, T],$$

follows a martingale under $\mathbb{P}^*$. We write $\Phi(\mathbb{P}^*)$ to denote the class of all $\mathbb{P}^*$ -admissible trading strategies. The triple $\mathcal{M}_{BS} = (S, B, \Phi(\mathbb{P}^*))$ is called the *arbitrage-free Black-Scholes model* of a financial market, or briefly, the *Black-Scholes model*.

For brevity, we shall say that ϕ is *admissible* rather than $\mathbb{P}^*$ -admissible if no confusion may arise. It is easy to check that by restricting our attention to the class of all admissible strategies, we have ensured the absence of arbitrage opportunities in the Black-Scholes model.

We say that a European contingent claim X that settles at time $T \leq T^*$ is *attainable* in the Black-Scholes model if it can be replicated by means of an admissible strategy. Given an attainable European contingent claim, we can uniquely define its arbitrage price in the Black-Scholes model.

Definition 13.5: Arbitrage Price Process

Let X be an attainable European contingent claim that settles at time $T \leq T^*$. Then its *arbitrage price process* in the Black-Scholes model $\mathcal{M}_{BS}$, denoted as $\pi_t(X)$, $t \in [0, T]$, is defined as the wealth process $V_t(\phi)$, $t \in [0, T]$, of any admissible trading strategy ϕ that replicates X , that is, satisfies the equality $V_T(\phi) = X$.

If no replicating admissible strategy exists², the arbitrage price of such a claim is not defined. Conforming to the definition of an arbitrage price, to value a derivative security, we typically first search for its replicating strategy. The risk-neutral valuation approach to the pricing problem is also possible, as the following simple result shows.

²This happens only if a claim is not integrable under $\mathbb{P}^*$

Corollary 13.3: Risk-Neutral Valuation Formula

Let X be an attainable European contingent claim that settles at time T . Then the arbitrage price $\pi_t(X)$ at time $t \in [0, T]$ in $\mathcal{M}_{BS}$ is given by the *risk-neutral valuation formula*

$$\pi_t(X) = B_t \mathbb{E}_{\mathbb{P}^*}(B_T^{-1} X \mid \mathcal{F}_t), \quad \forall t \in [0, T]. \quad (13.24)$$

13.5 Black-Scholes Option Pricing Formula

Two alternative justifications of the option valuation formula are provided. The first relies on the fact that the risk-free return can be replicated by holding a continuously adjusted position in the underlying stock and an option. In other words, if an option is not priced in accordance with the Black-Scholes formula, there is a sure profit to be made by some combination of either short or long sales of the option and the underlying asset. The second method of derivation is based on equilibrium arguments, which require, in particular, that the option earns an expected rate of return commensurate with the risk involved in holding the option as an asset. The first approach is usually referred to as the *risk-free portfolio* method, while the second is known as the *equilibrium derivation* of the Black-Scholes formula.

The replication approach presented below is based on the observation that in the Black-Scholes setting the option value can be mimicked by holding a continuously rebalanced position in the underlying stock and risk-free bonds. Recall that we assume throughout that the financial market we are dealing with is perfect (partially, this was already implicit in the definition of a self-financing trading strategy). We shall first consider a European call option written on a stock S , with expiry date T and strike price K . Hence the payoff (and the price) at time T equals $(S_T - K)^+$. Let the function $c : \mathbb{R}_+ \times (0, T] \rightarrow \mathbb{R}$ be given by the formula

$$c(s, t) = s\Phi(d_1(s, t)) - Ke^{-rt}\Phi(d_2(s, t)), \quad (13.25)$$

where

$$d_1(s, t) = \frac{\ln(s/K) + (r + \frac{1}{2}\sigma^2)t}{\sigma\sqrt{t}}, \quad (13.26)$$

and $d_2(s, t) = d_1(s, t) - \sigma\sqrt{t}$, or explicitly

$$d_2(s, t) = \frac{\ln(s/K) + (r - \frac{1}{2}\sigma^2)t}{\sigma\sqrt{t}}. \quad (13.27)$$

Furthermore, let Φ stand for the standard Gaussian cumulative distribution function

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-z^2/2} dz, \quad \forall x \in \mathbb{R}.$$

We adopt the following notational convention

$$d_{1,2}(s, t) = \frac{\ln(s/K) + (r \pm \frac{1}{2}\sigma^2)t}{\sigma\sqrt{t}}.$$

Let us denote by C_t the arbitrage price at time t of a European call option in the Black-Scholes model.

Theorem 13.1: Arbitrage Price of European Call Option

The arbitrage price at time $t < T$ of the European call option with expiry date T and strike price K in the Black-Scholes model is given by the formula

$$C_t = c(S_t, T - t), \quad \forall t \in [0, T), \quad (13.28)$$

where the function $c : \mathbb{R}_+ \times (0, T] \rightarrow \mathbb{R}$ is given by [eq. \(13.25\)](#). More explicitly, we have, for any $t < T$

$$C_t = S_t \Phi(d_1(S_t, T - t)) - KB(t, T) \Phi(d_2(S_t, T - t)),$$

where d_1, d_2 are given by eqs. (13.26) and (13.27). Moreover, the unique admissible replicating strategy $\phi = (\phi^1, \phi^2)$ of the call option satisfies, for every $t \in [0, T]$,

$$\phi_t^1 = \frac{\partial c}{\partial s}(S_t, T - t) = \Phi(d_1(S_t, T - t))$$

and

$$\phi_t^2 = e^{-rt}(c(S_t, T - t) - \phi_t^1 S_t).$$

13.6 Put-Call Parity for Spot Options

We shall show that if no arbitrage opportunities exist, puts and calls with the same strike and maturity must at all times during their lives obey the put-call parity relationship. Let us mention, in this regard, that empirical studies of the put-call parity have been reported (Gould and D. Galai, 1974; Klemkosky and Resnick, 1979).

A point worth stressing is that equality (13.29) does not rely on specific assumptions imposed on the stock price model. Indeed, it is satisfied in any arbitrage-free, continuous-time model of a security market, provided that the savings account B is given by the formula $B_t = e^{rt}$.

Proposition 13.2: Put-Call Parity

The arbitrage prices of European call and put options with the same expiry date T and strike price K satisfy the *put-call parity relationship*

$$C_t - P_t = S_t - Ke^{-r(T-t)} \quad (13.29)$$

for every $t \in [0, T]$.

The put-call parity can be used to derive a closed-form expression for the arbitrage price of a European put option. Let us denote by $p : \mathbb{R}_+ \times (0, T] \rightarrow \mathbb{R}$ the function

$$p(s, t) = Ke^{-rt}\Phi(-d_2(s, t)) - s\Phi(-d_1(s, t)), \quad (13.30)$$

with $d_1(s, t)$ and $d_2(s, t)$ given by eqs. (13.26) and (13.27). The following result is an immediate consequence of proposition 13.2 combined with theorem 13.1.

Corollary 13.4: Black-Scholes Price of European Put Option

The Black-Scholes price at time $t < T$ of a European put option with strike price K equals $P_t = p(S_t, T - t)$, where the function $p : \mathbb{R}_+ \times [0, T] \rightarrow \mathbb{R}$ is given by eq. (13.30). More explicitly, the price at time $t < T$ of a European put option equals

$$P_t = KB(t, T)\Phi(-d_2(S_t, T - t)) - S_t\Phi(-d_1(S_t, T - t)). \quad (13.31)$$

13.7 Black-Scholes PDE

Random payoffs of such a simple form are termed *path-independent* claims, as opposed to *path-dependent* contingent claims

Path-dependent payoffs correspond to the various kinds of OTC options, known under the generic name of *exotic options*.

Lemma 13.4: PDE for Brownian Functionals

Let W be a one-dimensional Brownian motion defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. For a Borel-measurable function $h : \mathbb{R} \rightarrow \mathbb{R}$, we define the function $u : \mathbb{R} \times [0, T] \rightarrow \mathbb{R}$ by setting

$$u(x, t) = \mathbb{E}_{\mathbb{P}}(e^{-r(T-t)}h(W_T) \mid W_t = x), \quad \forall (x, t) \in \mathbb{R} \times [0, T]. \quad (13.32)$$

Suppose that

$$\int_{-\infty}^{+\infty} e^{-ax^2}|h(x)| dx < \infty \quad (13.33)$$

for some $a > 0$. Then the function u is defined for $0 < T - t < 1/2a$ and $x \in \mathbb{R}$, and has derivatives of all orders. In particular, it belongs to the class $C^{2,1}(\mathbb{R} \times (0, T))$ and satisfies

the following PDE

$$-\frac{\partial u}{\partial t}(x, t) = \frac{1}{2} \frac{\partial^2 u}{\partial x^2}(x, t) - ru(x, t), \quad \forall (x, t) \in \mathbb{R} \times (0, T), \quad (13.34)$$

with the terminal condition $u(x, T) = h(x)$ for $x \in \mathbb{R}$.

14 A Dividend-paying Stock

Up to now, we have studied arbitrage pricing within the Black-Scholes framework under the assumption that the stock upon which an option is written pays no dividends during the option's lifetime. We assume that the dividends (or the dividend rate) that will be paid to the shareholders during an option's lifetime can be predicted with certainty.

14.1 Case of a Constant Dividend Yield

We will first examine the case when the dividend yield, rather than the dividend payoffs, is assumed to be known.

The dividend payments should be used in full, either to purchase additional shares of stock or to invest in risk-free bonds (however, intertemporal consumption or infusion of funds is not allowed).

Definition 14.1: Self-Financing Strategy

A trading strategy $\phi = (\phi^1, \phi^2)$ is said to be *self-financing* when its wealth process $V(\phi)$, which equals $V_t(\phi) = \phi_t^1 S_t + \phi_t^2 B_t$, satisfies

$$dV_t(\phi) = \phi_t^1 dS_t + \kappa \phi_t^1 S_t dt + \phi_t^2 dB_t. \quad (14.1)$$

In view of the postulated dynamics of the stock price, formula [eq. \(14.1\)](#) can also be represented as follows

$$dV_t(\phi) = \phi_t^1 (\mu + \kappa) S_t dt + \phi_t^1 \sigma S_t dW_t + \phi_t^2 dB_t. \quad (14.2)$$

We find it convenient to introduce an auxiliary process $\tilde{S}_t = e^{\kappa t} S_t$, whose dynamics are given by the SDE

$$d\tilde{S}_t = e^{\kappa t} (dS_t + \kappa S_t dt) = \mu_{\kappa} \tilde{S}_t dt + \sigma \tilde{S}_t dW_t, \quad (14.3)$$

where we write $\mu_\kappa = \mu + \kappa$. In terms of the process $\widetilde{S}$, we obtain

$$V_t(\phi) = \phi_t^1 e^{-\kappa t} \widetilde{S}_t + \phi_t^2 B_t \quad (14.4)$$

and

$$dV_t(\phi) = \phi_t^1 e^{-\kappa t} d\widetilde{S}_t + \phi_t^2 dB_t. \quad (14.5)$$

Also, it is not difficult to check that the discounted wealth $V^*(\phi)$ satisfies

$$dV_t^*(\phi) = \phi_t^1 e^{-\kappa t} d\widetilde{S}_t^*, \quad (14.6)$$

where we denote $\widetilde{S}_t^* = \widetilde{S}_t B_t^{-1}$. More explicitly, we have that

$$dV_t^*(\phi) = \sigma \phi_t^1 e^{-\kappa t} \widetilde{S}_t^* (dW_t + \sigma^{-1}(\mu_\kappa - r)dt). \quad (14.7)$$

14.2 Case of Known Dividends

We will now focus on valuing European options on a stock with declared dividends during the option's lifetime. We assume that the amount of dividends to be paid before the option's expiry and the dates of their payments are known in advance.

As usual, it is sufficient to consider the case $t = 0$. Assume that the dividends $\kappa_1, \dots, \kappa_m$ are to be paid at the dates $0 < T_1 < \dots < T_m < T$. We assume, in addition, that dividend payments are known in advance; that is, $\kappa_1, \dots, \kappa_m$ are real numbers. The present value of all future dividend payments equals

$$\widetilde{I}_t = \sum_{j=1}^m \kappa_j e^{-r(T_j-t)} \mathbb{1}_{[t,T]}(T_j), \quad \forall t \in [0, T], \quad (14.8)$$

and the value of all the dividends paid after time t and compounded at the risk-free rate to the option's expiry date T is given by the expression

$$I_t = \sum_{j=1}^m \kappa_j e^{r(T-T_j)} \mathbb{1}_{[t,T]}(T_j), \quad \forall t \in [0, T]. \quad (14.9)$$

In modelling the stock price, we will separate the capital gains process from the impact of dividend payments. The capital gains process G is assumed to follow the usual stochastic differential equation (Heath and Jarrow, 1988).

$$dG_t = \mu G_t dt + \sigma G_t dW_t, \quad G_0 > 0, \quad (14.10)$$

hence G follows a geometric Brownian motion. The process S , representing the price of a stock that pays dividends $\kappa_1, \dots, \kappa_m$ at times $T_1, \dots, T_m$ may be introduced by setting

$$S_t = G_t - \sum_{j=1}^m \kappa_j e^{r(t-T_j)} \mathbb{1}_{[T_j, T]}(t) = G_t - D_t, \quad (14.11)$$

where the process D , given by the equality

$$D_t = \sum_{j=1}^m \kappa_j e^{r(t-T_j)} \mathbb{1}_{[T_j, T]}(t), \quad (14.12)$$

accounts for the impact of ex-dividend stock price decline. Alternatively, one may put, for $t \in [0, T]$,

$$S_t = G_t + \sum_{j=1}^m \kappa_j e^{-r(T_j-t)} \mathbb{1}_{[0, T_j]}(t) = G_t + \tilde{D}_t, \quad (14.13)$$

where $\tilde{D}$ now satisfies

$$\tilde{D}_t = \sum_{j=1}^m \kappa_j e^{-r(T_j-t)} \mathbb{1}_{[0, T_j]}(t). \quad (14.14)$$

Note that in the first case we have $G_0 = S_0$ and $G_T = S_T + D_T = S_T + I_0$, while in the second case $G_0 = S_0 - \tilde{D}_0 = S_0 - \tilde{I}_0$ and $G_T = S_T$.

Not surprisingly, the option valuation formulas corresponding to various specifications of the stock price (that is, to (14.11) and (14.13)) are not identical.

As usual, the first step in option valuation is to introduce the notion of a self-financing trading strategy. Under the present assumptions, it is natural to assume that if $\phi = (\phi^1, \phi^2)$ is a self-financing trading strategy, then all dividends received during the option's lifetime are immediately reinvested in stocks or bonds. Hence the wealth process of any trading strategy ϕ , which equals $V_t(\phi) = \phi_t^1 S_t + \phi_t^2 B_t$, satisfies, as usual,

$$dV_t(\phi) = \phi_t^1 dG_t + \phi_t^2 dB_t. \quad (14.15)$$

Intuitively, since the decline of the stock price equals the dividend, the wealth of a portfolio is not influenced by dividend payments. It turns out that the resulting option valuation formula will not agree with the standard Black-Scholes result, however. In view of (14.15), the martingale measure $\mathbb{P}^*$ for the security market model corresponds to the unique martingale measure for the discounted capital gains process $G_t^* = G_t B_t^{-1}$. Hence $\mathbb{P}^*$ can be found in exactly the same way as in the proof of theorem 13.1. We can also price options using the standard risk-neutral valuation approach.

Proposition 14.1: Valuation with Deterministic Dividends

Consider a European call option with strike price K and expiry date T , written on a stock S that pays deterministic dividends $\kappa_1, \dots, \kappa_m$ at times $T_1, \dots, T_m$. Assume that the stock price S satisfies eqs. (14.10) and (14.11). Then the arbitrage price at time t of this option equals

$$C_t^K = S_t N(h_1(S_t, T - t)) - e^{-r(T-t)} (K + I_t) N(h_2(S_t, T - t)), \quad (14.16)$$

where I_t is given by expression (14.9), and

$$h_{1,2}(S_t, T - t) = \frac{\ln S_t - \ln(K + I_t) + (r \pm \frac{1}{2}\sigma^2)(T - t)}{\sigma\sqrt{T - t}}. \quad (14.17)$$

15 Robustness of the Black-Scholes Approach

15.1 Uncertain Volatility

15.2 European Call and Put Options

16 Discretisation Schemes

16.1 Stochastic Taylor Expansion

Deterministic Taylor expansions are well known. To begin, we shall consider the solution X_t of a 1-dimensional ordinary differential equation

$$\frac{d}{dt}X_t = a(X_t), \quad (16.1)$$

with initial value X_{t_0} , for $t \in [t_0, T]$ where $0 \leq t_0 < T$. We can write this in the equivalent integral equation form

$$X_t = X_{t_0} + \int_{t_0}^t a(X_s) ds. \quad (16.2)$$

To justify the following constructions, we shall require the function a to satisfy appropriate properties, for example, to be sufficiently smooth and to have a linear growth bound. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuously differentiable function. Then, by the chain rule, we have

$$\frac{d}{dt}f(X_t) = a(X_t) \frac{\partial}{\partial x}f(X_t), \quad (16.3)$$

which, using the operator

$$L = a \frac{\partial}{\partial x}, \quad (16.4)$$

we can express as the integral relation

$$f(X_t) = f(X_{t_0}) + \int_{t_0}^t Lf(X_s) ds \quad (16.5)$$

for all $t \in [t_0, T]$. When $f(x) \equiv x$ we have $Lf = a$, $L^2f = La, \dots$ and (16.5) reduces to

$$X_t = X_{t_0} + \int_{t_0}^t a(X_s) ds, \quad (16.6)$$

that is to equation (??).

If we now apply the relation (16.5) to the function $f = a$ in the integral in (16.6), we obtain

$$\begin{aligned} X_t &= X_{t_0} + \int_{t_0}^t \left(a(X_{t_0}) + \int_{t_0}^s La(X_z) dz \right) ds \\ &= X_{t_0} + a(X_{t_0}) \int_{t_0}^t ds + \int_{t_0}^t \int_{t_0}^s La(X_z) dz ds, \end{aligned} \quad (16.7)$$

which is the simplest nontrivial Taylor expansion for X_t . We can apply (16.5) again to the function $f = La$ in the double integral to derive

$$X_t = X_{t_0} + a(X_{t_0}) \int_{t_0}^t ds + La(X_{t_0}) \int_{t_0}^t \int_{t_0}^s dz ds + R_3 \quad (16.8)$$

with remainder

$$R_3 = \int_{t_0}^t \int_{t_0}^s \int_{t_0}^z L^2 a(X_u) du dz ds, \quad (16.9)$$

for $t \in [t_0, T]$. For a general $r + 1$ times continuously differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ this method gives the classical Taylor formula in integral form:

$$\begin{aligned} f(X_t) &= f(X_{t_0}) + \sum_{l=1}^r \frac{(t - t_0)^l}{l!} L^l f(X_{t_0}) \\ &\quad + \int_{t_0}^t \cdots \int_{t_0}^{s_2} L^{r+1} f(X_{s_1}) ds_1 \cdots ds_{r+1} \end{aligned} \quad (16.10)$$

for $t \in [t_0, T]$ and $r = 1, 2, 3, \dots$

The Taylor formula (16.10) has proven to be a very useful tool in both theoretical and practical investigations, particularly in numerical analysis. It allows the approximation of a suf-

ficiently smooth function in a neighbourhood of a given point to any desired order of accuracy. This expansion depends on the values of the function and some of its higher derivatives at the expansion point, weighted by corresponding multiple time integrals. In addition, there is a remainder term which contains the next multiple time integral, but now with a time-dependent integrand.

We shall indicate it here for the solution X_t of the 1-dimensional Ito stochastic differential equation in integral form

$$X_t = X_{t_0} + \int_{t_0}^t a(X_s) ds + \int_{t_0}^t b(X_s) dW_s \quad (16.11)$$

for $t \in [t_0, T]$, where the second integral in (16.11) is an Ito stochastic integral and the coefficients a and b are sufficiently smooth real-valued functions satisfying a linear growth bound. Then, for any twice continuously differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ the Ito formula [eq. \(16.12\)](#) gives

$$\begin{aligned} f(X_t) &= f(X_{t_0}) \\ &+ \int_{t_0}^t \left(a(X_s) \frac{\partial}{\partial x} f(X_s) + \frac{1}{2} b^2(X_s) \frac{\partial^2}{\partial x^2} f(X_s) \right) ds \\ &+ \int_{t_0}^t b(X_s) \frac{\partial}{\partial x} f(X_s) dW_s \\ &= f(X_{t_0}) + \int_{t_0}^t L^0 f(X_s) ds + \int_{t_0}^t L^1 f(X_s) dW_s, \end{aligned} \quad (16.12)$$

for $t \in [t_0, T]$. Here we have introduced the operators

$$L^0 = a \frac{\partial}{\partial x} + \frac{1}{2} b^2 \frac{\partial^2}{\partial x^2} \quad (16.13)$$

and

$$L^1 = b \frac{\partial}{\partial x}. \quad (16.14)$$

Obviously, for $f(x) \equiv x$ we have $L^0 f = a$ and $L^1 f = b$, in which case (16.12) reduces to the original Ito equation for X_t , that is to

$$X_t = X_{t_0} + \int_{t_0}^t a(X_s) ds + \int_{t_0}^t b(X_s) dW_s. \quad (16.15)$$

In analogy with the above deterministic expansions, if we apply the Ito formula (16.12) to the functions $f = a$ and $f = b$ in (16.15) we obtain

$$\begin{aligned} X_t &= X_{t_0} \\ &+ \int_{t_0}^t \left(a(X_{t_0}) + \int_{t_0}^s L^0 a(X_z) dz + \int_{t_0}^s L^1 a(X_z) dW_z \right) ds \\ &+ \int_{t_0}^t \left(b(X_{t_0}) + \int_{t_0}^s L^0 b(X_z) dz + \int_{t_0}^s L^1 b(X_z) dW_z \right) dW_s \\ &= X_{t_0} + a(X_{t_0}) \int_{t_0}^t ds + b(X_{t_0}) \int_{t_0}^t dW_s + R \end{aligned} \quad (16.16)$$

with remainder

$$\begin{aligned} R &= \int_{t_0}^t \int_{t_0}^s L^0 a(X_z) dz ds + \int_{t_0}^t \int_{t_0}^s L^1 a(X_z) dW_z ds \\ &+ \int_{t_0}^t \int_{t_0}^s L^0 b(X_z) dz dW_s + \int_{t_0}^t \int_{t_0}^s L^1 b(X_z) dW_z dW_s. \end{aligned} \quad (16.17)$$

This is the simplest nontrivial Ito-Taylor expansion. We can continue it, for instance, by applying the Ito formula (16.12) to $f = L^1 b$ in (16.16), in which case we get

$$\begin{aligned} X_t &= X_{t_0} + a(X_{t_0}) \int_{t_0}^t ds + b(X_{t_0}) \int_{t_0}^t dW_s \\ &+ L^1 b(X_{t_0}) \int_{t_0}^t \int_{t_0}^s dW_z dW_s + \bar{R} \end{aligned} \quad (16.18)$$

with remainder

$$\begin{aligned}
\bar{R} = & \int_{t_0}^t \int_{t_0}^s L^0 a(X_z) dz ds + \int_{t_0}^t \int_{t_0}^s L^1 a(X_z) dW_z ds \\
& + \int_{t_0}^t \int_{t_0}^s L^0 b(X_z) dz dW_s + \int_{t_0}^t \int_{t_0}^s \int_{t_0}^z L^0 L^1 b(X_u) du dW_z dW_s \\
& + \int_{t_0}^t \int_{t_0}^s \int_{t_0}^z L^1 L^1 b(X_u) dW_u dW_z dW_s.
\end{aligned} \tag{16.19}$$

We shall now consider another representation for the increments of a function of an Ito process X , which we shall call the *Stratonovich-Taylor expansion*. For this we start with the 1-dimensional Stratonovich stochastic differential equation in integral form

$$X_t = X_{t_0} + \int_{t_0}^t \underline{a}(X_s) ds + \int_{t_0}^t b(X_s) \circ dW_s \tag{16.20}$$

for $t \in [t_0, T]$, where the second integral in (16.20) is a Stratonovich stochastic integral and the coefficients $\underline{a}$ and b are sufficiently smooth real valued functions satisfying a linear growth bound. Consider that

$$\underline{a} = a - \frac{1}{2}bb'$$

the eq. (16.11) and the Stratonovich equation (16.20) are equivalent and have the same solutions. The solution of a Stratonovich SDE transforms according to the deterministic chain rule, so for any twice continuously differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ we have

$$\begin{aligned}
f(X_t) &= f(X_{t_0}) + \int_{t_0}^t \underline{a}(X_s) \frac{\partial}{\partial x} f(X_s) ds + \int_{t_0}^t b(X_s) \frac{\partial}{\partial x} f(X_s) \circ dW_s, \\
&= f(X_{t_0}) + \int_{t_0}^t \underline{L}^0 f(X_s) ds + \int_{t_0}^t \underline{L}^1 f(X_s) \circ dW_s,
\end{aligned} \tag{16.21}$$

for $t \in [t_0, T]$, with the operators

$$\underline{L}^0 = \underline{a} \frac{\partial}{\partial x} \quad (16.22)$$

and

$$\underline{L}^1 = b \frac{\partial}{\partial x}. \quad (16.23)$$

Obviously, for $f(x) \equiv x$ we have $\underline{L}^0 f = \underline{a}$ and $\underline{L}^1 f = b$, in which case (16.21) reduces to the original Stratonovich equation

$$X_t = X_{t_0} + \int_{t_0}^t \underline{a}(X_s) ds + \int_{t_0}^t b(X_s) \circ dW_s. \quad (16.24)$$

Analogously with the Ito case just considered, we can apply (16.21) to the integrand functions $f = \underline{a}$ and $f = b$ in (16.22). This gives

$$\begin{aligned} X_t &= X_{t_0} \\ &+ \int_{t_0}^t \left(\underline{a}(X_{t_0}) + \int_{t_0}^s \underline{L}^0 \underline{a}(X_z) dz + \int_{t_0}^s \underline{L}^1 \underline{a}(X_z) \circ dW_z \right) ds \\ &+ \int_{t_0}^t \left(b(X_{t_0}) + \int_{t_0}^s \underline{L}^0 b(X_z) dz + \int_{t_0}^s \underline{L}^1 b(X_z) \circ dW_z \right) \circ dW_s \\ &= X_{t_0} + \underline{a}(X_{t_0}) \int_{t_0}^t ds + b(X_{t_0}) \int_{t_0}^t \circ dW_s + R \end{aligned} \quad (16.25)$$

with remainder

$$\begin{aligned} R &= \int_{t_0}^t \int_{t_0}^s \underline{L}^0 \underline{a}(X_z) dz ds + \int_{t_0}^t \int_{t_0}^s \underline{L}^1 \underline{a}(X_z) \circ dW_z ds \\ &+ \int_{t_0}^t \int_{t_0}^s \underline{L}^0 b(X_z) dz \circ dW_s + \int_{t_0}^t \int_{t_0}^s \underline{L}^1 b(X_z) \circ dW_z \circ dW_s. \end{aligned} \quad (16.26)$$

This is the simplest nontrivial Stratonovich-Taylor expansion of $f(X_t)$. We can continue expanding, for instance, by applying (16.21) to the integrand $f = \underline{L}^1 b$ in (16.25) to obtain

$$\begin{aligned}
X_t = X_{t_0} &+ \underline{a}(X_{t_0}) \int_{t_0}^t ds + b(X_{t_0}) \int_{t_0}^t \circ dW_s \\
&+ \underline{L}^1 b(X_{t_0}) \int_{t_0}^t \int_{t_0}^s \circ dW_z \circ dW_s + \bar{R}
\end{aligned} \tag{16.27}$$

with remainder

$$\begin{aligned}
\bar{R} = & \int_{t_0}^t \int_{t_0}^s \underline{L}^0 \underline{a}(X_z) dz ds + \int_{t_0}^t \int_{t_0}^s \underline{L}^1 \underline{a}(X_z) \circ dW_z ds \\
& + \int_{t_0}^t \int_{t_0}^s \underline{L}^0 b(X_z) dz \circ dW_s \\
& + \int_{t_0}^t \int_{t_0}^s \int_{t_0}^z \underline{L}^0 \underline{L}^1 b(X_u) du \circ dW_z \circ dW_s \\
& + \int_{t_0}^t \int_{t_0}^s \int_{t_0}^z \underline{L}^1 \underline{L}^1 b(X_u) \circ dW_u \circ dW_z \circ dW_s.
\end{aligned} \tag{16.28}$$

Similar expansions hold for multi-dimensional Ito processes satisfying nonautonomous stochastic differential equations. We shall refer to the nonautonomous d -dimensional Ito equation

$$X_t = X_{t_0} + \int_{t_0}^t a(s, X_s) ds + \sum_{j=1}^m \int_{t_0}^t b^j(s, X_s) dW_s^j \tag{16.29}$$

for $t \in [t_0, T]$ and the equivalent Stratonovich equation

$$X_t = X_{t_0} + \int_{t_0}^t \underline{a}(s, X_s) ds + \sum_{j=1}^m \int_{t_0}^t b^j(s, X_s) \circ dW_s^j \tag{16.30}$$

for $t \in [t_0, T]$, where

$$\underline{a}^i = a^i - \frac{1}{2} \sum_{j=1}^m \sum_{k=1}^d b^{k,j} \frac{\partial b^{i,j}}{\partial x^k} \tag{16.31}$$

for $i = 1, 2, \dots, d$. In both of these SDEs $W = \{W_t, t \in [0, T]\}$ is a standard m -dimensional Wiener process adapted to an increasing family of sub- σ -algebras $\{\mathcal{A}_t, t \in [0, T]\}$.

16.2 Strong Convergence Criterion

In simulation, filtering, and estimation of Itô processes, the sample paths of the approximation must closely track those of the underlying process. The primary criterion for this proximity is the absolute error at the final time instant T .

$$\varepsilon(\delta) = \mathbb{E}(|X_T - Y_N|). \quad (16.32)$$

This error is bounded by the root mean square error via the *Lyapunov inequality*:

$$\varepsilon(\delta) = \mathbb{E}(|X_T - Y_N|) \leq \sqrt{\mathbb{E}(|X_T - Y_N|^2)}. \quad (16.33)$$

Definition 16.1: Strong Order of Convergence

An approximating process Y converges in the *strong sense with order* $\gamma \in (0, \infty]$ if there exists a finite constant K and a positive constant δ_0 such that

$$\mathbb{E}(|X_T - Y_N|) \leq K\delta^\gamma \quad (16.34)$$

for any time discretization with maximum step size $\delta \in (0, \delta_0)$.



In the deterministic limit where diffusion vanishes ($b \equiv 0$), the strong convergence criterion reduces to the standard error criterion for ordinary differential equations (ODEs). Stochastic schemes generically exhibit lower convergence orders than their deterministic counterparts because the regularity of the Wiener process dictates that increments ΔW_n scale as $\delta^{1/2}$ in the root mean square sense rather than δ . This intrinsic degradation results in the Euler approximation achieving a strong order of $\gamma = 0.5$ for SDEs in contrast to the order 1.0 obtained for ODEs.

16.3 Weak Convergence Criterion

In many practical scenarios, such as moment estimation or option pricing, a precise pathwise approximation of the Itô process is not required. Instead, the focus is often on the expectation of a function of the process at a final time T , such as the first two moments $\mathbb{E}(X_T)$ and $\mathbb{E}((X_T)^2)$, or more generally $\mathbb{E}(g(X_T))$ for some function g . In these cases, it suffices to approximate the probability distribution of the random variable X_T rather than the sample paths. This leads to a weaker convergence criterion than the strong sense.

Definition 16.2: Weak Order of Convergence

A time discrete approximation Y converges in the *weak sense with order* $\beta \in (0, \infty]$ if for any polynomial g there exists a finite constant K and a positive constant δ_0 such that

$$|\mathbb{E}(g(X_T)) - \mathbb{E}(g(Y_N))| \leq K\delta^\beta \quad (16.35)$$

for any time discretization with maximum step size $\delta \in (0, \delta_0)$.

In the limit of vanishing diffusion where $b \equiv 0$, the weak convergence criterion with $g(x) \equiv x$ collapses to the standard deterministic convergence criterion for ordinary differential equations. For stochastic systems, the weak order β frequently diverges from the strong order γ , exemplified by the Euler approximation achieving a higher weak order of $\beta = 1.0$ compared to its strong order of $\gamma = 0.5$. This distinction becomes critical when the objective involves estimating moments or probability distributions rather than simulating specific sample paths.



16.4 Euler-Maruyama

To apply the Monte Carlo method in option pricing, we generate a sequence $(Y_1, \dots, Y_N)$ of sample realisations of a random variable, such that the empirical mean

$$\mathbb{E}[\phi(X)] \approx \frac{1}{N} \sum_{i=1}^N \phi(Y_i) \quad (16.36)$$

converges to the expected value $\mathbb{E}[\phi(X)]$ by the Strong Law of Large Numbers.

Random samples for the solution of a stochastic differential equation defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ of the form

$$dX_t = b(t, X_t)dt + a(t, X_t)dW_t, \quad (16.37)$$

where $(W_t)_{t \in \mathbb{R}_+}$ is a standard Brownian motion, can be generated by time discretisation on a grid $\pi_\delta = \{0 = t_0 < t_1 < \dots < t_M = T\}$.

More precisely, the Euler-Maruyama discretisation scheme for the stochastic differential equation, [eq. \(16.37\)](#), is given by the recursive approximation Y^δ :

$$\begin{aligned} Y_{t_{k+1}}^\delta &= Y_{t_k}^\delta + \int_{t_k}^{t_{k+1}} b(t_k, Y_{t_k}^\delta) ds + \int_{t_k}^{t_{k+1}} a(t_k, Y_{t_k}^\delta) dW_s \\ &\approx Y_{t_k}^\delta + b(t_k, Y_{t_k}^\delta)(t_{k+1} - t_k) + a(t_k, Y_{t_k}^\delta)(W_{t_{k+1}} - W_{t_k}), \end{aligned} \quad (16.38)$$

where the increments satisfy $W_{t_{k+1}} - W_{t_k} \sim \mathcal{N}(0, t_{k+1} - t_k)$ for $k = 0, 1, \dots, M-1$.

Let $T > 0$ and consider the stochastic differential equation [eq. \(16.37\)](#) on the time interval $[0, T]$. Suppose that the initial data satisfies

$$\mathbb{E}(|X_0|^2) < \infty, \quad (16.39)$$

$$\mathbb{E}(|X_0 - Y_0^\delta|^2)^{1/2} \leq K_1 \delta^{1/2}, \quad (16.40)$$

and that the coefficients $a : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m}$ and $b : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ satisfy the global Lipschitz condition

$$|a(t, x) - a(t, y)| + |b(t, x) - b(t, y)| \leq K_2 |x - y| \quad (16.41)$$

and the linear growth condition

$$|a(t, x)| + |b(t, x)| \leq K_3(1 + |x|) \quad (16.42)$$

for all $t \in [0, T]$ and $x, y \in \mathbb{R}^d$. Furthermore, assume the coefficients satisfy the Hölder continuity condition in time:

$$|a(s, x) - a(t, x)| + |b(s, x) - b(t, x)| \leq K_4(1 + |x|)|s - t|^{1/2} \quad (16.43)$$

for all $s, t \in [0, T]$ and $x \in \mathbb{R}^d$. The constants $K_1, \dots, K_4$ must not depend on the discretisation step size δ .

Theorem 16.1: Strong Convergence of the Euler-Maruyama Scheme

Then, for the Euler-Maruyama approximation Y^δ defined on the grid π_δ , the following strong error estimate holds:

$$\mathbb{E} \left(|X_T - Y^\delta(T)| \right) \leq C\delta^{1/2}, \quad (16.44)$$

where the constant C depends only on $K_1, \dots, K_4$ and T , but does not depend on δ .

We illustrate Euler's method via the geometric Brownian motion SDE

`eulerMaruyama.py`

```
1 T, N, R = 1.0, 2**8, 4
2 dt = T / N
```



```
3 mu, sigma, X0 = 1.0, 0.5, 1.0
4
5 np.random.seed(4)
6 dB = np.sqrt(dt) * np.random.randn(N)
7
8 t = np.linspace(0, T, N + 1)
9 B = np.r_[0, np.cumsum(dB)]
10 Y = X0 * np.exp((mu - 0.5 * sigma**2) * t + sigma * B)
11
12 X = np.r_[X0, X0 * np.cumprod(1 + mu * dt + sigma * dB)]
13
14 dB_c = dB.reshape(-1, R).sum(axis=1)
15 X_c = np.r_[X0, X0 * np.cumprod(1 + mu*dt*R + sigma*dB_c)]
```

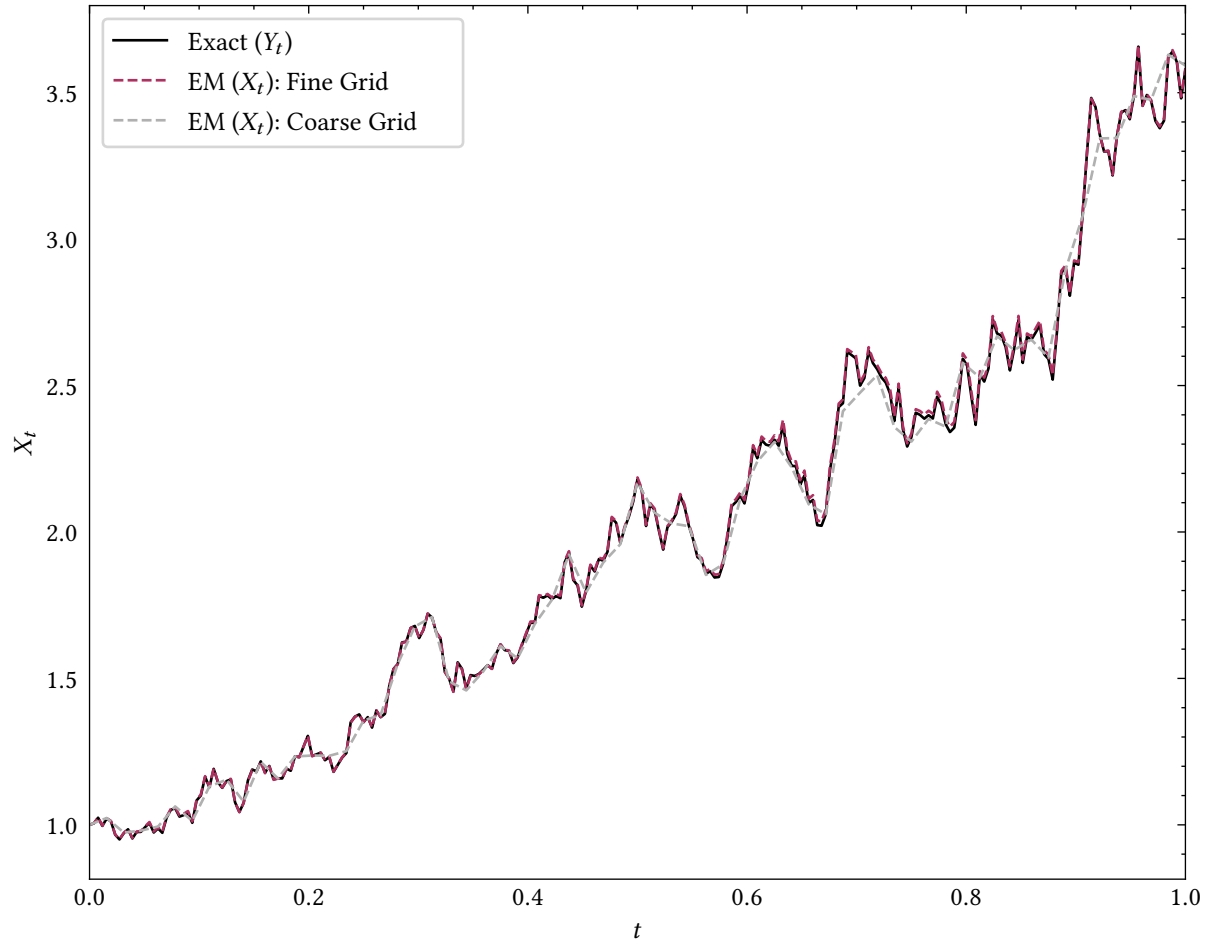


Figure 6: Comparison of Euler-Maruyama approximations against the exact analytical solution for a GBM. The simulation uses a fine grid step size of $\Delta t = 2^{-8}$ and a coarse grid step size of $\Delta t_{\text{coarse}} = 4\Delta t$. Parameters: $\mu = 1.0$, $\sigma = 0.5$, $X_0 = 1.0$.

16.5 Milstein

In the Milstein scheme we use [eq. \(16.37\)](#) to expand $a(X_s)$ as

$$\begin{aligned} a(X_s) &\simeq a(X_{t_k}) + a'(X_{t_k})(X_s - X_{t_k}) \\ &\simeq a(X_{t_k}) + a'(X_{t_k})(b(X_{t_k})(s - t_k) + a(X_{t_k})(W_s - W_{t_k})), \end{aligned}$$

$0 \leq t_k \leq s$. As a consequence, we get

$$\begin{aligned} \widehat{X}_{t_{k+1}}^N &= \widehat{X}_{t_k}^N + \int_{t_k}^{t_{k+1}} b(X_s) ds + \int_{t_k}^{t_{k+1}} a(X_s) dW_s \\ &\simeq \widehat{X}_{t_k}^N + \int_{t_k}^{t_{k+1}} b(X_s) ds + a(X_{t_k})(W_{t_{k+1}} - W_{t_k}) \\ &\quad + a'(X_{t_k})b(X_{t_k}) \int_{t_k}^{t_{k+1}} (s - t_k) dW_s \\ &\quad + a'(X_{t_k})a(X_{t_k}) \int_{t_k}^{t_{k+1}} (W_s - W_{t_k}) dW_s \\ &\simeq \widehat{X}_{t_k}^N + \int_{t_k}^{t_{k+1}} b(X_s) ds + a(X_{t_k})(W_{t_{k+1}} - W_{t_k}) \\ &\quad + a'(X_{t_k})a(X_{t_k}) \int_{t_k}^{t_{k+1}} (W_s - W_{t_k}) dW_s. \end{aligned}$$

Next, using Ito's formula, we note that

$$(W_{t_{k+1}} - W_{t_k})^2 = 2 \int_{t_k}^{t_{k+1}} (W_s - W_{t_k}) dW_s + \int_{t_k}^{t_{k+1}} ds,$$

hence

$$\int_{t_k}^{t_{k+1}} (W_s - W_{t_k}) dW_s = \frac{1}{2}((W_{t_{k+1}} - W_{t_k})^2 - (t_{k+1} - t_k))$$

and

$$\begin{aligned}
 \widehat{X}_{t_{k+1}}^N &\simeq \widehat{X}_{t_k}^N + \int_{t_k}^{t_{k+1}} b(X_s) ds + a(X_{t_k})(W_{t_{k+1}} - W_{t_k}) \\
 &\quad + \frac{1}{2} a'(X_{t_k}) a(X_{t_k}) ((W_{t_{k+1}} - W_{t_k})^2 - (t_{k+1} - t_k)) \\
 &\simeq \widehat{X}_{t_k}^N + b(X_{t_k})(t_{k+1} - t_k) + a(X_{t_k})(W_{t_{k+1}} - W_{t_k}) \\
 &\quad + \frac{1}{2} a'(X_{t_k}) a(X_{t_k}) ((W_{t_{k+1}} - W_{t_k})^2 - (t_{k+1} - t_k)).
 \end{aligned}$$

As a consequence, the Milstein scheme is written as

$$\begin{aligned}
 \widehat{X}_{t_{k+1}}^N &\simeq \widehat{X}_{t_k}^N + b(\widehat{X}_{t_k}^N)(t_{k+1} - t_k) + a(\widehat{X}_{t_k}^N)(W_{t_{k+1}} - W_{t_k}) \\
 &\quad + \frac{1}{2} a'(\widehat{X}_{t_k}^N) a(\widehat{X}_{t_k}^N) ((W_{t_{k+1}} - W_{t_k})^2 - (t_{k+1} - t_k)),
 \end{aligned}$$

i.e., in the Milstein scheme, we take into account the "small" difference

$$(W_{t_{k+1}} - W_{t_k})^2 - (t_{k+1} - t_k)$$

existing between $(\Delta W_t)^2$ and Δt . Taking $(\Delta W_t)^2$ equal to Δt brings us back to the Euler scheme.

Let $T > 0$. Suppose that the initial data satisfies

$$\mathbb{E}(|X_0|^2) < \infty, \tag{16.45}$$

$$\mathbb{E}\left(|X_0 - Y_0^\delta|^2\right)^{1/2} \leq K_1 \delta^{1/2}. \tag{16.46}$$

We assume higher-order regularity on the coefficients compared to the Euler scheme. Specifically, let the differential operator $\mathcal{L}^j$ be defined appropriately for the diffusion terms. Assume

the coefficients satisfy the global Lipschitz conditions:

$$\begin{aligned} |b(t, x) - b(t, y)| &\leq K_2 |x - y| \\ |a^j(t, x) - a^j(t, y)| &\leq K_2 |x - y| \\ |\mathcal{L}^{j_1} a^{j_2}(t, x) - \mathcal{L}^{j_1} a^{j_2}(t, y)| &\leq K_2 |x - y| \end{aligned} \tag{16.47}$$

and the linear growth conditions:

$$\begin{aligned} |b(t, x)| + |\mathcal{L}^j b(t, x)| &\leq K_3(1 + |x|) \\ |a^j(t, x)| + |\mathcal{L}^j a^{j_2}(t, x)| &\leq K_3(1 + |x|) \\ |\mathcal{L}^{j_1} \mathcal{L}^{j_2} a^{j_3}(t, x)| &\leq K_3(1 + |x|) \end{aligned} \tag{16.48}$$

Finally, assume the coefficients satisfy the Hölder continuity condition in time:

$$\begin{aligned} |b(s, x) - b(t, x)| &\leq K_4(1 + |x|)|s - t|^{1/2} \\ |a^j(s, x) - a^j(t, x)| &\leq K_4(1 + |x|)|s - t|^{1/2} \\ |\mathcal{L}^{j_1} a^{j_2}(s, x) - \mathcal{L}^{j_1} a^{j_2}(t, x)| &\leq K_4(1 + |x|)|s - t|^{1/2} \end{aligned} \tag{16.49}$$

for all $s, t \in [0, T]$, $x, y \in \mathbb{R}^d$, and indices $j, j_1, j_2, j_3 = 1, \dots, m$. The constants $K_1, \dots, K_4$ must not depend on δ .

Theorem 16.2: Strong Convergence of the Milstein Scheme

Under the assumptions above, for the Milstein approximation Y^δ , the following strong error estimate holds:

$$\mathbb{E} \left(|X_T - Y^\delta(T)| \right) \leq K_5 \delta, \tag{16.50}$$

where the constant K_5 does not depend on δ .

milstein.py

```
1 T, N, R = 1.0, 2**8, 4
2 dt = T / N
3 mu, sigma, X0 = 1.0, 0.5, 1.0
4
5 np.random.seed(4)
6 dB = np.sqrt(dt) * np.random.randn(N)
7
8 t = np.linspace(0, T, N + 1)
9 B = np.r_[0, np.cumsum(dB)]
10 Y = X0 * np.exp((mu - 0.5 * sigma**2) * t + sigma * B)
11
12 X = np.r_[X0, X0 * np.cumprod(1 + mu * dt + sigma * dB + 0.5 *
13     sigma**2 * (dB**2 - dt))]
14
15 dB_c = dB.reshape(-1, R).sum(axis=1)
16 dt_c = dt * R
17 X_c = np.r_[X0, X0 * np.cumprod(1 + mu * dt_c + sigma * dB_c +
18     0.5 * sigma**2 * (dB_c**2 - dt_c))]
```

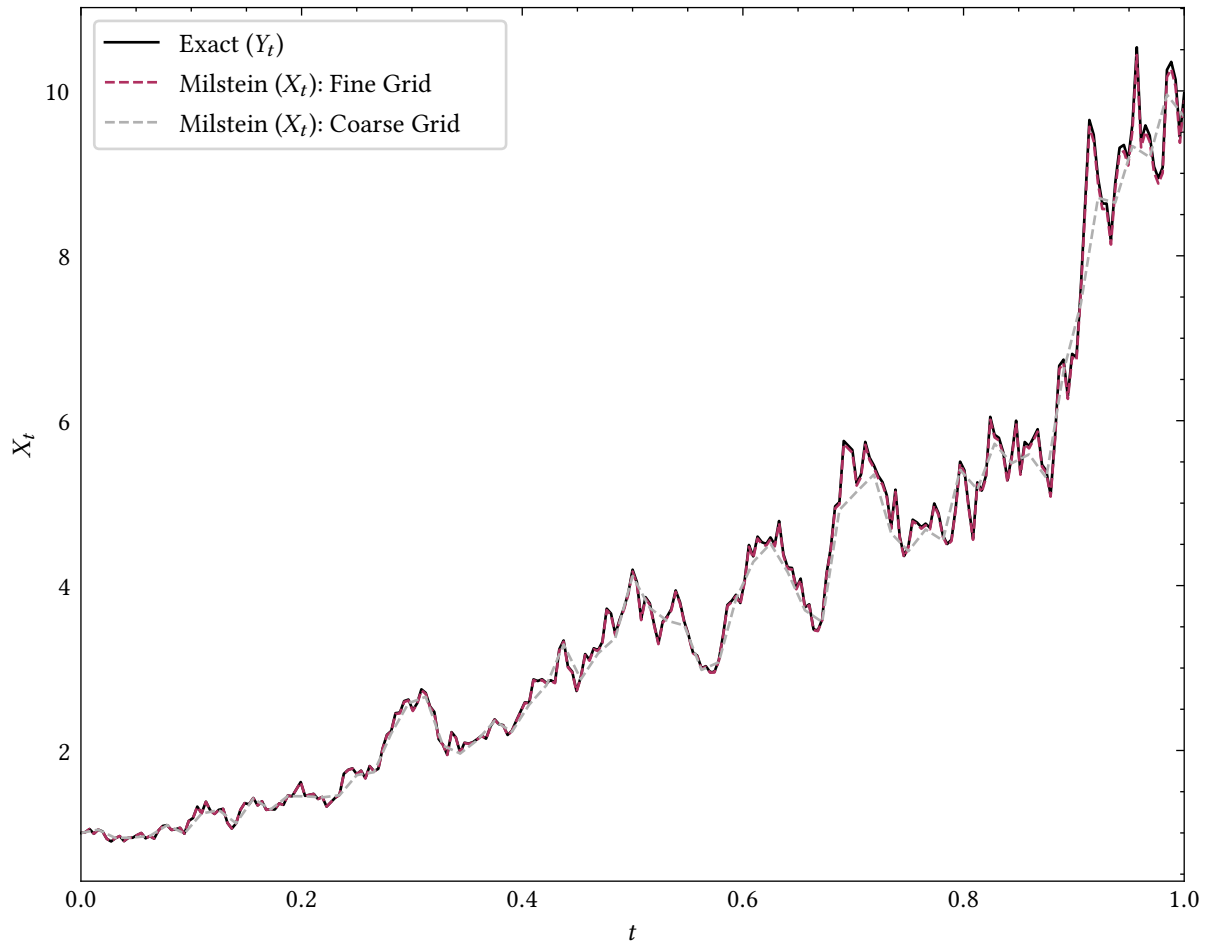


Figure 7: Comparison of Milstein approximations against the exact analytical solution for a GBM. The simulation uses a fine grid step size of $\Delta t = 2^{-8}$ and a coarse grid step size of $\Delta t_{\text{coarse}} = 4\Delta t$. Parameters: $\mu = 1.0$, $\sigma = 0.5$, $X_0 = 1.0$.

Part VI

Monte Carlo Methods

SIMULATION techniques are invaluable in managing complex financial scenarios that involve path-dependent payoffs or distributions without analytical expressions. One prominent application is pricing derivative securities, estimating value at risk, and emulating hedging strategies. Monte Carlo simulation, in particular, proves highly effective for such tasks, offering a blend of simplicity and adaptability as its primary advantages. However, the method's drawback lies in its considerable computational demands, which can be mitigated to some extent. Our approach employs Monte Carlo simulation to generate simulated option price values, which are then compared against outcomes derived from analytical formulas.

17 Generating Random Numbers and Random Variables

At the core of nearly all Monte Carlo simulations is a sequence of apparently random numbers used to drive the simulation. In analysing Monte Carlo methods, we will treat this driving sequence as though it were genuinely random. This is a convenient fiction that allows us to apply tools from probability and statistics to analyse Monte Carlo computations. It is convenient because modern Pseudo-random number generators are sufficiently good at mimicking genuine randomness to make this analysis informative. Nevertheless, we should be aware that the apparently random numbers at the heart of a simulation are, in fact, produced by completely deterministic algorithms.

Because a random number generator may be called thousands or even millions of times in a single simulation, it must be fast.

Before discussing sequences that appear to be random but are not, we should specify what we mean by a generator of genuinely random numbers: we mean a mechanism for producing a sequence of random variables $U_1, U_2, \dots$ with the property that

- (i) each U_i is uniformly distributed between 0 and 1;
- (ii) the U_i are mutually independent.

Property (i) is a convenient but arbitrary normalisation; values uniformly distributed between 0 and $1/2$ would be just as useful, as would values from nearly any other simple distribution. Uniform random variables on the unit interval can be transformed into samples from essentially any other distribution. Property (ii) is the more important one. It implies, in particular, that all pairs of values should be uncorrelated and, more generally, that the value of U_i should not be predictable from $U_1, \dots, U_{i-1}$.

A random number generator (often called a *pseudo-random* number generator to emphasise that it only mimics randomness) produces a finite sequence of numbers $u_1, u_2, \dots, u_K$ in the unit interval. Typically, the values generated depend in part on input parameters specified by

the user. Any such sequence constitutes a set of possible outcomes of independent uniforms $U_1, \dots, U_K$. A good random number generator satisfies the admittedly vague requirement that small (relative to K) segments of the sequence $u_1, \dots, u_K$ should be difficult to distinguish from a realisation of independent uniforms.

An effective generator, therefore, produces values that appear consistent with properties (i) and (ii) above. If the number of values K is large, the fraction of values falling in any sub-interval of the unit interval should be approximately the length of the sub-interval; this is uniformity. Independence suggests that there should be no discernible pattern among the values. To put this only slightly more precisely, statistical tests for independence should not easily reject segments of the sequence $u_1, \dots, u_K$.

Most random number generators at present are based on linear recurrence relations. One of the most important random number generators is the *multiple-recursive generator* (MRG) of order k , which generates a sequence of integers $X_k, X_{k+1}, \dots$ via the linear recurrence

$$X_t = (a_1 X_{t-1} + \dots + a_k X_{t-k}) \bmod m, \quad t = k, k+1, \dots \quad (17.1)$$

for some *modulus* m and *multipliers* $\{a_i, i = 1, \dots, k\}$. Here “mod” refers to the modulo operation: $n \bmod m$ is the remainder when n is divided by m . The recurrence is initialised by specifying k “seeds”, $X_0, \dots, X_{k-1}$. To yield fast algorithms, all but a few of the multipliers should be 0. When m is a large integer, one can obtain a stream of pseudo-random numbers $U_k, U_{k+1}, \dots$ between 0 and 1 from the sequence $X_k, X_{k+1}, \dots$, simply by setting $U_t = X_t/m$. It is also possible to set a small modulus, in particular $m = 2$. The output function for such *modulo 2 generators* is then typically of the form

$$U_t = \sum_{i=1}^w X_{t+w+i-1} 2^{-i} \quad (17.2)$$

for some $w \leq k$, e.g., $w = 32$ or 64 . Examples of modulo 2 generators are the *feedback shift register generators*, the most popular of which are the *Mersenne twisters*.

MRGs with excellent statistical properties can be implemented efficiently by combining several simpler MRGs and carefully choosing their respective moduli and multipliers. One of the most successful is L'Ecuyer's MRG32k3a generator, see (L'Ecuyer, 1999).

mrg32k5a.py

```

1 M1, M2, NORM = 4294949027.0, 4294934327.0, 2.3283163396834613e-10
2 A12, A14, A15 = 1154721.0, 1739991.0, 1108499.0
3 A21, A23, A25 = 1776413.0, 865203.0, 1641052.0
4
5 state = np.random.SeedSequence(42).generate_state(10).astype(np.
    float64)
6 s1, s2, out = state[:5], state[5:], np.empty(1000, dtype=np.float64)
7
8 for i in range(1000):
9     p1 = (A12 * s1[2] + A14 * s1[1] - A15 * s1[0]) % M1
10    s1[:-1], s1[-1] = s1[1:], p1
11
12    p2 = (A21 * s2[4] + A23 * s2[2] - A25 * s2[0]) % M2
13    s2[:-1], s2[-1] = s2[1:], p2
14
15    out[i] = (p1 - p2 + M1) * NORM if p1 <= p2 else (p1 - p2) * NORM

```

Simulating a random variable X from an arbitrary (that is, not necessarily uniform) distribution invariably involves the following two steps.

Algorithm 3 Simulation of Random Variable X

- 1: Simulate uniform random numbers $U_1, \dots, U_k$ on $(0, 1)$ for some $k = 1, 2, \dots$
 - 2: Return $X = g(U_1, \dots, U_k)$, where g is some real-valued function.
-

17.1 Simulating Random Variables

17.1.1 Inverse-Transform Method

Let X be a random variable with cumulative distribution function (cdf) F . Let F^{-1} denote the inverse of F and $U \sim \mathcal{U}(0, 1)$. Then,

$$\mathbb{P}[F^{-1}(U) \leq x] = \mathbb{P}[U \leq F(x)] = F(x). \quad (17.3)$$

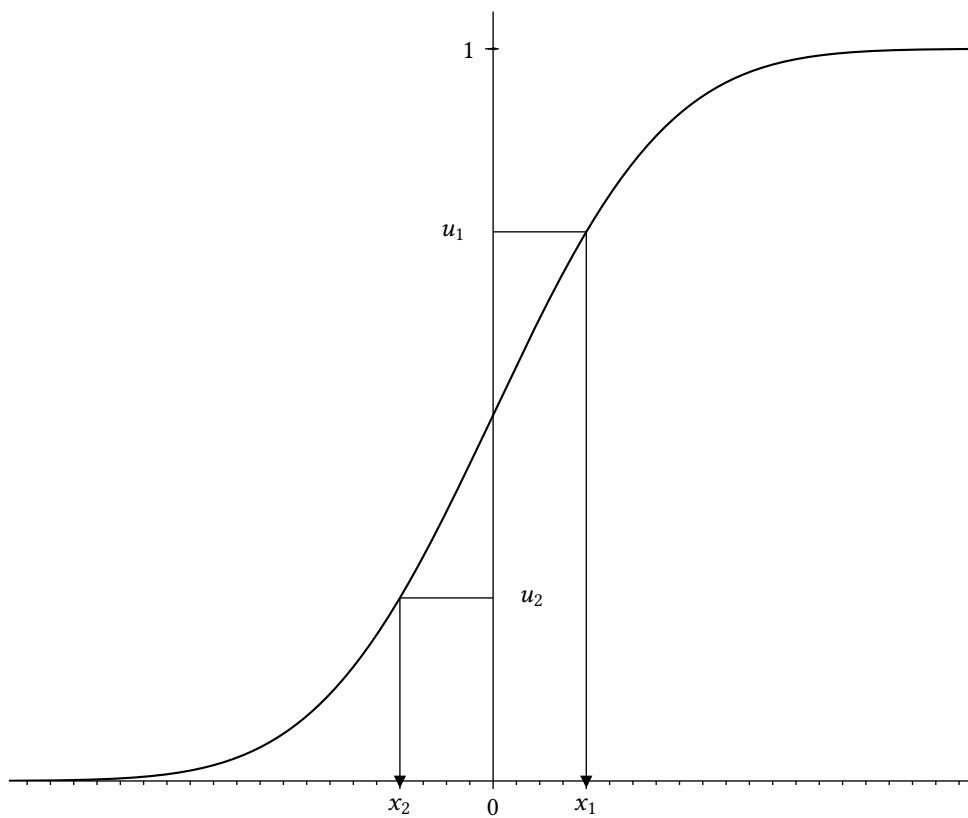


Figure 8: Inverse transform method.

This leads to the following method to simulate a random variable X with cdf F :

²Every cdf has a unique inverse function defined by $F^{-1}(u) = \inf\{x : F(x) \geq u\}$. If, for each u , the equation $F(x) = u$ has a unique solution x , this definition coincides with the usual interpretation of the inverse function.

Algorithm 4 Inverse-Transform Method

Require: Cumulative distribution function F .

Ensure: Random variable X distributed according to F .

- 1: Generate U from $\mathcal{U}(0, 1)$.
 - 2: $X \leftarrow F^{-1}(U)$
 - 3: **return** X
-



The inverse-transform method is applicable to both continuous and discrete distributions. After importing numpy as np, simulating numbers $0, \dots, k - 1$ according to probabilities $p_0, \dots, p_{k-1}$ can be done via `np.min(np.where(np.cumsum(p) > np.random.rand()))`, where `p` is the vector of the probabilities.

17.1.2 Acceptance-Rejection Method

The acceptance-rejection method is used to sample from a “difficult” probability density function (pdf) $f(x)$ by generating instead from an “easy” pdf $g(x)$ satisfying $f(x) \leq Cg(x)$ for some constant $C \geq 1$ (for example, via the inverse-transform method), and then accepting or rejecting the drawn sample with a certain probability. [algorithm 5](#) gives the pseudo-code.

The idea of the algorithm is to generate a point (X, Y) under the graph of the function Cg uniformly, by first drawing $X \sim g$ and then $Y \sim \mathcal{U}(0, Cg(X))$. If this point lies under the graph of f , then we accept X as a sample from f ; otherwise, we try again. The efficiency of the acceptance-rejection method is usually expressed in terms of the probability of acceptance, which is $1/C$.

Algorithm 5 Acceptance-Rejection Method

Require: Pdf g and constant C such that $Cg(x) \geq f(x)$ for all x .

Ensure: Random variable X distributed according to pdf f .

```

1: found  $\leftarrow$  false
2: while not found do
3:   Generate  $X$  from  $g$ .
4:   Generate  $U$  from  $\mathcal{U}(0, 1)$  independently of  $X$ .
5:    $Y \leftarrow UCg(X)$ 
6:   if  $Y \leq f(X)$  then
7:     found  $\leftarrow$  true
8:   end if
9: end while
10: return  $X$ 

```

18 Simulating Random Vectors and Processes

TECHNIQUES for generating random vectors and processes are as diverse as the class of random processes themselves. We highlight a few general scenarios.

When $X_1, \dots, X_n$ are *independent* random variables with pdfs $f_i, i = 1, \dots, n$, so that their joint pdf is $f(\mathbf{x}) = f_1(x_1) \cdots f_n(x_n)$, the random vector $X = [X_1, \dots, X_n]^\top$ can be simply simulated by drawing each component $X_i \sim f_i$ individually, for example, via the inverse-transform method or acceptance-rejection.

For *dependent* components $X_1, \dots, X_n$, we can, as a consequence of the *product rule* of probability, represent the joint pdf $f(\mathbf{x})$ as

$$f(\mathbf{x}) = f(x_1, \dots, x_n) = f_1(x_1)f_2(x_2|x_1) \cdots f_n(x_n|x_1, \dots, x_{n-1}), \quad (18.1)$$

where $f_1(x_1)$ is the marginal pdf of X_1 and $f_k(x_k|x_1, \dots, x_{k-1})$ is the conditional pdf of X_k given $X_1 = x_1, X_2 = x_2, \dots, X_{k-1} = x_{k-1}$. Provided the conditional pdfs are known, one can generate X by first generating X_1 , then, given $X_1 = x_1$, generate X_2 from $f_2(x_2|x_1)$, and so on, until generating X_n from $f_n(x_n|x_1, \dots, x_{n-1})$.

18.0.1 Normal Random Variables and Vectors

Normal random variables are the building blocks of many financial simulation models

$$\Phi(x) := \mathbb{P}(X \leq x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-y^2/2} dy, \quad x \in \mathbb{R},$$

denotes the standard Gaussian Cumulative Distribution Function (CDF) of a standard normal random variable $X \simeq \mathcal{N}(0, 1)$, which posits the relation

$$\Phi(-x) = 1 - \Phi(x), \quad x \in \mathbb{R}. \quad (18.2)$$

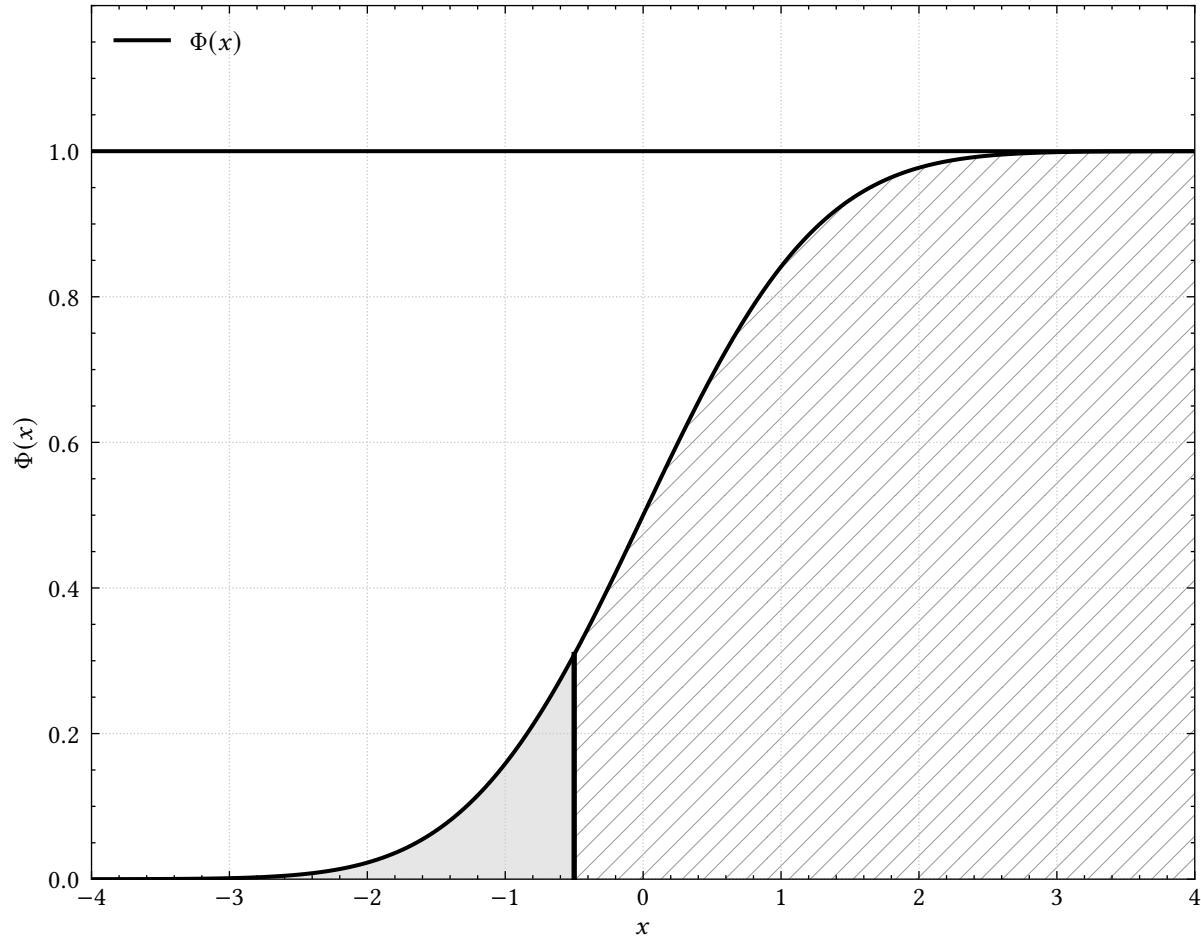


Figure 9: Visualisation of the Gaussian Cumulative Distribution Function.

Algorithm 6 Normal Random Vector Simulation

Require: Mean vector $\boldsymbol{\mu}$, Covariance matrix $\boldsymbol{\Sigma}$

Ensure: Random vector $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

- 1: Compute Cholesky decomposition $\boldsymbol{\Sigma} = \mathbf{B}\mathbf{B}^\top$
 - 2: Generate standard normal vector $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
 - 3: $\mathbf{X} \leftarrow \boldsymbol{\mu} + \mathbf{B}\mathbf{Z}$
 - 4: **return** $\mathbf{X}$
-

BivariateNormal.py

```
1 rng = np.random.default_rng(4)
2 N, rhos = 1000, [0.0, 0.8]
3 Z = rng.standard_normal((2, N))
4 Ls = [np.linalg.cholesky([[1, r], [r, 1]]) for r in rhos]
5 Xs = [L @ Z for L in Ls]
```

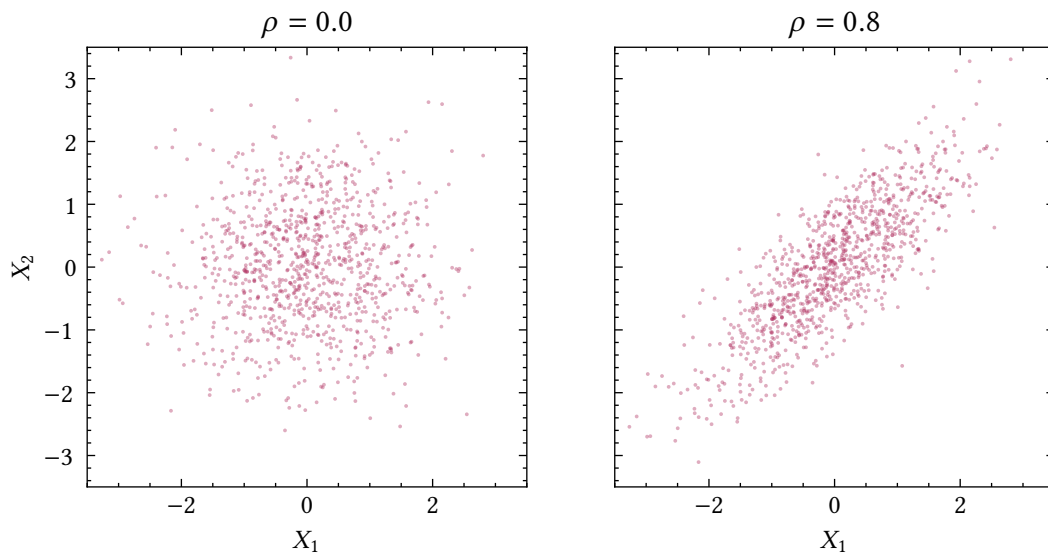


Figure 10: $N = 1000$ realisations of $X \sim \mathcal{N}(\mathbf{0}, \Sigma)$ with $\Sigma_{ii} = 1$ for $\rho = 0$ (left) and $\rho = 0.8$ (right).

19 Quasi Monte Carlo

IT is customary in this setting to focus on the problem of numerical integration over the unit hypercube.

The digital constructions are similar to feedback shift register random number generators, while lattice rules are similar to congruential generators. One useful way to think of QMC is that we are taking a small random number generator and using it in its entirety. The tools used to develop and analyse low-discrepancy methods are very different from those used in ordinary Monte Carlo, as they draw on number theory and abstract algebra rather than probability and statistics.

19.1 Discrepancy Measures

19.2 Low-discrepancy Sequences

19.2.1 Halton and Hammersley

19.2.2 Faure

19.2.3 Sobol

19.3 Koksma-Hlawka

20 Variance Reduction

The estimation of performance measures in Monte Carlo simulation can be made more efficient by utilising known information about the simulation model. Variance reduction techniques include antithetic variables, control variables, and importance sampling.

Suppose Y is the output of a simulation experiment. A random variable $\tilde{Y}$, obtained from the same simulation run, is called a *control variable* for Y if Y and $\tilde{Y}$ are correlated (negatively or positively) and the expectation of $\tilde{Y}$ is known. The use of control variables for variance reduction is based on the following theorem.

Theorem 20.1: Control Variable Estimation

Let $Y_1, \dots, Y_N$ be the output of N independent simulation runs and let $\tilde{Y}_1, \dots, \tilde{Y}_N$ be the corresponding control variables, with $\mathbb{E}\tilde{Y}_k = \tilde{\mu}$ known. Let $\varrho_{Y, \tilde{Y}}$ be the correlation coefficient between each Y_k and $\tilde{Y}_k$. For each $\alpha \in \mathbb{R}$ the estimator

$$\hat{\mu}^{(c)} = \frac{1}{N} \sum_{k=1}^N \left[Y_k - \alpha (\tilde{Y}_k - \tilde{\mu}) \right] \quad (20.1)$$

is an unbiased estimator for $\mu = \mathbb{E}Y$. The minimal variance of $\hat{\mu}^{(c)}$ is

$$\text{Var } \hat{\mu}^{(c)} = \frac{1}{N} (1 - \varrho_{Y, \tilde{Y}}^2) \text{Var } Y, \quad (20.2)$$

which is obtained for $\alpha = \varrho_{Y, \tilde{Y}} \sqrt{\text{Var } Y / \text{Var } \tilde{Y}}$.

20.0.1 Antithetics

When we use Monte Carlo averages of quantities $f(X_i)$, the randomness in the algorithm leads to some error cancellation. In antithetic sampling, we aim to achieve even greater cancellation. An antithetic sample is one that somehow gives the opposite value of $f(\mathbf{x})$, being low when

$f(\mathbf{x})$ is high and vice versa. Ordinarily, we get an opposite f by sampling at a point $\tilde{\mathbf{x}}$ that is somehow opposite to $\mathbf{x}$.

Let $\mu = \mathbb{E}(X)$ for $X \sim p$, where p is a symmetric density on the symmetric set $\mathcal{D}$. Here, symmetry is with respect to reflection through the center point $\mathbf{c}$ of $\mathcal{D}$. If we reflect $\mathbf{x} \in \mathcal{D}$ through $\mathbf{c}$ we get the point $\tilde{\mathbf{x}}$ with $\tilde{\mathbf{x}} - \mathbf{c} = -(\mathbf{x} - \mathbf{c})$, that is $\tilde{\mathbf{x}} = 2\mathbf{c} - \mathbf{x}$. Symmetry means that $p(\tilde{\mathbf{x}}) = p(\mathbf{x})$ including the constraint that $\mathbf{x} \in \mathcal{D}$ if and only if $\tilde{\mathbf{x}} \in \mathcal{D}$. For basic examples, when p is $\mathcal{N}(0, \Sigma)$ then $\tilde{\mathbf{x}} = -\mathbf{x}$, and when p is $\mathcal{U}(0, 1)^d$ we have $\tilde{\mathbf{x}} = 1 - \mathbf{x}$ componentwise. The antithetic counterpart of a random curve could be its reflection in the horizontal axis. From the symmetry it follows that $\tilde{\tilde{\mathbf{x}}} = \mathbf{x}$.

The *antithetic sampling* estimate of μ is

$$\hat{\mu}_{\text{anti}} = \frac{1}{n} \sum_{i=1}^{n/2} (f(X_i) + f(\tilde{X}_i)), \quad (20.3)$$

where $X_i \stackrel{\text{iid}}{\sim} p$, and n is an even number.

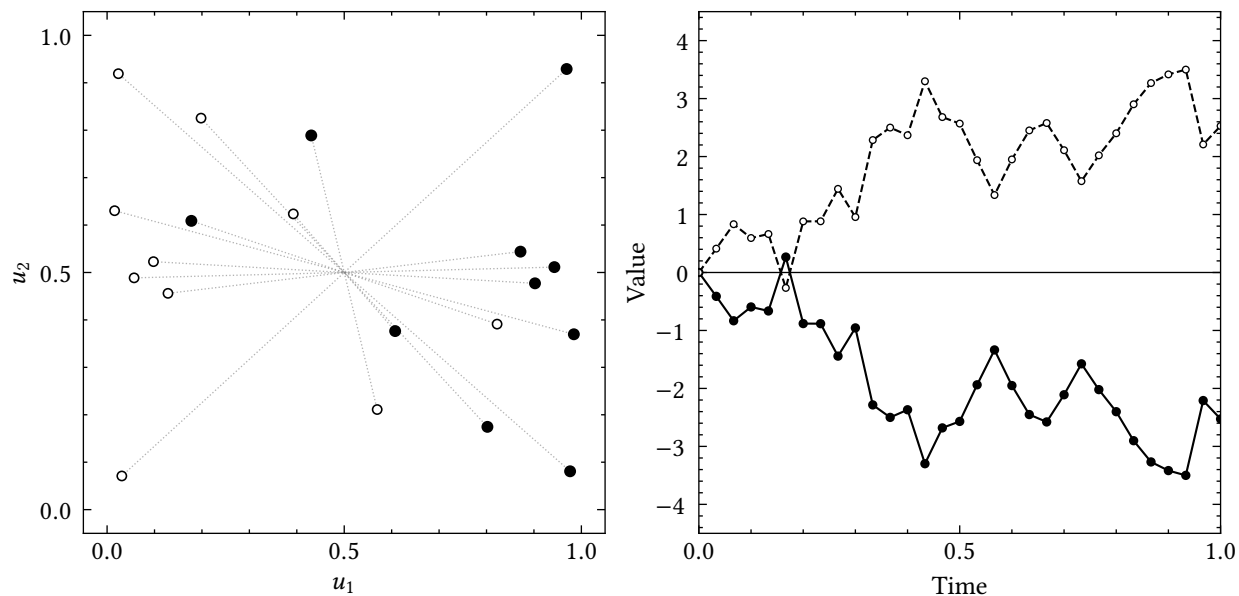


Figure 11: The left panel shows 10 points $\mathbf{u}_i \in [0, 1]^2$ as solid points, connected to their antithetic counterparts $\tilde{\mathbf{u}}_i = 1 - \mathbf{u}_i$, shown as open circles. The right panel shows one random trajectory of 30 points joined by solid lines and connected to the origin, along with its antithetic mirror image in open points.

The rationale for antithetic sampling is that each value of $\mathbf{x}$ is balanced by its opposite $\tilde{\mathbf{x}}$ satisfying $(\mathbf{x} + \tilde{\mathbf{x}})/2 = \mathbf{c}$. Whether this balance is helpful depends on f . Clearly, if f is nearly linear, we could obtain a large improvement. Suppose that $\sigma^2 = \mathbb{E}((f(\mathbf{X}) - \mu)^2) < \infty$. Then the variance in antithetic sampling is

$$\begin{aligned}
\text{Var}(\hat{\mu}_{\text{anti}}) &= \text{Var}\left(\frac{1}{n} \sum_{i=1}^{n/2} f(\mathbf{X}_i) + f(\tilde{\mathbf{X}}_i)\right) \\
&= \frac{n/2}{n^2} \text{Var}(f(\mathbf{X}) + f(\tilde{\mathbf{X}})) \\
&= \frac{1}{2n} \left(\text{Var}(f(\mathbf{X})) + \text{Var}(f(\tilde{\mathbf{X}})) + 2 \text{Cov}(f(\mathbf{X}), f(\tilde{\mathbf{X}})) \right) \\
&= \frac{\sigma^2}{n} (1 + \rho)
\end{aligned} \tag{20.4}$$

where $\rho = \text{Corr}(f(\mathbf{X}), f(\tilde{\mathbf{X}}))$.

From $-1 \leq \rho \leq 1$ we obtain $0 \leq \sigma^2(1 + \rho) \leq 2\sigma^2$. In the best case, antithetic sampling gives the exact answer from just one pair of function evaluations. In the worst case, it doubles the variance. Both cases do arise.

It is clear that a negative correlation is favourable. If f happens to be monotone in all d components of $\mathbf{x}$, then it is known that $\rho < 0$. Monotonicity of f is a safe harbour, wherein if f is monotone, then we're sure antithetic sampling will reduce the variance. We can often establish monotonicity theoretically, for example, by differentiating f . But $\rho < 0$ can hold without f being monotone in any of its inputs. Conversely, ρ can be just barely negative when f is monotone. As a result, monotonicity alone is not a reliable guide to determining whether antithetic sampling will yield a significant gain.

To get a qualitative understanding of antithetic sampling, break f into even and odd parts via

$$\begin{aligned}
f(\mathbf{x}) &= \frac{f(\mathbf{x}) + f(\tilde{\mathbf{x}})}{2} + \frac{f(\mathbf{x}) - f(\tilde{\mathbf{x}})}{2} \\
&\equiv f_{\text{E}}(\mathbf{x}) + f_{\text{O}}(\mathbf{x}).
\end{aligned} \tag{20.5}$$

The even part satisfies $f_E(\mathbf{x}) = f_E(\tilde{\mathbf{x}})$ and $\int_{\mathcal{D}} f_E(\mathbf{x})p(\mathbf{x})d\mathbf{x} = \mu$. The odd part satisfies $f_O(\mathbf{x}) = -f_O(\tilde{\mathbf{x}})$ and $\int_{\mathcal{D}} f_O(\mathbf{x})p(\mathbf{x})d\mathbf{x} = 0$.

The even and odd parts of f are orthogonal. This is not a surprise, because the product $f_O(\mathbf{x})f_E(\mathbf{x})$ is an odd function. But to be careful and rule out $\mathbb{E}(|f_O(\mathbf{X})f_E(\mathbf{X})|) = \infty$, we compute directly that

$$\begin{aligned} \int_{\mathcal{D}} f_E(\mathbf{x})f_O(\mathbf{x})p(\mathbf{x})d\mathbf{x} &= \int_{\mathcal{D}} \left(\frac{f(\mathbf{x}) + f(\tilde{\mathbf{x}})}{2} \right) \left(\frac{f(\mathbf{x}) - f(\tilde{\mathbf{x}})}{2} \right) p(\mathbf{x})d\mathbf{x} \\ &= \frac{1}{4} \int_{\mathcal{D}} (f(\mathbf{x})^2 - f(\tilde{\mathbf{x}})^2)p(\mathbf{x})d\mathbf{x} = 0. \end{aligned}$$

Now it follows easily that $\sigma^2 = \sigma_E^2 + \sigma_O^2$ where $\sigma_E^2 = \int_{\mathcal{D}} (f_E(\mathbf{x}) - \mu)^2 p(\mathbf{x})d\mathbf{x}$ and $\sigma_O^2 = \int_{\mathcal{D}} f_O(\mathbf{x})^2 p(\mathbf{x})d\mathbf{x}$.

Reworking [eq. \(20.5\)](#) yields $\hat{\mu}_{\text{anti}} = (2/n) \sum_{i=1}^{n/2} f_E(\mathbf{X}_i)$. Therefore $\text{Var}(\hat{\mu}_{\text{anti}}) = 2\sigma_E^2/n$ and we can combine this with the variance of ordinary Monte Carlo sampling as follows

$$\begin{pmatrix} V(\hat{\mu}) \\ V(\hat{\mu}_{\text{anti}}) \end{pmatrix} = \frac{1}{n} \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} \sigma_E^2 \\ \sigma_O^2 \end{pmatrix}. \quad (20.6)$$

We see from [eq. \(20.6\)](#) that antithetic sampling eliminates the variance contribution of f_O but doubles the contribution from f_E . Antithetic sampling is extremely beneficial for integrands that are primarily odd functions of their inputs, having $\sigma_O^2 \gg \sigma_E^2$. The connection to correlation is via $\rho = (\sigma_E^2 - \sigma_O^2)/(\sigma_E^2 + \sigma_O^2)$.

The analysis above shows that antithetic sampling reduces variance if $\rho = \text{Corr}(f(\mathbf{X}), f(\tilde{\mathbf{X}})) < 0$, or equivalently, if $\sigma_O^2 > \sigma_E^2$. That analysis is appropriate when the most of the computation is in evaluating f and there is no economy in evaluating both $f(\mathbf{X})$ and $f(\tilde{\mathbf{X}})$.

Variance reduction is only part of the story because the cost of antithetic sampling using n points could well be smaller than the cost of plain Monte Carlo with n points. That will happen if it is expensive to generate $\mathbf{X}$, compared to the cost of computing f , but inexpensive to generate $\tilde{\mathbf{X}}$. For example, $\mathbf{X}$ might be a carefully constructed and expensive sample path from a Gaussian process while $\tilde{\mathbf{X}} = -\mathbf{X}$.

We can explore this effect by letting c_x be the cost of generating X , and c_f be the cost of computing $f(X)$ once we have X . We also let $\widetilde{c}_x$ and $\widetilde{c}_f$ be the corresponding costs for the antithetic sample. For illustration, suppose that to a reasonable approximation $\widetilde{c}_x = 0$ and $\widetilde{c}_f = c_f$. In special circumstances $\widetilde{c}_f < c_f$ because it may be possible to reuse some computation.

Under the assumptions we are exploring, the efficiency of antithetic sampling relative to plain Monte Carlo is

$$E_{\text{anti}} = \frac{2c_x + 2c_f}{c_x + 2c_f} \cdot \frac{\sigma_O^2 + \sigma_E^2}{2\sigma_E^2}. \quad (20.7)$$

Then antithetic sampling is more efficient than plain Monte Carlo if

$$\frac{\sigma_O^2}{\sigma_E^2} > \frac{c_f}{c_x + c_f}.$$

If generating x costs ten times as much as computing f then antithetic sampling pays off when $\sigma_O^2/\sigma_E^2 > 1/11$.

Because antithetic samples have dependent values within pairs, the usual variance estimate must be modified. The most straightforward approach is to analyze the data as a sample of size $m = n/2$ values of $f_E(X)$. Let $Y_i = f_E(X_i) = (f(X_i) + f(\widetilde{X}_i))/2$ for $i = 1, \dots, m = n/2$. Then take

$$\widehat{\mu}_{\text{anti}} = \frac{1}{m} \sum_{i=1}^m Y_i, \quad \text{and}$$

$$s_{\text{anti}}^2 = \frac{1}{m-1} \sum_{i=1}^m (Y_i - \widehat{\mu}_{\text{anti}})^2,$$

and use s_{anti}^2/m as the estimate of $\text{Var}(\widehat{\mu}_{\text{anti}})$.

20.0.2 Common Random Numbers

Suppose that f and g are closely related functions and that we want to find $\mathbb{E}(f(X) - g(X))$ for $X \sim p$. Perhaps $f(x) = h(x, \theta)$ for a parameter $\theta \in \mathbb{R}^p$, and then to study the effect of θ we look at $g(x) = h(x, \widetilde{\theta})$ for some $\widetilde{\theta} \neq \theta$. We assume at first that neither f nor g (nor h) makes any use of random numbers other than X . Later we relax that assumption.

Because $\mathbb{E}(f(X) - g(X)) = \mathbb{E}(f(X)) - \mathbb{E}(g(X))$ we clearly have two different ways to go. We could estimate the difference by

$$\widehat{D}_{\text{com}} = \frac{1}{n} \sum_{i=1}^n f(X_i) - g(X_i), \quad (20.8)$$

for $X_i \stackrel{\text{iid}}{\sim} p$, or by differencing averages

$$\widehat{D}_{\text{ind}} = \frac{1}{n_1} \sum_{i=1}^{n_1} f(X_{i1}) - \frac{1}{n_2} \sum_{i=1}^{n_2} g(X_{i2}) \quad (20.9)$$

for $X_{ij} \stackrel{\text{iid}}{\sim} p$. Taking $n = n_1 = n_2$ makes the computing costs in (20.8) and (20.9) comparable, assuming that costs of computing f and g dominate those of generating X .

The sampling variances of these methods are

$$\begin{aligned} \text{Var}(\widehat{D}_{\text{com}}) &= \frac{1}{n} \left(\sigma_f^2 + \sigma_g^2 - 2\rho\sigma_f\sigma_g \right) \\ \text{Var}(\widehat{D}_{\text{ind}}) &= \frac{1}{n} \left(\sigma_f^2 + \sigma_g^2 \right), \end{aligned} \quad (20.10)$$

where σ_f^2 and σ_g^2 are individual function variances and $\rho = \text{Corr}(f(X), g(X))$. When $\rho > 0$ we are better off using common random numbers. There is no guarantee that $\rho > 0$. When f and g compute similar quantities then we anticipate that $\rho > 0$, and if so, then $\widehat{D}_{\text{com}}$ is more effective than $\widehat{D}_{\text{ind}}$.

Most people would instinctively use the common variates. So at first sight, the method looks more like avoiding a variance increase than engineering a variance decrease. Later, when we relax the rule forbidding f , g , and h to use other sources of randomness, we will find that retaining some common random numbers requires considerable care in synchronization. The added complexity might well tip the balance against using common random numbers.

Much the same problem arises if we are comparing $\mathbb{E}(f(X))$ for $X \sim p$ and $\mathbb{E}(f(\tilde{X}))$ for $\tilde{X} \sim \tilde{p}$. Sometimes we can rewrite that problem in terms of common random variables that get transformed to a different distribution before f is applied. For instance, if the first simulation

has $X_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ and the second has $\tilde{X}_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\tilde{\mu}, \tilde{\sigma}^2)$ then we can sample $Z_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ and use

$$\hat{D}_{\text{com}} = \frac{1}{n} \sum_{i=1}^n f(\mu + \sigma Z_i) - f(\tilde{\mu} + \tilde{\sigma} Z_i).$$

More generally, when X_i is generated via a transformation $\Psi(U_i; \theta)$ of $U_i \sim \mathcal{U}(0, 1)^s$ then we can average $f(\Psi(U_i; \theta)) - f(\Psi(U_i; \tilde{\theta}))$. Acceptance-rejection sampling of X does not fit cleanly into this framework, because the number s of needed uniform random variables is not fixed and may vary with θ .

The construction above is a *coupling* of the random vectors X and $\tilde{X}$. Any joint distribution on $(X, \tilde{X})$ with $X \sim p$ and $\tilde{X} \sim \tilde{p}$ is a coupling. Common random numbers provide a particularly close coupling between X and $\tilde{X}$.

We want to estimate $\mu_j = \mathbb{E}(h(X, \theta_j))$ for $j = 1, \dots, m$ using n random inputs X_i , for $i = 1, \dots, n$. The content uniformity example had a large value of m but in the simplest case, $m = 2$ and we're interested in $\mu_1 - \mu_2$. We still assume that h really is a function of X and θ and in particular our implementation of h does not cause more random numbers to be generated.

Algorithm 7 Common Random Numbers Algorithm I

```

1: SETSEED(seed)
2:  $\hat{\mu}_j \leftarrow 0, \quad 1 \leq j \leq m$ 
3: for  $i \leftarrow 1, n$  do
4:    $X_i \sim p$ 
5:    $\hat{\mu}_j \leftarrow \hat{\mu}_j + h(X_i, \theta_j), \quad 1 \leq j \leq m$ 
6: end for
7:  $\hat{\mu}_j \leftarrow \hat{\mu}_j / n, \quad 1 \leq j \leq m$ 
8: return  $\hat{\mu}_1, \dots, \hat{\mu}_m$ 

```

We can run a nested loop over samples indexed by i and parameter values indexed by j . There are two main approaches that we can take, depending on which is the outer loop.

[algorithm 7](#) shows common random numbers with the outer loop over X_i for $i = 1, \dots, n$. When X_i is multi-dimensional we have to make sure that every component of X needed for any value of θ_j is provided. In the content uniformity problem (page 18) we generate Z_{11} for every simulated batch even though some only use $Z_1, \dots, Z_{10}$.

A vectorized implementation of [algorithm 7](#) is advantageous. It uses a function H that takes X and a list $\Theta = (\theta_1, \dots, \theta_m)$ of parameter values. This H returns a list $(h(X, \theta_1), \dots, h(X, \theta_m))$ and the simulation computes

$$(\hat{\mu}_1, \dots, \hat{\mu}_m) = \frac{1}{n} \sum_{i=1}^n H(X_i, \Theta). \quad (20.11)$$

This vectorized H makes it easier to separate the code that creates Θ from that which evaluates h .

20.0.3 Control Variates

Control variates provide another way to exploit closed-form results. With control variates, we use some other problem, quite similar to our given one, but for which an exact answer is known. The precise meaning of 'similar' depends on how we intend to use this other problem, and more than one method is provided below. As for 'exact', we will mean it literally, but in practice it may just mean known with an error negligible compared to Monte Carlo errors.

Suppose first that we want to find $\mu = \mathbb{E}(f(X))$ and that we know the value $\theta = \mathbb{E}(h(X))$ where $h(X) \approx f(X)$. Letting $\hat{\mu} = (1/n) \sum_{i=1}^n f(X_i)$ and $\hat{\theta} = (1/n) \sum_{i=1}^n h(X_i)$ we can estimate μ by the *difference estimator*

$$\hat{\mu}_{\text{diff}} = \frac{1}{n} \sum_{i=1}^n (f(X_i) - h(X_i)) + \theta = \hat{\mu} - \hat{\theta} + \theta. \quad (20.12)$$

The expected value of $\hat{\mu}_{\text{diff}}$ is μ . The variance of $\hat{\mu}_{\text{diff}}$ is

$$\text{Var}(\hat{\mu}_{\text{diff}}) = \frac{1}{n} \text{Var}(f(X) - h(X)). \quad (20.13)$$

So if h is similar to f in the sense that the difference $f(X) - h(X)$ has smaller variance than $f(X)$ has, we will get reduced variance by using $\widehat{\mu}_{\text{diff}}$. In this setting $h(X)$, the random variable whose mean is known, is the *control variate*. The difference estimator is not the only way to use a control variate. The ratio and product estimators

$$\widehat{\mu}_{\text{ratio}} = \widehat{\mu}/\widehat{\theta}, \quad \text{and} \quad (20.14)$$

$$\widehat{\mu}_{\text{prod}} = \widehat{\mu}\widehat{\theta}/\theta \quad (20.15)$$

respectively, are also used. These estimators are undefined when $\theta = 0$, but otherwise they generally converge to μ as $n \rightarrow \infty$. The ratio and product estimators are usually biased because $\mathbb{E}(\widehat{\mu}/\widehat{\theta}) \neq \mathbb{E}(\widehat{\mu})/\mathbb{E}(\widehat{\theta})$ in general. It is possible to generalize the control variate method in very complicated ways. Maybe we could use $\widehat{\mu} \cos(\widehat{\theta} - \theta)$ or some more imaginative quantity. But we don't. By far the most common way of using a control variate is through the regression estimator, considered next. For a value $\beta \in \mathbb{R}$, the *regression estimator* of μ is

$$\widehat{\mu}_{\beta} = \frac{1}{n} \sum_{i=1}^n (f(X_i) - \beta h(X_i)) + \beta \theta. \quad (20.16)$$

Taking $\beta = 0$ yields the simple MC estimator $\widehat{\mu}$ and $\beta = 1$ gives the difference estimator. The regression estimator is unbiased: $\mathbb{E}(\widehat{\mu}_{\beta}) = \mu$ for all β because $\mathbb{E}(h(X) - \theta) = 0$.

The variance of the regression estimator is

$$\text{Var}(\widehat{\mu}_{\beta}) = \frac{1}{n} (\text{Var}(f(X)) - 2\beta \text{Cov}(f(X), h(X)) + \beta^2 \text{Var}(h(X))). \quad (20.17)$$

By differentiating, we find that the best value of β is

$$\beta_{\text{opt}} = \frac{\text{Cov}(f(X), h(X))}{\text{Var}(h(X))} = \frac{\mathbb{E}((f(X) - \mu)(h(X) - \theta))}{\mathbb{E}((h(X) - \theta)^2)}, \quad (20.18)$$

and after some algebra, the resulting minimal variance is

$$\text{Var}(\widehat{\mu}_{\beta_{\text{opt}}}) = \frac{\sigma^2}{n}(1 - \rho^2), \quad (20.19)$$

where $\rho = \text{Corr}(f(X), h(X))$. In the regression estimator, any control variate that correlates with f is helpful, even one that correlates negatively.

Algorithm 8 Control Variates by Regression

Require: Given $f(X_i), h_j(X_i), \theta_j$ for $i = 1, \dots, n, j = 1, \dots, J$

- 1: $\bar{f} \leftarrow \frac{1}{n} \sum f(X_i); \quad \bar{h}_j \leftarrow \frac{1}{n} \sum h_j(X_i)$
 - 2: $Y_i \leftarrow f(X_i); \quad Z_{ij} \leftarrow h_j(X_i) - \bar{h}_j \quad \text{for all } i, j$
 - 3: $\widehat{\mu} \leftarrow$ estimated intercept from MLR
 - 4: $\widehat{\beta} \leftarrow$ estimated slope vector from MLR
 - 5: $\widehat{\mu} \leftarrow \widehat{\mu} + \widehat{\beta}^\top (\boldsymbol{\theta} - \bar{\mathbf{h}})$
 - 6: **return** $\widehat{\mu}$ and intercept standard error from MLR
-

This algorithm demonstrates how to utilise linear regression software for control variate computation. It is *essential* to centre the control variates. It may be necessary to drop one or more control variates if they are linearly dependent in the sample.

To minimize [eq. \(20.17\)](#) with respect to β is a least squares problem and the solution vector β satisfies $\text{Var}(h(X))\beta = \text{Cov}(h(X), f(X))$. If the J by J matrix $\text{Var}(h(X))$ is singular, then one of the h_j is a linear combination of the other $J - 1$ control variates. In the former case, the best value of β is

$$\beta_{\text{opt}} = \text{Var}(h(X))^{-1} \text{Cov}(h(X), f(X)) - \text{Var}(h(X))^{-1} \mathbb{E}(h(X) - \boldsymbol{\theta}) \mathbb{E}(f(X)). \quad (20.20)$$

In applications, we ordinarily do not know β_{opt} . The usual way to estimate it is by replacing expectations by sample averages:

$$\hat{\beta} = \left(\frac{1}{n} \sum_{i=1}^n (h(\mathbf{X}_i) - \bar{\mathbf{H}})(h(\mathbf{X}_i) - \bar{\mathbf{H}})^\top \right)^{-1} \frac{1}{n} \sum_{i=1}^n (h(\mathbf{X}_i) - \bar{\mathbf{H}})f(\mathbf{X}_i). \quad (20.21)$$

Equation [eq. \(20.21\)](#) is the least squares estimate of β_{opt} . The usual estimate of μ with control variates is $\hat{\mu}_{\hat{\beta}}$. The estimated variance is

$$\widehat{\text{Var}}(\hat{\mu}_{\hat{\beta}}) = \frac{1}{n^2} \sum_{i=1}^n (f(\mathbf{X}_i) - \hat{\mu}_{\hat{\beta}} - \hat{\beta}^\top (h(\mathbf{X}_i) - \bar{\mathbf{h}}))^2, \quad (20.22)$$

and its standard error $\sqrt{\widehat{\text{Var}}(\hat{\mu}_{\hat{\beta}})}$, can be computed using standard multiple linear regression software. See [algorithm 8](#). The key insight is to treat μ as the intercept in a multiple linear regression relating $f(\mathbf{X})$ to predictors $h_j(\mathbf{X}) - \theta_j$. The regression formula is $f(\mathbf{X}) \approx \mu + \beta^\top (h(\mathbf{X}) - \theta)$. It is crucial to subtract θ_j from the random variable in order to make $\mathbb{E}(f(\mathbf{X})) = \mu$ match the regression intercept.

The error of the regression estimator using $\hat{\beta} = \hat{\beta}$ is

$$\hat{\mu}_{\hat{\beta}} - \mu = (\hat{\beta} - \beta_{\text{opt}})^\top (\theta - \bar{\mathbf{H}}) + \hat{\mu}_{\beta_{\text{opt}}} - \mu. \quad (20.23)$$

The first term in [eq. \(20.23\)](#) is the product of two components of mean zero, while the second term is the error in the unknown optimal regression estimator. The second term has a mean of zero, but the first does not in general, because the expected value of a product is not necessarily the same as the product of the expected values. As a result, the control variate estimator is usually biased.

Although estimating β from the sample data brings a bias, that bias is ordinarily negligible. Each of the factors $\hat{\beta} - \beta_{\text{opt}}$ and $\bar{\mathbf{h}} - \theta$ is of order $O(n^{-1/2})$. For applications with $J \ll \sqrt{n}$, it is common to neglect the bias from using estimated control variate coefficients.

When an unbiased estimator is required, then we can get one by using an estimate of β_{opt} that is independent of the X_i used in $\widehat{\mu}_{\widehat{\beta}}$. For example $\widehat{\beta}$ can be computed from [eq. \(20.21\)](#) using only a pilot sample $\widetilde{X}_1, \dots, \widetilde{X}_m \sim p$ independent of the X_i . Then $\widehat{\mu}_{\widehat{\beta}}$ can be computed by [eq. \(20.16\)](#) using $X_1, \dots, X_n$ and taking $\beta = \widehat{\beta}$. Now $\mathbb{E}(\widehat{\mu}_{\widehat{\beta}}) = \mu$ and

$$\text{Var}(\widehat{\mu}_{\widehat{\beta}}) = \mathbb{E}(\text{Var}(\widehat{\mu}_{\widehat{\beta}} \mid \widetilde{X}_1, \dots, \widetilde{X}_m)) = \frac{1}{n} \mathbb{E}(\sigma_{\widehat{\beta}}^2). \quad (20.24)$$

If $f(X_i)$ and $h(X_i)$ have finite fourth moments then $\widehat{\beta} = \beta_{\text{opt}} + O(1/\sqrt{m})$. The variance with a control variate is $\sigma^2(1 - \rho^2)/n$, which is never worse than σ^2/n and usually better. Whether the control variate is helpful ultimately depends on how much it costs to use it. Suppose that the total cost of generating X_i and then computing $f(X_i)$ is, on average, c_f . Let c_h be the average extra cost incurred by the control variate. That includes the cost to evaluate $h(X_i)$ but not the cost of sampling X_i . We will suppose that the cost to compute $\widehat{\beta}$ is small. If not, then c_h should be increased to reflect it. Control variates improve efficiency when $(1 - \rho^2)(c_f + c_h) < c_f$, that is when $|\rho| > \sqrt{c_h/(c_f + c_h)}$. For illustration, if $c_h = c_f$ then we need $|\rho| > \sqrt{1/2} \approx 0.71$ in order to benefit from the control variate.

The estimated variance of $\widehat{\mu}_{\widehat{\beta}}$ is

$$\widehat{\text{Var}}(\widehat{\mu}_{\widehat{\beta}}) = \frac{1}{n^2} \sum_{i=1}^n (f(X_i) - \widehat{\mu}_{\widehat{\beta}} - \widehat{\beta}^\top (h(X_i) - \bar{h}))^2, \quad (20.25)$$

and a 99% confidence interval is $\widehat{\mu}_{\widehat{\beta}} \pm 2.58 \sqrt{\widehat{\text{Var}}(\widehat{\mu}_{\widehat{\beta}})}$.

20.0.4 Importance Sampling

Suppose that our problem is to find $\mu = \mathbb{E}(f(X)) = \int_{\mathcal{D}} f(x)p(x) dx$ where p is a probability density function on $\mathcal{D} \subseteq \mathbb{R}^d$ and f is the integrand. We take $p(x) = 0$ for all $x \notin \mathcal{D}$. If q is a

positive probability density function on $\mathbb{R}^d$, then

$$\mu = \int_{\mathcal{D}} f(\mathbf{x})p(\mathbf{x}) \, d\mathbf{x} = \int_{\mathcal{D}} \frac{f(\mathbf{x})p(\mathbf{x})}{q(\mathbf{x})} q(\mathbf{x}) \, d\mathbf{x} = \mathbb{E}_q \left(\frac{f(\mathbf{X})p(\mathbf{X})}{q(\mathbf{X})} \right), \quad (20.26)$$

where $\mathbb{E}_q(\cdot)$ denotes expectation for $X \sim q$. We also write $\text{Var}_q(\cdot)$, $\text{Cov}_q(\cdot, \cdot)$, and $\text{Corr}_q(\cdot, \cdot)$ for variance, covariance and correlation when $X \sim q$. Our original goal then is to find $\mathbb{E}_p(f(\mathbf{X}))$. By making a multiplicative adjustment to f we compensate for sampling from q instead of p . The adjustment factor $p(\mathbf{x})/q(\mathbf{x})$ is called the *likelihood ratio*. The distribution q is the *importance distribution* and p is the *nominal distribution*.

The importance distribution q does not have to be positive everywhere. It is enough to have $q(\mathbf{x}) > 0$ whenever $f(\mathbf{x})p(\mathbf{x}) \neq 0$. That is, for $Q = \{\mathbf{x} \mid q(\mathbf{x}) > 0\}$ we have $\mathbf{x} \in Q$ whenever $f(\mathbf{x})p(\mathbf{x}) \neq 0$. So if $\mathbf{x} \in \mathcal{D} \cap Q^c$ we know that $f(\mathbf{x}) = 0$, while if $\mathbf{x} \in Q \cap \mathcal{D}^c$ we have $p(\mathbf{x}) = 0$. Now

$$\begin{aligned} \mathbb{E}_q \left(\frac{f(\mathbf{X})p(\mathbf{X})}{q(\mathbf{X})} \right) &= \int_Q \frac{f(\mathbf{x})p(\mathbf{x})}{q(\mathbf{x})} q(\mathbf{x}) \, d\mathbf{x} = \int_Q f(\mathbf{x})p(\mathbf{x}) \, d\mathbf{x} \\ &= \int_{\mathcal{D}} f(\mathbf{x})p(\mathbf{x}) \, d\mathbf{x} + \int_{Q \cap \mathcal{D}^c} f(\mathbf{x})p(\mathbf{x}) \, d\mathbf{x} - \int_{\mathcal{D} \cap Q^c} f(\mathbf{x})p(\mathbf{x}) \, d\mathbf{x} \\ &= \int_{\mathcal{D}} f(\mathbf{x})p(\mathbf{x}) \, d\mathbf{x} = \mu. \end{aligned} \quad (20.27)$$

It is natural to wonder what happens for $\mathbf{x}$ with $q(\mathbf{x}) = 0$ in the denominator. The answer is that there are no such points $\mathbf{x} \in Q$ and we will never see one when sampling $X \sim q$. Later we will see examples where $q(\mathbf{x})$ close to 0 causes extreme difficulties, but $q(\mathbf{x}) = 0$ is not a problem if $f(\mathbf{x})p(\mathbf{x}) = 0$ too.

When we want q to work for many different functions f_j then we need $q(\mathbf{x}) > 0$ at every $\mathbf{x}$ where any $f_j(\mathbf{x})p(\mathbf{x}) \neq 0$. Then, a density q with $q(\mathbf{x}) > 0$ whenever $p(\mathbf{x}) > 0$ will suffice, allowing us to add new functions f_j to our list after the sample has been drawn.

The *importance sampling estimate* of $\mu = \mathbb{E}_p(f(X))$ is

$$\hat{\mu}_q = \frac{1}{n} \sum_{i=1}^n \frac{f(X_i)p(X_i)}{q(X_i)}, \quad X_i \sim q. \quad (20.28)$$

To use (20.28) we must be able to compute $f p/q$. Assuming that we can compute f , this estimate requires that we can compute $p(x)/q(x)$ at any x we might sample. When p or q has an unknown normalization constant, then we will resort to a ratio estimate. For now, we assume that p/q is computable, and study the variance of $\hat{\mu}_q$. Exponential tilting is one way to choose q with computable p/q even when p is unnormalized.

Theorem 20.2: Variance of Importance Sampling

Let $\hat{\mu}_q$ be given by (20.28) where $\mu = \int_{\mathcal{D}} f(x)p(x) dx$ and $q(x) > 0$ whenever $f(x)p(x) \neq 0$. Then $\mathbb{E}_q(\hat{\mu}_q) = \mu$, and $\text{Var}_q(\hat{\mu}_q) = \sigma_q^2/n$ where

$$\sigma_q^2 = \int_Q \frac{(f(x)p(x))^2}{q(x)} dx - \mu^2 = \int_Q \frac{(f(x)p(x) - \mu q(x))^2}{q(x)} dx, \quad (20.29)$$

where $Q = \{x \mid q(x) > 0\}$. The domain Q can be replaced by $\mathcal{D}$ in the above integrals.

Proof. That $\mathbb{E}_q(\hat{\mu}_q) = \mu$ follows directly from (20.27). Next,

$$\text{Var}_q(\hat{\mu}_q) = \frac{1}{n} \left[\int_Q \left(\frac{f(x)p(x)}{q(x)} \right)^2 q(x) dx - \mu^2 \right].$$

The first expression in equation (20.29) follows by the definition of σ_q^2 and the second by simple rearrangement. Because $q > 0$ whenever $f p \neq 0$, we can replace Q by $\mathcal{D}$ in the integrals. $\square$

To form a confidence interval for μ , we need to estimate σ_q^2 . From the second expression in (20.29) we find that

$$\sigma_q^2 = \mathbb{E}_q((f(X)p(X) - \mu q(X))^2 / q(X)^2).$$

Because $\mathbf{x}_i$ are sampled from q , the natural variance estimate is

$$\widehat{\sigma}_q^2 = \frac{1}{n} \sum_{i=1}^n \left(\frac{f(\mathbf{x}_i)p(\mathbf{x}_i)}{q(\mathbf{x}_i)} - \widehat{\mu}_q \right)^2 = \frac{1}{n} \sum_{i=1}^n (w_i f(\mathbf{x}_i) - \widehat{\mu}_q)^2, \quad (20.30)$$

where $w_i = p(\mathbf{x}_i)/q(\mathbf{x}_i)$. Then an approximate 99% confidence interval for μ is $\widehat{\mu}_q \pm 2.58\widehat{\sigma}_q/\sqrt{n}$.

Theorem 20.2 guides us in selecting a good importance sampling rule. The first expression in (20.29) is simpler to study. A better q is one that gives a smaller value of $\int_{\mathcal{D}} (fp)^2/q \, d\mathbf{x}$. When we want upper bounds on σ_q^2 we can bound $\int_{\mathcal{D}} (fp)^2/q \, d\mathbf{x}$.

The second integral expression in (20.29) illustrates how importance sampling can succeed or fail. The numerator in the integrand at the right of (20.29) is small when $f(\mathbf{x})p(\mathbf{x}) - \mu q(\mathbf{x})$ is close to zero, that is, when $q(\mathbf{x})$ is nearly proportional to $f(\mathbf{x})p(\mathbf{x})$. From the denominator, we see that regions with small values of $q(\mathbf{x})$ greatly magnify whatever lack of proportionality appears in the numerator.

Suppose that $f(\mathbf{x}) \geq 0$ for all $\mathbf{x}$, and to rule out a trivial case, assume that $\mu > 0$ too. Then $q_{\text{opt}}(\mathbf{x}) = f(\mathbf{x})p(\mathbf{x})/\mu$ is a probability density, and it has $\sigma_{q_{\text{opt}}}^2 = 0$. It is optimal, but not really usable because $\widehat{\mu}_{q_{\text{opt}}}$ becomes an average of $f(\mathbf{x}_i)p(\mathbf{x}_i)/q(\mathbf{x}_i) = \mu$ meaning that we could compute μ directly from f , p , and q without any sampling. Likewise, for $f(\mathbf{x}) \leq 0$ with $\mu < 0$, a zero variance is obtained for the density $q = -fp/\mu$.

Although zero-variance importance sampling densities are not usable, they provide valuable insight into the design of an effective importance sampling scheme. Suppose that $f(\mathbf{x}) \geq 0$. It may be beneficial for q to have spikes in the same places as f or p , but it is even better to have them where fp does. Moreover, the best q is proportional to fp , not $\sqrt{fp}$ or f^2p or some other combination.

Choosing a good importance sampling distribution requires some educated guessing and possibly numerical search. In many applications, there is domain knowledge about where the spikes are. In a financial setting, we may know which stock fluctuations will cause an option to

go to its maximum value. For a queuing system, it may be easy to determine which combination of arrivals will cause the system to become overloaded.

In general, the density q^* that minimizes σ_q^2 is proportional to $|f(\mathbf{x})|p(\mathbf{x})$ (Kahn and Marshall, 1953), outside of trivial cases where $\int |f(\mathbf{x})|p(\mathbf{x})d\mathbf{x} = 0$. This optimal density does not have $\sigma_{q^*}^2 = 0$ on problems where fp can be positive for some $\mathbf{x}$ and negative for other $\mathbf{x}$. When fp takes both positive and negative values, a zero variance method is still possible, but it requires sampling at two points.

To prove that $q^*(\mathbf{x}) = |f(\mathbf{x})|p(\mathbf{x})/\mathbb{E}_p(|f(\mathbf{X})|)$ is optimal, let q be any density that is positive when $fp \neq 0$. Then

$$\begin{aligned}\mu^2 + \sigma_{q^*}^2 &= \int \frac{f(\mathbf{x})^2 p(\mathbf{x})^2}{q^*(\mathbf{x})} d\mathbf{x} = \int \frac{f(\mathbf{x})^2 p(\mathbf{x})^2}{|f(\mathbf{x})|p(\mathbf{x})/\mathbb{E}_p(|f(\mathbf{X})|)} d\mathbf{x} \\ &= \mathbb{E}_p(|f(\mathbf{X})|)^2 = \mathbb{E}_q(|f(\mathbf{X})|p(\mathbf{X})/q(\mathbf{X}))^2 \\ &\leq \mathbb{E}_q(f(\mathbf{X})^2 p(\mathbf{X})^2 / q(\mathbf{X})^2) = \mu^2 + \sigma_q^2,\end{aligned}$$

and so $\sigma_{q^*}^2 \leq \sigma_q^2$. This proof is a straightforward consequence of the Cauchy-Schwarz inequality. However, it requires that we know the optimal density before applying the Cauchy-Schwarz inequality. For a principled way to find candidates for q^* in problems like importance sampling, we can turn to the calculus of variations.

We may use the likelihood ratio $w(\mathbf{x}) = p(\mathbf{x})/q(\mathbf{x})$ as a means to understand which importance sampling densities are good or bad choices. The first term in σ_q^2 is $\int f(\mathbf{x})^2 p(\mathbf{x})^2 / q(\mathbf{x}) d\mathbf{x}$. We may write this term as $\mathbb{E}_p(f(\mathbf{X})^2 w(\mathbf{X})) = \mathbb{E}_q(f(\mathbf{X})^2 w(\mathbf{X})^2)$. The appearance of q in the denominator of w means that light-tailed importance densities q are dangerous. If we are clever or lucky, then f might be small just where it needs to be to offset the small denominator. But we often need to use the same sample with multiple integrands f , and so, as a rule, q should have tails at least as heavy as p does.

When p is a Gaussian distribution, then a common tactic is to take q to be a Student's t distribution. Such a q has heavier tails than p . It even has heavier tails than fp for integrands of the form $f(\mathbf{x}) = \exp(\mathbf{x}^\top \boldsymbol{\theta})$, because then fp is proportional to another Gaussian density.

The reverse practice, of using a Gaussian importance distribution q for a student's t nominal distribution, can easily lead to $\sigma_q^2 = \infty$. Even when q is nearly proportional to fp we may still have $\sigma_q^2 = \infty$. The irony is that the infinite variance could be largely attributable to the unimportant region of $\mathcal{D}$ where $|f|p$ is small (but q is extremely small).

Bounded weights are especially valuable. If $w(\mathbf{x}) \leq c$ the $\sigma_q^2 \leq c\sigma_p^2$.

The likelihood ratio also reveals a dimension effect for importance sampling. Suppose that $\mathbf{x} \in \mathbb{R}^d$ and to keep things simple, that the components of $\mathbf{X}$ are independent under both p and q . Then $w(\mathbf{X}) = \prod_{j=1}^d p_j(X_j)/q_j(X_j)$. Though f plays a role, the value of σ_q^2 is also driven largely by $\mathbb{E}_q(w(\mathbf{X})^2) = \prod_{j=1}^d \mathbb{E}_q(w_j(X_j)^2)$ where $w_j = p_j/q_j$. Each $\mathbb{E}_q(w_j(X_j)) = 1$ and $\mathbb{E}_q(w(\mathbf{X})^2) = \prod_{j=1}^d (1 + \text{Var}_q(w_j(X_j)))$. This quantity will grow exponentially with d unless we arrange for the variances of $w_j(X_j)$ to eventually decrease sharply as j increases.

Although we can get a lot of insight from the likelihood ratio $w(\mathbf{x})$, the efficiency of importance sampling compared to alternatives still depends on f . We will see another example of that phenomenon in §9.3 where we discuss some measures of effective sample size for importance sampling.

Sometimes we can only compute an unnormalized version of p , $p_u(\mathbf{x}) = cp(\mathbf{x})$ where $c > 0$ is unknown. The same may be true of q . Suppose that we can compute $q_u(\mathbf{x}) = bq(\mathbf{x})$ where $b > 0$ might be unknown. If we are fortunate or clever enough to have $b = c$, then $p(\mathbf{x})/q(\mathbf{x}) = p_u(\mathbf{x})/q_u(\mathbf{x})$ and we can still use (20.28). Otherwise we may compute the ratio $w_u(\mathbf{x}) = p_u(\mathbf{x})/q_u(\mathbf{x}) = (c/b)p(\mathbf{x})/q(\mathbf{x})$ and consider the *self-normalized importance sampling estimate*

$$\tilde{\mu}_q = \frac{\sum_{i=1}^n f(\mathbf{X}_i) w_u(\mathbf{X}_i)}{\sum_{i=1}^n w_u(\mathbf{X}_i)} \quad (20.31)$$

where $\mathbf{X}_i \sim q$ are independent. The factor c/b cancels from numerator and denominator in (20.31), leading to the same estimate as if we had used the desired ratio $w(\mathbf{x}) = p(\mathbf{x})/q(\mathbf{x})$

instead of the computable alternative $w_u(\mathbf{x})$. As a result we can write

$$\tilde{\mu}_q = \frac{\frac{1}{n} \sum_{i=1}^n f(\mathbf{X}_i) w(\mathbf{X}_i)}{\frac{1}{n} \sum_{i=1}^n w(\mathbf{X}_i)}. \quad (20.32)$$

Theorem 20.3: Consistency of Self-Normalized Importance Sampling

Let p be a probability density function on $\mathbb{R}^d$ and let $f(\mathbf{x})$ be a function such that $\mu = \int f(\mathbf{x})p(\mathbf{x})d\mathbf{x}$ exists. Suppose that $q(\mathbf{x})$ is a probability density function on $\mathbb{R}^d$ with $q(\mathbf{x}) > 0$ whenever $p(\mathbf{x}) > 0$. Let $X_1, \dots, X_n \sim q$ be independent and let $\tilde{\mu}_q$ be the self-normalized importance sampling estimate (20.31). Then

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} \tilde{\mu}_q = \mu\right) = 1.$$

Proof. We may use expression (20.32) for $\tilde{\mu}_q$. The numerator in (20.32) is the average $\hat{\mu}_q$ of $f(\mathbf{X}_i)p(\mathbf{X}_i)/q(\mathbf{X}_i)$ under sampling from q . These are iid with mean μ by (20.27). The strong law of large numbers gives $\mathbb{P}(\lim_{n \rightarrow \infty} \hat{\mu}_q = \mu) = 1$. The denominator of (20.32) converges to 1, using the argument for the numerator on the function f with $f(\mathbf{x}) \equiv 1$. $\square$

Importance sampling uses unequally weighted observations. The weights are $w_i = p(\mathbf{X}_i)/q(\mathbf{X}_i) \geq 0$ for $i = 1, \dots, n$. In extreme settings, one of the w_i may be vastly larger than all the others and then we have effectively only got one observation. We would like to have a diagnostic to tell when the weights are problematic. It is even possible that $w_1 = w_2 = \dots = w_n = 0$. In that case, importance sampling has clearly failed and we don't need a diagnostic to tell us so. Hence, we may assume that $\sum_{i=1}^n w_i > 0$.

Consider a hypothetical linear combination

$$S_w = \frac{\sum_{i=1}^n w_i Z_i}{\sum_{i=1}^n w_i} \quad (20.33)$$

where Z_i are independent random variables with a common mean and common variance $\sigma^2 > 0$, and $\mathbf{w} \in [0, \infty)^n$ is the vector of weights. The unweighted average of n_e independent random

variables Z_i has variance σ^2/n_e . Setting $\text{Var}(S_w) = \sigma^2/n_e$ and solving for n_e yields the *effective sample size*

$$n_e = \frac{(\sum_{i=1}^n w_i)^2}{\sum_{i=1}^n w_i^2} = \frac{n\bar{w}^2}{\overline{w^2}} \quad (20.34)$$

where $\bar{w} = (1/n) \sum_{i=1}^n w_i$ and $\overline{w^2} = (1/n) \sum_{i=1}^n w_i^2$. If the weights are too imbalanced, then the result is similar to averaging only $n_e \ll n$ observations and might therefore be unreliable. The point at which n_e becomes alarmingly small is hard to specify, because it is application-specific.

The effective sample size [eq. \(20.34\)](#) is computed from the generated values. For simple enough distributions, we can obtain a population version of n_e , as

$$n_e^* = \frac{n\mathbb{E}_q(w)^2}{\mathbb{E}_q(w^2)} = \frac{n}{\mathbb{E}_q(w^2)} = \frac{n}{\mathbb{E}_p(w)}. \quad (20.35)$$

If n_e^* is too small, then we know q will produce imbalanced weights. Another effective sample size formula is

$$\frac{n}{1 + \text{cv}(\mathbf{w})^2} \quad (20.36)$$

where $\text{cv}(\mathbf{w}) = ((n-1)^{-1} \sum_{i=1}^n (w_i - \bar{w})^2)^{1/2} / \bar{w}$ is the coefficient of variation of the weights. If we replace the $n-1$ by n in $\text{cv}(\mathbf{w})$ and substitute into $n/(1 + \text{cv}(\mathbf{w})^2)$, the result is n_e of [eq. \(20.34\)](#), so the two formulas are essentially the same.

Instead of using an effective sample size, we might just estimate the variance of $\hat{\mu}_q$ (or of $\tilde{\mu}_q$) and use the variance as a diagnostic. When that variance is quite large, we would conclude that the importance sampling had not worked well. Unfortunately, the variance estimate is based on the same weights as the estimate itself. Badly skewed weights can lead to a badly estimated mean, along with a poor variance estimate that masks the problem.

The variance estimate for $\tilde{\mu}_q$ uses a weighted average of $(f(\mathbf{x}_i) - \tilde{\mu}_q)^2$ with weights equal to w_i^2 . Similarly, for $\hat{\mu}_q$ we have $\hat{\sigma}_q^2 = (1/n) \sum_{i=1}^n w_i^2 f(\mathbf{x}_i)^2 - \hat{\mu}_q^2$ which again depends on weights

w_i^2 . We can compute an effective sample size for variance estimates by

$$n_{e,\sigma} = \frac{(\sum_{i=1}^n w_i^2)^2}{\sum_{i=1}^n w_i^4}. \quad (20.37)$$

If $n_{e,\sigma}$ is small, then we cannot trust the variance estimate. Estimating a variance well is typically a harder problem than estimating a mean well, and here we have $n_{e,\sigma} \leq n_{e,\mu}$.

The accuracy of the central limit theorem for a mean depends on the skewness of the random variables being summed. To gauge the reliability of the CLT for $\hat{\mu}_q$ and $\tilde{\mu}_q$ we find that the skewness of S_w matches that of the average of

$$n_{e,\gamma} = \frac{(\sum w_i^2)^3}{(\sum w_i^3)^2}$$

equally weighted observations.

Effective sample sizes are imperfect diagnostics. When they are too small, then we have a sign that the importance sampling weights are problematic. When they are large, we still cannot conclude that importance sampling has worked. It remains possible that some important region was missed by all of $X_1, \dots, X_n$.

The diagnostics above used the weights w_i only through their relative values $w_i/\sum_j w_j$. In cases where w_i is computable, it is the observed value of a random variable with mean $\mathbb{E}_q(p(X)/q(X)) = 1$. If the sample mean of the weights is far from 1, then that is a sign that q was poorly chosen. That is, $\bar{w} = (1/n) \sum_{i=1}^n w_i$ is another diagnostic.

These weight-based diagnostics are the same for every function f . They are convenient summaries, but the effectiveness of importance sampling also depends on f . If every $f(X_i)p(X_i)/q(X_i) = 0$, it is likely that our importance sampling has failed, and no further diagnostic is needed. Otherwise, for a diagnostic that is specific to f let

$$\tilde{w}_i(f) = \frac{|f(X_i)p(X_i)/q(X_i)|}{\sum_{j=1}^n |f(X_j)p(X_j)/q(X_j)|}.$$

After some algebra, the sample coefficient of variation of these weights is $\text{cv}(\mathbf{w}, f) = \sqrt{n \sum_{i=1}^n \widetilde{w}_i^2 - 1}$.

An effective sample size customised to f is

$$n_e(f) = \frac{n}{1 + \text{cv}(\mathbf{w}, f)^2} = \frac{1}{\sum_{i=1}^n \widetilde{w}_i(f)^2}. \quad (20.38)$$

Part VII

Grid & Lattice Methods

21 Lattice Methods

(Cox, Ross, and Rubinstein, 1979)

Lattice methods are computational techniques used to value options and other derivatives by constructing a lattice, essentially a grid-like representation of the underlying asset's possible price movements over discrete time intervals. In a lattice, each node represents a possible price level of the underlying asset at a specific point in time. Starting from the initial price, the lattice evolves by considering possible upward and downward movements based on the asset's volatility.

Pricing barrier options can be challenging due to the added complexity introduced by the barrier conditions. In a lattice, the barrier levels can be incorporated into the calculations by adjusting the probabilities of price movements, which will be analysed with the Bino-Trinomial model.

At each step of the tree, the asset price can over the time step Δt either move up by a fixed amount u with a probability p_u or decrease by a fixed amount d with a probability $p_d = 1 - p_u$. We assume there are n steps in the tree.

We define i and j as the number of price and time steps until the considered node (j, i) . As a consequence, the number of paths that lead to the node is $\binom{j}{i}$ with probability $\binom{j}{i} p_u^i p_d^{j-i}$. At each time point j , the random variable S_n (describing the stock price) takes value on $\{S_0 u^j, S_0 u^{j-1} d, \dots, S_0 d^j\}$. The discrete stock price process follows

$$S_n = S_0 \prod_{i=1}^n \xi_i$$

where ξ_i are iid random variables defined as follows:

$$\xi_i = \begin{cases} u & \text{with probability } q_u \\ d & \text{with probability } q_d \end{cases}$$

A simple approach uses backward induction to price the option with a binomial tree.

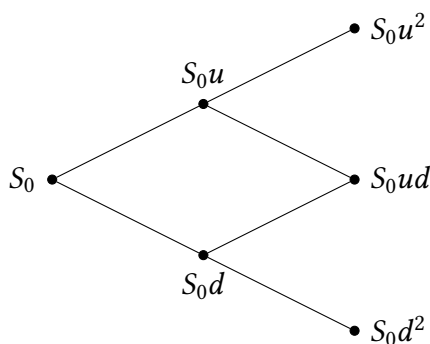


Figure 12: A two-step Binomial Tree

Theorem 21.1: Backward Induction

Consider $V^j = e^{-r_d(n-j)\Delta t} \mathbb{E}[g(S_n) | \mathcal{F}_j]$, the time- j fair value of the European call option.

We have the following recursive algorithm:

$$V^j = \begin{cases} g(S_n) & j = n \\ e^{-r_d\Delta t} \mathbb{E}[V^{j+1} | \mathcal{F}_j] & j \leq n-1 \end{cases}$$

The up and down factors u and d , along with their corresponding probabilities, must match the first two moments of the price distribution. There are only two equations and three parameters to determine (knowing one probability p_u also makes it possible to know $p_d = 1 - p_u$). Cox-Ross-Rubinstein (CRR) set the additional equation to $ud = 1$. The CRR model has parameters:

$$u = e^{\sigma\sqrt{\Delta t}} \quad (21.1)$$

$$d = e^{-\sigma\sqrt{\Delta t}} \quad (21.2)$$

$$p_u = \frac{e^{(r-q)\Delta t} - d}{u - d} \quad (21.3)$$

Under this configuration and with the previous theorem, we define the following recursive formula:

$$V_i^j = e^{-r_d \Delta t} \left[p_u V_{i+1}^{j+1} + p_d V_i^{j+1} \right] \quad (21.4)$$

At the end of the recursive algorithm, we obtain the time-zero option value.

In the standard CRR tree, the local volatility is constant for each time step and converges to the input volatility as n becomes large:

$$\sigma_i^j = \frac{1}{\sqrt{\Delta t}} \sqrt{p_u p_d} \ln u^2 \quad (21.5)$$

Theorem 21.2: Closed-Form Binomial Price for Vanilla Option

The price of a Vanilla option for an $n \in \mathbb{N}$ step tree is given by,

$$V = e^{-r_d n \Delta t} \left[\sum_{j=0}^n \binom{n}{j} p_u^j p_d^{n-j} V^j \right].$$



One remark to make with that CRR model is that in some cases where $\sigma < |(r - q)/\sqrt{\Delta t}|$, negative probabilities can occur. First, it is inconsistent from a probability theory perspective, but additionally, it may prevent the model from capturing relevant events.

Proposition 21.1: Cox-Ross-Rubinstein Valuation

For every $m = 1, 2, \dots, T$, the arbitrage price of a European call option at time $t = T - m$ is given by the Cox-Ross-Rubinstein valuation formula

$$C_{T-m} = S_{T-m} \sum_{j=a}^m \binom{m}{j} \bar{p}^j (1 - \bar{p})^{m-j} - \frac{K}{\bar{r}^m} \sum_{j=a}^m \binom{m}{j} p_*^j (1 - p_*)^{m-j}, \quad (21.6)$$

where $a = a_m(S_{T-m})$, $p_* = (\widehat{r} - d)/(u - d)$ and $\bar{p} = p_* u / \widehat{r}$. At time $t = T - m - 1$, the unique replicating strategy $\phi_{T-m-1} = (\alpha_{T-m-1}, \beta_{T-m-1})$ equals

$$\alpha_{T-m-1} = \sum_{j=a^d}^m \binom{m}{j} \bar{p}^j (1 - \bar{p})^{m-j} + \frac{\delta \Delta_m(uS_{T-m-1}, a^u)}{S_{T-m-1}(u - d)},$$

$$\beta_{T-m-1} = -\frac{K}{\widehat{r}^{m+1}} \sum_{j=a^d}^m \binom{m}{j} p_*^j (1 - p_*)^{m-j} - \frac{\delta d \Delta_m(uS_{T-m-1}, a^u)}{\widehat{r}(u - d)},$$

where $a^d = a_m(dS_{T-m-1})$, $a^u = a_m(uS_{T-m-1})$ and $\delta = a^d - a^u$.

22 Finite Difference Methods

Finite Difference Methods (FDM) are essential numerical techniques employed in option pricing to solve partial differential equations (PDEs). These methods discretise the PDE into a grid of points in both time and space, allowing us to approximate option prices efficiently. By breaking down complex derivative pricing problems into manageable steps, FDMs play a crucial role in the valuation of exotic options.

23 Contract Sensitivities

23.0.1 Finite Differences

23.0.2 Likelihood Ratio

23.0.3 Pathwise Derivatives

24 Finite-Difference Approximations

Consider a model that depends on a parameter θ ranging over some interval of the real line. Suppose that for each value of θ we have a mechanism for generating a random variable $Y(\theta)$, representing the output of the model at parameter θ . Let

$$\alpha(\theta) = \mathbb{E}[Y(\theta)]$$

The derivative estimation problem consists of finding a way to estimate $\alpha'(\theta)$, the derivative of α with respect to θ .

In the application to option pricing, $Y(\theta)$ is the discounted payoff of an option, $\alpha(\theta)$ is its price, and θ could be any of the many model or market parameters that influence the price. When θ is the initial price of an underlying asset, then $\alpha'(\theta)$ is the option's delta (with respect to that asset). The second derivative $\alpha''(\theta)$ is the option's gamma. When θ is a volatility parameter, $\alpha'(\theta)$ is often called "vega." For interest rate derivatives, the sensitivities of prices to the initial term structure (as represented by, e.g., a yield curve or forward curve) are important.

24.1 Bias and Variance

An obvious approach to derivative estimation proceeds as follows. Simulate independent replications $Y_1(\theta), \dots, Y_n(\theta)$ of the model at parameter θ and n additional replications $Y_1(\theta + h), \dots, Y_n(\theta + h)$ at $\theta + h$, for some $h > 0$. Average each set of replications to get $\bar{Y}_n(\theta)$ and $\bar{Y}_n(\theta + h)$ and form the forward-difference estimator.

$$\widehat{\Delta}_F = \widehat{\Delta}_F(n, h) = \frac{\bar{Y}_n(\theta + h) - \bar{Y}_n(\theta)}{h} \quad (24.1)$$

This estimator has an expectation.

$$\mathbb{E}[\widehat{\Delta}_F] = h^{-1}[\alpha(\theta + h) - \alpha(\theta)]. \quad (24.2)$$

We have not specified what relation, if any, holds between the outcomes $Y_i(\theta)$ and $Y_i(\theta + h)$; this will be important when we consider variance, but Equation (24.2) is purely a property of the marginal distributions of the outcomes at θ and $\theta + h$.

If α is twice differentiable at θ , then

$$\alpha(\theta + h) = \alpha(\theta) + \alpha'(\theta)h + \frac{1}{2}\alpha''(\theta)h^2 + O(h^2).$$

In this case, it follows from Equation (24.2) that the bias in the forward-difference estimator is

$$\text{Bias}(\widehat{\Delta}_F) = \mathbb{E}[\widehat{\Delta}_F - \alpha'(\theta)] = \frac{1}{2}\alpha''(\theta)h + O(h). \quad (24.3)$$

By simulating at $\theta - h$ and $\theta + h$, we can form a central-difference estimator

$$\widehat{\Delta}_C \equiv \widehat{\Delta}_C(n, h) = \frac{\bar{Y}_n(\theta + h) - \bar{Y}_n(\theta - h)}{2h}. \quad (24.4)$$

This estimator is often more costly than the forward-difference estimator in the following sense. If we are ultimately interested in estimating $\alpha(\theta)$ and $\alpha'(\theta)$, then we would ordinarily simulate at θ to estimate $\alpha(\theta)$, so the forward-difference estimator requires simulating at just one additional point $\theta + h$. Conversely, the central-difference estimator requires simulating at two additional points. However, this additional computational effort improves the convergence rate of the bias. If α is at least twice differentiable in a neighbourhood of θ , then

$$\alpha(\theta + h) = \alpha(\theta) + \alpha'(\theta)h + \alpha''(\theta)h^2/2 + O(h^2)$$

$$\alpha(\theta - h) = \alpha(\theta) - \alpha'(\theta)h + \alpha''(\theta)h^2/2 + O(h^2)$$

so subtraction eliminates the second-order terms, leading

$$\text{Bias}(\widehat{\Delta}_C) = \frac{\alpha(\theta + h) - \alpha(\theta - h)}{2h} - \alpha'(\theta) = \mathcal{O}(h), \quad (24.5)$$

which is of smaller order than Equation (24.3). If α'' is itself differentiable at θ , we can refine Equation (24.5) to

$$\text{Bias}(\widehat{\Delta}_C) = \frac{1}{6}\alpha'''(\theta)h^2 + \mathcal{O}(h^3). \quad (24.6)$$

Variance The form of the bias of the forward- and central-difference estimators would lead us to take ever smaller values of h to improve accuracy, at least if we ignore the limits of machine precision. But the effect of h on bias must be weighed against its effect on variance.

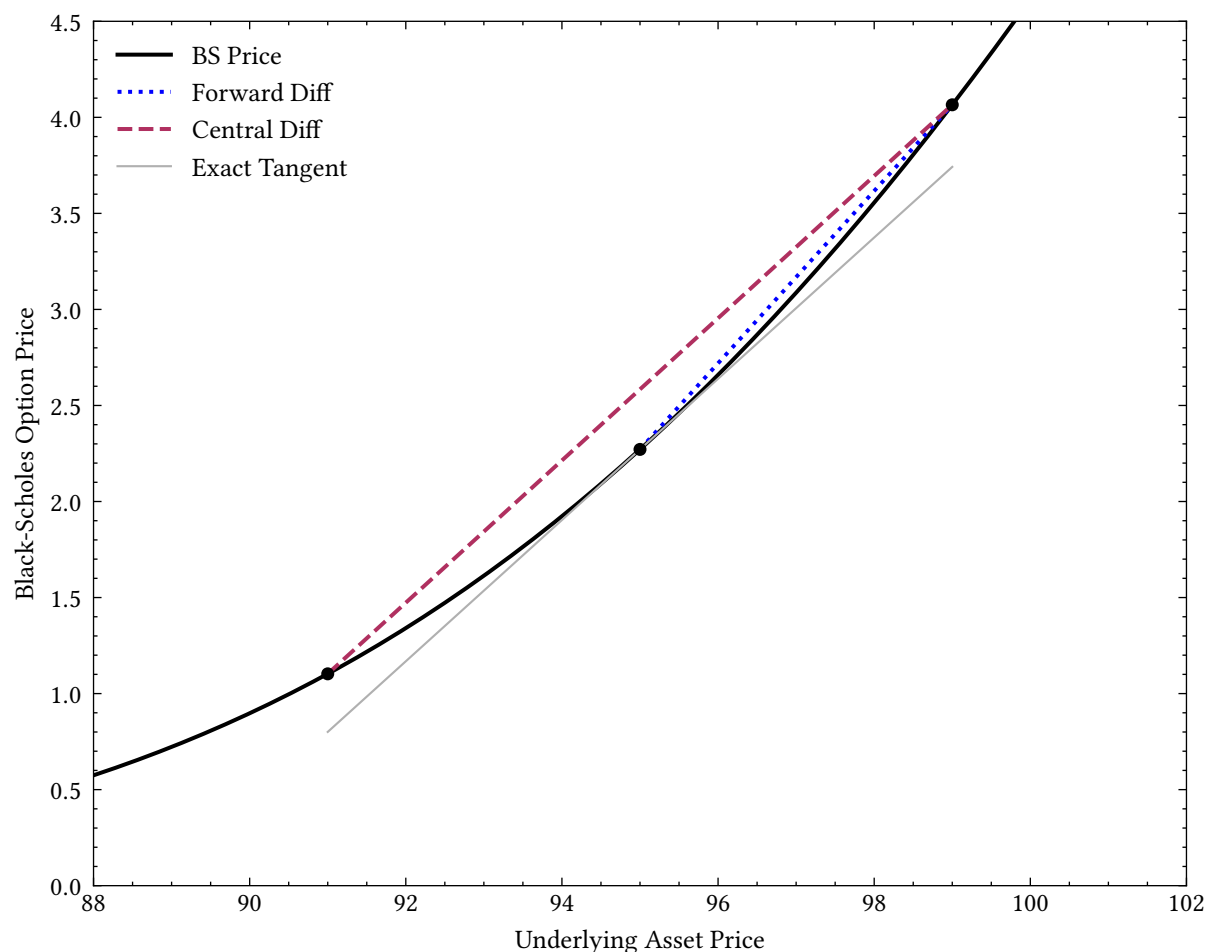


Figure 13: Comparison of central-difference approximations (dashed) and forward-difference approximation (dotted) with exact tangent value to the Black-Scholes formula.

24.2 Optimal Mean Square Error

Part VIII

Financial Contracts

25 European Options

26 Digital Options

27 Barrier Options

The generic term "barrier options" refers to the class of options whose payoff depends on whether or not the underlying price hits a prespecified barrier during the option's lifetime.

27.1 Knock-out Barrier

27.1.1 Up-and-out barrier call option

Let us consider an up-and-out barrier call option with maturity T , strike price K , barrier (or call level) B , and payoff

$$C = (S_T - K)^+ \mathbb{1}_{\left\{ \max_{0 \leq t \leq T} S_t < B \right\}} = \begin{cases} S_T - K & \text{if } \max_{0 \leq t \leq T} S_t \leq B, \\ 0 & \text{if } \max_{0 \leq t \leq T} S_t > B, \end{cases}$$

with $B \geq K$.

Proposition 27.1: Up-and-Out Barrier Call Option Pricing

When $K < B$, the price

$$e^{-(T-t)r} \mathbb{1}_{\{M_0^t < B\}} \mathbb{E}^* \left[\left(\frac{S_{T-t}}{S_0} - K \right)^+ \mathbb{1}_{\left\{ x \max_{0 \leq u \leq T-t} \frac{S_u}{S_0} < B \right\}} \middle| x = S_t \right] \quad (27.1)$$

of the up-and-out barrier call option with maturity T , strike price K , and barrier level B is given by the following formula, where $\tau = T - t$:

$$\begin{aligned}
 C_{UO}(t) = S_t \mathbb{1}_{\{M_0^t < B\}} & \left\{ [\Phi(d_1(S_t, K)) - \Phi(d_1(S_t, B))] \right. \\
 & \left. - \left(\frac{B}{S_t}\right)^{1+\frac{2r}{\sigma^2}} \left[\Phi\left(d_1\left(\frac{B^2}{S_t}, K\right)\right) - \Phi\left(d_1\left(\frac{B^2}{S_t}, B\right)\right) \right] \right\} \\
 & - e^{-r\tau} K \mathbb{1}_{\{M_0^t < B\}} \left\{ [\Phi(d_2(S_t, K)) - \Phi(d_2(S_t, B))] \right. \\
 & \left. - \left(\frac{S_t}{B}\right)^{1-\frac{2r}{\sigma^2}} \left[\Phi\left(d_2\left(\frac{B^2}{S_t}, K\right)\right) - \Phi\left(d_2\left(\frac{B^2}{S_t}, B\right)\right) \right] \right\}, \quad (27.2)
 \end{aligned}$$

If $K \geq B$:

$$C_{UO}(t, S_t) = 0. \quad (27.3)$$

where $d_{1,2}(S, X)$ corresponds to the definitions in (27.1) with spot price S , strike price K , and time to maturity τ . The price of the up-and-out barrier call option vanishes when $B \leq K$.

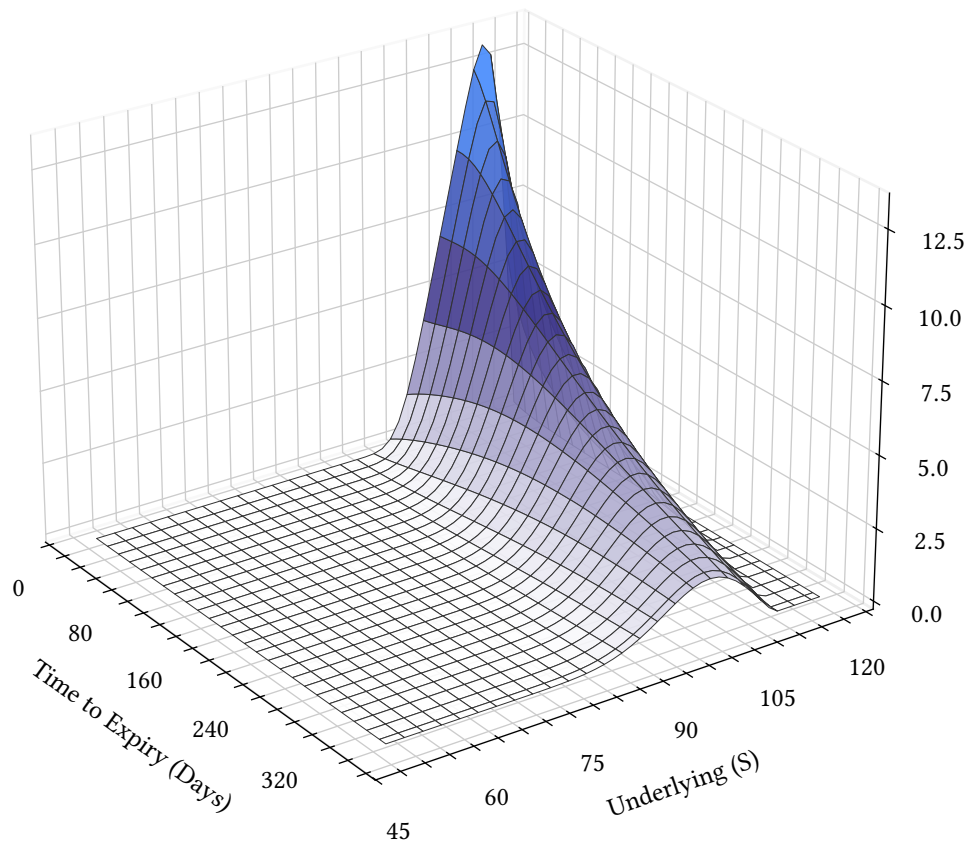


Figure 14: Graph of the up-and-out call option price

27.1.2 Up-and-out barrier put option

This option is also known as a *Callable Bear Contract*, or a Bear CBBC with no residual value, or a turbo warrant with no rebate. In this context, B denotes the call level for pricing Bear CBBCs (up-and-out barrier put options). The payoff is given by

$$P = (K - S_T)^+ \mathbb{1}_{\left\{\max_{0 \leq t \leq T} S_t < B\right\}} = \begin{cases} K - S_T & \text{if } \max_{0 \leq t \leq T} S_t \leq B, \\ 0 & \text{if } \max_{0 \leq t \leq T} S_t > B. \end{cases}$$

Proposition 27.2: Up-and-Out Barrier Put Option Pricing

The price of the up-and-out barrier put option with maturity T , strike price K , and barrier level B is given by the following formulas, where $\tau = T - t$.

If $B < K$, the price is given by

$$\begin{aligned} P_{UO}(t) = S_t \mathbb{1}_{\{M_0^t < B\}} & \left\{ -\Phi(-d_1(S_t, B)) + \left(\frac{B}{S_t}\right)^{1+\frac{2r}{\sigma^2}} \Phi\left(-d_1\left(\frac{B^2}{S_t}, B\right)\right) \right\} \\ & + K e^{-r\tau} \mathbb{1}_{\{M_0^t < B\}} \left\{ \Phi(-d_2(S_t, B)) - \left(\frac{S_t}{B}\right)^{1-\frac{2r}{\sigma^2}} \Phi\left(-d_2\left(\frac{B^2}{S_t}, B\right)\right) \right\}. \end{aligned} \quad (27.4)$$

If $B \geq K$, the price is given by

$$\begin{aligned} P_{UO}(t) = S_t \mathbb{1}_{\{M_0^t < B\}} & \left\{ -\Phi(-d_1(S_t, K)) + \left(\frac{B}{S_t}\right)^{1+\frac{2r}{\sigma^2}} \Phi\left(-d_1\left(\frac{B^2}{S_t}, K\right)\right) \right\} \\ & + K e^{-r\tau} \mathbb{1}_{\{M_0^t < B\}} \left\{ \Phi(-d_2(S_t, K)) - \left(\frac{S_t}{B}\right)^{1-\frac{2r}{\sigma^2}} \Phi\left(-d_2\left(\frac{B^2}{S_t}, K\right)\right) \right\}. \end{aligned} \quad (27.5)$$

This corresponds to the value of a vanilla put option minus the value of the "knock-out" component.

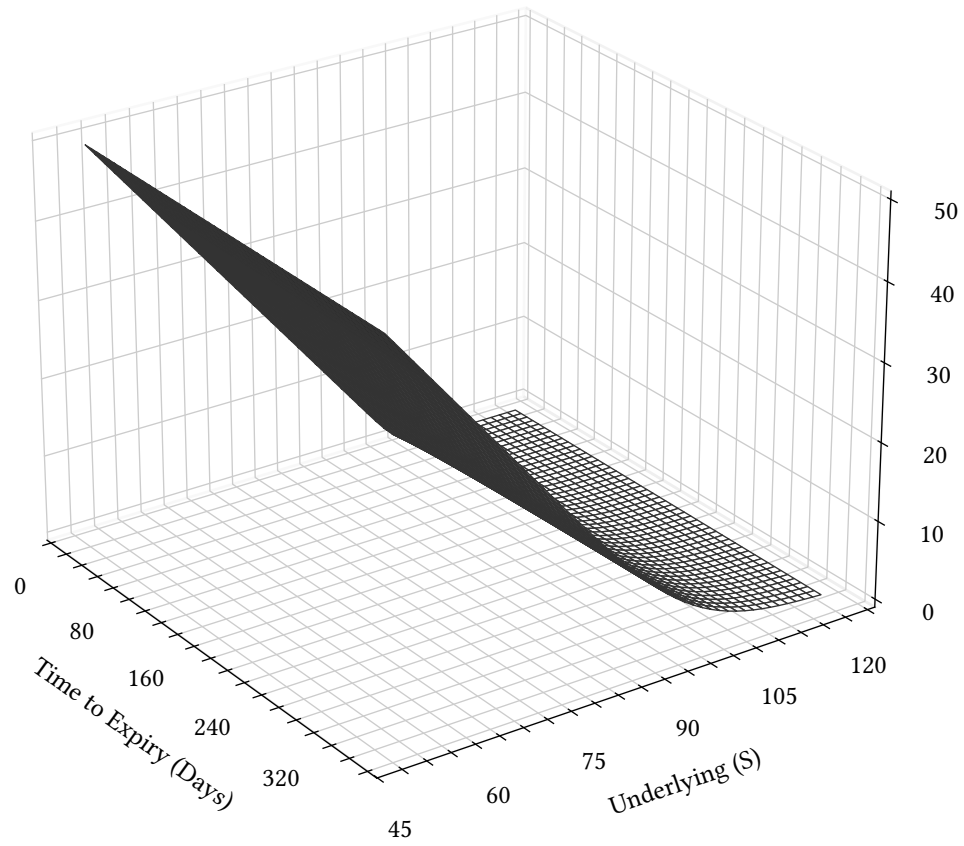


Figure 15: Graph of the up-and-out put option price

27.1.3 Down-and-out barrier call option

Let us now consider a down-and-out barrier call option on the underlying asset price S_t with exercise date T , strike price K , barrier level B , and payoff

$$C = (S_T - K)^+ \mathbb{1}_{\left\{ \min_{0 \leq t \leq T} S_t > B \right\}} = \begin{cases} S_T - K & \text{if } \min_{0 \leq t \leq T} S_t > B, \\ 0 & \text{if } \min_{0 \leq t \leq T} S_t \leq B, \end{cases} \quad (27.6)$$

with $0 \leq B \leq K$. The down-and-out barrier call option is also called a *Callable Bull Contract*, or a Bull CBBC with no residual value, or a turbo warrant with no rebate, in which B denotes the call level.

Proposition 27.3: Down-and-Out Barrier Call Option Pricing

The price of the down-and-out barrier call option with maturity T , strike price K , and barrier level B is given by the following formulas, where $\tau = T - t$.

When $B \leq K$, we have:

$$\begin{aligned} C_{DO}(t, S_t) &= \mathbb{1}_{\{m_0^t > B\}} \text{Bl}(S_t, K, r, \tau, \sigma) \\ &\quad - B \mathbb{1}_{\{m_0^t > B\}} \left(\frac{B}{S_t} \right)^{1+2r/\sigma^2} \Phi \left(d_1 \left(\frac{B^2}{KS_t}, \tau \right) \right) \\ &\quad + K e^{-r\tau} \mathbb{1}_{\{m_0^t > B\}} \left(\frac{B}{S_t} \right)^{-1+2r/\sigma^2} \Phi \left(d_2 \left(\frac{B^2}{KS_t}, \tau \right) \right). \end{aligned} \quad (27.7)$$

When $B \geq K$, we find:

$$\begin{aligned} C_{DO}(t, S_t) &= S_t \mathbb{1}_{\{m_0^t > B\}} \Phi \left(d_1 \left(\frac{S_t}{B}, \tau \right) \right) - K e^{-r\tau} \mathbb{1}_{\{m_0^t > B\}} \Phi \left(d_2 \left(\frac{S_t}{B}, \tau \right) \right) \\ &\quad - B \mathbb{1}_{\{m_0^t > B\}} \left(\frac{B}{S_t} \right)^{2r/\sigma^2} \Phi \left(d_1 \left(\frac{B}{S_t}, \tau \right) \right) \\ &\quad + K e^{-r\tau} \mathbb{1}_{\{m_0^t > B\}} \left(\frac{S_t}{B} \right)^{1-2r/\sigma^2} \Phi \left(d_2 \left(\frac{B}{S_t}, \tau \right) \right). \end{aligned} \quad (27.8)$$

Here B denotes the standard Black-Scholes call price, and $d_{1,2}(x, \tau)$ are defined as in ??? with the spot price term replaced by the ratio argument x . If $m_0^t \leq B$, the price is 0.

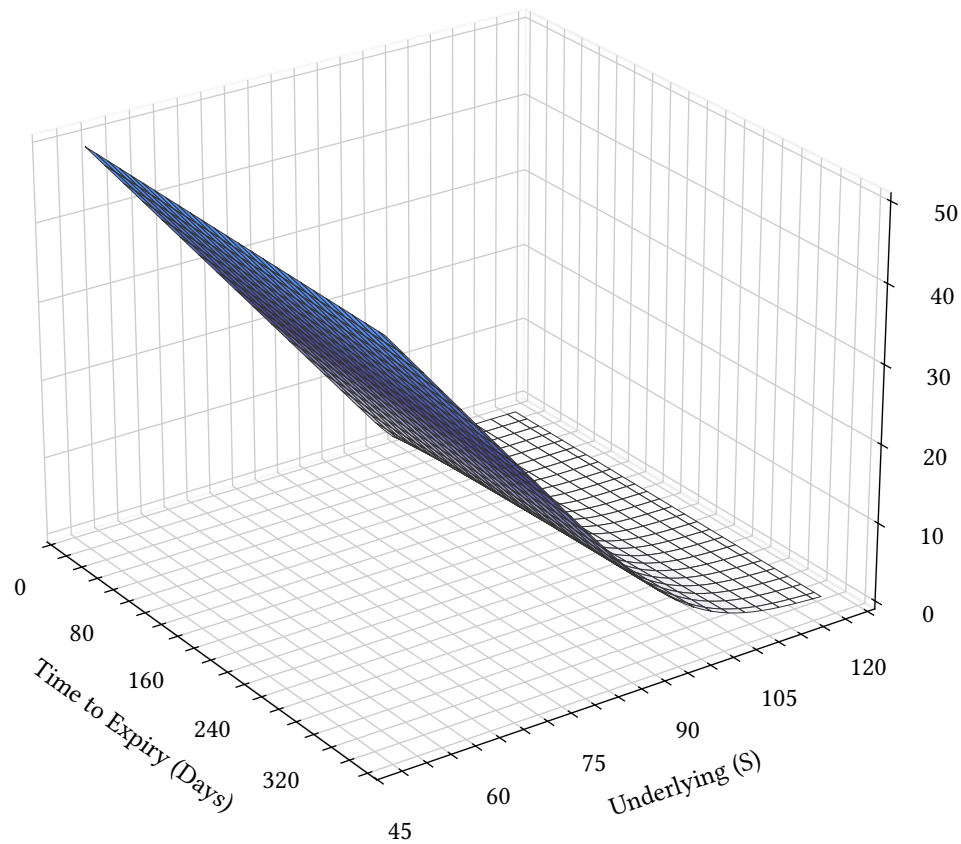


Figure 16: Graph of the down-and-out call option price

27.1.4 Down-and-out barrier put option

The down-and-out barrier put option with exercise date T , strike price K , and barrier level B has the payoff:

$$P = (K - S_T)^+ \mathbb{1}_{\{\min_{0 \leq t \leq T} S_t > B\}} = \begin{cases} K - S_T & \text{if } \min_{0 \leq t \leq T} S_t > B, \\ 0 & \text{if } \min_{0 \leq t \leq T} S_t \leq B. \end{cases} \quad (27.9)$$

For this option to have non-zero value, we typically require $S_0 > B$. Since it is a put option, the intrinsic value is positive when $S_T < K$. If $B \geq K$, the condition $S_T < K$ combined with the knock-out condition $\min S_t > B \geq K$ makes the payoff always zero (as $S_T > B \geq K \implies K - S_T < 0$). Thus, we only consider the case $B < K$.

Proposition 27.4: Down-and-Out Barrier Put Option Pricing

The price of the down-and-out barrier put option with maturity T , strike price K , and barrier level B (where $B \leq K$) is given by

$$\begin{aligned} P_{DO}(t, S_t) = & K e^{-r\tau} \mathbb{1}_{\{m_0^t > B\}} \left[\Phi \left(-d_2 \left(\frac{S_t}{K}, \tau \right) \right) - \Phi \left(-d_2 \left(\frac{S_t}{B}, \tau \right) \right) \right] \\ & - S_t \mathbb{1}_{\{m_0^t > B\}} \left[\Phi \left(-d_1 \left(\frac{S_t}{K}, \tau \right) \right) - \Phi \left(-d_1 \left(\frac{S_t}{B}, \tau \right) \right) \right] \\ & - K e^{-r\tau} \mathbb{1}_{\{m_0^t > B\}} \left(\frac{S_t}{B} \right)^{1-2r/\sigma^2} \left[\Phi \left(d_2 \left(\frac{B^2}{K S_t}, \tau \right) \right) - \Phi \left(d_2 \left(\frac{B}{S_t}, \tau \right) \right) \right] \\ & + S_t \mathbb{1}_{\{m_0^t > B\}} \left(\frac{B}{S_t} \right)^{1+2r/\sigma^2} \left[\Phi \left(d_1 \left(\frac{B^2}{K S_t}, \tau \right) \right) - \Phi \left(d_1 \left(\frac{B}{S_t}, \tau \right) \right) \right]. \end{aligned} \quad (27.10)$$

Here $\tau = T - t$, and $d_{1,2}(x, \tau)$ correspond to the definitions with the spot price term replaced by the ratio argument x .

If $B > K$, the price vanishes: $P_{DO}(t, S_t) = 0$.

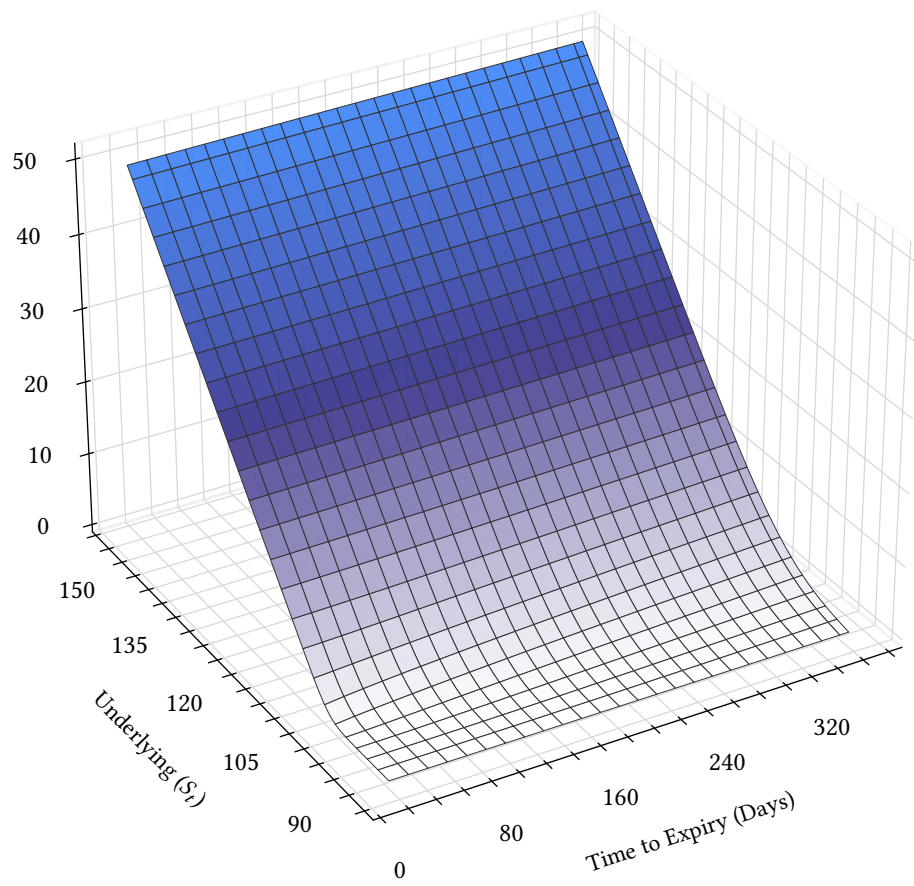


Figure 17: Graph of the down-and-out put option price

27.2 Knock-in Barrier

28 American Options

Whereas a European option can be exercised only at a fixed date, an American option can be exercised any time up to its expiration. The value of an American option is the value achieved by exercising optimally. Finding this value entails finding the optimal exercise rule by solving an optimal stopping problem and computing the expected discounted payoff of the option under this rule. The embedded optimisation problem makes this a difficult problem for simulation.

28.1 Approximate Continuation Values

Regression-based methods posit an expression for the continuation value of the form

$$\mathbb{E}[V_{i+1}(X_{i+1}) \mid X_i = x] = \sum_{r=1}^M \beta_{ir} \psi_r(x), \quad (28.1)$$

for some basis functions $\psi_r : \mathbb{R}^d \rightarrow \mathbb{R}$ and constants β_{ir} , $r = 1, \dots, M$. Using the notation C_i for the continuation value at time i , we may equivalently write [eq. \(28.1\)](#) as

$$C_i(x) = \beta_i^\top \psi(x), \quad (28.2)$$

with

$$\beta_i^\top = (\beta_{i1}, \dots, \beta_{iM}), \quad \psi(x) = (\psi_1(x), \dots, \psi_M(x))^\top.$$

One could use different basis functions at different exercise dates, but to simplify notation, we suppress any dependence of ψ on i .

Assuming a relation of the form [eq. \(28.1\)](#) holds, the vector β_i is given by

$$\beta_i = (\mathbb{E}[\psi(X_i)\psi(X_i)^\top])^{-1} \mathbb{E}[\psi(X_i)V_{i+1}(X_{i+1})] \equiv B_\psi^{-1} B_{\psi V}. \quad (28.3)$$

Here, B_ψ is the indicated $M \times M$ matrix (assumed nonsingular) and $B_{\psi V}$ the indicated vector of length M . The variables (X_i, X_{i+1}) inside the expectations have a joint distribution corresponding to the state of the underlying Markov chain at dates i and $i + 1$.

The coefficients β_{ir} could be estimated from observations of pairs $(X_{ij}, V_{i+1}(X_{i+1,j}))$, $j = 1, \dots, b$, each consisting of the state at time i and the corresponding option value at time $i + 1$. Consider, in particular, independent paths $(X_{1j}, \dots, X_{mj})$, $j = 1, \dots, b$, and suppose for a moment that the values $V_{i+1}(X_{i+1,j})$ are known. The least-squares estimate of β_i is then given by

$$\widehat{\beta}_i = \widehat{B}_\psi^{-1} \widehat{B}_{\psi V},$$

where $\widehat{B}_\psi$ and $\widehat{B}_{\psi V}$ are the sample counterparts of B_ψ and $B_{\psi V}$. More explicitly, $\widehat{B}_\psi$ is the $M \times M$ matrix with qr entry

$$\frac{1}{b} \sum_{j=1}^b \psi_q(X_{ij}) \psi_r(X_{ij})$$

and $\widehat{B}_{\psi V}$ is the M -vector with r th entry

$$\frac{1}{b} \sum_{k=1}^b \psi_r(X_{ik}) V_{i+1}(X_{i+1,k}). \quad (28.4)$$

All of these quantities can be calculated from function values at pairs of consecutive nodes $(X_{ij}, X_{i+1,j})$, $j = 1, \dots, b$. In practice, V_{i+1} is unknown and must be replaced by estimated values $\widehat{V}_{i+1}$ at downstream nodes. The estimate $\widehat{\beta}_i$ then defines an estimate

$$\widehat{C}_i(x) = \widehat{\beta}_i^\top \psi(x), \quad (28.5)$$

of the continuation value at an arbitrary point x in the state space $\mathbb{R}^d$. This, in turn, defines a procedure for estimating the option value.

Algorithm 9 Regression-Based Pricing Algorithm

- 1: Simulate b independent paths $\{X_{1j}, \dots, X_{mj}\}$, $j = 1, \dots, b$, of the Markov chain.
- 2: At terminal nodes, set $\widehat{V}_{mj} = h_m(X_{mj})$ for $j = 1, \dots, b$.
- 3: **for** $i = m - 1$ **down to** 1 **do**
- 4: Given estimated values $\widehat{V}_{i+1,j}$, $j = 1, \dots, b$, use regression to calculate $\widehat{\beta}_i = \widehat{B}_\psi^{-1} \widehat{B}_\psi V$.
- 5: Compute estimated continuation values $\widehat{C}_i(X_{ij})$ using [eq. \(28.5\)](#).
- 6: **for** $j = 1, \dots, b$ **do**
- 7: Set value based on immediate exercise versus estimated continuation:

$$\widehat{V}_{ij} = \max \left\{ h_i(X_{ij}), \widehat{C}_i(X_{ij}) \right\}. \quad (28.6)$$

8: **end for**

9: **end for**

10: Set $\widehat{V}_0 = (\widehat{V}_{11} + \dots + \widehat{V}_{1b})/b$.

Definition 28.1: American Contingent Claim

An *American contingent claim* $X^a = (X, \mathcal{T}_{[0,T]})$ expiring at time T consists of a sequence of payoffs $(X_t)_{t=0}^T$, where X_t is an $\mathcal{F}_t$ -measurable random variable for $t = 0, 1, \dots, T$, and of the set $\mathcal{T}_{[0,T]}$ of admissible exercise policies.

Proposition 28.1: Arbitrage Price of American Put

The arbitrage price P_t^a of an American put option equals

$$P_t^a = \max_{\tau \in \mathcal{T}_{[t,T]}} \mathbb{E}_{\mathbb{P}^*} \left(\widehat{r}^{-(\tau-t)} (K - S_\tau)^+ \mid \mathcal{F}_t \right), \quad \forall t \leq T. \quad (28.7)$$

Moreover, for any $t \leq T$ the stopping time τ_t^* that realizes the maximum in [eq. \(28.7\)](#) is given by the expression (by convention $\min \emptyset = T$)

$$\tau_t^* = \min \left\{ u \in \{t, t+1, \dots, T\} \mid (K - S_u)^+ \geq P_u^a \right\}. \quad (28.8)$$

Proposition 28.2: Arbitrage Price of American Claim in CRR

For every $t \leq T$, the arbitrage price $\pi(X^a)$ of an arbitrary American claim X^a in the CRR model equals

$$\pi_t(X^a) = \max_{\tau \in \mathcal{T}_{[t,T]}} \mathbb{E}_{\mathbb{P}^*} \left(\widehat{r}^{-(\tau-t)} X_\tau \mid \mathcal{F}_t \right). \quad (28.9)$$

The price process $\pi(X^a)$ can be determined using the following recurrence relation, for every $t \leq T - 1$,

$$\pi_t(X^a) = \max \left\{ X_t, \mathbb{E}_{\mathbb{P}^*} \left(\widehat{r}^{-1} \pi_{t+1}(X^a) \mid \mathcal{F}_t \right) \right\} \quad (28.10)$$

subject to the terminal condition $\pi_T(X^a) = X_T$. In the case of a path-independent American claim X^a with the reward function h we have that, for every $t \leq T - 1$,

$$\pi_t(X^a) = \max \left\{ h(S_t, t), \widehat{r}^{-1} \left(p_* \pi_{t+1}^u(X^a) + (1 - p_*) \pi_{t+1}^d(X^a) \right) \right\}, \quad (28.11)$$

where, for a generic stock price value S_t at time t , we write $\pi_{t+1}^u(X^a)$ and $\pi_{t+1}^d(X^a)$ to denote the values of the price process $\pi(X^a)$ at time $t + 1$ in the nodes that correspond to the upward and downward movements of the stock price during the time-period $[t, t + 1]$, that is, for the values uS_t and dS_t of the stock price at time $t + 1$, respectively.

29 Lookback Options

Lookback call (resp. put) options are financial derivatives that allow their holders to exercise the option by setting the strike price at the minimum (resp. maximum) of the underlying asset price process $(S_t)_{t \in [0, T]}$ over the time interval $[0, T]$. Lookback options can be priced by PDE arguments or by computing the discounted expected values of their claim payoff C , namely $C = S_T - \min_{0 \leq t \leq T} S_t$ in the case of call options, and $C = \max_{0 \leq t \leq T} S_t - S_T$ in the case of put options.

29.1 Maximum of Brownian Motion

Figure 18 represents the running maximum process

$$M_0^t := \max_{s \in [0, t]} W_s, \quad t \geq 0,$$

of Brownian motion $(W_t)_{t \in \mathbb{R}_+}$.

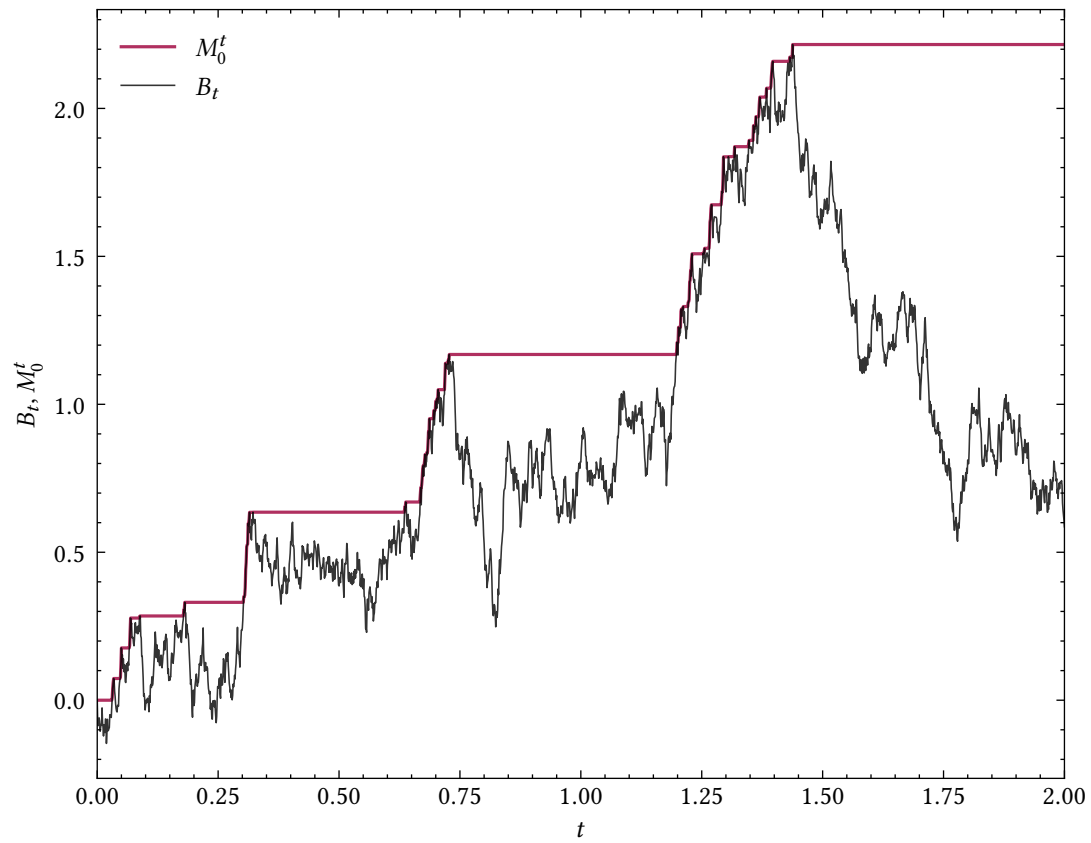


Figure 18: A Wiener process path B_t and its running maximum M_0^t .

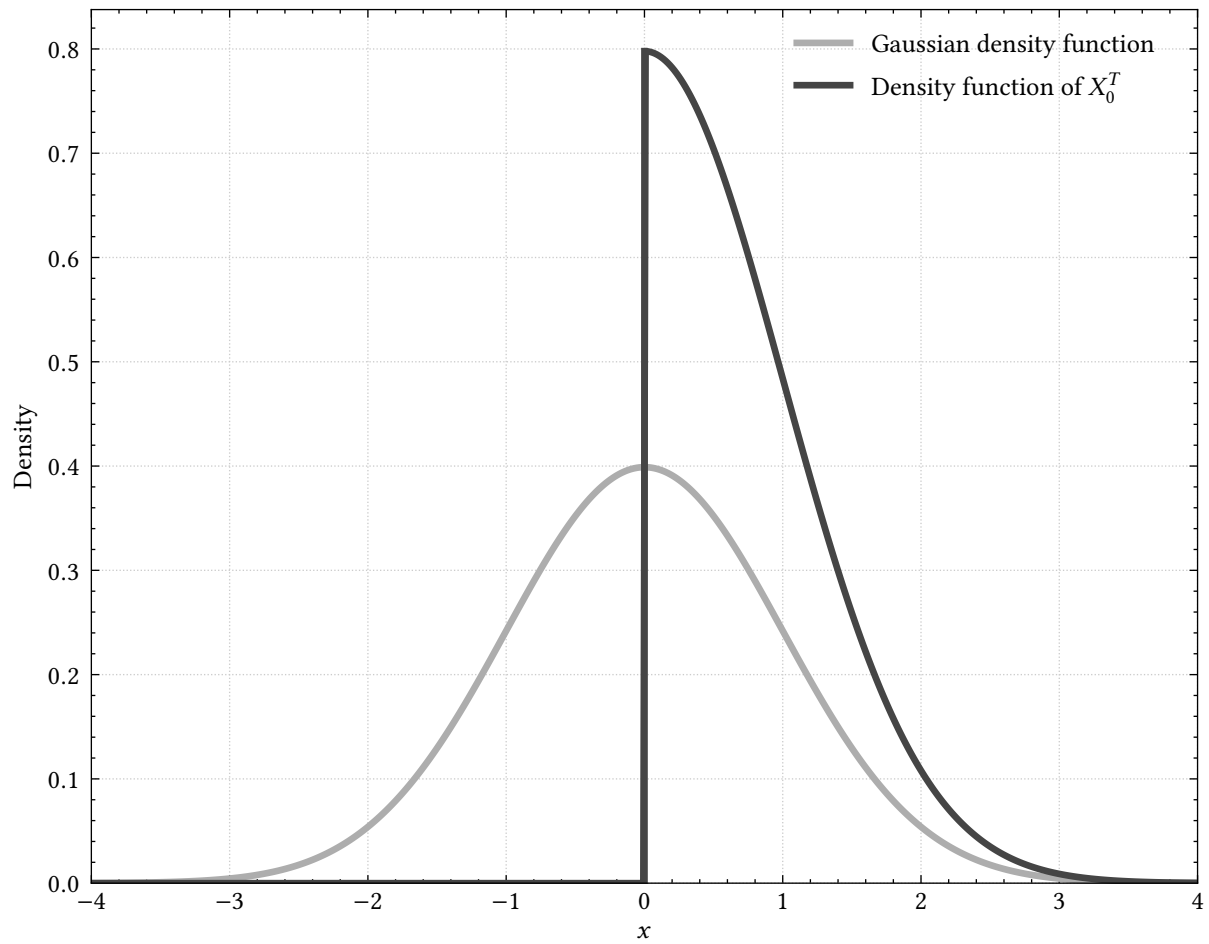


Figure 19: Probability density of the maximum X_0^1 of Brownian motion over $[0, 1]$

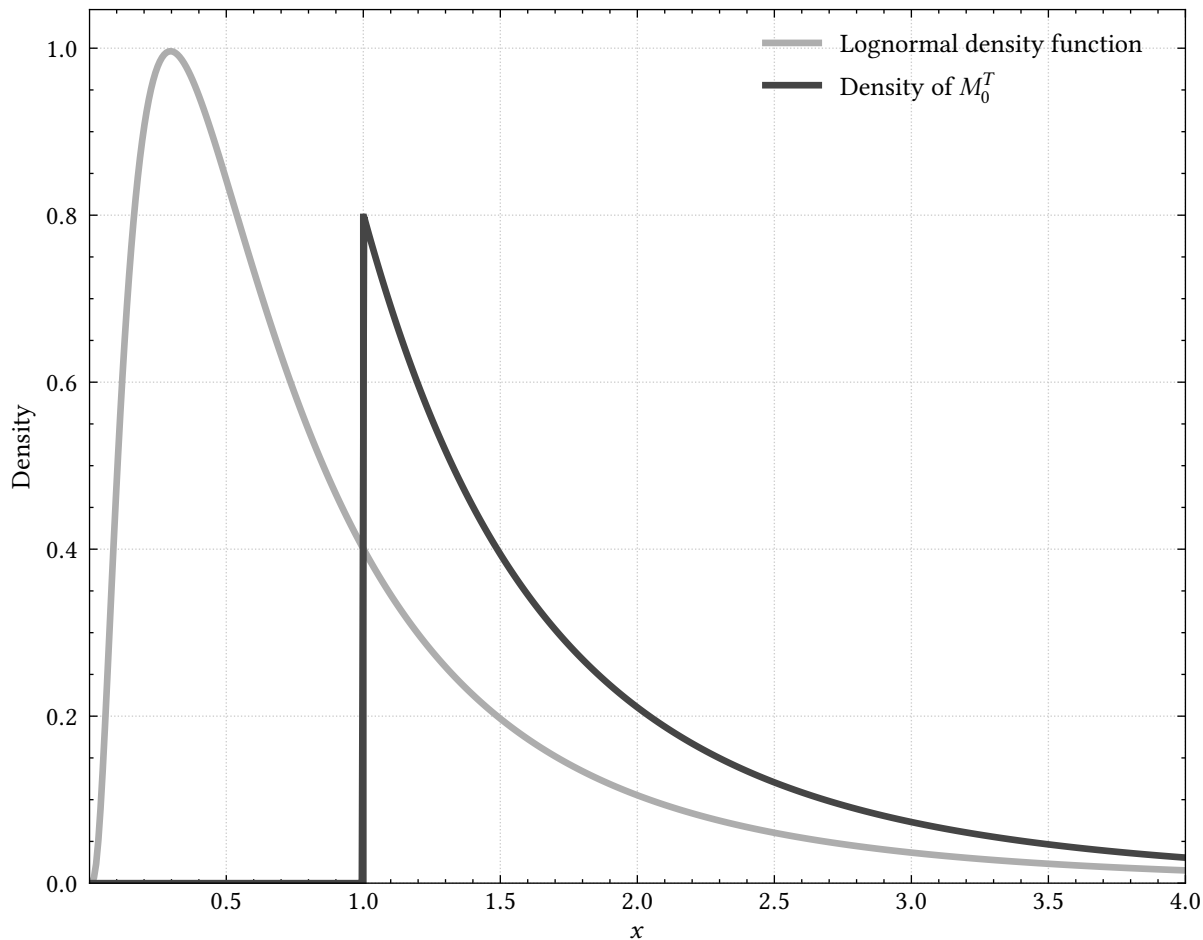


Figure 20: Density of the maximum $M_0^T = \text{Max}_{t \in [0, T]} S_t$ of geometric Brownian motion with $S_0 = 1$.

29.2 The Lookback Put Option

The standard lookback put option gives its holder the right to sell the underlying asset at its historically highest price. In this case, the floating strike price is M_0^T and the payoff is given by the terminal value

$$C = M_0^T - S_T \quad (29.1)$$

of the drawdown process $(M_0^T - S_t)_{t \in [0, T]}$. The following pricing formula for lookback put options is a direct consequence of Proposition 10.9.

Proposition 29.1: The Price Of The Lookback Put Option

The price at time $t \in [0, T]$ of the lookback put option with payoff $M_0^T - S_T$ is given by

$$\begin{aligned}
 P_{LB}(t) &= e^{-r\tau} \mathbb{E}^* [M_0^T - S_T \mid \mathcal{F}_t] \\
 &= M_0^t e^{-r\tau} \Phi(-d_2(S_t, M_0^t)) \\
 &\quad + S_t \left(1 + \frac{\sigma^2}{2r}\right) \Phi(d_1(S_t, M_0^t)) \\
 &\quad - S_t e^{-r\tau} \frac{\sigma^2}{2r} \left(\frac{M_0^t}{S_t}\right)^{2r/\sigma^2} \Phi(-d_2(M_0^t, S_t)) - S_t,
 \end{aligned} \tag{29.2}$$

where $\tau = T - t$, and $d_{1,2}(\cdot, \cdot)$ correspond to the definitions in ??? with the fixed strike K replaced by the running maximum M_0^t (or S_t where arguments are inverted).

30 Asian Options

An *Asian option* (or *average option*) defines a class of options, whether European or American style, whose terminal payoff depends on the average asset value over a specific period within the option's life. Due to this averaging feature, Asian options are particularly suitable for thinly traded assets or commodities. Unlike standard options, they are typically less expensive and offer greater robustness against price manipulation near the expiry date.

Asian options were first traded in Tokyo in 1987, and have become particularly popular in commodities trading.

Let T denote the exercise date, and let $0 \leq T_0 < T$ represent the commencement of the averaging period. The payoff at expiry for an Asian call option is defined as

$$C_T^A \stackrel{\text{def}}{=} (A_S(T_0, T) - K)^+, \quad (30.1)$$

where

$$A_S(T_0, T) = \frac{1}{T - T_0} \int_{T_0}^T S_u \, du \quad (30.2)$$

represents the arithmetic average of the asset price over the time interval $[T_0, T]$. Here, K denotes the fixed strike price, and the asset price S is assumed to follow a geometric Brownian motion.

The main difficulty in pricing and hedging Asian options is due to the fact that the random variable $A(T_0, T)$ does not have a lognormal distribution. This feature makes the task of finding an explicit formula for the price of an Asian option surprisingly involved.

For this reason, early studies of Asian options were based either on approximations or on the direct application of the Monte Carlo method. The numerical approach to the valuation of Asian options, is based on the approximation of the arithmetic average using the geometric average. Note first that it is natural to substitute for the continuous-time average $A(T_0, T)$ its

discrete-time counterpart

$$A_S^n(T_0, T) = \frac{1}{n} \sum_{i=0}^{n-1} S_{T_i}, \quad (30.3)$$

where $T_i = T_0 + i(T - T_0)/n$. Furthermore, the arithmetic average $A_S^n(T_0, T)$ can be replaced with the geometric average, denoted by $G_S^n(T_0, T)$, in what follows. Recall that the random variables $S_{T_i}, i = 1, \dots, n$ are given explicitly by the expression

$$S_{T_i} = S_{T_0} \exp \left(\sigma (W_{T_i}^* - W_{T_0}^*) + \left(r - \frac{1}{2} \sigma^2 \right) (T_i - T_0) \right), \quad (30.4)$$

where W^* is a standard Brownian motion under the martingale measure $\mathbb{P}^*$. Hence the geometric average admits the following representation

$$G_S^n(T_0, T) = \left(\prod_{i=0}^{n-1} S_{T_i} \right)^{1/n} = c S_{T_0} \exp \left(\frac{\sigma}{n} \sum_{i=0}^{n-1} (n - i - 1) (W_{T_{i+1}}^* - W_{T_i}^*) \right) \quad (30.5)$$

for a strictly positive constant c . In view of the independence of increments of the Brownian motion, the last formula makes clear that the geometric average $G_S^n(T_0, T)$ has a lognormal distribution under $\mathbb{P}^*$. The approximate Black-Scholes-like formula for the price of an Asian call option can thus be easily found by the direct evaluation of the conditional expectation

$$\tilde{C}_{T_0}^n = e^{-r(T-T_0)} \mathbb{E}_{\mathbb{P}^*} \left((G_S^n(T_0, T) - K)^+ \mid \mathcal{F}_{T_0} \right). \quad (30.6)$$

It appears, however, that such an approach significantly underprices Asian call options. To overcome this deficiency, one may directly approximate the true distribution of the arithmetic average using an approximate distribution, typically a lognormal distribution with the appropriate parameters

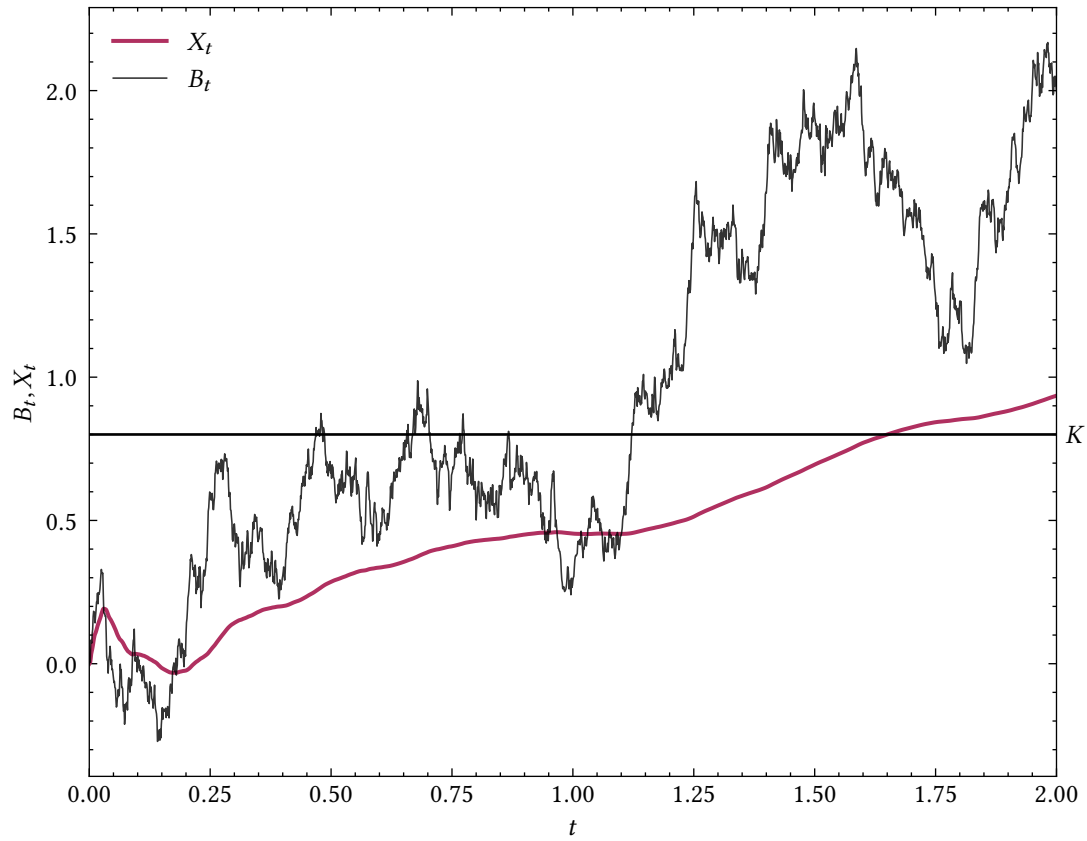


Figure 21: Brownian motion B_t and its moving average X_t

31 Basket Options

A *basket option*, as its name suggests, is a type of option contract that serves to hedge against the risk exposure of a basket of assets, i.e., a predetermined portfolio of assets. Generally speaking, a basket option is more cost-effective than a portfolio of single options, as the latter over-hedges the exposure and incurs higher costs than a basket option. An intuitive explanation for this feature is that the basket option considers the correlation between different risk factors. For instance, in the case of a strong negative correlation between two or more underlying assets, the total risk exposure may almost vanish; however, this beneficial feature is not reflected in the payoffs and prices of single options.

From the analytical viewpoint, there is a close analogy between basket options and Asian options. Let us denote by S^i , $i = 1, \dots, k$, the price processes of k underlying assets; they will be referred to as stocks in what follows. In this case, it seems natural to refer to such a basket option as the *stock index option* (in market practice, options on a basket of currencies are also quite common). The payoff at expiry of a basket call option is defined in the following way

$$C_T^B \stackrel{\text{def}}{=} \left(\sum_{i=1}^k w_i S_T^i - K \right)^+ = (A_T - K)^+, \quad (31.1)$$

where $w_i \geq 0$ is the weight of the i^{th} asset, so that $\sum_{i=1}^k w_i = 1$. Note that A_T stands for the weighted arithmetic average $A_T = \sum_{i=1}^k w_i S_T^i$. We assume that each stock price S^i follows a geometric Brownian motion. More explicitly, under the martingale measure $\mathbb{P}^*$, we have

$$dS_t^i = S_t^i (r dt + \widehat{\sigma}_i \cdot dW_t^*) \quad (31.2)$$

for some non-zero vectors $\widehat{\sigma}_i \in \mathbb{R}^k$, where $W^* = (W^{1*}, \dots, W^{k*})$ stands for a k -dimensional Brownian motion under $\mathbb{P}^*$, and r is the risk-free interest rate. Observe that for any fixed i , we

can find a standard one-dimensional Brownian motion $\widetilde{W}^i$ such that

$$dS_t^i = S_t^i(r dt + \sigma_i d\widetilde{W}_t^i) \quad (31.3)$$

and $\sigma_i = |\widehat{\sigma}_i|$, where $|\widehat{\sigma}_i|$ is the Euclidean norm of $\widehat{\sigma}_i$. Let us denote by $\rho_{i,j}$ the instantaneous correlation coefficient

$$\rho_{i,j} = \frac{\widehat{\sigma}_i \cdot \widehat{\sigma}_j}{\sigma_i \sigma_j} = \frac{\widehat{\sigma}_i \cdot \widehat{\sigma}_j}{|\widehat{\sigma}_i| |\widehat{\sigma}_j|}.$$

We may thus alternatively assume that the dynamics of price processes S^i are given by (31.3), where $\widetilde{W}^i$, $i = 1, \dots, k$ are one-dimensional Brownian motions, whose cross-variations satisfy $\langle \widetilde{W}^i, \widetilde{W}^j \rangle_t = \rho_{i,j} t$ for every $i, j = 1, \dots, k$. Let us return to the problem of valuation of basket options. For similar reasons as those applying to Asian options, basket options are rather intractable analytically. developed a simple technique of pricing basket options on a bivariate binomial lattice, thus generalising the standard Cox-Ross-Rubinstein methodology. Unfortunately, this numerical method is very time-consuming, especially where there are several underlying assets. To overcome this, a proposed valuation of a basket using an approximation of the weighted arithmetic mean in the form of its geometric counterpart (this follows the approach of Asian options). For a fixed $t \leq T$, let us denote by $\widehat{w}_i$ the modified weights

$$\widehat{w}_i = \frac{w_i S_t^i}{\sum_{j=1}^k w_j S_t^j} = \frac{w_i F_{S^i}(t, T)}{\sum_{j=1}^k w_j F_{S^j}(t, T)}, \quad (31.4)$$

where $F_{S^i}(t, T)$ is the forward price at time t of the i^{th} asset for the settlement date T .

We may rewrite (31.1) as follows³

$$\begin{aligned} C_T^B &= \left(\sum_{j=1}^k w_j F_{S^j}(t, T) \right) \left(\sum_{i=1}^k \widehat{w}_i \widetilde{S}_T^i - \widetilde{K} \right)^+ \\ &= \left(\sum_{j=1}^k w_j F_{S^j}(t, T) \right) (\widetilde{A}_T - \widetilde{K})^+, \end{aligned} \quad (31.5)$$

³This representation is introduced because it appears to give a better approximation of the price of a basket option than formula (31.1).

where $\widetilde{S}_T^i = S_T^i / F_{S^i}(t, T)$, $\widetilde{A}_T = \sum_{i=1}^k \widehat{w}_i \widetilde{S}_T^i$, and

$$\widetilde{K} = \frac{K}{\sum_{j=1}^k w_j F_{S^j}(t, T)} = \frac{e^{-r(T-t)} K}{\sum_{j=1}^k w_j S_t^j}. \quad (31.6)$$

The arbitrage price at time t of a basket call option thus equals

$$C_t^B = e^{-r(T-t)} \left(\sum_{j=1}^k w_j F_{S^j}(t, T) \right) \mathbb{E}_{\mathbb{P}^*}((\widetilde{A}_T - \widetilde{K})^+ \mid \mathcal{F}_t),$$

or equivalently,

$$C_t^B = \left(\sum_{j=1}^k w_j S_t^j \right) \mathbb{E}_{\mathbb{P}^*}((\widetilde{A}_T - \widetilde{K})^+ \mid \mathcal{F}_t). \quad (31.7)$$

The next step relies on an approximation of the weighted arithmetic mean $\sum_{i=1}^k \widehat{w}_i \widetilde{S}_T^i$ using a similarly weighted geometric mean. More specifically, we approximate the price C_0^B of the basket option using the number $\widehat{C}_0^B$ given by the formula (for ease of notation, we set $t = 0$ in what follows)

$$\widehat{C}_0^B = \left(\sum_{j=1}^k w_j S_0^j \right) \mathbb{E}_{\mathbb{P}^*}(\widetilde{G}_T - \widehat{K})^+, \quad (31.8)$$

where $\widetilde{G}_T = \prod_{i=1}^k (\widetilde{S}_T^i)^{\widehat{w}_i}$ and

$$\widehat{K} = \widetilde{K} + \mathbb{E}_{\mathbb{P}^*}(\widetilde{G}_T - \widetilde{A}_T). \quad (31.9)$$

In view of (31.2), we have (recall that $F_{S^i}(0, T) = e^{rT} S_0^i$)

$$\widetilde{S}_T^i = S_T^i / F_{S^i}(0, T) = e^{\widehat{\sigma}_i \cdot W_T^* - \sigma_i^2 T / 2},$$

and thus the weighted geometric average $\widetilde{G}_T$ equals

$$\widetilde{G}_T = e^{c_1 \cdot W_T^* - c_2 T / 2} = e^{\eta_T - c_2 T / 2}, \quad (31.10)$$

with $\eta_T = c_1 \cdot W_T^*$, where we write $c_1 = \sum_{i=1}^k \widehat{w}_i \widehat{\sigma}_i$ and $c_2 = \sum_{i=1}^k \widehat{w}_i \sigma_i^2$. We conclude that the random variable $\widetilde{G}_T$ is log-normally distributed under $\mathbb{P}^*$.

More precisely, the random variable η_T in (31.10) has Gaussian distribution with expected value 0 and variance

$$\begin{aligned} \text{Var}_{\mathbb{P}^*}(\eta_T) &= \mathbb{E}_{\mathbb{P}^*} \left\{ \left(\sum_{i,l=1}^k \widehat{w}_i \widehat{\sigma}_{il} W_T^{l*} \right) \left(\sum_{j,m=1}^k \widehat{w}_j \widehat{\sigma}_{jm} W_T^{m*} \right) \right\} \\ &= \sum_{i,j,l,m=1}^k \widehat{w}_i \widehat{w}_j \widehat{\sigma}_{il} \widehat{\sigma}_{jm} \mathbb{E}_{\mathbb{P}^*}(W_T^{l*} W_T^{m*}) = \sum_{i,j=1}^k \widehat{w}_i \widehat{w}_j \widehat{\sigma}_i \cdot \widehat{\sigma}_j T = v^2 T, \end{aligned}$$

where we write $v^2 = \sum_{i,j=1}^k \rho_{i,j} \widehat{w}_i \widehat{w}_j \sigma_i \sigma_j$. We have used here the equality $\mathbb{E}_{\mathbb{P}^*}(W_T^{l*} W_T^{m*}) = \delta_{lm} T$, where δ_{lm} stands for Kronecker's delta; that is, δ_{lm} equals 1 if $l = m$ and zero otherwise. Note that

$$\mathbb{E}_{\mathbb{P}^*}(\widetilde{A}_T) = \sum_{j=1}^k \widehat{w}_j S_0^j \mathbb{E}_{\mathbb{P}^*}(e^{-rT} S_T^j) = \sum_{j=1}^k \widehat{w}_j = 1,$$

and

$$\mathbb{E}_{\mathbb{P}^*}(\widetilde{G}_T) = e^{(v^2 - c_2)T/2} \mathbb{E}_{\mathbb{P}^*}(e^{\eta_T - \frac{1}{2} \text{Var}(\eta_T)}) = e^{(v^2 - c_2)T/2} \stackrel{\text{def}}{=} c.$$

We conclude that $\widehat{K} = \widetilde{K} + c - 1$. The expectation in (??) can now be evaluated explicitly.

Lemma 31.1: Gaussian Expectation

Let ξ be a Gaussian random variable on $(\Omega, \mathcal{F}, \mathbb{P})$ with expected value 0 and variance $\sigma^2 > 0$. For any real numbers $a, b > 0$, we have

$$\mathbb{E}_{\mathbb{P}}(ae^{\xi - \frac{1}{2}\sigma^2} - b)^+ = aN(h) - bN(h - \sigma), \quad (31.11)$$

where $h = \sigma^{-1} \ln(a/b) + \frac{1}{2}\sigma$.

We have

$$\mathbb{E}_{\mathbb{P}^*}(\widetilde{G}_T - \widehat{K})^+ = \mathbb{E}_{\mathbb{P}^*}(ce^{\eta_T - \frac{1}{2} \text{Var}(\eta_T)} - (\widetilde{K} + c - 1))^+,$$

so that we may set $\xi = \eta_T$, $a = c$ and $b = \tilde{K} + c - 1$. In view of [lemma 31.1](#), the following result is straightforward.

Proposition 31.1: Approximate Basket Option Valuation

The approximate value $\widehat{C}_t^B$ of the price C_t^B of a basket call option with strike price K and expiry date T equals

$$\widehat{C}_t^B = \left(\sum_{j=1}^k w_j S_t^j \right) (cN(l_1(T-t)) - (\tilde{K} + c - 1)N(l_2(T-t))), \quad (31.12)$$

where

$$c = \exp \left\{ \left(\frac{1}{2} \sum_{i,j=1}^k \rho_{i,j} \widehat{w}_i \widehat{w}_j \sigma_i \sigma_j - \sum_{j=1}^k \widehat{w}_j \sigma_j^2 \right) (T-t) \right\}, \quad (31.13)$$

and where the modified weights $\widehat{w}_i$ are given by [\(31.4\)](#), $\tilde{K}$ is given by [\(31.6\)](#), and

$$l_{1,2}(t) = \frac{\ln c - \ln(\tilde{K} + c - 1) \pm \frac{1}{2}v^2 t}{v\sqrt{t}}. \quad (31.14)$$

Suppose that $k = 1$. In this case, $w_1 = 1$, and the arithmetic average agrees with the geometric one. Consequently, $c = 1$ and [\(31.12\)](#) reduces to the standard Black-Scholes formula. In this approach, the weighted average $\widetilde{A}_T$ in [\(31.7\)](#) is directly replaced by a log-normally distributed random variable. Then the series expansion of the true distribution of $\widetilde{A}_T$ is derived in terms of an approximating log-normal distribution. An estimate of the basket option price can thus be obtained by direct integration.

32 Quantile Options

Part IX

Hedging

33 Delta Hedging

References

- Bensoussan, Alain, Michel Crouhy, and Dan Galai (Sept. 1, 1994). “Stochastic Equity Volatility Related to the Leverage Effect: I Equity Volatility Behaviour”. In: *Applied Mathematical Finance* 1.1, pp. 63–85. ISSN: 1350-486X. DOI: [10.1080/13504869400000004](https://doi.org/10.1080/13504869400000004). URL: <https://doi.org/10.1080/13504869400000004> (visited on 11/27/2025).
- Bielecki, Tomasz R., Monique Jeanblanc, and Marek Rutkowski (2004). “Hedging of Defaultable Claims”. In: *Paris-Princeton Lectures on Mathematical Finance 2003*. Ed. by Tomasz R. Bielecki et al. Berlin, Heidelberg: Springer, pp. 1–132. ISBN: 978-3-540-44468-8. DOI: [10.1007/978-3-540-44468-8_1](https://doi.org/10.1007/978-3-540-44468-8_1). URL: https://doi.org/10.1007/978-3-540-44468-8_1 (visited on 11/27/2025).
- Black, Fischer and Myron Scholes (1973). “The Pricing of Options and Corporate Liabilities”. In: *Journal of Political Economy* 81.3, pp. 637–654. ISSN: 0022-3808. DOI: [10.1086/260062](https://www.jstor.org/stable/1831029). JSTOR: 1831029. URL: <https://www.jstor.org/stable/1831029> (visited on 11/24/2025).
- Brennan, Michael J. and Eduardo S. Schwartz (July 1, 1979). “A Continuous Time Approach to the Pricing of Bonds”. In: *Journal of Banking & Finance* 3.2, pp. 133–155. ISSN: 0378-4266. DOI: [10.1016/0378-4266\(79\)90011-6](https://www.sciencedirect.com/science/article/pii/0378426679900116). URL: <https://www.sciencedirect.com/science/article/pii/0378426679900116> (visited on 12/02/2025).
- Cox, John C., Stephen A. Ross, and Mark Rubinstein (Sept. 1, 1979). “Option Pricing: A Simplified Approach”. In: *Journal of Financial Economics* 7.3, pp. 229–263. ISSN: 0304-405X. DOI: [10.1016/0304-405X\(79\)90015-1](https://www.sciencedirect.com/science/article/pii/0304405X79900151). URL: <https://www.sciencedirect.com/science/article/pii/0304405X79900151> (visited on 11/18/2025).
- Gould, J. P. and D. Galai (July 1, 1974). “Transactions Costs and the Relationship between Put and Call Prices”. In: *Journal of Financial Economics* 1.2, pp. 105–129. ISSN: 0304-405X. DOI: [10.1016/0304-405X\(74\)90001-4](https://www.sciencedirect.com/science/article/pii/0304405X74900014). URL: <https://www.sciencedirect.com/science/article/pii/0304405X74900014> (visited on 12/10/2025).

- Heath, David C. and Robert A. Jarrow (1988). “Ex-Dividend Stock Price Behavior and Arbitrage Opportunities”. In: *The Journal of Business* 61.1, pp. 95–108. ISSN: 0021-9398. DOI: [10.1086/296421](https://doi.org/10.1086/296421). JSTOR: [2352981](https://www.jstor.org/stable/2352981). URL: <https://www.jstor.org/stable/2352981> (visited on 12/11/2025).
- Heston, Steven L. (1993). “A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options”. In: *The Review of Financial Studies* 6.2, pp. 327–343. ISSN: 0893-9454. DOI: [10.1093/rfs/6.2.327](https://doi.org/10.1093/rfs/6.2.327). JSTOR: [2962057](https://www.jstor.org/stable/2962057). URL: <https://www.jstor.org/stable/2962057> (visited on 11/27/2025).
- Kahn, H. and A. W. Marshall (1953). “Methods of Reducing Sample Size in Monte Carlo Computations”. In: *Journal of the Operations Research Society of America* 1.5, pp. 263–278. ISSN: 0096-3984. DOI: [10.1287/opre.1.5.263](https://doi.org/10.1287/opre.1.5.263). JSTOR: [166789](https://www.jstor.org/stable/166789). URL: <https://www.jstor.org/stable/166789> (visited on 12/06/2025).
- Klemkosky, Robert C. and Bruce G. Resnick (1979). “Put-Call Parity and Market Efficiency”. In: *The Journal of Finance* 34.5, pp. 1141–1155. ISSN: 0022-1082. DOI: [10.2307/2327240](https://doi.org/10.2307/2327240). JSTOR: [2327240](https://www.jstor.org/stable/2327240). URL: <https://www.jstor.org/stable/2327240> (visited on 12/10/2025).
- L’Ecuyer, Pierre (1999). “Good Parameters and Implementations for Combined Multiple Recursive Random Number Generators”. In: *Operations Research* 47.1, pp. 159–164. ISSN: 0030-364X. DOI: [10.1287/opre.47.1.159](https://doi.org/10.1287/opre.47.1.159). JSTOR: [222902](https://www.jstor.org/stable/222902). URL: <https://www.jstor.org/stable/222902> (visited on 11/29/2025).
- Merton, Robert C. (1973). “Theory of Rational Option Pricing”. In: *The Bell Journal of Economics and Management Science* 4.1, pp. 141–183. ISSN: 0005-8556. DOI: [10.2307/3003143](https://doi.org/10.2307/3003143). JSTOR: [3003143](https://www.jstor.org/stable/3003143). URL: <https://www.jstor.org/stable/3003143> (visited on 11/24/2025).

Appendices

A Existence of Lebesgue Measure

B Linear Algebra and Functional Analysis

B.1 Vector Spaces, Bases, and Matrices

Linear algebra is the study of vector spaces and linear mappings. Vectors are, by definition, elements of some *vector space* $\mathcal{V}$ and satisfy the usual rules of addition and scalar multiplication, e.g.,

$$\text{if } \mathbf{x} \in \mathcal{V} \text{ and } \mathbf{y} \in \mathcal{V}, \text{ then } \alpha\mathbf{x} + \beta\mathbf{y} \in \mathcal{V} \text{ for all } \alpha, \beta \in \mathbb{R} \text{ (or } \mathbb{C}\text{)}.$$

We will be dealing mostly with vectors in the Euclidean vector space $\mathbb{R}^n$ for some n . That is, we view the points of $\mathbb{R}^n$ as objects that can be added up and multiplied by a scalar, e.g., $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1, x_2 + y_2)$ for points in $\mathbb{R}^2$. Sometimes it is convenient to work with the complex vector space $\mathbb{C}^n$ instead of $\mathbb{R}^n$.

Vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are called *linearly independent* if none of them can be expressed as a linear combination of the others; that is, if $\alpha_1\mathbf{v}_1 + \dots + \alpha_n\mathbf{v}_n = \mathbf{0}$, then it must hold that $\alpha_i = 0$ for all $i = 1, \dots, n$.

Definition B.1: Basis of a Vector Space

A set of vectors $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is called a *basis* of the vector space $\mathcal{V}$ if every vector $\mathbf{x} \in \mathcal{V}$ can be written as a unique linear combination of the vectors in $\mathcal{B}$.

$$\mathbf{x} = \alpha_1\mathbf{v}_1 + \dots + \alpha_n\mathbf{v}_n.$$

The (possibly infinite) number n is called the *dimension* of $\mathcal{V}$.

Using a basis $\mathcal{B}$ of $\mathcal{V}$, we can thus represent each vector $\mathbf{x} \in \mathcal{V}$ as a row or column of numbers

$$[\alpha_1, \dots, \alpha_n] \quad \text{or} \quad \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}. \quad (\text{B.1})$$

Typically, vectors in $\mathbb{R}^n$ are represented via the *standard basis*, consisting of unit vectors (points) $\mathbf{e}_1 = (1, 0, \dots, 0), \dots, \mathbf{e}_n = (0, 0, \dots, 0, 1)$. As a consequence, any point $(x_1, \dots, x_n) \in \mathbb{R}^n$ can be represented, using the standard basis, as a row or column vector of the form eq. (B.1) above, with $\alpha_i = x_i, i = 1, \dots, n$. We will also write $[x_1, x_2, \dots, x_n]^\top$, for the corresponding column vector, where $^\top$ denotes the *transpose*.



To avoid confusion, we will use the convention from now on that a generic vector $\mathbf{x}$ is always represented via the standard basis as a column vector. The corresponding row vector is denoted by $\mathbf{x}^\top$.

A *matrix* can be viewed as an array of m rows and n columns that defines a *linear transformation* from $\mathbb{R}^n$ to $\mathbb{R}^m$ (or for complex matrices, from $\mathbb{C}^n$ to $\mathbb{C}^m$). The matrix is said to be *square* if $m = n$. If $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ are the columns of A , that is, $A = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n]$, and if $\mathbf{x} = [x_1, \dots, x_n]^\top$, then $A\mathbf{x} = x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n$. In particular, the standard basis vector $\mathbf{e}_k$ is mapped to the vector $\mathbf{a}_k, k = 1, \dots, n$. We sometimes use the notation $A = [a_{ij}]$, to denote a matrix whose (i, j) -th element is a_{ij} . When we wish to emphasize that a matrix A is real-valued with m rows and n columns, we write $A \in \mathbb{R}^{m \times n}$. The *rank* of a matrix is the number of linearly independent rows or, equivalently, the number of linearly independent columns.

The *transpose* of a matrix $A = [a_{ij}]$ is the matrix $A^\top = [a_{ji}]$; that is, the (i, j) -th element of $A^\top$ is the (j, i) -th element of A . The *trace* of a square matrix is the sum of its diagonal elements. A useful result is the following cyclic property.

Theorem B.1: Cyclic Property

The trace is invariant under cyclic permutations: $\text{tr}(ABC) = \text{tr}(BCA) = \text{tr}(CAB)$.

Proof. It suffices to show that $\text{tr}(DE)$ is equal to $\text{tr}(ED)$ for any $m \times n$ matrix $D = [d_{ij}]$ and $n \times m$ matrix $E = [e_{ij}]$. The diagonal elements of DE are $\sum_{j=1}^n d_{ij}e_{ji}, i = 1, \dots, m$ and the diagonal elements of ED are $\sum_{i=1}^m e_{ji}d_{ij}, j = 1, \dots, n$. They sum up to the same number $\sum_{i=1}^m \sum_{j=1}^n d_{ij}e_{ji}$. □

A square matrix has an inverse if and only if its columns (or rows) are *linearly independent*. This is the same as the matrix being of *full rank*; that is, its rank is equal to the number of columns. An equivalent statement is that its determinant is not zero. The *determinant* of an $n \times n$ matrix $A = [a_{i,j}]$ is defined as

$$\det(A) := \sum_{\pi} (-1)^{\zeta(\pi)} \prod_{i=1}^n a_{\pi_i, i}, \quad (\text{A.3})$$

where the sum is over all permutations $\pi = (\pi_1, \dots, \pi_n)$ of $(1, \dots, n)$, and $\zeta(\pi)$ is the number of pairs (i, j) for which $i < j$ and $\pi_i > \pi_j$. For example, $\zeta(2, 3, 4, 1) = 3$ for the pairs $(1, 4), (2, 4), (3, 4)$. The determinant of a *diagonal matrix*, a matrix with only zero elements off the diagonal, is simply the product of its diagonal elements.

Geometrically, the determinant of a square matrix $A = [\mathbf{a}_1, \dots, \mathbf{a}_n]$ is the (signed) *volume* of the parallelepiped (n -dimensional parallelogram) defined by the columns $\mathbf{a}_1, \dots, \mathbf{a}_n$; that is, the set of points $\mathbf{x} = \sum_{i=1}^n \alpha_i \mathbf{a}_i$, where $0 \leq \alpha_i \leq 1, i = 1, \dots, n$.

The easiest way to compute the determinant of a general matrix is to apply simple operations to the matrix that potentially reduce its complexity (as in the number of non-zero elements, for example), while retaining its determinant:

- Adding a multiple of one column (or row) to another does not change the determinant.
- Multiplying a column (or row) by a number multiplies the determinant by the same number.
- Swapping two rows changes the sign of the determinant.

By applying these rules repeatedly, one can reduce any matrix to a diagonal matrix. It follows then that the determinant of the original matrix is equal to the product of the diagonal elements of the resulting diagonal matrix multiplied by a known constant.

Theorem B.2: Useful Matrix Rules

1. $(AB)^\top = B^\top A^\top$
2. $(AB)^{-1} = B^{-1}A^{-1}$
3. $(A^{-1})^\top = (A^\top)^{-1} := A^{-\top}$
4. $\det(AB) = \det(A)\det(B)$
5. $\mathbf{x}^\top A \mathbf{x} = \text{tr}(A \mathbf{x} \mathbf{x}^\top)$
6. $\det(A) = \prod_i a_{ii}$ if $A = [a_{ij}]$ is triangular

B.2 summarises a number of useful matrix rules for the concepts that we have discussed so far.

Next, consider an $n \times p$ matrix A for which the matrix inverse fails to exist. That is, A is either non-square ($n \neq p$) or its determinant is 0. Instead of the inverse, we can use its so-called pseudo-inverse, which always exists.

Definition B.2: Moore–Penrose Pseudo-Inverse

The *Moore–Penrose pseudo-inverse* of a real matrix $A \in \mathbb{R}^{n \times p}$ is defined as the unique matrix $A^+ \in \mathbb{R}^{p \times n}$ that satisfies the conditions:

1. $AA^+A = A$
2. $A^+AA^+ = A^+$
3. $(AA^+)^\top = AA^+$
4. $(A^+A)^\top = A^+A$

We can write A^+ explicitly in terms of A when A has a full column or row rank. For example, we always have

$$A^\top AA^+ = A^\top (AA^+)^\top = ((AA^+)A)^\top = (A)^\top = A^\top. \quad (\text{B.2})$$

If $\mathbf{A}$ has a full column rank p , then $(\mathbf{A}^\top \mathbf{A})^{-1}$ exists, so that from (B.2) it follows that $\mathbf{A}^+ = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top$. This is referred to as the *left pseudo-inverse*, as $\mathbf{A}^+ \mathbf{A} = \mathbf{I}_p$. Similarly, if $\mathbf{A}$ has a full row rank n , that is, $(\mathbf{A} \mathbf{A}^\top)^{-1}$ exists, then it follows from

$$\mathbf{A}^+ \mathbf{A} \mathbf{A}^\top = (\mathbf{A}^+ \mathbf{A})^\top \mathbf{A}^\top = (\mathbf{A} (\mathbf{A}^+ \mathbf{A}))^\top = \mathbf{A}^\top$$

that $\mathbf{A}^+ = \mathbf{A}^\top (\mathbf{A} \mathbf{A}^\top)^{-1}$. This is the *right pseudo-inverse*, as $\mathbf{A} \mathbf{A}^+ = \mathbf{I}_n$. Finally, if $\mathbf{A}$ is of full rank and square, then $\mathbf{A}^+ = \mathbf{A}^{-1}$.

B.2 Inner Product

The (Euclidean) *inner product* of two real vectors $\mathbf{x} = [x_1, \dots, x_n]^\top$ and $\mathbf{y} = [y_1, \dots, y_n]^\top$ is defined as the number

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n x_i y_i = \mathbf{x}^\top \mathbf{y}.$$

Here $\mathbf{x}^\top \mathbf{y}$ is the matrix multiplication of the $(1 \times n)$ matrix $\mathbf{x}^\top$ and the $(n \times 1)$ matrix $\mathbf{y}$. The inner product induces a geometry on the linear space $\mathbb{R}^n$, allowing for the definition of length, angle, and so on. The inner product satisfies the following properties:

1. $\langle \alpha \mathbf{x} + \beta \mathbf{y}, \mathbf{z} \rangle = \alpha \langle \mathbf{x}, \mathbf{z} \rangle + \beta \langle \mathbf{y}, \mathbf{z} \rangle$;
2. $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$;
3. $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$;
4. $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ if and only if $\mathbf{x} = \mathbf{0}$.

Vectors $\mathbf{x}$ and $\mathbf{y}$ are called *perpendicular* (or *orthogonal*) if $\langle \mathbf{x}, \mathbf{y} \rangle = 0$. The *Euclidean norm* (or length) of a vector $\mathbf{x}$ is defined as

$$\|\mathbf{x}\| = \sqrt{x_1^2 + \dots + x_n^2} = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}.$$

If $\mathbf{x}$ and $\mathbf{y}$ are perpendicular, then *Pythagoras' theorem* holds:

$$\|\mathbf{x} + \mathbf{y}\|^2 = \langle \mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{x} \rangle + 2\langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{y}, \mathbf{y} \rangle = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2. \quad (\text{A.5})$$

A basis $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ of $\mathbb{R}^n$ in which all the vectors are pairwise perpendicular and have norm 1 is called an *orthonormal* (short for orthogonal and normalized) basis. For example, the standard basis is orthonormal.

Theorem B.3: Orthonormal Basis Representation

If $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is an orthonormal basis of $\mathbb{R}^n$, then any vector $\mathbf{x} \in \mathbb{R}^n$ can be expressed as

$$\mathbf{x} = \langle \mathbf{x}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \dots + \langle \mathbf{x}, \mathbf{v}_n \rangle \mathbf{v}_n. \quad (\text{A.6})$$

Proof. Observe that, because the $\{\mathbf{v}_i\}$ form a basis, there exist unique $\alpha_1, \dots, \alpha_n$ such that $\mathbf{x} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n$. By the linearity of the inner product and the orthonormality of the $\{\mathbf{v}_i\}$ it follows that $\langle \mathbf{x}, \mathbf{v}_j \rangle = \langle \sum_i \alpha_i \mathbf{v}_i, \mathbf{v}_j \rangle = \alpha_j$. $\square$

An $n \times n$ matrix V whose columns form an orthonormal basis is called an *orthogonal matrix*.⁴

Note that for an orthogonal matrix $V = [\mathbf{v}_1, \dots, \mathbf{v}_n]$, we have

$$V^T V = \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_n^T \end{bmatrix} [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n] = \begin{bmatrix} \mathbf{v}_1^T \mathbf{v}_1 & \mathbf{v}_1^T \mathbf{v}_2 & \dots & \mathbf{v}_1^T \mathbf{v}_n \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{v}_n^T \mathbf{v}_1 & \mathbf{v}_n^T \mathbf{v}_2 & \dots & \mathbf{v}_n^T \mathbf{v}_n \end{bmatrix} = I_n.$$

Hence, $V^{-1} = V^T$. Note also that an orthogonal transformation is *length preserving*; that is, $V\mathbf{x}$ has the same length as $\mathbf{x}$. This follows from

$$\|V\mathbf{x}\|^2 = \langle V\mathbf{x}, V\mathbf{x} \rangle = \mathbf{x}^T V^T V \mathbf{x} = \mathbf{x}^T \mathbf{x} = \|\mathbf{x}\|^2.$$

⁴The qualifier “orthogonal” for such matrices has been fixed by history. A better term would have been “orthonormal”.

B.3 Complex Vectors and Matrices

Instead of the vector space $\mathbb{R}^n$ of n -dimensional real vectors, it is sometimes useful to consider the vector space $\mathbb{C}^n$ of n -dimensional complex vectors. In this case the *adjoint* or *conjugate transpose* operation ($*$) replaces the transpose operation ($\top$). This involves the usual transposition of the matrix or vector with the additional step that any complex number $z = x + iy$ is replaced by its complex conjugate $\bar{z} = x - iy$. For example, if

$$\mathbf{x} = \begin{bmatrix} a_1 + ib_1 \\ a_2 + ib_2 \end{bmatrix} \quad \text{and} \quad \mathbf{A} = \begin{bmatrix} a_{11} + ib_{11} & a_{12} + ib_{12} \\ a_{21} + ib_{21} & a_{22} + ib_{22} \end{bmatrix},$$

then

$$\mathbf{x}^* = [a_1 - ib_1, a_2 - ib_2] \quad \text{and} \quad \mathbf{A}^* = \begin{bmatrix} a_{11} - ib_{11} & a_{21} - ib_{21} \\ a_{12} - ib_{12} & a_{22} - ib_{22} \end{bmatrix}.$$

The (Euclidean) inner product of $\mathbf{x}$ and $\mathbf{y}$ (viewed as column vectors) is now defined as

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{y}^* \mathbf{x} = \sum_{i=1}^n x_i \bar{y}_i,$$

which is no longer symmetric: $\langle \mathbf{x}, \mathbf{y} \rangle = \overline{\langle \mathbf{y}, \mathbf{x} \rangle}$. Note that this generalizes the real-valued inner product. The determinant of a complex matrix $\mathbf{A}$ is defined exactly as in (??). As a consequence, $\det(\mathbf{A}^*) = \overline{\det(\mathbf{A})}$.

A complex matrix is said to be *Hermitian* or *self-adjoint* if $\mathbf{A}^* = \mathbf{A}$, and *unitary* if $\mathbf{A}^* \mathbf{A} = \mathbf{I}$ (that is, if $\mathbf{A}^* = \mathbf{A}^{-1}$). For real matrices “Hermitian” is the same as “symmetric”, and “unitary” is the same as “orthogonal”.

B.4 Orthogonal Projections

Let $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ be a set of linearly independent vectors in $\mathbb{R}^n$. The set

$$\mathcal{V} = \text{Span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\} = \{\alpha_1 \mathbf{u}_1 + \dots + \alpha_k \mathbf{u}_k, \alpha_1, \dots, \alpha_k \in \mathbb{R}\},$$

is called the *linear subspace* spanned by $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$. The *orthogonal complement* of $\mathcal{V}$, denoted by $\mathcal{V}^\perp$, is the set of all vectors $\mathbf{w}$ that are orthogonal to $\mathcal{V}$, in the sense that $\langle \mathbf{w}, \mathbf{v} \rangle = 0$ for all $\mathbf{v} \in \mathcal{V}$. The matrix P such that $P\mathbf{x} = \mathbf{x}$, for all $\mathbf{x} \in \mathcal{V}$, and $P\mathbf{x} = \mathbf{0}$, for all $\mathbf{x} \in \mathcal{V}^\perp$ is called the *orthogonal projection matrix* onto $\mathcal{V}$. Suppose that $U = [\mathbf{u}_1, \dots, \mathbf{u}_k]$ has full rank, in which case $U^\top U$ is an invertible matrix. The orthogonal projection matrix P onto $\mathcal{V} = \text{Span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ is then given by

$$P = U(U^\top U)^{-1}U^\top.$$

Namely, since $PU = U$, the matrix P projects any vector in $\mathcal{V}$ onto itself. Moreover, P projects any vector in $\mathcal{V}^\perp$ onto the zero vector. Using the pseudo-inverse, it is possible to specify the projection matrix also for the case where U not of full rank, leading to the following theorem.

Theorem B.4: Orthogonal Projection

Let $U = [\mathbf{u}_1, \dots, \mathbf{u}_k]$. Then, the orthogonal projection matrix P onto $\mathcal{V} = \text{Span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ is given by

$$P = UU^+, \tag{A.7}$$

where U^+ is the (right) pseudo-inverse of U .

Proof. By Property 1 of B.2 we have $PU = UU^+U = U$, so that P projects any vector in $\mathcal{V}$ onto itself. Moreover, P projects any vector in $\mathcal{V}^\perp$ onto the zero vector. $\square$

Note that in the special case where $\mathbf{u}_1, \dots, \mathbf{u}_k$ above form an orthonormal basis of $\mathcal{V}$, then the projection onto $\mathcal{V}$ is very simple to describe, namely we have

$$P\mathbf{x} = UU^\top \mathbf{x} = \sum_{i=1}^k \langle \mathbf{x}, \mathbf{u}_i \rangle \mathbf{u}_i. \quad (\text{A.8})$$

For any point $\mathbf{x} \in \mathbb{R}^n$, the point in $\mathcal{V}$ that is closest to $\mathbf{x}$ is its orthogonal projection $P\mathbf{x}$, as the following theorem shows.

Theorem B.5: Orthogonal Projection and Minimal Distance

Let $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ be an orthonormal basis of subspace $\mathcal{V}$ and let P be the orthogonal projection matrix onto $\mathcal{V}$. The solution to the minimization program

$$\min_{\mathbf{y} \in \mathcal{V}} \|\mathbf{x} - \mathbf{y}\|^2$$

is $\mathbf{y} = P\mathbf{x}$. That is, $P\mathbf{x} \in \mathcal{V}$ is closest to $\mathbf{x}$.

Proof. We can write each point $\mathbf{y} \in \mathcal{V}$ as $\mathbf{y} = \sum_{i=1}^k \alpha_i \mathbf{u}_i$. Consequently,

$$\|\mathbf{y} - \mathbf{x}\|^2 = \left\| \mathbf{x} - \sum_{i=1}^k \alpha_i \mathbf{u}_i, \mathbf{x} - \sum_{i=1}^k \alpha_i \mathbf{u}_i \right\|^2 = \|\mathbf{x}\|^2 - 2 \sum_{i=1}^k \alpha_i \langle \mathbf{x}, \mathbf{u}_i \rangle + \sum_{i=1}^k \alpha_i^2.$$

Minimizing this with respect to the $\{\alpha_i\}$ gives $\alpha_i = \langle \mathbf{x}, \mathbf{u}_i \rangle$, $i = 1, \dots, k$. In view of eq. (A.8), the optimal $\mathbf{y}$ is thus $P\mathbf{x}$. $\square$

B.5 Eigenvalues and Eigenvectors

Let A be an $n \times n$ matrix. If $A\mathbf{v} = \lambda\mathbf{v}$ for some number λ and non-zero vector $\mathbf{v}$, then λ is called an *eigenvalue* of A with *eigenvector* $\mathbf{v}$.

If $(\lambda, \mathbf{v})$ is an (eigenvalue, eigenvector) pair, the matrix $\lambda I - A$ maps any multiple of $\mathbf{v}$ to the zero vector. Consequently, the columns of $\lambda I - A$ are linearly dependent, and hence their determinant is 0. This provides a way to identify the eigenvalues, namely as the $r \leq n$ different

roots $\lambda_1, \dots, \lambda_r$ of the *characteristic polynomial*

$$\det(\lambda \mathbf{I} - \mathbf{A}) = (\lambda - \lambda_1)^{\alpha_1} \cdots (\lambda - \lambda_r)^{\alpha_r},$$

where $\alpha_1 + \cdots + \alpha_r = n$. The integer α_i is called the *algebraic multiplicity* of λ_i . The eigenvectors that correspond to an eigenvalue λ_i lie in the *kernel* or *null space* of the matrix $\lambda_i \mathbf{I} - \mathbf{A}$; that is, the linear space of vectors $\mathbf{v}$ such that $(\lambda_i \mathbf{I} - \mathbf{A})\mathbf{v} = \mathbf{0}$. This space is called the *eigenspace* of λ_i . Its dimension, $d_i \in \{1, \dots, n\}$, is called the *geometric multiplicity* of λ_i . It always holds that $d_i \leq \alpha_i$. If $\sum_i d_i = n$, then we can construct a basis for $\mathbb{R}^n$ consisting of eigenvectors.

A matrix for which the algebraic and geometric multiplicities of all its eigenvalues are the same is called *semi-simple*. This is equivalent to the matrix being *diagonalizable*, meaning that there is a matrix $\mathbf{V}$ and a diagonal matrix $\mathbf{D}$ such that

$$\mathbf{A} = \mathbf{V}\mathbf{D}\mathbf{V}^{-1}.$$

To see that this so-called *eigen-decomposition* holds, suppose $\mathbf{A}$ is a semi-simple matrix with eigenvalues

$$\underbrace{\lambda_1, \dots, \lambda_1}_{d_1}, \dots, \underbrace{\lambda_r, \dots, \lambda_r}_{d_r}.$$

Let $\mathbf{D}$ be the diagonal matrix whose diagonal elements are the eigenvalues of $\mathbf{A}$, and let $\mathbf{V}$ be a matrix whose columns are linearly independent eigenvectors corresponding to these eigenvalues. Then, for each (eigenvalue, eigenvector) pair $(\lambda, \mathbf{v})$, we have $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$. Hence, in matrix notation, we have $\mathbf{A}\mathbf{V} = \mathbf{V}\mathbf{D}$, and so $\mathbf{A} = \mathbf{V}\mathbf{D}\mathbf{V}^{-1}$.

B.6 Matrix Decompositions

Matrix decompositions are frequently employed in linear algebra to simplify proofs, mitigate numerical instability, and expedite computations. We mention three important matrix decompositions: (P)LU, QR, and SVD.

B.6.1 (P)LU Decomposition

Every invertible matrix A can be written as the product of three matrices:

$$A = PLU, \tag{B.3}$$

where L is a lower triangular matrix, U an upper triangular matrix, and P a *permutation matrix*. A permutation matrix is a square matrix with a single 1 in each row and column, and zeros otherwise. The matrix product PB simply permutes the rows of a matrix B and, likewise, BP permutes its columns. A decomposition of the form eq. (B.3) is called a *PLU decomposition*. As a permutation matrix is orthogonal, its transpose is equal to its inverse, and so we can write eq. (B.3) as

$$P^T A = LU.$$

The decomposition is not unique, and in many cases P can be taken to be the identity matrix, in which case we speak of the *LU decomposition* of A , also called the LR for left–right (triangular) decomposition.

A PLU decomposition of an invertible $n \times n$ matrix A_0 can be obtained recursively as follows. The first step is to swap the rows of A_0 such that the element in the first column and first row of the pivoted matrix is as large as possible in absolute value. Write the resulting matrix as

$$\tilde{P}_0 A_0 = \begin{bmatrix} a_1 & \mathbf{b}_1^T \\ \mathbf{c}_1 & D_1 \end{bmatrix},$$

where $\tilde{P}_0$ is the permutation matrix that swaps the first and k -th row, where k is the row that contains the largest element in the first column. Next, add the matrix $-\mathbf{c}_1[1, \mathbf{b}_1^T/a_1]$ to the last

$n - 1$ rows of $\widetilde{P}_0 A_0$, to obtain the matrix

$$\begin{bmatrix} a_1 & \mathbf{b}_1^\top \\ \mathbf{0} & D_1 - \mathbf{c}_1 \mathbf{b}_1^\top / a_1 \end{bmatrix} := \begin{bmatrix} a_1 & \mathbf{b}_1^\top \\ \mathbf{0} & A_1 \end{bmatrix}.$$

In effect, we add some multiple of the first row to each of the remaining rows in order to obtain zeros in the first column, except for the first element.

We now apply the same procedure to A_1 as we did to A_0 and then to subsequent smaller matrices $A_2, \dots, A_{n-1}$:

1. Swap the first row with the row having the maximal absolute value element in the first column.
2. Make every other element in the first column equal to 0 by adding appropriate multiples of the first row to the other rows.

Suppose that A_t has a PLU decomposition $P_t L_t U_t$. Then it is easy to check that

$$\underbrace{\widetilde{P}_{t-1}^\top \begin{bmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & P_t \end{bmatrix}}_{P_{t-1}} \underbrace{\begin{bmatrix} 1 & \mathbf{0}^\top \\ P_t^\top \mathbf{c}_t / a_t & L_t \end{bmatrix}}_{L_{t-1}} \underbrace{\begin{bmatrix} a_t & \mathbf{b}_t^\top \\ \mathbf{0} & U_t \end{bmatrix}}_{U_{t-1}} \quad (\text{A.10})$$

is a PLU decomposition of A_{t-1} . Since the PLU decomposition for the scalar A_{n-1} is trivial, by working backwards, we obtain a PLU decomposition $P_0 L_0 U_0$ of A .

B.6.2 Woodbury Identity

LU (or more generally PLU) decompositions can also be applied to block matrices. A starting point is the following LU decomposition for a general 2×2 matrix:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & 0 \\ c & d - bc/a \end{bmatrix} \begin{bmatrix} 1 & b/a \\ 0 & 1 \end{bmatrix},$$

which holds as long as $a \neq 0$; this can be seen by simply writing out the matrix product. The block matrix generalization for matrices $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times k}$, $C \in \mathbb{R}^{k \times n}$, $D \in \mathbb{R}^{k \times k}$ is

$$\Sigma := \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & O_{n \times k} \\ C & D - CA^{-1}B \end{bmatrix} \begin{bmatrix} I_n & A^{-1}B \\ O_{k \times n} & I_k \end{bmatrix}, \quad (\text{B.4})$$

provided that A is invertible (again, write out the block matrix product). Here, we use the notation $O_{p \times q}$ to denote the $p \times q$ matrix of zeros. We can further rewrite this as:

$$\Sigma = \begin{bmatrix} I_n & O_{n \times k} \\ CA^{-1} & I_k \end{bmatrix} \begin{bmatrix} A & O_{n \times k} \\ O_{k \times n} & D - CA^{-1}B \end{bmatrix} \begin{bmatrix} I_n & A^{-1}B \\ O_{k \times n} & I_k \end{bmatrix}.$$

Thus, inverting both sides, we obtain

$$\Sigma^{-1} = \begin{bmatrix} I_n & A^{-1}B \\ O_{k \times n} & I_k \end{bmatrix}^{-1} \begin{bmatrix} A & O_{n \times k} \\ O_{k \times n} & D - CA^{-1}B \end{bmatrix}^{-1} \begin{bmatrix} I_n & O_{n \times k} \\ CA^{-1} & I_k \end{bmatrix}^{-1}.$$

Inversion of the above block matrices gives (again write out)

$$\Sigma^{-1} = \begin{bmatrix} I_n & -A^{-1}B \\ O_{k \times n} & I_k \end{bmatrix} \begin{bmatrix} A^{-1} & O_{n \times k} \\ O_{k \times n} & (D - CA^{-1}B)^{-1} \end{bmatrix} \begin{bmatrix} I_n & O_{n \times k} \\ -CA^{-1} & I_k \end{bmatrix}. \quad (\text{B.5})$$

Assuming that D is invertible, we could also perform a block UL (as opposed to LU) decomposition:

$$\Sigma = \begin{bmatrix} A - BD^{-1}C & B \\ O_{k \times n} & D \end{bmatrix} \begin{bmatrix} I_n & O_{n \times k} \\ D^{-1}C & I_k \end{bmatrix}, \quad (\text{B.6})$$

which, after a similar calculation as the one above, yields

$$\Sigma^{-1} = \begin{bmatrix} I_n & O_{n \times k} \\ -D^{-1}C & I_k \end{bmatrix} \begin{bmatrix} (A - BD^{-1}C)^{-1} & O_{n \times k} \\ O_{k \times n} & D^{-1} \end{bmatrix} \begin{bmatrix} I_n & -BD^{-1} \\ O_{k \times n} & I_k \end{bmatrix}. \quad (\text{B.7})$$

The upper-left block of Σ^{-1} from eq. (B.7) must be the same as the upper-left block of Σ^{-1} from eq. (B.5), leading to the *Woodbury identity*:

$$(A - BD^{-1}C)^{-1} = A^{-1} + A^{-1}B(D - CA^{-1}B)^{-1}CA^{-1}. \quad (\text{B.8})$$

From eq. (B.4) and the fact that the determinant of a product is the product of the determinants, we see that $\det(\Sigma) = \det(A) \det(D - CA^{-1}B)$. Similarly, from eq. (B.6) we have $\det(\Sigma) = \det(A - BD^{-1}C) \det(D)$, leading to the identity

$$\det(A - BD^{-1}C) \det(D) = \det(A) \det(D - CA^{-1}B). \quad (\text{B.9})$$

B.6.3 Cholesky Decomposition

If A is a real-valued positive definite matrix (and therefore symmetric), e.g., a covariance matrix, then an LU decomposition can be achieved with matrices L and $U = L^\top$.

Theorem B.6: Cholesky Decomposition

A real-valued positive definite matrix $A = [a_{ij}] \in \mathbb{R}^{n \times n}$ can be decomposed as

$$A = LL^\top, \quad (\text{B.10})$$

where the real $n \times n$ lower triangular matrix $L = [l_{kj}]$ satisfies the recursive formula

$$l_{kj} = \frac{a_{kj} - \sum_{i=1}^{j-1} l_{ji}l_{ki}}{\sqrt{a_{jj} - \sum_{i=1}^{j-1} l_{ji}^2}}, \quad \text{where} \quad \sum_{i=1}^0 l_{ji}l_{ki} := 0 \quad (\text{B.11})$$

for $k = 1, \dots, n$ and $j = 1, \dots, k$.

Proof. The proof is by inductive construction. For $k = 1, \dots, n$, let A_k be the left-upper $k \times k$ submatrix of $A = A_n$. With $\mathbf{e}_1 := [1, 0, \dots, 0]^\top$, we have $A_1 = a_{11} = \mathbf{e}_1^\top A \mathbf{e}_1 > 0$ by the positive-definiteness of A . It follows that $l_{11} = \sqrt{a_{11}}$. Suppose that A_{k-1} has a Cholesky factorization

$L_{k-1}L_{k-1}^\top$ with L_{k-1} having strictly positive diagonal elements, we can construct a Cholesky factorization of A_k as follows. First write

$$A_k = \begin{bmatrix} L_{k-1}L_{k-1}^\top & \mathbf{a}_{k-1} \\ \mathbf{a}_{k-1}^\top & a_{kk} \end{bmatrix} \quad (\text{B.12})$$

and propose L_k to be of the form

$$L_k = \begin{bmatrix} L_{k-1} & \mathbf{0} \\ \mathbf{l}_k^\top & l_{kk} \end{bmatrix} \quad (\text{B.13})$$

for some vector $\mathbf{l}_k \in \mathbb{R}^{k-1}$ and scalar l_{kk} , for which it must hold that

$$\begin{bmatrix} L_{k-1}L_{k-1}^\top & \mathbf{a}_{k-1} \\ \mathbf{a}_{k-1}^\top & a_{kk} \end{bmatrix} = \begin{bmatrix} L_{k-1} & \mathbf{0} \\ \mathbf{l}_k^\top & l_{kk} \end{bmatrix} \begin{bmatrix} L_{k-1}^\top & \mathbf{l}_k \\ \mathbf{0}^\top & l_{kk} \end{bmatrix}. \quad (\text{B.14})$$

To establish that such an $\mathbf{l}_k$ and l_{kk} exist, we must verify that the set of equations

$$\begin{aligned} L_{k-1}\mathbf{l}_k &= \mathbf{a}_{k-1} \\ \mathbf{l}_k^\top \mathbf{l}_k + l_{kk}^2 &= a_{kk} \end{aligned} \quad (\text{B.15})$$

has a solution. The system $L_{k-1}\mathbf{l}_k = \mathbf{a}_{k-1}$ has a unique solution, because (by assumption) L_{k-1} is lower diagonal with strictly positive entries down the main diagonal and we can solve for $\mathbf{l}_k$ using forward substitution: $\mathbf{l}_k = L_{k-1}^{-1}\mathbf{a}_{k-1}$. We can solve the second equation as $l_{kk} = \sqrt{a_{kk} - \|\mathbf{l}_k\|^2}$, provided that the term within the square root is positive. We demonstrate this using the fact that A is a positive definite matrix. In particular, for $\mathbf{x} \in \mathbb{R}^n$ of the form $[\mathbf{x}_1^\top, x_2, \mathbf{0}^\top]^\top$, where $\mathbf{x}_1$ is a non-zero $(k-1)$ -dimensional vector and x_2 a non-zero number, we have

$$0 < \mathbf{x}^\top A \mathbf{x} = [\mathbf{x}_1^\top, x_2] \begin{bmatrix} L_{k-1}L_{k-1}^\top & \mathbf{a}_{k-1} \\ \mathbf{a}_{k-1}^\top & a_{kk} \end{bmatrix} \begin{bmatrix} \mathbf{x}_1 \\ x_2 \end{bmatrix} = \|L_{k-1}^\top \mathbf{x}_1\|^2 + 2\mathbf{x}_1^\top \mathbf{a}_{k-1}x_2 + a_{kk}x_2^2. \quad (\text{B.16})$$

Now take $\mathbf{x}_1 = -x_2 \mathbf{L}_{k-1}^{-\top} \mathbf{l}_k$ to obtain $0 < \mathbf{x}^\top \mathbf{A} \mathbf{x} = x_2^2 (a_{kk} - \|\mathbf{l}_k\|^2)$. Therefore, (B.15) can be uniquely solved. As we have already solved it for $k = 1$, we can solve it for any $k = 1, \dots, n$, leading to the recursive formula (B.11) and Algorithm 10 below. $\square$

An implementation of Cholesky's decomposition, using the notation from the proof of B.6, is as follows. The running cost of this algorithm is $O(n^3)$.

Algorithm 10 Cholesky Decomposition

Require: Positive-definite $n \times n$ matrix $\mathbf{A}_n$ with entries $\{a_{ij}\}$.

Ensure: Lower triangular $\mathbf{L}_n$ such that $\mathbf{L}_n \mathbf{L}_n^\top = \mathbf{A}_n$.

```

1:  $\mathbf{L}_1 \leftarrow \sqrt{a_{11}}$ 
2: for  $k = 2, \dots, n$  do
3:    $\mathbf{a}_{k-1} \leftarrow [a_{1k}, \dots, a_{k-1,k}]^\top$ 
4:    $\mathbf{l}_{k-1} \leftarrow \mathbf{L}_{k-1}^{-1} \mathbf{a}_{k-1}$ 
5:    $l_{kk} \leftarrow \sqrt{a_{kk} - \mathbf{l}_{k-1}^\top \mathbf{l}_{k-1}}$ 
6:    $\mathbf{L}_k \leftarrow \begin{bmatrix} \mathbf{L}_{k-1} & \mathbf{0} \\ \mathbf{l}_{k-1}^\top & l_{kk} \end{bmatrix}$ 
7: end for
8: return  $\mathbf{L}_n$ 
    
```

B.7 Functional Analysis

Much of the previous theory on Euclidean vector spaces can be generalized to vector spaces of *functions*. Every element of a (real-valued) *function space* $\mathcal{H}$ is a function from some set X to $\mathbb{R}$, and elements can be added and scalar multiplied as if they were vectors. In other words, if $f \in \mathcal{H}$ and $g \in \mathcal{H}$, then $\alpha f + \beta g \in \mathcal{H}$ for all $\alpha, \beta \in \mathbb{R}$. On $\mathcal{H}$ we can impose an inner product as a mapping $\langle \cdot, \cdot \rangle$ from $\mathcal{H} \times \mathcal{H}$ to $\mathbb{R}$ that satisfies

1. $\langle \alpha f_1 + \beta f_2, g \rangle = \alpha \langle f_1, g \rangle + \beta \langle f_2, g \rangle;$
2. $\langle f, g \rangle = \langle g, f \rangle;$

3. $\langle f, f \rangle \geq 0$;
4. $\langle f, f \rangle = 0$ if and only if $f = 0$ (the zero function).

We focus on real-valued function spaces, although the theory for complex-valued function spaces is similar (and sometimes easier), under suitable modifications (e.g., $\langle f, g \rangle = \overline{\langle g, f \rangle}$).

We say that two elements f and g in $\mathcal{H}$ are *orthogonal* to each other with respect to this inner product if $\langle f, g \rangle = 0$. Given an inner product, we can measure distances between elements of the function space $\mathcal{H}$ using the *norm*

$$\|f\| := \sqrt{\langle f, f \rangle}.$$

For example, the distance between two functions f_m and f_n is given by $\|f_m - f_n\|$. The space $\mathcal{H}$ is said to be *complete* if every sequence of functions $f_1, f_2, \dots \in \mathcal{H}$ for which

$$\|f_m - f_n\| \rightarrow 0 \text{ as } m, n \rightarrow \infty, \quad (\text{B.17})$$

converges to some $f \in \mathcal{H}$; that is, $\|f - f_n\| \rightarrow 0$ as $n \rightarrow \infty$. A sequence that satisfies (B.17) is called a *Cauchy sequence*.

A complete inner product space is called a *Hilbert space*. The most fundamental Hilbert space of functions is the space L^2 . An in-depth introduction to L^2 requires some measure theory [6]. For our purposes, it suffices to assume that $\mathcal{X} \subseteq \mathbb{R}^d$ and that on $\mathcal{X}$ a *measure* μ is defined which assigns to each suitable⁵ set A a positive number $\mu(A) \geq 0$ (e.g., its volume). In many cases of interest μ is of the form

$$\mu(A) = \int_A w(x) dx \quad (\text{B.18})$$

where $w \geq 0$ is a positive function on $\mathcal{X}$ which is called the *density* of μ with respect to the Lebesgue measure (the natural volume measure on $\mathbb{R}^d$). We write $\mu(dx) = w(x)dx$ to indicate

⁵Not all sets have a measure. Suitable sets are Borel sets, which can be thought of as countable unions of rectangles.

that μ has density w . Another important case is where

$$\mu(A) = \sum_{x \in A \cap \mathbb{Z}^d} w(x), \quad (\text{B.19})$$

where $w \geq 0$ is again called the density of μ , but now with respect to the counting measure on $\mathbb{R}^d$ (which counts the points of $\mathbb{Z}^d$). Integrals with respect to measures μ in (B.18) and (B.19) can now be defined as

$$\int f(x) \mu(dx) = \int f(x) w(x) dx,$$

and

$$\int f(x) \mu(dx) = \sum_x f(x) w(x),$$

respectively. We assume for simplicity that μ has the form (B.18). For measures of the form (B.19) (so-called discrete measures), replace integrals by sums in what follows.

Definition B.3: L^2 Space

Let X be a subset of $\mathbb{R}^d$ with measure $\mu(dx) = w(x)dx$. The Hilbert space $L^2(X, \mu)$ is the linear space of functions from X to $\mathbb{R}$ that satisfy

$$\int_X f(x)^2 w(x) dx < \infty, \quad (\text{B.20})$$

and with inner product

$$\langle f, g \rangle = \int_X f(x) g(x) w(x) dx. \quad (\text{B.21})$$

Let $\mathcal{H}$ be a Hilbert space. A set of functions $\{f_i, i \in \mathcal{I}\}$ is called an *orthonormal system* if

1. the norm of every f_i is 1; that is, $\langle f_i, f_i \rangle = 1$ for all $i \in \mathcal{I}$,
2. the $\{f_i\}$ are orthogonal; that is, $\langle f_i, f_j \rangle = 0$ for $i \neq j$.

It follows then that the $\{f_i\}$ are linearly independent; that is, the only linear combination $\sum_j \alpha_j f_j(x)$ that is equal to $f_i(x)$ for all x is the one where $\alpha_i = 1$ and $\alpha_j = 0$ for $j \neq i$. An

orthonormal system $\{f_i\}$ is called an *orthonormal basis* if there is no other $f \in \mathcal{H}$ that is orthogonal to all the $\{f_i, i \in \mathcal{I}\}$ (other than the zero function). Although the general theory allows for uncountable bases, in practice the set $\mathcal{I}$ is taken to be countable.

We conclude with a number of key results in functional analysis. The first one is the celebrated *Cauchy–Schwarz inequality*.

Theorem B.7: Cauchy–Schwarz

Let $\mathcal{H}$ be a Hilbert space. For every $f, g \in \mathcal{H}$ it holds that

$$|\langle f, g \rangle| \leq \|f\| \|g\|.$$

Proof. The inequality is trivially true for $g = 0$ (zero function). For $g \neq 0$, we can write $f = \alpha g + h$, where $h \perp g$ and $\alpha = \langle f, g \rangle / \|g\|^2$. Consequently, $\|f\|^2 = |\alpha|^2 \|g\|^2 + \|h\|^2 \geq |\alpha|^2 \|g\|^2$. The result follows after rearranging this last inequality. $\square$

Let $\mathcal{V}$ and $\mathcal{W}$ be two linear vector spaces (for example, Hilbert spaces) on which norms $\|\cdot\|_{\mathcal{V}}$ and $\|\cdot\|_{\mathcal{W}}$ are defined. Suppose $A : \mathcal{V} \rightarrow \mathcal{W}$ is a mapping from $\mathcal{V}$ to $\mathcal{W}$. When $\mathcal{W} = \mathcal{V}$, such a mapping is often called an *operator*; when $\mathcal{W} = \mathbb{R}$ it is called a *functional*. Mapping A is said to be *linear* if $A(\alpha f + \beta g) = \alpha A(f) + \beta A(g)$. In this case we write Af instead of $A(f)$. If there exists $\gamma < \infty$ such that

$$\|Af\|_{\mathcal{W}} \leq \gamma \|f\|_{\mathcal{V}}, \quad f \in \mathcal{V}, \quad (\text{B.22})$$

then A is said to be a *bounded mapping*. The smallest γ for which (B.22) holds is called the *norm* of A ; denoted by $\|A\|$. A (not necessarily linear) mapping $A : \mathcal{V} \rightarrow \mathcal{W}$ is said to be *continuous* at f if for any sequence $f_1, f_2, \dots$ converging to f the sequence $A(f_1), A(f_2), \dots$ converges to $A(f)$. That is, if

$$\forall \varepsilon > 0, \exists \delta > 0 : \forall g \in \mathcal{V}, \|f - g\|_{\mathcal{V}} < \delta \implies \|A(f) - A(g)\|_{\mathcal{W}} < \varepsilon. \quad (\text{B.23})$$

If the above property holds for every $f \in \mathcal{V}$, then the mapping A itself is called *continuous*.

Theorem B.8: Continuity and Boundedness for Linear Mappings

For a linear mapping, continuity and boundedness are equivalent.

Proof. Let A be linear and bounded. We may assume that A is non-zero (otherwise the statement holds trivially), and that therefore $0 < \|A\| < \infty$. Taking $\delta < \varepsilon/\|A\|$ in (B.23) now ensures that $\|Af - Ag\|_{\mathcal{W}} \leq \|A\|\|f - g\|_{\mathcal{V}} < \|A\|\delta < \varepsilon$. This shows that A is continuous. Conversely, suppose A is continuous. In particular, it is continuous at $f = 0$ (the zero element of $\mathcal{V}$). Thus, take $f = 0$ and let ε and δ be as in (B.23). For any $g \neq 0$, let $h = \delta/(2\|g\|_{\mathcal{V}})g$. As $\|h\|_{\mathcal{V}} = \delta/2 < \delta$, it follows from (B.23) that

$$\|Ah\|_{\mathcal{W}} = \frac{\delta}{2\|g\|_{\mathcal{V}}} \|Ag\|_{\mathcal{W}} < \varepsilon.$$

Rearranging the last inequality gives $\|Ag\|_{\mathcal{W}} < 2\varepsilon/\delta\|g\|_{\mathcal{V}}$, showing that A is bounded. $\square$

Theorem B.9: Riesz Representation Theorem

Any bounded linear functional ϕ on a Hilbert space $\mathcal{H}$ can be represented as $\phi(h) = \langle h, g \rangle$, for some $g \in \mathcal{H}$ (depending on ϕ).

Proof. Let P be the projection of $\mathcal{H}$ onto the nullspace $\mathcal{N}$ of ϕ ; that is, $\mathcal{N} = \{g \in \mathcal{H} : \phi(g) = 0\}$. If ϕ is not the 0-functional, then there exists a $g_0 \neq 0$ with $\phi(g_0) \neq 0$. Let $g_1 = g_0 - Pg_0$. Then $g_1 \perp \mathcal{N}$ and $\phi(g_1) = \phi(g_0)$. Take $g_2 = g_1/\phi(g_1)$. For any $h \in \mathcal{H}$, $f := h - \phi(h)g_2$ lies in $\mathcal{N}$. As $g_2 \perp \mathcal{N}$ it holds that $\langle f, g_2 \rangle = 0$, which is equivalent to $\langle h, g_2 \rangle = \phi(h)\|g_2\|^2$. By defining $g = g_2/\|g_2\|^2$ we have found our representation. $\square$

B.8 Fourier Transforms

A central concern in mathematical physics and engineering mathematics is the transformation of equations into a coordinate system where expressions simplify, decouple, and become amenable to computation and analysis.

We will now briefly introduce the Fourier transform. Before doing so, we will extend the concept of L^2 space of real-valued functions as follows.

Definition B.4: L^p Space

Let $\mathcal{X}$ be a subset of $\mathbb{R}^d$ with measure $\mu(dx) = w(x)dx$ and $p \in [1, \infty)$. Then $L^p(\mathcal{X}, \mu)$ is the linear space of functions from $\mathcal{X}$ to $\mathbb{C}$ that satisfy

$$\int_{\mathcal{X}} |f(x)|^p w(x) dx < \infty. \quad (\text{B.24})$$

When $p = 2$, $L^2(\mathcal{X}, \mu)$ is in fact a Hilbert space equipped with inner product

$$\langle f, g \rangle = \int_{\mathcal{X}} f(x) \overline{g(x)} w(x) dx. \quad (\text{B.25})$$

Definition B.5: (Multivariate) Fourier Transform

The *Fourier transform* $\mathcal{F}[f]$ of a (real- or complex-valued) function $f \in L^1(\mathbb{R}^d)$ is the function $\tilde{f}$ defined as

$$\tilde{f}(t) := \int_{\mathbb{R}^d} e^{-i2\pi t^\top x} f(x) dx, \quad t \in \mathbb{R}^d.$$

The Fourier transform $\tilde{f}$ is continuous, uniformly bounded (since $f \in L^1(\mathbb{R}^d)$ implies that $|\tilde{f}(t)| \leq \int_{\mathbb{R}^d} |f(x)| dx < \infty$), and satisfies $\lim_{\|t\| \rightarrow \infty} \tilde{f}(t) = 0$ (a result known as the *Riemann–Lebesgue lemma*). However, $|\tilde{f}|$ does not necessarily have a finite integral. A simple example in $\mathbb{R}^1$ is the Fourier transform of $f(x) = \mathbb{1}_{\{-1/2 < x < 1/2\}}$. Then $\tilde{f}(t) = \sin(\pi t)/(\pi t) = \text{sinc}(\pi t)$, which is not absolutely integrable.

Definition B.6: (Multivariate) Inverse Fourier Transform

The *inverse Fourier transform* $\mathcal{F}^{-1}[\tilde{f}]$ of a (real- or complex-valued) function $\tilde{f} \in L^1(\mathbb{R}^d)$ is the function $\check{f}$ defined as

$$\check{f}(x) := \int_{\mathbb{R}^d} e^{i2\pi t^\top x} \tilde{f}(t) dt, \quad x \in \mathbb{R}^d.$$

As one would hope, it holds that if f and $\mathcal{F}[f]$ are both in $L^1(\mathbb{R}^d)$, then $f = \mathcal{F}^{-1}[\mathcal{F}[f]]$ almost everywhere. The Fourier transform enjoys many interesting and useful properties, some of which we list below.

1. *Linearity:* For $f, g \in L^1(\mathbb{R}^d)$ and constants $a, b \in \mathbb{R}$,

$$\mathcal{F}[af + bg] = a\mathcal{F}[f] + b\mathcal{F}[g].$$

2. *Space Shifting and Scaling:* Let $A \in \mathbb{R}^{d \times d}$ be an invertible matrix and $\mathbf{b} \in \mathbb{R}^d$ a constant vector. Let $f \in L^1(\mathbb{R}^d)$ and define $h(x) := f(Ax + \mathbf{b})$. Then

$$\mathcal{F}[h](t) = e^{i2\pi(A^{-1}t)^\top \mathbf{b}} \tilde{f}(A^{-\top} t) / |\det(A)|,$$

where $A^{-\top} := (A^\top)^{-1} = (A^{-1})^\top$.

3. *Frequency Shifting and Scaling:* Let $A \in \mathbb{R}^{d \times d}$ be an invertible matrix and $\mathbf{b} \in \mathbb{R}^d$ a constant vector. Let $f \in L^1(\mathbb{R}^d)$ and define

$$h(x) := e^{-i2\pi \mathbf{b}^\top A^{-\top} x} f(A^{-\top} x) / |\det(A)|.$$

Then $\mathcal{F}[h](t) = \tilde{f}(At + \mathbf{b})$.

4. *Differentiation:* Let $f \in L^1(\mathbb{R}^d) \cap C^1(\mathbb{R}^d)$ and let $f_k := \partial f / \partial x_k$ be the partial derivative of f with respect to x_k . If $f_k \in L^1(\mathbb{R}^d)$ for $k = 1, \dots, d$, then

$$\mathcal{F}[f_k](t) = (i2\pi t_k) \widetilde{f}(t).$$

5. *Convolution:* Let $f, g \in L^1(\mathbb{R}^d)$ be real or complex valued functions. Their convolution, $f * g$, is defined as

$$(f * g)(x) = \int_{\mathbb{R}^d} f(y)g(x - y)dy,$$

and is also in $L^1(\mathbb{R}^d)$. Moreover, the Fourier transform satisfies

$$\mathcal{F}[f * g] = \mathcal{F}[f]\mathcal{F}[g].$$

6. *Duality:* Let f and $\mathcal{F}[f]$ both be in $L^1(\mathbb{R}^d)$. Then $\mathcal{F}[\mathcal{F}[f]](t) = f(-t)$.
7. *Product Formula:* Let $f, g \in L^1(\mathbb{R}^d)$ and denote by $\widetilde{f}, \widetilde{g}$ their respective Fourier transforms. Then $\widetilde{f}g, f\widetilde{g} \in L^1(\mathbb{R}^d)$, and

$$\int_{\mathbb{R}^d} \widetilde{f}(z)g(z)dz = \int_{\mathbb{R}^d} f(z)\widetilde{g}(z)dz.$$

There are many additional properties which hold if $f \in L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d)$. In particular, if $f, g \in L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d)$, then $\widetilde{f}, \widetilde{g} \in L^2(\mathbb{R}^d)$ and $\langle \widetilde{f}, \widetilde{g} \rangle = \langle f, g \rangle$, a result often known as *Parseval's formula*. Putting $g = f$ gives the result often referred to as *Plancherel's theorem*.

The Fourier transform can be extended in several ways, in the first instance to functions in $L^2(\mathbb{R}^d)$ by continuity. A substantial extension of the theory is realized by replacing integration with respect to the Lebesgue measure (i.e., $\int_{\mathbb{R}^d} \cdots dx$) with integration with respect to a (finite Borel) measure μ (i.e., $\int_{\mathbb{R}^d} \cdots \mu(dx)$). Moreover, there is a close connection between the Fourier transform and characteristic functions arising in probability theory. Indeed, if X is a random

vector with pdf f , then its characteristic function ψ satisfies

$$\psi(t) := \mathbb{E}e^{it^\top X} = \mathcal{F}[f](-t/(2\pi)).$$

B.8.1 Discrete Fourier Transform

Here, we introduce the (univariate) discrete Fourier transform, which can be viewed as a special case of the Fourier transform introduced in Definition A.6, where $d = 1$, integration is with respect to the counting measure, and $f(x) = 0$ for $x < 0$ and $x > (n - 1)$.

Definition B.7: Discrete Fourier Transform

The *discrete Fourier transform* (DFT) of a vector $\mathbf{x} = [x_0, \dots, x_{n-1}]^\top \in \mathbb{C}^n$ is the vector $\tilde{\mathbf{x}} = [\tilde{x}_0, \dots, \tilde{x}_{n-1}]^\top$ whose elements are given by

$$\tilde{x}_t = \sum_{s=0}^{n-1} \omega^{st} x_s, \quad t = 0, \dots, n-1, \quad (\text{B.26})$$

where $\omega = \exp(-i2\pi/n)$.

In other words, $\tilde{\mathbf{x}}$ is obtained from $\mathbf{x}$ via the linear transformation

$$\tilde{\mathbf{x}} = F\mathbf{x},$$

where

$$F = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \dots & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & \dots & \omega^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{n-1} & \omega^{2(n-1)} & \dots & \omega^{(n-1)^2} \end{bmatrix}.$$

The matrix F is a so-called *Vandermonde matrix*, and is clearly symmetric (i.e., $F = F^\top$). Moreover, $F/\sqrt{n}$ is in fact a unitary matrix and hence its inverse is simply its complex conjugate

$\bar{F}/\sqrt{n}$. Thus, $F^{-1} = \bar{F}/n$ and we have that the *inverse discrete Fourier transform* (IDFT) is given by

$$x_t = \frac{1}{n} \sum_{s=0}^{n-1} \omega^{-st} \tilde{x}_s, \quad t = 0, \dots, n-1, \quad (\text{B.27})$$

or in terms of matrices and vectors,

$$\mathbf{x} = \bar{F} \tilde{\mathbf{x}}/n.$$

Observe that the IDFT of a vector $\mathbf{y}$ is related to the DFT of its complex conjugate $\bar{\mathbf{y}}$, since

$$\bar{F} \mathbf{y}/n = \overline{F \bar{\mathbf{y}}}/n.$$

Consequently, an IDFT can be computed via a DFT. There is a close connection between circulant matrices C and the DFT. To make this connection concrete, let C be the circulant matrix corresponding to the vector $\mathbf{c} \in \mathbb{C}^n$ and denote by $\mathbf{f}_t$ the t -th column of the discrete Fourier matrix F , $t = 0, 1, \dots, n-1$. Then, the s -th element of $C \mathbf{f}_t$ is

$$\sum_{k=0}^{n-1} c_{(s-k) \bmod n} \omega^{tk} = \sum_{y=0}^{n-1} c_y \omega^{t(s-y)} = \underbrace{\omega^{ts}}_{s\text{-th element of } \mathbf{f}_t} \underbrace{\sum_{y=0}^{n-1} c_y \omega^{-ty}}_{\lambda_t}.$$

Hence, the eigenvalues of C are

$$\lambda_t = \mathbf{c}^\top \bar{\mathbf{f}}_t, \quad t = 0, 1, \dots, n-1,$$

with corresponding eigenvectors $\mathbf{f}_t$. Collecting the eigenvalues into the vector $\boldsymbol{\lambda} = [\lambda_0, \dots, \lambda_{n-1}]^\top = \bar{F} \mathbf{c}$, we therefore have the eigen-decomposition

$$C = F \text{diag}(\boldsymbol{\lambda}) \bar{F}/n.$$

Consequently, one can compute the *circular convolution* of a vector $\mathbf{a} = [a_1, \dots, a_n]^\top$ and $\mathbf{c} = [c_0, \dots, c_{n-1}]^\top$ by a series of DFTs as follows. Construct the circulant matrix C corresponding to $\mathbf{c}$. Then, the circular convolution of $\mathbf{a}$ and $\mathbf{c}$ is given by $\mathbf{y} = C\mathbf{a}$. Proceed in four steps:

1. Compute $\mathbf{z} = \overline{F}\mathbf{a}/n$.
2. Compute $\boldsymbol{\lambda} = \overline{F}\mathbf{c}$.
3. Compute $\mathbf{p} = \mathbf{z} \odot \boldsymbol{\lambda} = [z_1\lambda_0, \dots, z_n\lambda_{n-1}]^\top$.
4. Compute $\mathbf{y} = F\mathbf{p}$.

Steps 1 and 2 are (up to constants) in the form of an IDFT, and step 4 is in the form of a DFT. These are computable via the FFT in $O(n \ln n)$ time. Step 3 is a dot product computable in $O(n)$ time. Thus, the circular convolution can be computed with aid of the FFT in $O(n \ln n)$ time.

One can also efficiently compute the product of an $n \times n$ Toeplitz matrix T and an $n \times 1$ vector $\mathbf{a}$ in $O(n \ln n)$ time by embedding T into a circulant matrix C of size $2n \times 2n$. Namely, define

$$C = \begin{bmatrix} T & B \\ B & T \end{bmatrix},$$

where

$$B = \begin{bmatrix} 0 & t_{n-1} & \dots & t_2 & t_1 \\ t_{-(n-1)} & 0 & t_{n-1} & \dots & t_2 \\ \vdots & t_{-(n-1)} & 0 & \ddots & \vdots \\ t_{-2} & \ddots & \ddots & \ddots & t_{n-1} \\ t_{-1} & t_{-2} & \dots & t_{-(n-1)} & 0 \end{bmatrix}.$$

Then a product of the form $\mathbf{y} = T\mathbf{a}$ can be computed in $O(n \ln n)$ time, since we may write

$$C \begin{bmatrix} \mathbf{a} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} T & B \\ B & T \end{bmatrix} \begin{bmatrix} \mathbf{a} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} T\mathbf{a} \\ B\mathbf{a} \end{bmatrix}.$$

The left-hand side is a product of a $2n \times 2n$ circulant matrix with vector of length $2n$, and so can be computed in $O(n \ln n)$ time via the FFT, as previously discussed.

Conceptually, one can also solve equations of the form $C\mathbf{x} = \mathbf{b}$ for a given vector $\mathbf{b} \in \mathbb{C}^n$ and circulant matrix C (corresponding to $\mathbf{c} \in \mathbb{C}^n$, assuming all its eigenvalues are non-zero) via the following four steps:

1. Compute $\mathbf{z} = \overline{\mathbf{F}}\mathbf{b}/n$.
2. Compute $\boldsymbol{\lambda} = \overline{\mathbf{F}}\mathbf{c}$.
3. Compute $\mathbf{p} = \mathbf{z}/\boldsymbol{\lambda} = [z_1/\lambda_0, \dots, z_n/\lambda_{n-1}]^\top$.
4. Compute $\mathbf{x} = \mathbf{F}\mathbf{p}$.

Once again, Steps 1 and 2 are (up to constants) in the form of an IDFT, and Step 4 is in the form of a DFT, all of which are computable via the FFT in $O(n \ln n)$ time, and Step 3 is computable in $O(n)$ time, meaning the solution $\mathbf{x}$ can be computed using the FFT in $O(n \ln n)$ time.

B.8.2 Fast Fourier Transform

The *fast Fourier transform* (FFT) is a numerical algorithm for the fast evaluation of (B.26) and (B.27). By using a divide-and-conquer strategy, the algorithm reduces the computational complexity from $O(n^2)$ (for the naïve evaluation of the linear transformation) to $O(n \ln n)$.

The essence of the algorithm lies in the following observation. Suppose $n = r_1 r_2$. Then one can express any index t appearing in (B.26) via a pair (t_0, t_1) , with $t = t_1 r_1 + t_0$, where $t_0 \in \{0, 1, \dots, r_1 - 1\}$ and $t_1 \in \{0, 1, \dots, r_2 - 1\}$. Similarly, one can express any index s appearing in (B.26) via a pair (s_0, s_1) , with $s = s_1 r_2 + s_0$, where $s_0 \in \{0, 1, \dots, r_2 - 1\}$ and $s_1 \in \{0, 1, \dots, r_1 - 1\}$.

Identifying $\tilde{x}_t \equiv \tilde{x}_{t_1, t_0}$ and $x_s \equiv x_{s_1, s_0}$, we may re-express (B.26) as

$$\tilde{x}_{t_1, t_0} = \sum_{s_0=0}^{r_2-1} \omega^{s_0 t} \sum_{s_1=0}^{r_1-1} \omega^{s_1 r_2 t} x_{s_1, s_0}, \quad t_0 = 0, 1, \dots, r_1 - 1, t_1 = 0, 1, \dots, r_2 - 1. \quad (\text{B.28})$$

Observe that $\omega^{s_1 r_2 t} = \omega^{s_1 r_2 t_0}$ (because $\omega^{r_1 r_2} = 1$), so that the inner sum over s_1 depends only on s_0 and t_0 . Define

$$y_{t_0, s_0} := \sum_{s_1=0}^{r_1-1} \omega^{s_1 r_2 t_0} x_{s_1, s_0}, \quad t_0 = 0, 1, \dots, r_1 - 1, s_0 = 0, 1, \dots, r_2 - 1.$$

Computing each y_{t_0, s_0} requires $O(nr_1)$ operations. In terms of the $\{y_{t_0, s_0}\}$, (B.28) can be written as

$$\widetilde{x}_{t_1, t_0} = \sum_{s_0=0}^{r_2-1} \omega^{s_0 t} y_{t_0, s_0}, \quad t_1 = 0, 1, \dots, r_2 - 1, t_0 = 0, 1, \dots, r_1 - 1,$$

requiring $O(nr_2)$ operations to compute. Thus, calculating the DFT using this two-step procedure requires $O(n(r_1 + r_2))$ operations, rather than $O(n^2)$.

Now supposing $n = r_1 r_2 \cdots r_m$, repeated application the above divide-and-conquer idea yields an m -step procedure requiring $O(n(r_1 + r_2 + \cdots + r_m))$ operations. In particular, if $r_k = r$ for all $k = 1, 2, \dots, m$, we have that $n = r^m$ and $m = \ln_r n$, so that the total number of operations is $O(rnm) \equiv O(rn \ln_r(n))$. Typically, the radix r is a small (not necessarily prime) number, for instance $r = 2$.

C Multivariate Differentiation and Optimisation

The purpose of this appendix is to review various aspects of multivariate differentiation and optimisation. We assume the reader is familiar with differentiating a real-valued function.

C.1 Multivariate Differentiation

For a multivariate function f that maps a vector $\mathbf{x} = [x_1, \dots, x_n]^\top$ to a real number $f(\mathbf{x})$, the *partial derivative* with respect to x_i , denoted $\frac{\partial f}{\partial x_i}$, is the derivative taken with respect to x_i while all other variables are held constant. We can write all the n partial derivatives neatly using the “scalar/vector” derivative notation:

$$\text{scalar/vector:} \quad \frac{\partial f}{\partial \mathbf{x}} := \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}. \quad (\text{C.1})$$

This vector of partial derivatives is known as the *gradient* of f at $\mathbf{x}$ and is sometimes written as $\nabla f(\mathbf{x})$.

Next, suppose that $\mathbf{f}$ is a multivalued (vector-valued) function taking values in $\mathbb{R}^m$, defined by

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \mapsto \begin{bmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ \vdots \\ f_m(\mathbf{x}) \end{bmatrix} =: \mathbf{f}(\mathbf{x}).$$

We can compute each of the partial derivatives $\partial f_i / \partial x_j$ and organize them neatly in a “vector/vector” derivative notation:

$$\text{vector/vector: } \frac{\partial \mathbf{f}}{\partial \mathbf{x}} := \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_2}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_1} \\ \frac{\partial f_1}{\partial x_2} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_2} \\ \vdots & \vdots & \cdots & \vdots \\ \frac{\partial f_1}{\partial x_n} & \frac{\partial f_2}{\partial x_n} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}. \quad (\text{C.2})$$

The transpose of this matrix is known as the *matrix of Jacobi* of $\mathbf{f}$ at $\mathbf{x}$ (sometimes called the *Fréchet derivative* of $\mathbf{f}$ at $\mathbf{x}$); that is,

$$J_f(\mathbf{x}) := \left[\frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right]^\top = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \cdots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}. \quad (\text{C.3})$$

If we define $\mathbf{g}(\mathbf{x}) := \nabla f(\mathbf{x})$ and take the “vector/vector” derivative of $\mathbf{g}$ with respect to $\mathbf{x}$, we obtain the matrix of second-order partial derivatives of f :

$$\mathbf{H}_f(\mathbf{x}) := \frac{\partial \mathbf{g}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_m} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_m} \\ \vdots & \vdots & \cdots & \vdots \\ \frac{\partial^2 f}{\partial x_m \partial x_1} & \frac{\partial^2 f}{\partial x_m \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_m^2} \end{bmatrix}, \quad (\text{C.4})$$

which is known as the *Hessian matrix* of f at $\mathbf{x}$, also denoted as $\nabla^2 f(\mathbf{x})$. If these second-order partial derivatives are *continuous* in a region around $\mathbf{x}$, then $\frac{\partial f}{\partial x_i \partial x_j} = \frac{\partial f}{\partial x_j \partial x_i}$ and, hence, the Hessian matrix $\mathbf{H}_f(\mathbf{x})$ is symmetric.

Finally, note that we can also define a “scalar/matrix” derivative of y with respect to $X \in \mathbb{R}^{m \times n}$ with (i, j) -th entry x_{ij} :

$$\frac{\partial y}{\partial X} := \begin{bmatrix} \frac{\partial y}{\partial x_{11}} & \frac{\partial y}{\partial x_{12}} & \cdots & \frac{\partial y}{\partial x_{1n}} \\ \frac{\partial y}{\partial x_{21}} & \frac{\partial y}{\partial x_{22}} & \cdots & \frac{\partial y}{\partial x_{2n}} \\ \vdots & \vdots & \cdots & \vdots \\ \frac{\partial y}{\partial x_{m1}} & \frac{\partial y}{\partial x_{m2}} & \cdots & \frac{\partial y}{\partial x_{mn}} \end{bmatrix}$$

and a “matrix/scalar” derivative:

$$\frac{\partial X}{\partial y} := \begin{bmatrix} \frac{\partial x_{11}}{\partial y} & \frac{\partial x_{12}}{\partial y} & \cdots & \frac{\partial x_{1n}}{\partial y} \\ \frac{\partial x_{21}}{\partial y} & \frac{\partial x_{22}}{\partial y} & \cdots & \frac{\partial x_{2n}}{\partial y} \\ \vdots & \vdots & \cdots & \vdots \\ \frac{\partial x_{m1}}{\partial y} & \frac{\partial x_{m2}}{\partial y} & \cdots & \frac{\partial x_{mn}}{\partial y} \end{bmatrix}.$$

C.2 Optimisation Theory

Optimisation is concerned with finding minimal or maximal solutions of a real-valued *objective function* f in some set $\mathcal{X}$:

$$\min_{x \in \mathcal{X}} f(x) \quad \text{or} \quad \max_{x \in \mathcal{X}} f(x). \quad (\text{C.5})$$

Since any maximisation problem can easily be converted into a minimisation problem via the equivalence $\max_x f(x) = -\min_x -f(x)$, we focus only on minimisation problems. We use the following terminology. A *local minimiser* of $f(x)$ is an element $x^* \in \mathcal{X}$ such that $f(x^*) \leq f(x)$ for all x in some neighborhood of x^* . If $f(x^*) \leq f(x)$ for all $x \in \mathcal{X}$, then x^* is called a *global minimiser* or *global solution*. The set of global minimiser is denoted by

$$\operatorname{argmin}_{x \in \mathcal{X}} f(x).$$

The function value $f(x^*)$ corresponding to a local/global minimiser x^* is referred to as the *local/global minimum* of $f(x)$.

Optimisation problems may be classified by the set $\mathcal{X}$ and the objective function f . If $\mathcal{X}$ is countable, the optimisation problem is called *discrete* or *combinatorial*. If instead $\mathcal{X}$ is a nondenumerable set such as $\mathbb{R}^n$ and f takes values in a nondenumerable set, then the problem is said to be *continuous*. Optimisation problems that are neither discrete nor continuous are said to be *mixed*.

The search set $\mathcal{X}$ is often defined by means of *constraints*. A standard setting for *constrained optimisation* (minimisation) is the following:

$$\begin{aligned} & \min_{x \in \mathcal{Y}} f(x) \\ & \text{subject to: } h_i(\mathbf{x}) = 0, \quad i = 1, \dots, m, \\ & \quad \quad \quad g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, k. \end{aligned} \tag{C.6}$$

Here, f is the objective function, and $\{g_i\}$ and $\{h_i\}$ are given functions so that $h_i(\mathbf{x}) = 0$ and $g_i(\mathbf{x}) \leq 0$ represent the *equality* and *inequality constraints*, respectively. The region $\mathcal{X} \subseteq \mathcal{Y}$ where the objective function is defined and where all the constraints are satisfied is called the *feasible region*. An optimisation problem without constraints is said to be an *unconstrained problem*.

For an unconstrained continuous optimisation problem, the search space $\mathcal{X}$ is often taken to be (a subset of) $\mathbb{R}^n$, and f is assumed to be a C^k function for sufficiently high k (typically $k = 2$ or 3 suffices); that is, its k -th order derivative is continuous. For a C^1 function the standard approach to minimising $f(\mathbf{x})$ is to solve the equation

$$\nabla f(\mathbf{x}) = \mathbf{0}, \tag{C.7}$$

where $\nabla f(\mathbf{x})$ is the *gradient* of f at $\mathbf{x}$. The solutions $\mathbf{x}^*$ to [eq. \(C.7\)](#) are called *stationary points*. Stationary points can be local/global minimisers, local/global maximisers, or *saddle*

points (which are neither). If, in addition, the function is C^2 , the condition

$$\mathbf{y}^\top (\nabla^2 f(\mathbf{x}^*)) \mathbf{y} > 0 \quad \text{for all } \mathbf{y} \neq \mathbf{0} \quad (\text{C.8})$$

ensures that the stationary point $\mathbf{x}^*$ is a local minimiser of f . The condition [eq. \(C.8\)](#) states that the *Hessian matrix* of f at $\mathbf{x}^*$ is *positive definite*. Recall that we write $\mathbf{H} \succ 0$ to indicate that a matrix $\mathbf{H}$ is positive definite.

C.3 Numerical Root-Finding and Minimisation

In order to minimize a C^1 function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ one may solve

$$\nabla f(\mathbf{x}) = \mathbf{0},$$

which gives a stationary point of f . As a consequence, any technique for root-finding can be transformed into an unconstrained optimization method by attempting to locate roots of the gradient. However, as noted in [section C.2](#), not all stationary points are minima, and so additional information (such as is contained in the Hessian, if f is C^2) needs to be considered in order to establish the type of stationary point.

Alternatively, a root of a continuous function $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ may be found by minimizing the norm of $\mathbf{g}(\mathbf{x})$ over all $\mathbf{x}$; that is, by solving $\min_{\mathbf{x}} f(\mathbf{x})$, with $f(\mathbf{x}) := \|\mathbf{g}(\mathbf{x})\|_p$, where for $p \geq 1$ the p -norm of $\mathbf{y} = [y_1, \dots, y_n]^\top$ is defined as

$$\|\mathbf{y}\|_p := \left(\sum_{i=1}^n |y_i|^p \right)^{1/p}.$$

Hence, any (un)constrained optimization method can be transformed into a technique for locating the roots of a function.

Starting with an initial guess $\mathbf{x}_0$, most minimization and root-finding algorithms create a sequence $\mathbf{x}_0, \mathbf{x}_1, \dots$ using the iterative updating rule:

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \alpha_t \mathbf{d}_t, \quad t = 0, 1, 2, \dots, \quad (\text{C.9})$$

where $\alpha_t > 0$ is a (typically small) step size, called the *learning rate*, and the vector $\mathbf{d}_t$ is the *search direction* at step t . The iteration [eq. \(C.9\)](#) continues until the sequence $\{\mathbf{x}_t\}$ is deemed to have converged to a solution, or a computational budget has been exhausted. The performance of all such iterative methods depends crucially on the quality of the initial guess $\mathbf{x}_0$.

There are two broad categories of iterative optimization algorithms of the form [eq. \(C.9\)](#):

- Those of *line search* type, where at iteration t we first compute a direction $\mathbf{d}_t$ and then determine a reasonable step size α_t along this direction. For example, in the case of minimization, $\alpha_t > 0$ may be chosen to approximately minimize $f(\mathbf{x}_t + \alpha \mathbf{d}_t)$ for fixed $\mathbf{x}_t$ and $\mathbf{d}_t$.
- Those of *trust region* type, where at each iteration t we first determine a suitable step size α_t and then compute an approximately optimal direction $\mathbf{d}_t$.

In the following sections, we review several widely-used root-finding and optimization algorithms of the line search type.

C.3.1 Newton-Like Methods

Suppose we wish to find roots of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$. If f is in C^1 , we can approximate f around a point $\mathbf{x}_t$ as

$$f(\mathbf{x}) \approx f(\mathbf{x}_t) + J_f(\mathbf{x}_t)(\mathbf{x} - \mathbf{x}_t),$$

where J_f is the *matrix of Jacobi*, the matrix of partial derivatives of f ; see [eq. \(C.3\)](#). When $J_f(\mathbf{x}_t)$ is invertible, this linear approximation has root $\mathbf{x}_t - J_f^{-1}(\mathbf{x}_t)f(\mathbf{x}_t)$. This gives the iterative

updating formula eq. (C.9) for finding roots of f with direction $\mathbf{d}_t = -J_f^{-1}(\mathbf{x}_t)f(\mathbf{x}_t)$ and learning rate $\alpha_t = 1$. This is known as *Newton's method* (or the *Newton–Raphson method*) for root-finding.

Instead of a unit learning rate, sometimes it is more effective to use an α_t that satisfies the *Armijo inexact line search condition*:

$$\|f(\mathbf{x}_t + \alpha_t \mathbf{d}_t)\| < (1 - \varepsilon_1 \alpha_t) \|f(\mathbf{x}_t)\|,$$

where ε_1 is a small heuristically chosen constant, say $\varepsilon_1 = 10^{-4}$. For C^1 functions, such an α_t always exists by continuity and can be computed as in the following algorithm.

Algorithm 11 Newton–Raphson for Finding Roots of $f(\mathbf{x}) = \mathbf{0}$

Require: An initial guess $\mathbf{x}$ and stopping error $\varepsilon > 0$.

Ensure: The approximate root of $f(\mathbf{x}) = \mathbf{0}$.

```

1: while  $\|f(\mathbf{x})\| > \varepsilon$  and budget is not exhausted do
2:   Solve the linear system  $J_f(\mathbf{x})\mathbf{d} = -f(\mathbf{x})$ .
3:    $\alpha \leftarrow 1$ 
4:   while  $\|f(\mathbf{x} + \alpha\mathbf{d})\| > (1 - 10^{-4}\alpha)\|f(\mathbf{x})\|$  do
5:      $\alpha \leftarrow \alpha/2$ 
6:   end while
7:    $\mathbf{x} \leftarrow \mathbf{x} + \alpha\mathbf{d}$ 
8: end while
9: return  $\mathbf{x}$ 

```

We can adapt a root-finding Newton-like method in order to minimise a differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$. We simply try to locate a zero of the gradient of f . When f is a C^2 function, the function $\nabla f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is continuous, and so the root of ∇f leads to the search direction

$$\mathbf{d}_t = -H_t^{-1} \nabla f(\mathbf{x}_t), \tag{C.10}$$

where $\mathbf{H}_t$ is the Hessian matrix at $\mathbf{x}_t$ (the matrix of Jacobi of the gradient is the Hessian). When the learning rate α_t is equal to 1, the update $\mathbf{x}_t - \mathbf{H}_t^{-1} \nabla f(\mathbf{x}_t)$ can alternatively be derived by assuming that $f(\mathbf{x})$ is approximately quadratic and convex in the neighborhood of $\mathbf{x}_t$, that is,

$$f(\mathbf{x}) \approx f(\mathbf{x}_t) + (\mathbf{x} - \mathbf{x}_t)^\top \nabla f(\mathbf{x}_t) + \frac{1}{2} (\mathbf{x} - \mathbf{x}_t)^\top \mathbf{H}_t (\mathbf{x} - \mathbf{x}_t), \quad (\text{C.11})$$

and then minimizing the right-hand side of [eq. \(C.11\)](#) with respect to $\mathbf{x}$.

C.4 Constrained Minimisation via Penalty Functions

D Probability and Statistics

We review basic concepts, including the sum and product rules of probability, random variables and their probability distributions, expectations, independence, conditional probability, transformation rules, limit theorems, and Markov chains. The properties of the multivariate normal distribution are discussed in more detail. The main ideas from statistics are also reviewed, including estimation techniques (such as maximum likelihood estimation), confidence intervals, and hypothesis testing.

D.1 Random Experiments and Probability Spaces

The basic notion in probability theory is that of a *random experiment*: an experiment whose outcome cannot be determined in advance. Mathematically, a random experiment is modeled via a triplet $(\Omega, \mathcal{H}, \mathbb{P})$, where:

- Ω is the set of all possible outcomes of the experiment, called the *sample space*.
- $\mathcal{H}$ is the collection of all subsets of Ω to which a probability can be assigned; such subsets are called *events*.
- $\mathbb{P}$ is a *probability measure*, which assigns to each event A a number $\mathbb{P}[A]$ between 0 and 1, indicating the likelihood that the outcome of the random experiment lies in A .

Any probability measure $\mathbb{P}$ must satisfy the following *Kolmogorov axioms*:

1. $\mathbb{P}[A] \geq 0$ for every event A .
2. $\mathbb{P}[\Omega] = 1$.
3. For any sequence $A_1, A_2, \dots$ of events,

$$\mathbb{P}\left[\bigcup_i A_i\right] \leq \sum_i \mathbb{P}[A_i], \quad (\text{D.1})$$

with strict equality whenever the events are *disjoint* (that is, non-overlapping).

In many applications the sample space is *countable*; that is, $\Omega = \{a_1, a_2, \dots\}$. In this case the easiest way to specify a probability measure $\mathbb{P}$ is to first assign a number p_i to each *elementary event* $\{a_i\}$, with $\sum_i p_i = 1$, and then to define

$$\mathbb{P}[A] = \sum_{i: a_i \in A} p_i \quad \text{for all } A \subseteq \Omega.$$

Here the collection of events $\mathcal{H}$ can be taken to be equal to the collection of *all subsets* of Ω . The triple $(\Omega, \mathcal{H}, \mathbb{P})$ is called a *discrete probability space*. This idea is graphically represented in [fig. 22](#). Each element a_i , represented by a dot, is assigned a weight (that is, probability) p_i , indicated by the size of the dot. The probability of the event A is simply the sum of the weights of all the outcomes in A .

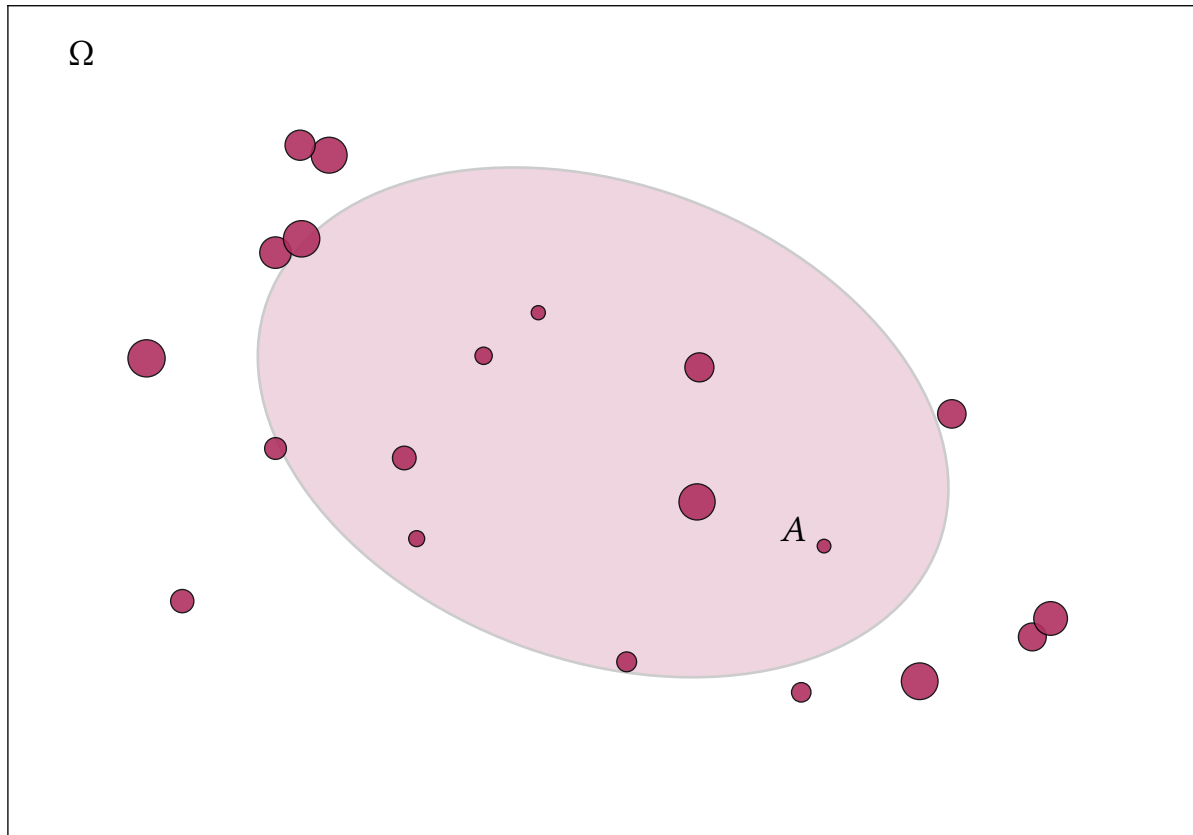


Figure 22: A discrete probability space.

D.2 Random Variables and Probability Distributions

It is often convenient to describe a random experiment via “random variables”, representing numerical measurements of the experiment. Random variables are usually denoted by capital letters from the last part of the alphabet. From a mathematical point of view, a *random variable* X is a function from Ω to $\mathbb{R}$ such that sets of the form $\{a < X \leq b\} := \{\omega \in \Omega : a < X(\omega) \leq b\}$ are events (and so can be assigned a probability).

A cdf F_c is called *continuous*⁶, if there exists a positive function f such that for all x

$$F_c(x) = \int_{-\infty}^x f(u)du. \quad (\text{D.2})$$

Note that such an F_c is differentiable (and hence continuous) with derivative f . The function f is called the *probability density function* (continuous pdf). By the fundamental theorem of integration, we have

$$\mathbb{P}[a < X \leq b] = F(b) - F(a) = \int_a^b f(x)dx.$$

Thus, calculating probabilities reduces to integration, as illustrated in ??.

⁶In advanced probability, we would say “*absolutely continuous with respect to the Lebesgue measure*”.

Table 1: Commonly used discrete distributions.

Name	Notation	$f(x)$	$x \in$	Parameters
Bernoulli	$\text{Ber}(p)$	$p^x(1-p)^{1-x}$	$\{0, 1\}$	$0 \leq p \leq 1$
Binomial	$\text{Bin}(n, p)$	$\binom{n}{x} p^x (1-p)^{n-x}$	$\{0, 1, \dots, n\}$	$0 \leq p \leq 1, n \in \mathbb{N}$
Discrete uniform	$\mathcal{U}\{1, \dots, n\}$	$\frac{1}{n}$	$\{1, \dots, n\}$	$n \in \{1, 2, \dots\}$
Geometric	$\text{Geom}(p)$	$p(1-p)^{x-1}$	$\{1, 2, \dots\}$	$0 \leq p \leq 1$
Poisson	$\text{Poi}(\lambda)$	$e^{-\lambda} \frac{\lambda^x}{x!}$	$\mathbb{N}$	$\lambda > 0$

Table 2: Commonly used continuous distributions.

Name	Notation	$f(x)$	$x \in$	Parameters
Uniform	$\mathcal{U}[\alpha, \beta]$	$\frac{1}{\beta - \alpha}$	$[\alpha, \beta]$	$\alpha < \beta$
Normal	$\mathcal{N}(\mu, \sigma^2)$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$	$\mathbb{R}$	$\sigma > 0, \mu \in \mathbb{R}$
Gamma	$\text{Gamma}(\alpha, \lambda)$	$\frac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}$	$\mathbb{R}_+$	$\alpha, \lambda > 0$
Inverse Gamma	$\text{InvGamma}(\alpha, \lambda)$	$\frac{\lambda^\alpha x^{-\alpha-1} e^{-\lambda x^{-1}}}{\Gamma(\alpha)}$	$\mathbb{R}_+$	$\alpha, \lambda > 0$
Exponential	$\text{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$\mathbb{R}_+$	$\lambda > 0$
Beta	$\text{Beta}(\alpha, \beta)$	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$	$[0, 1]$	$\alpha, \beta > 0$
Weibull	$\text{Weib}(\alpha, \lambda)$	$\alpha \lambda (\lambda x)^{\alpha-1} e^{-(\lambda x)^\alpha}$	$\mathbb{R}_+$	$\alpha, \lambda > 0$
Pareto	$\text{Pareto}(\alpha, \lambda)$	$\alpha \lambda (1 + \lambda x)^{-(\alpha+1)}$	$\mathbb{R}_+$	$\alpha, \lambda > 0$
Student	t_ν	$\frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi}\Gamma(\frac{\nu}{2})} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2}$	$\mathbb{R}$	$\nu > 0$
F	$F(m, n)$	$\frac{\Gamma(\frac{m+n}{2})(m/n)^{m/2} x^{(m-2)/2}}{\Gamma(\frac{m}{2})\Gamma(\frac{n}{2})[1+(m/n)x]^{(m+n)/2}}$	$\mathbb{R}_+$	$m, n \in \mathbb{N}_+$

D.3 Expectation

It is often useful to consider different kinds of numerical characteristics of a random variable. One such quantity is the expectation, which measures the “average” value of the distribution.

The *expectation* (or expected value or mean) of a random variable X with pdf f , denoted by $\mathbb{E}X$ or⁷ $\mathbb{E}[X]$ (and sometimes μ), is defined by

$$\mathbb{E}X = \begin{cases} \sum_x x f(x) & \text{discrete case,} \\ \int_{-\infty}^{\infty} x f(x) dx & \text{continuous case.} \end{cases}$$

If X is a random variable, then a function of X , such as X^2 or $\sin(X)$, is again a random variable. Moreover, the expected value of a function of X is simply a weighted average of the possible values that this function can take. That is, for any real function h

$$\mathbb{E}h(X) = \begin{cases} \sum_x h(x) f(x) & \text{discrete case,} \\ \int_{-\infty}^{\infty} h(x) f(x) dx & \text{continuous case,} \end{cases}$$

provided that the sum or integral are well-defined.

The *variance* of a random variable X , denoted by $\text{Var } X$ (and sometimes σ^2), is defined by

$$\text{Var } X = \mathbb{E}(X - \mathbb{E}[X])^2 = \mathbb{E}X^2 - (\mathbb{E}X)^2.$$

The square root of the variance is called the *standard deviation*. [table 3](#) lists the expectations and variances for some well-known distributions. Both variance and standard deviation measure the spread or dispersion of the distribution. Note, however, that the standard deviation measures the dispersion in the same units as the random variable, unlike the variance, which uses squared units.

⁷We only use brackets in an expectation if it is unclear with respect to which random variable the expectation is taken.

Table 3: Expectations and variances for some well-known distributions.

Dist.	$\mathbb{E}X$	$\text{Var } X$	Dist.	$\mathbb{E}X$	$\text{Var } X$
Bin(n, p)	np	$np(1 - p)$	Gamma(α, λ)	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$
Geom(p)	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\mathcal{N}(\mu, \sigma^2)$	μ	σ^2
Poi(λ)	λ	λ	Beta(α, β)	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(1+\alpha+\beta)}$
$\mathcal{U}[\alpha, \beta]$	$\frac{\alpha+\beta}{2}$	$\frac{(\beta-\alpha)^2}{12}$	Weib(α, λ)	$\frac{\Gamma(1/\alpha)}{\alpha\lambda}$	$\frac{2\Gamma(2/\alpha)}{\alpha\lambda^2} - \left(\frac{\Gamma(1/\alpha)}{\alpha\lambda}\right)^2$
Exp(λ)	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	F(m, n)	$\frac{n}{n-2} \ (n > 2)$	$\frac{2n^2(m+n-2)}{m(n-2)^2(n-4)} \ (n > 4)$
t_ν	$0 \ (\nu > 1)$	$\frac{\nu}{\nu-2} \ (\nu > 2)$			

It is sometimes useful to consider the *moment generating function* of a random variable X . This is the function M defined by

$$M(s) = \mathbb{E}e^{sX}, \quad s \in \mathbb{R}. \quad (\text{D.3})$$

The moment generating functions of two random variables coincide if and only if the random variables have the same distribution

D.4 Joint Distributions

Distributions for random vectors and stochastic processes can be specified in much the same way as for random variables. In particular, the distribution of a random vector $\mathbf{X} = [X_1, \dots, X_n]^\top$ is completely determined by specifying the *joint cdf* F , defined by

$$F(x_1, \dots, x_n) = \mathbb{P}[X_1 \leq x_1, \dots, X_n \leq x_n], \quad x_i \in \mathbb{R}, i = 1, \dots, n.$$

Similarly, the distribution of a *stochastic process*, that is, a collection of random variables $\{X_t, t \in \mathcal{T}\}$, for some index set $\mathcal{T}$, is completely determined by its finite-dimensional distributions; specifically, the distributions of the random vectors $[X_{t_1}, \dots, X_{t_n}]^\top$ for every choice of n and $t_1, \dots, t_n$.

By analogy to the one-dimensional case, a random vector $\mathbf{X} = [X_1, \dots, X_n]^\top$ taking values in $\mathbb{R}^n$ is said to have a pdf f if, in the continuous case,

$$\mathbb{P}[\mathbf{X} \in B] = \int_B f(\mathbf{x}) d\mathbf{x}, \tag{D.4}$$

for all n -dimensional rectangles B . Replace the integral with a sum for the discrete case. The pdf is also called the *joint pdf* of $X_1, \dots, X_n$. The pdfs of the individual components, called *marginal pdfs*, can be recovered from the joint pdf by “integrating out the other variables”. For example, for a continuous random vector $[X, Y]^\top$ with pdf f , the pdf f_X of X is given by

$$f_X(x) = \int f(x, y) dy.$$

D.5 Conditioning and Independence

Conditional probabilities and conditional distributions are used to model additional information on a random experiment. Independence is used to model *lack* of such information.

D.5.1 Conditional Probability

Suppose some event $B \subseteq \Omega$ occurs. Given this fact, event A will occur if and only if $A \cap B$ occurs, and the relative chance of A occurring is therefore $\mathbb{P}[A \cap B]/\mathbb{P}[B]$, provided $\mathbb{P}[B] > 0$. This leads to the definition of the *conditional probability* of A given B :

$$\mathbb{P}[A \mid B] = \frac{\mathbb{P}[A \cap B]}{\mathbb{P}[B]}, \quad \text{if } \mathbb{P}[B] > 0. \quad (\text{D.5})$$

The above definition breaks down if $\mathbb{P}[B] = 0$. Such conditional probabilities must be treated with more care [11]. Three important consequences of the definition of conditional probability are:

1. *Product rule:* For any sequence of events $A_1, A_2, \dots, A_n$,

$$\mathbb{P}[A_1 \cdots A_n] = \mathbb{P}[A_1] \mathbb{P}[A_2 \mid A_1] \mathbb{P}[A_3 \mid A_1 A_2] \cdots \mathbb{P}[A_n \mid A_1 \cdots A_{n-1}], \quad (\text{D.6})$$

using the abbreviation $A_1 A_2 \cdots A_k := A_1 \cap A_2 \cap \cdots \cap A_k$.

2. *Law of total probability:* If $\{B_i\}$ forms a partition of Ω (that is, $B_i \cap B_j = \emptyset, i \neq j$ and $\cup_i B_i = \Omega$), then for any event A

$$\mathbb{P}[A] = \sum_i \mathbb{P}[A \mid B_i] \mathbb{P}[B_i]. \quad (\text{D.7})$$

3. *Bayes' rule:* Let $\{B_i\}$ form a partition of Ω . Then, for any event A with $\mathbb{P}[A] > 0$,

$$\mathbb{P}[B_j \mid A] = \frac{\mathbb{P}[A \mid B_j] \mathbb{P}[B_j]}{\sum_i \mathbb{P}[A \mid B_i] \mathbb{P}[B_i]}. \quad (\text{D.8})$$

D.5.2 Independence

Two events A and B are said to be *independent* if the knowledge that B has occurred does not change the probability that A occurs. That is, A, B independent $\iff \mathbb{P}[A \mid B] = \mathbb{P}[A]$. Since

$\mathbb{P}[A | B]\mathbb{P}[B] = \mathbb{P}[A \cap B]$, an alternative definition of independence is

$$A, B \text{ independent} \iff \mathbb{P}[A \cap B] = \mathbb{P}[A]\mathbb{P}[B].$$

This definition covers the case where $\mathbb{P}[B] = 0$ and can be extended to arbitrarily many events: events $A_1, A_2, \dots$ are said to be (mutually) independent if for any k and any choice of distinct indices $i_1, \dots, i_k$,

$$\mathbb{P}[A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}] = \mathbb{P}[A_{i_1}]\mathbb{P}[A_{i_2}] \cdots \mathbb{P}[A_{i_k}].$$

The concept of independence can also be formulated for random variables. Random variables $X_1, X_2, \dots$ are said to be *independent* if the events $\{X_{i_1} \leq x_{i_1}\}, \dots, \{X_{i_n} \leq x_{i_n}\}$ are independent for all finite choices of n distinct indices $i_1, \dots, i_n$ and values $x_{i_1}, \dots, x_{i_n}$.

An important characterization of independent random variables is the following.

Theorem D.1: Independence Characterization

Random variables $X_1, \dots, X_n$ with marginal pdfs $f_{X_1}, \dots, f_{X_n}$ and joint pdf f are independent if and only if

$$f(x_1, \dots, x_n) = f_{X_1}(x_1) \cdots f_{X_n}(x_n) \quad \text{for all } x_1, \dots, x_n. \quad (\text{D.9})$$

Many probabilistic models involve random variables $X_1, X_2, \dots$ that are *independent and identically distributed*, abbreviated as iid .

D.5.3 Expectation and Covariance

Similar to the univariate case, the expected value of a real-valued function h of a random vector $\mathbf{X} \sim f$ is a weighted average of all values that $h(\mathbf{X})$ can take. Specifically, in the continuous case, $\mathbb{E}h(\mathbf{X}) = \int h(\mathbf{x})f(\mathbf{x})d\mathbf{x}$. In the discrete case replace this multidimensional integral with a sum. Using this result, it is not difficult to show that for any collection of dependent or independent

random variables $X_1, \dots, X_n$,

$$\mathbb{E}[a + b_1X_1 + b_2X_2 + \dots + b_nX_n] = a + b_1\mathbb{E}X_1 + \dots + b_n\mathbb{E}X_n \quad (\text{D.10})$$

for all constants $a, b_1, \dots, b_n$. Moreover, for *independent* random variables,

$$\mathbb{E}[X_1X_2 \dots X_n] = \mathbb{E}X_1\mathbb{E}X_2 \dots \mathbb{E}X_n. \quad (\text{D.11})$$

We leave the proofs as an exercise.

The *covariance* of two random variables X and Y with expectations μ_X and μ_Y , respectively, is defined as

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]. \quad (\text{D.12})$$

This measures the degree of linear dependency between the variables. Let $\sigma_X^2 = \text{Var } X$ and $\sigma_Y^2 = \text{Var } Y$. A scaled version of the covariance is given by the *correlation coefficient*,

$$\varrho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}. \quad (\text{D.13})$$

The following properties follow directly from the definitions of variance and covariance.

1. $\text{Var } X = \mathbb{E}X^2 - \mu_X^2$.
2. $\text{Var}[aX + b] = a^2\sigma_X^2$.
3. $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mu_X\mu_Y$.
4. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$.
5. $-\sigma_X\sigma_Y \leq \text{Cov}(X, Y) \leq \sigma_X\sigma_Y$.
6. $\text{Cov}(aX + bY, Z) = a\text{Cov}(X, Z) + b\text{Cov}(Y, Z)$.
7. $\text{Cov}(X, X) = \sigma_X^2$.

$$8. \text{Var}[X + Y] = \sigma_X^2 + \sigma_Y^2 + 2 \text{Cov}(X, Y).$$

$$9. \text{ If } X \text{ and } Y \text{ are independent, then } \text{Cov}(X, Y) = 0.$$

As a consequence of Properties 2 and 8 we have that for any sequence of *independent* random variables $X_1, \dots, X_n$ with variances $\sigma_1^2, \dots, \sigma_n^2$,

$$\text{Var}[a_1X_1 + a_2X_2 + \dots + a_nX_n] = a_1^2\sigma_1^2 + a_2^2\sigma_2^2 + \dots + a_n^2\sigma_n^2, \quad (\text{D.14})$$

for any choice of constants $a_1, \dots, a_n$.

For random column vectors, such as $\mathbf{X} = [X_1, \dots, X_n]^\top$, it is convenient to write the expectations and covariances in vector and matrix notation. For a random vector $\mathbf{X}$ we define its *expectation vector* as the vector of expectations

$$\boldsymbol{\mu} = [\mu_1, \dots, \mu_n]^\top = [\mathbb{E}X_1, \dots, \mathbb{E}X_n]^\top. \quad (\text{D.15})$$

The *covariance matrix* Σ is defined as the matrix whose (i, j) -th element is

$$\text{Cov}(X_i, X_j) = \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)]. \quad (\text{D.16})$$

If we define the expectation of a vector (matrix) to be the vector (matrix) of expectations, then we can compactly write

$$\boldsymbol{\mu} = \mathbb{E}\mathbf{X} \quad (\text{D.17})$$

and

$$\Sigma = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top]. \quad (\text{D.18})$$

A useful application of the cyclic property of the trace of a matrix (see [B.1](#)) is the following.

Theorem D.2: Expectation of a Quadratic Form

Let A be an $n \times n$ matrix and X an n -dimensional random vector with expectation vector μ and covariance matrix Σ . The random variable $Y := X^\top A X$ has expectation $\text{tr}(A\Sigma) + \mu^\top A \mu$.

Proof. Since Y is a scalar, it is equal to its trace. Now, using the cyclic property: $\mathbb{E}Y = \mathbb{E}\text{tr}(Y) = \mathbb{E}\text{tr}(X^\top A X) = \mathbb{E}\text{tr}(A X X^\top) = \text{tr}(A \mathbb{E}[X X^\top]) = \text{tr}(A(\Sigma + \mu \mu^\top)) = \text{tr}(A\Sigma) + \text{tr}(A \mu \mu^\top) = \text{tr}(A\Sigma) + \mu^\top A \mu. \quad \square$

D.6 Markov Chains

The index t in X_t is usually seen as a “time” or “step” parameter. The index set $\{0, 1, 2, \dots\}$ in the definition above was chosen out of convenience. It can be replaced by any countable index set. We restrict ourselves to *time-homogeneous Markov chains*, Markov chains for which the conditional pdfs $f_{X_{t+1}|X_t}(y|x)$ do not depend on t ; we abbreviate these as $q(y|x)$. The $\{q(y|x)\}$ are called the (one-step) *transition densities* of the Markov chain. Note that the random variables or vectors $\{X_t\}$ may be *discrete* (e.g., taking values in some set $\{1, \dots, r\}$) or *continuous* (e.g., taking values in an interval $[0, 1]$ or $\mathbb{R}^d$). In particular, in the *discrete* case, each $q(y|x)$ is a probability: $q(y|x) = \mathbb{P}[X_{t+1} = y | X_t = x]$.

Definition D.1: Markov Chain

A *Markov chain* is a collection $\{X_t, t = 0, 1, 2, \dots\}$ of random variables (or random vectors) whose futures are conditionally independent of their pasts given their present values. That is,

$$\mathbb{P}[X_{t+1} \in A | X_s, s \leq t] = \mathbb{P}[X_{t+1} \in A | X_t] \quad \text{for all } t. \quad (\text{D.19})$$

In other words, the conditional distribution of the future variable X_{t+1} , given the entire past $\{X_s, s \leq t\}$, is the same as the conditional distribution of X_{t+1} given only the present X_t . eq. (D.19) is called the *Markov property*.

The distribution of X_0 is called the *initial distribution* of the Markov chain. The one-step transition densities and the initial distribution completely specify the distribution of the random vector $[X_0, X_1, \dots, X_t]^\top$. Namely, we have by the product rule [eq. \(18.1\)](#) and the Markov property that the joint pdf is given by

$$\begin{aligned} f_{X_0, \dots, X_t}(x_0, \dots, x_t) &= f_{X_0}(x_0) f_{X_1|X_0}(x_1|x_0) \cdots f_{X_t|X_{t-1}, \dots, X_0}(x_t|x_{t-1}, \dots, x_0) \\ &= f_{X_0}(x_0) f_{X_1|X_0}(x_1|x_0) \cdots f_{X_t|X_{t-1}}(x_t|x_{t-1}) \\ &= f_{X_0}(x_0) q(x_1|x_0) q(x_2|x_1) \cdots q(x_t|x_{t-1}). \end{aligned}$$

A Markov chain is said to be *ergodic* if the probability distribution of X_t converges to a fixed distribution as $t \rightarrow \infty$. Ergodicity is a property of many Markov chains. Intuitively, the probability of encountering the Markov chain in a state x at a time t far into the future should not depend on the t , provided that the Markov chain can reach every state from any other state, such Markov chains are said to be *irreducible*, and does not “escape” to infinity. Thus, for an ergodic Markov chain the pdf $f_{X_t}(x)$ converges to a fixed *limiting pdf* $f(x)$ as $t \rightarrow \infty$, irrespective of the starting state. For the discrete case, $f(x)$ corresponds to the long-run fraction of times that the Markov process visits x .

Under mild conditions (such as irreducibility) the limiting pdf $f(x)$ can be found by solving the *global balance equations*:

$$f(x) = \begin{cases} \sum_y f(y) q(x|y) & \text{(discrete case),} \\ \int f(y) q(x|y) dy & \text{(continuous case).} \end{cases} \quad (\text{D.20})$$

For the discrete case the rationale behind this is as follows. Since $f(x)$ is the long-run proportion of time that the Markov chain spends in x , the proportion of transitions *out* of x is $f(x)$. This should be balanced with the proportion of transitions *into* state x , which is $\sum_y f(y) q(x|y)$.

One is often interested in a stronger type of balance equations. Imagine that we have taken a video of the evolution of the Markov chain, which we may run in forward and reverse time. If

we cannot determine whether the video is running forward or backward (we cannot determine any systematic “looping”, which would indicate in which direction time is flowing), the chain is said to be time-reversible or simply *reversible*.

Although not every Markov chain is reversible, each ergodic Markov chain, when run backwards, gives another Markov chain, the *reverse Markov chain*, with transition densities $\tilde{q}(y|x) = f(y)q(x|y)/f(x)$. To see this, first observe that $f(x)$ is the long-run proportion of time spent in x for both the original and reverse Markov chain. Secondly, the “probability flux” from x to y in the reversed chain must be equal to the probability flux from y to x in the original chain, meaning $f(x)\tilde{q}(y|x) = f(y)q(x|y)$, which yields the stated transition probabilities for the reversed chain. In particular, for a *reversible* Markov chain we have

$$f(x)q(y|x) = f(y)q(x|y) \quad \text{for all } x, y. \quad (\text{D.21})$$

These are the *detailed* (or *local*) *balance equations*. Note that the detailed balance equations imply the global balance equations. Hence, if a Markov chain is irreducible and there exists a pdf such that [eq. \(D.21\)](#) holds, then $f(x)$ must be the limiting pdf. In the discrete state space case an additional condition is that the chain must be *aperiodic*, meaning that the return times to the same state cannot always be a multiple of some integer ≥ 2 .

E Python Primer

P^{ROVIDED}